Two hundred seventy conjectures in mathematics

Fourteen parts across arithmetic, combinatorics, geometry and probability

The Conjecture Engine*

27 July 2026

Summary sheet: the two hundred seventy conjectures

Part I: conjectures from the local–global random model, in importance–novelty order.

1. **Prime-power contamination calculus (Conjecture 1).** For a balanced pair race $(n, n+d) \bmod m$, the surviving prime-square orientations (q^2-d, q^2), (q^2, q^2+d) occupy provably computable residue classes, and the operator $\mathcal{C}_{d,m}(a;x) = \sum_o \sum_q \Lambda(q^2)\Lambda(\text{prime member})$ determines the drift vector: for twins, $\mathcal{M}_x(D_1(t)\log^2 t/\sqrt{t}) \to c_T = \frac{1}{2}C(x, x^2-2) \pmod 5$ and $\to 2c_T$ on class $7 \pmod 8$, symmetric differences $\to 0$; in general each class deficit satisfies $\mathcal{M}_x(\text{deficit}\cdot\log^2 t/\sqrt{t}) \to \lim \mathcal{C}_{d,m}(a;x)/\sqrt{x}$. Convergence of these logarithmic means at drift scale is part of the conjecture (Rubinstein–Sarnak-type structure for pattern races).

2. **Pinned singular-series variance (Conjecture 2).** With $R(h) = C_4(0,2,h,h+2)/\mathfrak{S}(2)^2$ and $G(H) = \sum_{h\le H}(1-h/H)(R(h)-1)$: (i) [conditional proposition] the moving-window twin-pair count obeys $\operatorname{Var}/\mathbb{E} = 1 + \mathfrak{S}(2)(2G(H)-1)/\log^2 x + o(\log^{-2} x)$; (ii) $-G(H)/\log H \to \infty$: pair counts are more sub-Poisson than prime counts.

3. **Least-prime ordering deficit (Conjecture 3).** With $U(a,q) = \mathrm{Li}(p(a,q))/\varphi(q)$: (i) $U \Rightarrow \mathrm{Exp}(1)$ marginally (attributed), Gumbel maximum under joint independence; (ii) the dyadic averages of $\Theta(q) = (1-\mathbb{E}_a[U])\log q$ over prime $q \in [Q, 2Q]$ converge to a limit $\Theta > 0$: the ordered primes fill residue classes faster than any exchangeable model, in excess of the derived discreteness and injectivity baselines.

4. **Goldbach lane race (Conjecture 4).** For $n \equiv 2 \pmod 4$, prime squares enter ordered Goldbach representations only through the $(1,1)$ lane mod 4, so with $D = R_3 - R_1$ and the explicit weighted census D_{sys}: $\mathbb{E}_x[D - D_{\text{sys}}] = o(\mathbb{E}_x[D_{\text{sys}}])$ under logarithmic sampling; the drift vanishes on the internal null subprogression $n \equiv 10 \pmod{12}$; the sign density of D tends to $\frac{1}{2}$.

*This paper was produced by an automated conjecture engine (LLM tools), operated as an exercise in disciplined conjecture-making. All constants, counts, and verification results are reproducible from the accompanying repository (`conjecture-engine`). `run_all.py` regenerates Part I, while the per-part scripts in `verify/` regenerate the later computations. Primality claims are stratified between deterministic verification and Baillie–PSW probable primes. The exact certification boundary and the affected counts are recorded in the computational-check standards section.

5. **Least Goldbach summand (Conjecture 5).** With $s(n)$ the least prime $p \nmid n$ with $n-p$ prime, $U(n) = \mathfrak{S}(n) \sum_{p \le s(n),\, p \nmid n} 1/\log(n-p)$, and the dyadic-log ensemble $\mathbb{E}_X$: (i) $U \Rightarrow \mathrm{Exp}(1)$; (ii) $\Theta_G(X) = (1 - \mathbb{E}_X U) \log X \to \Theta_G > 0$, the Goldbach sibling of the ordering deficit.

6. **Polynomial–exponential entanglement (Conjecture 6).** Primes $n^2 + 2^n$ force $n \equiv 3 \pmod 6$; with $\kappa_S = 3\, D_S / \prod_{p \in S}(1 - 1/p)$ built from the CRT-exact survivor density D_S over the joint period of S and the lane bonus 3 at the primes 2 and 3: κ_{S_z} converges along the canonical exhaustion $S_z = \{5 \le p \le z\}$ to κ, the counting function is $\sim \sum_{n \equiv 3(6),\, n \le N} \kappa / \log(n^2 + 2^n) \asymp \log N$; whether the limit factors exactly (as observed through $p \le 19$) is an open question.

7. **Multibase Fermat quotients (Conjecture 7).** For $p \nmid a$: the Eisenstein–Lerch homomorphism confines multiplicatively dependent bases to a rational subtorus, and (iv) for multiplicatively independent bases $(q_p(a_1), \dots, q_p(a_r))/p$ equidistributes on the full torus as p varies; (v) at most finitely many simultaneous Wieferich primes; plus single-base clauses: KS-discrepancy LIL at constant $1/\sqrt{2}$, shrinking-target law $\#\{p \le x : q_p < K\} \sim \sum \min(1, K/p)$, Wieferich-count LIL and an almost-sure CLT at the du/u weighting.

8. **Uniform de Polignac covariance field (Conjecture 8).** $\pi_d(x) = \mathfrak{S}(d)\, \mathrm{Li}_2(x)(1 + o(1))$ uniformly for even $d \le (\log x)^A$; the moving-window residual field has covariance $K(d, d')H/\log^3 x + (1/\log^4 x) \sum_{|h| \le H} (H - |h|) K_4(d, d'; h)$, triple term dominant on the polylogarithmic window class, with Poisson/Gaussian regimes as finite-x corrections to independence.

9. **Race sub-diffusivity (Conjecture 9).** Balanced-race steps at event index have negative autocorrelations obeying $\rho_k(x) = -(c_k \log\log x + d_k)/\log x\,(1 + o(1))$; under uniformity in k, tail summability, and $\sum c_k, \sum |d_k| < \infty$, the diffusivity is $\sigma^2(x) = 1 - 2(\sum c_k \log\log x + \sum d_k)/\log x\,(1 + o(1))$, those hypotheses being model-layer, not asserted asymptotically; the asymptotic diffusive-vs-rigid dichotomy is registered open, with the rigid branch selected only at Conjecture 1(i-b) for its contaminated races.

10. **Uniform quadratic de Polignac and its constants (Conjecture 10).** $\#\{n \le N : n^2 + 1,\ n^2 + 1 + d \text{ prime}\} = C(d) I_d(N)(1 + o(1))$ uniformly for even $d \le (\log N)^B$; a separate power-saving rate clause; and the family law: moments of $C(d)$ converge to derived Euler products (mean $2.7456\ldots$, sd $1.6840\ldots$), the empirical law converging to the random product with CRT-independent local factors.

11. **First occurrences of prime gaps (Conjecture 11).** On realized gaps: $\log p(g)/\sqrt{g}$ bounded between positive constants; $\liminf = \sqrt{e^{\gamma}/2}$ conditional on the realization hypothesis; and $\log p(g) = \sqrt{g} + \frac{1}{2}\log g - \frac{1}{2}\log \mathfrak{S}^*(g) + E_g$ with $\Pr[E_g \le t] \to 1 - \exp(-e^{2(t-\mu)})$, the min-type Gumbel at scale $\frac{1}{2}$ forced by the slope-2 hazard expansion.

12. **Stern lane race (Conjecture 12).** In $n = p + 2k^2$, square contamination $n = q^2 + 2k^2$ sits in the k-even lane iff $n \equiv 1 \pmod 8$, k-odd iff $n \equiv 3$, and is impossible for $n \equiv 5, 7$ (provable null classes); on contaminated classes the clean-minus-contaminated difference obeys $\mathbb{E}_X[D - D_{\mathrm{sys}}] = o(\mathbb{E}_X[D_{\mathrm{sys}}])$ with the $\Lambda(q^2)$-weighted census D_{sys}.

13. **Microscopic variance law (Conjecture 13).** [Conditional on quantitative Hardy–Littlewood] For windows of length $H = \lambda(\log x)^{\alpha}$, $\mathrm{Var}\, X/\mathbb{E}X = 1 - (\log H + \gamma + \log 2\pi -$

$1)/\log x + o(1/\log x)$ uniformly in $1 \le \alpha \le A$, with Gallagher's Poisson law at the boundary $\alpha = 1$.

14. **Null-mechanism race (Conjecture 14).** For (n^2+1, n^2+3), square contamination is algebraically impossible (single bounded exception $(n,q) = (1,2)$); the race mod 5 is driftless at the contamination scale, $\mathcal{M}_x(D(t)\log^2 t/\sqrt{t}) \to 0$—the negative control for item 1—and the logarithmic occupation of leadership tends to $\frac{1}{2}$ with Gaussian fluctuations $\sqrt{\log 2/\log x}$, conjectured directly (the event-index invariance principle is a separately falsifiable local hypothesis, which does not imply it).

15. **Cubic-shift constants (Conjecture 15).** For non-cube a, the three-state local law gives $\mathbb{E}_a[\omega(p)] = 1$ exactly, so the limit law of $C(a)$ has mean exactly 1 (an L^2-martingale argument), derived sd $0.2762\ldots$; counts are uniform over non-cube $a \le (\log N)^B$.

16. **Hardy–Littlewood's Conjecture F as a family (Conjecture 16).** $Q_A(N) = C(A)I_A(N)(1+o(1))$ uniformly over odd $A \le (\log N)^B$; same-index window covariance $[C(A,A') - C(A)C(A')]H/4\log^2 N$ (negative for locally exclusive pairs), with the off-index pinned sum at the same $1/4\log^2 N$ order left open.

17. **Twin-member Goldbach, orientation-resolved (Conjecture 17).** Under n-uniform Hardy–Littlewood for the four role systems, $R_T(n) = [\sum_o \mathfrak{S}_4^{(o)}(n)] \int_5^{n-5} dt/(\log^2 t \log^2(n - t))\,(1+o(1))$, with the exact identities $\mathfrak{S}_4^{(\mathrm{lu})} = \mathfrak{S}_4^{(\mathrm{ul})}$ and $\mathfrak{S}_4^{(\mathrm{uu})}(n) = \mathfrak{S}_4^{(\mathrm{ll})}(n-4)$.

18. **Triplet contamination (Conjecture 18).** For $(n, n+2, n+6)$ mod 5, the unique surviving square configuration is the doubly-thinned (q^2-2, q^2, q^2+4) (the other two configurations dying for every prime $q > 3$), contaminating class 2: $\mathcal{M}_x((\pi_3(t;5,1) - \pi_3(t;5,2))\log^3 t/\sqrt{t}) \to c_3$, the Bateman–Horn constant of (q, q^2-2, q^2+4).

19. **Sexy-pair contamination matrix (Conjecture 19).** For $(n, n+6)$ both orientations survive on complementary q-classes, feeding start classes 3 and 1 (mod 5 and mod 8) with independent constants c_A, c_B (the $(q, q^2 \mp 6)$ Bateman–Horn constants) in the drift-scale sense; class 2 (mod 5) and classes $5, 7$ (mod 8) are provably clean.

20. **Fibonacci–Lucas twins (Conjecture 20).** The odd prime-divisor pools of F_p and L_p are disjoint (ranks p vs $2p$; $\gcd = 2$ iff $3 \mid p$); only finitely many p have both prime; the naive joint accounting assigns prior mass about 1.4×10^{-2} to the index range beyond 10^4, in which the catalogued index 148091 falls, recorded descriptively and conditionally on the probable-prime status of both F_{148091} and L_{148091}, and fatal to any completeness list.

21. **Factorial twins (Conjecture 21).** Window rigidity: $n! + a$ is composite for $2 \le |a| \le n$ (theorem), so ± 1 are the only bounded offsets; $n = 3$ uniqueness is attributed; the joint-fluctuation clause is a declared-model CLT for $F_+ - F_-$ under explicit covariance and higher-cumulant hypotheses.

22. **Boundary trichotomy (Conjecture 22).** For $\deg F \ge 2$ with positive leading coefficient, $F(m) - F(j)$ prime forces $m - j = 1$ outside a bounded region; for $x^3 + cx$ the boundary lane $3m^2 - 3m + 1 + c$ is dead-parity for odd c, dead-3-adic for even $c \equiv 2 \pmod 3$, admissible exactly for $c \equiv 0, 4 \pmod 6$, with uniform Bateman–Horn counts over admissible $c \le (\log M)^B$.

23. **Power-obstruction ladder (Conjecture 23).** $m^k = p + j^k$ is impossible for composite k (theorem); for prime k it forces $j = m - 1$ and reduces to primality of the irreducible $D_k(m) = m^k - (m-1)^k$, which follows Bateman–Horn uniformly over the prime-k lanes.

24. **Alternating cyclotomic chain (Conjecture 24).** Infinitely many primes p with $\Phi_3(p)$ and $\Phi_6(\Phi_3(p))$ prime, at the derived Bateman–Horn rate; alternation is the unique admissible continuation in the anchored construction space.

25. **Twin cyclotomic bases (Conjecture 25).** Of the $\varphi(k) = 2$ family, only $k = 3$ survives ($k = 4$ parity-dead, $k = 6$ a translate), and $\Phi_3(n)$, $\Phi_3(n+1)$ are simultaneously prime infinitely often with the computed Bateman–Horn constant.

Part II: structural conjectures, in programme order.

26. **Connected motif generating functional (Conjecture 26).** A full connected diagram calculus for all joint cumulants of prime motifs.

27. **Complete overlap-renormalization filtration (Conjecture 27).** A classification of every possible covariance scale and rigid eigenspace in a finite motif family.

28. **Regularized mesoscopic local–spectral trace formula (Conjecture 28).** Independently mollified Euler-product and multiple-zero functionals agree after named counterterms, and the limit is regularization independent.

29. **Anchored arithmetic polymer expansion (Conjecture 29).** Palm cluster activities are graded by the number of new prime constraints beyond one anchored motif.

30. **Topological expansion of non-Gaussianity (Conjecture 30).** Each independent cycle in a connected diagram's overlap graph costs one logarithmic order, so the leading non-Gaussian odd-cumulant term is carried by the tree-like diagrams.

31. **Connected first-arrival functional (Conjecture 31).** The entire least-prime point process has connected kernels obtained from same-class prime correlations.

32. **Tested Gauss-polyspectral reciprocity (Conjecture 32).** Scalar tensor observables have exact Gauss-transport identities under the random-modulus law and dual arithmetic limits.

33. **Rubinstein–Sarnak terminal chaos dichotomy (Conjecture 33).** Terminal extremes are directed by the limiting low-zero chaos measure, with an explicit self-averaging versus Cox criterion.

34. **Nonlinear spectral response calculus (Conjecture 34).** Low-zero and exceptional-zero perturbations pass through first-arrival statistics by universal Volterra response operators.

35. **Local-information capacity of odd class groups (Conjecture 35).** Cohen–Lenstra independence has a sharp two-resource boundary governed by cell entropy and conductor complexity.

36. **Kummer–Haar law on relation tori (Conjecture 36).** Saturated multiplicative relations give the exact horizontal support and Haar law of Fermat-quotient vectors.

37. **Rank-dimensional finite-logarithm large sieve (Conjecture 37).** A positive power range of frequencies obeys a large sieve whose dual dimension is saturated rank.

38. **Mesoscopic-to-lattice shrinking-target transition (Conjecture 38).** Haar universality persists for growing targets, while bounded lattice targets acquire a separate arithmetic regulator intensity.

39. **Global-lattice p-adic regulator-matrix law (Conjecture 39).** A fixed integral Galois-relation lattice controls horizontal regulator matrices and their determinantal rare events.

40. **Functorial horizontal logarithms on tori (Conjecture 40).** A finitely generated global subgroup of a torus has a functorial matrix-valued finite-logarithm Haar law.

41. **Entropy–conductor profile of arboreal resolution (Conjecture 41).** Exact class-measure entropy and Artin-conductor profiles determine observable depth and finite-index shifts.

42. **Arboreal family large sieve and cumulant independence (Conjecture 42).** Growing preimage quotients satisfy square-root trace bounds and higher connected Frobenius factorization over parameters.

43. **Coloured dynatomic Galois-factor process (Conjecture 43).** Exact-period components generate independent coloured, Frobenius-marked factor processes, with growing-degree support restrictions.

44. **Complexity-uniform primitive valuation process (Conjecture 44).** The complete primitive valuation vector has an adelic product law under an explicit height–discriminant complexity regime and uniform squarefull tails.

45. **Divisor-sensitive dynamical gcd classification (Conjecture 45).** Positive normalized gcd height is exactly eventual entry into a preperiodic subvariety carrying the relevant common divisor.

46. **Frobenius-marked Poisson–Dirichlet process (Conjecture 46).** Macroscopic polynomial-value factors carry fixed-point-size-biased Galois marks.

47. **Three-scale polynomial factorization (Conjecture 47).** Small valuations, mesoscopic scale-invariant factors, and macroscopic marked factors form one conservation-corrected product structure.

48. **Adelic Gibbs gluing for reducible values (Conjecture 48).** Reducible polynomial factor processes are an archimedean–p-adic Gibbs mixture of independent component processes.

49. **Full multivariate adelic saddle law (Conjecture 49).** Joint smoothness is governed by a complete joint Perron action, saddle displacement, and Hessian, not a scalar Euler correction.

50. **Buchstab–Bateman–Horn component flow (Conjecture 50).** Rough prime, semiprime, and higher-almost-prime polynomial values follow universal Buchstab components with Galois marks.

Part III: six conjectures across fields, in statement order.

51. **Relative polynomial pretentiousness inverse principle (Conjecture 51).** A persistently nonzero logarithmic correlation of completely multiplicative unimodular functions along multiplicatively independent irreducible polynomials forces one existential block of primitive characters and archimedean parameters with finite relative pretentious distances, the balance identity $\sum_j t_j \deg P_j = 0$, and a nonvanishing joint character average at a common modulus.

52. **Monotone threshold purification of vector paths (Conjecture 52).** Every monotone fractional activation schedule of m vectors of norm at most one in $\mathbb{R}^d$ admits a rounding to distinct irreversible activation times whose path error is $O(\sqrt{d})$ uniformly in m, containing the open Euclidean Steinitz prefix-sum problem and asserting removal of the logarithm from Banaszczyk's $O(\sqrt{d + \log m})$.

53. **Entropy dimension of random-free graphons (Conjecture 53).** If the row space of a $\{0, 1\}$-valued graphon is Ahlfors s-regular then $H(G(n, W))/(n \log n) \to s$, the hypothesis class being nonvacuous at every $s \in (0, 2]$ by Cantor-type threshold constructions.

54. **A Hessian principle for isolated microcanonical graph phases (Conjecture 54).** Under isolated-phase, transversally Lipschitz, and nonemptiness hypotheses, the tangent block averages of a microcanonical coloured graph ensemble fluctuate at scale n^{-1} around their mean with centred Gaussian limit of covariance A^{-1}, the inverse of the contracted entropy Hessian.

55. **Sharp stability of entropy idempotence on finite abelian groups (Conjecture 55).** The squared total variation distance from a probability measure to the nearest uniform coset measure is at most $2/\log 2$ times the convolution entropy defect $H(\mu * \mu) - H(\mu)$, sharply, the constant pinned by discretized Gaussians on $\mathbb{Z}/p\mathbb{Z}$; some universal constant already follows from the Green–Manners–Tao entropic inverse theorem, so the sharp constant and the extremal geometry are the content.

56. **Stationary convolution-entropy rigidity (Conjecture 56).** An ergodic stationary process over a finite abelian group preserves entropy under independent self-convolution exactly when its quotient by the translational stabilizer has zero entropy, so in particular every ergodic binary process of intermediate entropy gains entropy under independent XOR.

Part IV: twenty conjectures on the class-counting polynomial, in statement order.

57. **Irreducibility and connectivity (Conjecture 57).** $H_G(U, Y) = C_G(1 + U, Y)$ is irreducible in $\mathbb{Z}[U, Y]$ exactly when G is connected, the disconnected direction being immediate from Rossmann's multiplicativity and the connected direction verified by exact factorization for all 996 connected graphs on at most seven vertices.

58. **Recovery of the subgraph-component polynomial (Conjecture 58).** Equality of class-counting polynomials forces equality of Tittmann–Averbouch–Makowsky subgraph-component polynomials, though the two arrays, $(|S|, |N[S]| - c(G[S]))$ against $(|S|, c(G[S]))$, admit no apparent functional relation.

59. **Two-field rigidity (Conjecture 59).** The adjoint-rank distributions of the graphical Lie algebras over $\mathbb{F}_2$ and $\mathbb{F}_3$ together determine them over every finite field.

60. **Hybrid rigidity (Conjecture 60).** The $\mathbb{F}_2$ rank distribution together with the subgraph-component polynomial determines the class-counting polynomial, while the degree sequence in place of Q_G does not suffice.

61. **Binary edge-deck reconstruction (Conjecture 61).** The multiset of binary specializations $C_{G-e}(2, Y)$ over the edges determines a graph with at least four edges up to isomorphism, a statement in the orbit of the edge reconstruction conjecture and the high-risk member of the part.

62. **Bipartite binary rigidity (Conjecture 62).** Within bipartite graphs the single specialization $C_G(2, Y)$ determines C_G.

63. **Connected-pseudoforest binary rigidity (Conjecture 63, resolved false).** The proposed one-field rigidity for trees and connected unicyclic graphs is refuted by an explicit seven-vertex unicyclic pair sharing the binary specialization with distinct degree multisets, the unique violating class through nine vertices, and the tree case survives exhaustively through fourteen vertices.

64. **Chordal binary rigidity (Conjecture 64).** The same one-field rigidity holds within chordal graphs.

65. **Recovery of the domination number (Conjecture 65).** Equality of class-counting polynomials forces equality of domination numbers, a statement strictly between Rossmann's connected-domination theorem and the false total-domination analogue.

66. **Recovery of the independent domination number (Conjecture 66).** Equality of class-counting polynomials forces equality of independent domination numbers.

67. **Global double log-concavity (Conjecture 67).** For every graph the coefficient sequence a of $f_G(1 + Z)$ is log-concave and remains log-concave after one application of $L(a)_i = a_i^2 - a_{i-1}a_{i+1}$, the third iterate of L failing for a thirteen-vertex graph.

68. **Strict log-concavity for connected graphs (Conjecture 68).** For connected graphs with at least one edge the inequality $a_i^2 > a_{i-1}a_{i+1}$ is strict at every interior position of the support.

69. **Strict second-order log-concavity (Conjecture 69).** Under the same hypotheses $L(a)$ is strictly log-concave at every interior position of its support.

70. **Rank-slice strict log-concavity (Conjecture 70).** For connected graphs every non-monomial rank slice $[Y^r]F_G(1 + Z, Y)$ has strictly log-concave coefficients at interior positions, the second-order strengthening of the slice statement failing for an eight-vertex graph.

71. **Interval support at fixed rank (Conjecture 71, resolved false).** The proposed gaplessness is refuted by a connected eight-vertex graph whose support sizes attaining rank five skip size two, the unique violating order, while the transposed statement was already false.

72. **Coefficientwise tree extremality (Conjecture 72).** Among n-vertex trees the shifted polynomial $f_T(1+Z)$ is sandwiched coefficientwise between path and star, with equality only at the extremes.

73. **Star rank-majorization (Conjecture 73).** For $n \geq 5$ and every prime power q the decreasingly ordered adjoint-rank distribution of the star majorizes that of every n-vertex tree, the hypothesis being sharp since the star fails to majorize the path at $(n,q) = (4,2)$.

74. **Tree rank-variance extremality (Conjecture 74).** For $n \geq 4$ and every prime power the variance of $\operatorname{rank}(\operatorname{ad}_x)$ over a uniformly random element is uniquely minimized by the path and uniquely maximized by the star among n-vertex trees.

75. **Leaf-count rigidity (Conjecture 75).** Equality in the lemma's bound $s(T) \geq n - \ell(T) + 2$ holds exactly for paths and stars.

76. **Diameter rigidity (Conjecture 76).** Equality in the lemma's bound $s(T) \geq \operatorname{diam}(T) + 1$ holds exactly for paths and stars.

Part V: twenty conjectures on pattern Lie algebras of posets, in statement order.

77. **Adjoint–Kirillov determination (Conjecture 77).** Equality of adjoint enumerators $A_{P,q}$ forces equality of Kirillov enumerators $K_{P,q}$, the two contractions of the same bracket, verified across 64 adjoint-collision classes.

78. **Lower-central recovery (Conjecture 78).** The adjoint enumerator determines the vector of lower-central factor dimensions of L_P, which was constant on every adjoint-collision class in testing.

79. **Ordinary Poincaré recovery (Conjecture 79).** The adjoint enumerator is conjectured to determine the ordinary cohomological Poincaré polynomial, tested only for $\dim H^2$ and $\dim H^3$ and which rests on the deposited cohomology data.

80. **Derivation recovery (Conjecture 80).** The adjoint enumerator determines $\dim \operatorname{Der}(L_P)$, which was constant on every adjoint-collision class.

81. **Adjoint Poincaré recovery (Conjecture 81).** The adjoint enumerator is conjectured to determine every adjoint-cohomology dimension $\dim H^i(L_P, L_P)$, tested only for $\dim H^1$, $\dim H^2$, and $\dim H^3$ from the deposited data.

82. **Centroid-dimension recovery (Conjecture 82).** The adjoint enumerator determines $\dim \operatorname{Cent}(L_P)$, which was constant on every adjoint-collision class.

83. **Adjoint spectrum characteristic independence (Conjecture 83).** The set of attained adjoint ranks is independent of q, agreeing across $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ in every case.

84. **Kirillov spectrum characteristic independence (Conjecture 84).** The set of attained Kirillov ranks is independent of q, agreeing across $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ in every case.

85. **Two-field arithmetic rigidity (Conjecture 85).** Equality of adjoint enumerators over $\mathbb{F}_2$ and $\mathbb{F}_3$ is conjectured to force equality over every field, tested only at the single holdout $q = 5$ on 16 three-field collision classes.

86. **Adjoint stochastic field monotonicity (Conjecture 86).** For $q < q'$ the adjoint-rank distribution over $\mathbb{F}_{q'}$ first-order stochastically dominates that over $\mathbb{F}_q$, held in all 221 field-pair tests, while the likelihood-ratio strengthening is false.

87. **Kirillov likelihood-ratio field monotonicity (Conjecture 87, resolved false).** The proposed monotone likelihood-ratio domination of the normalized even-rank distributions is refuted by a seven-point poset whose likelihood ratio from $\mathbb{F}_2$ to $\mathbb{F}_3$ already decreases across the even ranks, while the first-order dominance of Conjecture 86 stands.

88. **Centre-one reverse determination (Conjecture 88).** When $\dim Z(L_P) = 1$ the Kirillov enumerator determines the adjoint enumerator, verified on all 43 centre-one records.

89. **Centre-one Laplace positivity (Conjecture 89).** When $\dim Z(L_P) = 1$ the incidence quotient $R_{P,q}$ has nonnegative integer coefficients, on all 43 centre-one records.

90. **Centre-one quotient shape (Conjecture 90, resolved false).** The proposed unimodality of $R_{P,q}$ under a line centre is refuted by a seven-point poset whose quotient has a strict interior valley, while the positivity and interval support of Conjecture 89 stand.

91. **Hasse-forest reverse determination (Conjecture 91).** When the Hasse cover graph is a forest the Kirillov enumerator determines the adjoint enumerator, across 44 forest-collision classes.

92. **Hasse-forest coadjoint saturation (Conjecture 92).** On a Hasse forest the Kirillov support is the full even range $\{0, 2, \dots, b_K\}$, filled in every case.

93. **Hasse-forest Kirillov unimodality (Conjecture 93).** On a Hasse forest the even-rank sequence of $K_{P,q}$ is unimodal, strict log-concavity being false.

94. **Hasse-forest breadth–orbit bound (Conjecture 94).** On a Hasse forest $b_K \le 2b_A$, the unrestricted bound failing fourteen times off the forest class.

95. **Unitriangular Kirillov strict log-concavity (Conjecture 95).** For ut_n the even-rank sequence of $K_{L,q}$ is strictly log-concave at every nontrivial interior index, confirmed through ut_6 over $\mathbb{F}_2$.

96. **Adjoint–coadjoint incidence anticorrelation (Conjecture 96).** Sampling on the incidence variety gives negative covariance between adjoint and Kirillov ranks for every nonabelian L_P and zero exactly in the abelian case.

Part VI: twenty conjectures on local arithmetic structure, in statement order.

97. **Local-sieve limit for polygonal compasses (Conjecture 97).** The nearest-prime direction of the polygonal numbers P_s has a limit equal to a deep local-sieve hazard, whose primes through 47 already predict the order-level bias with correlation 0.896.

98. **Pentagonal and pronic compasses (Conjecture 98).** The pentagonal compass is right-biased in $(0.56, 0.62)$ and the pronic compass left-biased in $(0.42, 0.48)$, opposite directions measured at 0.583 and 0.450.

99. **Opposite polygonal orders (Conjecture 99).** The 60-gonal compass exceeds 0.58 and the 69-gonal compass falls below 0.49, at 0.609 and 0.468 with both signs replicated above 10^9.

100. **Triangular and hexagonal equality (Conjecture 100).** The triangular and hexagonal compass constants are equal, by $P_6(n) = P_3(2n-1)$, agreeing to three places at about 0.536.

101. **Square compass by root class (Conjecture 101).** The nearest-prime direction of n^2 splits sharply by $n \bmod 6$, below 0.45 for $n \equiv 1, 5$ and above 0.54 for $n \equiv 0, 2, 3, 4$.

102. **Two-sided order spectrum (Conjecture 102).** As the number of sides grows the polygonal compass straddles one half, so infinitely many orders are left-biased and infinitely many right-biased.

103. **Permanent nonuniformity from length five (Conjecture 103).** Every fixed-length conditional order-pattern law now has a proved limit, while nonuniformity remains open separately for every $d \geq 5$, with the length-five chi-square about 2.1×10^7.

104. **Thirteen-fold pattern separation (Conjecture 104).** At window length five the density ratio of the patterns 30241 and 13240 lies in $(12, 15)$, measured at 13.31 through 10^9.

105. **Abundancy order constant (Conjecture 105).** Adjacent triples of $\sigma(n)/n$ are monotone with a density in $(0.094, 0.098)$, far below the independent value $1/3$, measured at 0.0958.

106. **Totient order constant (Conjecture 106).** Adjacent triples of $\phi(n)/n$ are monotone with a density in $(0.019, 0.022)$, measured at 0.0202.

107. **Five-term totient barrier (Conjecture 107, resolved false).** The proposed barrier is refuted by a theorem of Martin [53], which yields strictly monotone runs of $\phi(n)/n$ of every length on positive lower density; the runs are astronomically rare, none below 4×10^9, whereas the abundancy analogue already fails at $n = 36{,}721{,}681$.

108. **Primorial-stride order constant (Conjecture 108).** Along primorial strides the abundancy order density tends to a limit in $(0.16, 0.19)$, decreasing from 0.202 at stride 30 toward 0.177.

109. **Dyadic completion overshoot (Conjecture 109).** Raising only the dyadic exponent of an even squarefree six-divisible stride lifts the abundancy order density past $1/3$ and holds it there, isolated by two controls to the exponents rather than the prime support.

110. **Binomial-slice negative shocks (Conjecture 110, resolved true).** For every $k \geq 3$ the single-step Ω-jump of $\binom{kn}{n}$ is unbounded below, now proved in Proposition 11 by a Chinese-remainder forcing construction that loads the denominator forms with prime factors while a first-moment bound keeps the numerator forms typical.

111. **Binomial-slice positive shocks (Conjecture 111, resolved true).** For every $k \geq 2$ the same jump is unbounded above, so the prime-factor count is a two-sided oscillating process; the same forcing construction proves it with the two form families exchanged.

112. **Fuss–Catalan two-sided shocks (Conjecture 112, resolved true).** For every $r \geq 1$ the Ω-jump of the Fuss–Catalan numbers is unbounded in both directions, extending the shock phenomenon beyond the binomial coefficient; Proposition 11 proves both directions.

113. **Negative shock covariance (Conjecture 113).** In every fixed binomial-slice, central-multinomial, and Fuss–Catalan family successive Ω-jumps have a strictly negative lag-one covariance, negative in all 22 tested families; the prime-truncated central-binomial covariance is now exactly $-\frac{5}{3}$ at every cutoff, the predicted limit for that family.

114. **Dyadic multinomial slope (Conjecture 114).** The dyadic Ω-jump of $M_{k,n}$ is conjectured to have slope $-(k-1)$, with the $k = 2$ case now proved by cyclotomic factorisation.

115. **Eventual sign balance (Conjecture 115, resolved true).** In each binomial-slice and Fuss–Catalan family the positive- and negative-jump densities each tend to $\frac{1}{2}$ with vanishing zero mass, now proved through the multivariate Erdős–Kac theorem of [171] applied to the exact jump identities.

116. **Boundary Euler sum (Conjecture 116).** The mean prime-factor balance of the four composites bracketing a fixed prime gap equals an explicit ℓ-adic Euler sum, matched at correlation 0.998 across 39 gaps.

Part VII: twenty conjectures on topological invariants of geometric families, in statement order.

117. **Quadric-ladder ratio law (Conjecture 117, resolved true).** The all-quadric Calabi–Yau Euler indices satisfy $Q_{n+1}/Q_n < 8$ with strictly increasing ratios for $n \geq 5$, tending to 8; since resolved true, all three clauses proved with the sharpening $8 - Q_{n+1}/Q_n \sim 4/n$, Proposition 15, the limit $8 = 2^3$ being the $d = 2$ member of the Euler family d^{d+1}.

118. **Normalized merge monotonicity (Conjecture 118).** The degree-normalized Euler index E_n/D strictly increases under every degree merge, while the raw version fails at $E_4(2, 2, 4) = 1632 > 1476 = E_4(3, 4)$, the retained control.

119. **Maximal-degree merge monotonicity (Conjecture 119).** Every merge involving a largest degree strictly increases the unnormalized index E_n, in all 377,356 tested merges through dimension 30.

120. **Extremal multidegrees (Conjecture 120).** The all-quadric and hypersurface configurations are the unique Euler extremes at every dimension, flagged for elevated attribution risk as likely accessible by coefficient inequalities.

121. **Dimension-wise Euler gcd law (Conjecture 121).** The gcd of Calabi–Yau complete-intersection Euler characteristics at dimension n equals $24/\gcd(24, n)$, doubled exactly when $n \equiv 0, 2 \pmod 8$, verified in every dimension 2 to 30.

122. **Five-link cap extremum (Conjecture 122).** Among Fano Brieskorn 4-tuples with cap $A \geq 9$ the middle Betti number is at most $A - 1$, with equality only at $(2, 2, A, A)$, exact through cap 50.

123. **Seven-link cap extremum (Conjecture 123).** For 5-tuples with cap $A \geq 13$ the bound is $(A - 1)(A - 2)$, uniquely at $(2, 2, A, A, A)$, the transition at $A = 12$ peaking at 222.

124. **Connected gcd-graph positivity (Conjecture 124).** A connected gcd graph on a Brieskorn 4-tuple forces positive middle homology, a dimension-specific law false at $N = 5$, flagged for elevated attribution risk.

125. **Common-scale monotonicity (Conjecture 125, resolved true).** Scaling all exponents of a Brieskorn tuple by a common factor never decreases the middle Betti number, now proved in Proposition 17: the scale sequence is a sum of interior Ehrhart counts of integral weighted hypersimplices, with every forward difference through order $N - 1$ nonnegative.

126. **Scale convexity and log-concavity (Conjecture 126).** The same scale sequences are simultaneously discretely convex and log-concave; the convexity clause is now proved by the Ehrhart model, and the log-concavity clause survives an exact stress of 24,950,000 inequalities.

127. **Hochster interval support (Conjecture 127).** Every aggregated Hochster strand of a flag moment-angle complex has gapless support, though the ordinary Betti sequence of $K_{2,3}$ is not even unimodal.

128. **Hochster-strand log-concavity (Conjecture 128).** Every strand sequence is log-concave in the subset size, while ultra-log-concavity fails at a connected seven-vertex graph, the retained boundary.

129. **Strand Hurwitz stability (Conjecture 129).** After removing the monomial factor, every strand polynomial has all zeros in the open left half-plane, verified by exact Routh–Hurwitz tests on 50,045 polynomials; real-rootedness is false at four vertices.

130. **Cycle extremality at connectivity two (Conjecture 130).** Among 2-connected graphs the cycle dominates every component-strand coefficient, with coefficientwise equality only for the cycle itself.

131. **Strict component-strand log-concavity (Conjecture 131).** On connected noncomplete graphs the component-strand log-concavity is everywhere strict on positive triples.

132. **Parity-strand unimodality (Conjecture 132).** The even and odd exterior-invariant profiles of every orientation-preserving signed permutation are unimodal, log-concavity failing at dimension 10.

133. **Mapping-torus Betti unimodality (Conjecture 133).** The full Betti sequence of the associated torus mapping torus is unimodal, log-concavity failing already at $-I_4$ with $(1, 1, 6, 6, 1, 1)$.

134. **Sparse-cycle minimizer structure (Conjecture 134).** Some total-Betti minimizer in every dimension $d \geq 8$ uses pairwise distinct negative cycles and at most the parity-forced positive cycle, as at $(5, 7)$ for $d = 12$.

135. **Square-root-two compression exponent (Conjecture 135, resolved false).** The claimed exponent $\sqrt{2}$ is refuted by the Landau floor $M_d \geq 2^d/g(d)$ of Proposition 18: the true exponent is 2, and the square-root fit was a finite-size illusion, since $\log_2 g(d) \approx d/2$ throughout the tested range.

136. **Torsion budget inequality (Conjecture 136).** The 2-primary torsion generators of these mapping tori never outnumber the total rational Betti number, equivalently negative orbits are at most twice the positive orbits, with equality attained 109 times through dimension 12.

Part VIII: twenty conjectures on shape laws for enumerative geometric invariants, in statement order.

137. **Strict dominance law (Conjecture 137).** The signed Euler index of anticanonical hypersurfaces in products of projective spaces strictly increases along every dominance move on the ambient partition, in all 42,691 tested moves through dimension 22.

138. **Strict refinement contraction (Conjecture 138).** Splitting one ambient factor always strictly decreases the signed Euler index, the mirror on ambients of Part VII's merge monotonicity on defining equations.

139. **The $8/9$ refinement barrier (Conjecture 139).** The least destructive ambient split asymptotically retains exactly eight ninths of the Euler index, with $S_{22} = 0.88947$ against $8/9 = 0.88889$; the limit clause is flagged.

140. **Positive interaction of disjoint refinements (Conjecture 140).** For dimension at least six the log Euler index is strictly supermodular on disjoint split squares, the threshold genuine with four violating squares at dimension five.

141. **Strict Betti log-concavity (Conjecture 141).** The Betti profiles of Göttsche's product are strictly log-concave in the cohomological degree for every seed $b \geq 3$, covering every $\mathrm{Hilb}^n(\mathrm{K3})$, in 179,200 exact Turán inequalities.

142. **Cohomology-seed total positivity (Conjecture 142).** Adding one seed generator concentrates the normalized profile toward the Lefschetz centre in likelihood-ratio order, in all 53,070 adjacent minors.

143. **Kurtosis trichotomy (Conjecture 143).** Over the proved exact mean n and variance $2n/(b+2)$, the profile kurtosis follows a complete phase diagram in b with a unique exceptional peak at $(6, 9)$, the equality $\kappa_7(1) = \kappa_7(2) = 9/2$, and limit $21/5$.

144. **Fixed-charge log-concavity (Conjecture 144).** At fixed displacement from middle cohomology the profiles are strictly log-concave in the number of points, making the two-parameter array log-concave along both axes.

145. **Sharp low-dimensional age unimodality (Conjecture 145).** Twisted-sector age histograms of isolated cyclic Calabi–Yau quotients are unimodal in dimensions four and five, sharply: $\frac{1}{3}(1,1,1,1,1,1)$ gives $(0,1,0,1,0)$ in dimension six; flagged for attribution risk.

146. **Gaussian bulk age law (Conjecture 146).** In the random prime-order model the standardized age distribution tends to a Gaussian, a random-model statement flagged as likely accessible to classical equidistribution machinery.

147. **Local central-sector law (Conjecture 147).** Individual age sectors carry lattice-Gaussian mass, with central density $\sqrt{6/(\pi d)}$, the local upgrade of the bulk law with the same flags.

148. **High-sampling restoration of unimodality (Conjecture 148).** Once $r/d^3 \to \infty$ the whole histogram is unimodal with probability tending to one, the exponent three deliberately exposed to refinement.

149. **Exterior torsion-entropy log-concavity (Conjecture 149).** The exterior-degree profile of torsion growth rates of hyperbolic toral mapping tori is log-concave, in 81,000 inequalities on 16,200 spectra.

150. **Equality rigidity (Conjecture 150).** Interior equality in the profile forces a two-valued spectrum, in all 222 equality cases of the exact integer corpus.

151. **Sharp binomial upper envelope (Conjecture 151, resolved true).** Convexity on the product of sign simplices proves the sharp upper bound, attained by $(1, 0, \dots, 0, -1)$.

152. **Two-level lower envelope (Conjecture 152, resolved true).** Same-sign averaging proves that a two-level spectrum attains every lower envelope.

153. **Nullity-strand interval support (Conjecture 153, resolved false).** The cycle matroid of $K_{3,3}$ has nullity-one strand $(9, 0, 6)$ at sizes four to six.

154. **Row-and-column log-concavity (Conjecture 154, resolved false).** The same strand gives the horizontal failure $0^2 < 9 \cdot 6$, while the vertical direction remains open separately.

155. **Nearest-rhombus log-concavity (Conjecture 155, resolved false).** A ten-edge graphic matroid has $H_{2,6} = 2$, $H_{1,5} = 2$, and $H_{3,7} = 3$, giving $4 < 6$.

156. **Hurwitz stability of nullity strands (Conjecture 156, resolved false).** The $K_{3,3}$ strand reduces to $9 + 6z^2$, whose zeros lie on the imaginary axis.

Part IX: twenty conjectures on spectra and filtrations of combinatorial geometries, in statement order.

157. **Universal negative real roots (Conjecture 157).** The h-polynomial of every graph associahedron has all zeros real and strictly negative, interpolating the Narayana and Eulerian endpoints, in all 878 deposited root sets and an independent scan of all 771 connected graphs through five vertices; flagged for attribution risk.

158. **Connected-deletion interlacing (Conjecture 158).** Deleting a vertex that keeps the graph connected weakly interlaces the h-zeros, the recursion that would prove the root law, in 3,621 deposited deletion pairs.

159. **Strict Mahler growth under edge addition (Conjecture 159).** Every edge added to a connected graph strictly increases the logarithmic Mahler measure of h_G, in 3,956 pairs with minimum increment 0.0263.

160. **Edge addition centralizes the spectrum (Conjecture 160).** Every edge addition moves the normalized even-Betti distribution of the toric variety toward middle degree in the symmetric tail order, every tested proper comparison strict.

161. **Path–star Mahler extremality (Conjecture 161).** Among trees the path minimizes and the star maximizes the Mahler measure, each uniquely, the spectral shadow of known coefficientwise orders; flagged for attribution risk.

162. **One-crest component law (Conjecture 162).** The component census of the age filtration of an isolated cyclic quotient with distinct weights is unimodal, in 3,135 deposited profiles, distinctness being genuine: a repeated-weight control has census $(1, 2, 1, 2, 1, 1, 1, 1)$.

163. **Early component crest (Conjecture 163).** Fragmentation always peaks by half age, so the second half of the filtration only consolidates, by the age duality mechanism.

164. **Half-age connectivity (Conjecture 164).** For $d \geq 5$ the half-age sublevel complex is already connected, in all 2,905 eligible profiles, with a disconnected repeated-weight control; flagged for attribution risk.

165. **One-crest first homology (Conjecture 165).** The $\mathbf{F}_2$ first-Betti census of the filtration is unimodal, the simplest persistence-type law the object could satisfy.

166. **One-crest second homology (Conjecture 166).** The second-Betti census is unimodal as well, and the degree bound is exact: a distinct-weight quotient with two-crest β_3 census is retained as a control.

167. **Positive-temperature log-concavity (Conjecture 167).** Lens-space sine-torsion partition functions are strictly log-concave across the sector index at every positive temperature, in 40,932 triples with minimum curvature 0.0079.

168. **Concavity of mean log torsion (Conjecture 168).** The zero-temperature sector means are strictly concave outright, in 6,822 triples with minimum curvature 0.0770, over the exactly pinned global mean.

169. **The $\zeta(3)$ central parabola (Conjecture 169).** In the iterated limit the centred sector means follow the parabola $\frac{3\zeta(3)}{\pi^2}(1-x^2)$, the constant forced by the proved moment identities; flagged as a limit claim and for attribution risk.

170. **Age–torsion decoupling (Conjecture 170).** In the random model the age and the log-torsion sum are asymptotically jointly Gaussian and independent, strengthening the annealed age law of Conjecture 146; flagged as a random-model claim and for attribution risk.

171. **Negative-temperature reversal (Conjecture 171).** On $-1 \leq s < 0$ the sector shape reverses to strict log-convexity, in 20,466 triples, the window calibrated by an $s = -2$ control of curvature $+0.4980$.

172. **Log-concavity of boundary entropy (Conjecture 172).** The log pseudodeterminants of a simplicial sphere's boundary Laplacians are log-concave across the degree, in 262 triples with minimum margin 51.8.

173. **Entropy per rank decreases (Conjecture 173).** The mean log eigenvalue at each degree weakly decreases with the degree, in all 392 deposited adjacent comparisons.

174. **Simplex rigidity of equality (Conjecture 174).** Any adjacent equality of entropy per rank forces the boundary of a simplex, where all equalities hold, in 130 rigidity tests.

175. **Strict stellar growth (Conjecture 175).** Stellar subdivision of a facet strictly increases every entry of the determinant profile, in 900 comparisons with minimum increment 1.42.

176. **Log-concave stellar increments (Conjecture 176).** The entropy created by a facet subdivision is itself log-concave across the degree, in 450 triples with minimum margin 17.8.

Part X: twenty conjectures on flux cohomology and protected index laws, in statement order.

177. **Dense discrete flux saturation (Conjecture 177).** Random sign flux on the torus attains the generic exterior-algebra cohomology profile with probability tending to one, exponentially fast, in 142 of 144 deposited and 71 of 72 independent trials; the random model and the exponential clause are flagged.

178. **Sparse-flux threshold at the coverage scale (Conjecture 178).** Saturation has a coarse threshold at $(\log n)/n^2$, the coverage scale of random 3-uniform hypergraphs, the transition observed across 960 deposited trials; the subcritical direction is now proved for every fixed $c < 2$ at the coverage scale and the supercritical one is flagged.

179. **Linear-complexity sparse witnesses (Conjecture 179).** The minimum saturating support grows linearly, now proved from below in the sharp form $\mu_n \geq \lceil (n-1)/3 \rceil$, with conjectured linear-hypergraph witnesses of size $n + C$, the explicit witnesses using $1, 1, 2, 2, 5, 5, 8, 7$ triples for $n = 3$ to 10.

180. **Coefficient-law universality (Conjecture 180).** Inside the critical window the limiting saturation statistics are the same for every coefficient law, the two deposited laws tracking to 0.2125 against 0.2125; the corresponding statement conditional on the support is false and a five-coordinate witness is retained as a boundary control; random model flagged.

181. **Finite-field resonance power law (Conjecture 181).** Conditional on the displayed target being the true generic profile, Proposition 28 proves the Frobenius-dependent power law by Lang–Weil. The open content is the generic-profile identification and the geometry of the top resonance components.

182. **Central-binomial law for the complete flux (Conjecture 182, resolved true).** The complete flux has cohomology totals F_n, a central binomial, over the rationals and P_n in characteristic two, now proved from a symplectic normal form that reduces the complex to Boolean inclusion matrices.

183. **Prime torsion staircase (Conjecture 183, resolved false).** The staircase clause is now proved for every prime, first p-torsion being exactly one $\mathbb{Z}/p$ at $n = 4p - 1$, but the elementary-abelian clause is refuted at $n = 15$ by a summand $\mathbb{Z}/4$ in degree nine, the least dimension at which one can occur.

184. **Parity-locked torsion count (Conjecture 184).** Saturating sign flux carries exactly $(P_n - G_n)/2$ even torsion summands, forced by the proved parity calibration together with Conjecture 177 and the now-proved Conjecture 182, while the orders fluctuate, with factors 4, 40, 80 on record; retained as a corollary rather than an independent law.

185. **Coefficientwise endpoint extremality (Conjecture 185).** The all-quadric and hypersurface configurations bound every interior coefficient of the signed χ_y-profile, in 7,124 exact comparisons over 680 configurations through dimension 14.

186. **Degree-normalized merge monotonicity (Conjecture 186).** Every coefficient density a_p/D strictly increases under every equation merge, in 110,777 exact comparisons, extending Conjecture 118's Euler law to the full protected profile.

187. **Maximal-degree raw monotonicity (Conjecture 187).** Merges through a maximal degree raise every interior coefficient without normalization, in 47,433 exact comparisons.

188. **Real-rooted protected polynomial (Conjecture 188).** The reduced signed χ_y-polynomial has only simple strictly negative real zeros, in 369 numerically checked polynomials with smallest normalized separation 1.276×10^{-3}; numerical evidence flagged.

189. **Strict ultra-log-concavity (Conjecture 189).** The binomial-normalized profile is strictly log-concave, in 6,540 exact inequalities, the first layer of the real-rootedness hierarchy; flagged for attribution risk beside Hodge–Riemann machinery.

190. **Degree-normalized Mahler growth (Conjecture 190).** The Mahler measure per unit degree strictly increases under every merge, in 4,267 numerically checked tests, returning Conjecture 159's Mahler order on the protected polynomial.

191. **Spectral-radius growth with low-dimensional rigidity (Conjecture 191).** Maximal merges never contract the root radius, with equality exactly in complex dimensions 2, 3, and 5 and strict growth elsewhere, in 2,102 checked tests.

192. **Signature endpoint extremality (Conjecture 192).** The all-quadric and hypersurface configurations bound the normalized signature in every even dimension, with endpoint rigidity, in 8,190 exact configurations through dimension 26.

193. **Degree-normalized signature monotonicity (Conjecture 193).** The signature density S/D strictly increases under every merge, in 302,130 exact comparisons.

194. **Maximal-degree signature monotonicity (Conjecture 194).** Merges through a maximal degree strictly raise the signature itself, in 80,294 exact tests.

195. **The quadric ratio law with limit 27 (Conjecture 195, resolved true).** Successive all-quadric signatures grow by strictly increasing ratios below 27 converging to 27 with correction $27/n$, exact through $n = 60$ where $R_{58} = 26.55045876$; since resolved true, all three clauses proved from an algebraic generating function, Proposition 30, whose mechanism extends to every repeated block and identifies 27 as $\cot^6(\pi/6)$.

196. **Dimension-wise signature gcd law (Conjecture 196).** The gcd of all signatures in even dimension n is 2 for $n \equiv 0 \pmod 4$ and $2^{\max(4,1+\nu_2(n+2))}$ for $n \equiv 2 \pmod 4$, the spike 32 at $n = 14$ confirmed, the signature sibling of Conjecture 121's Euler gcd law.

Part XI: twenty conjectures on spectral entropies and multibase carry fields, in statement order.

197. **All-order path maximality (Conjecture 197).** Among trees of a fixed order the path uniquely maximizes every Rényi entropy of the Laplacian density matrix simultaneously, the order-two and min-entropy cases proved, in 8,775 deposited comparisons through twelve vertices and 3,031 independent ones with smallest margin 2.23×10^{-4}; flagged for attribution risk beside the star-minimum conjecture.

198. **Leaf-pair closure law (Conjecture 198).** Closing the path between two leaves of a tree into a cycle raises the Laplacian entropy at every order, in 86,982 deposited and 14,252 independent comparisons with smallest margin 1.66×10^{-3}.

199. **Edge-subdivision law (Conjecture 199).** Subdividing any tree edge raises the Laplacian entropy at every order, in 70,126 deposited and 11,249 independent comparisons with the programme's most comfortable margin, 6.12×10^{-2}.

200. **Leaf-compression window (Conjecture 200).** Moving a leaf to a no-smaller neighbour lowers the Laplacian entropy throughout the window from order one half to two, the order-two case proved, in 15,385 deposited window comparisons, while an explicit twelve-vertex tree violates the direction at order 0.1 by 7.65×10^{-7}; the boundary control is retained.

201. **Unicyclic endpoint law (Conjecture 201).** Among unicyclic graphs the cycle uniquely maximizes and the star-plus-edge uniquely minimizes the Laplacian entropy at every finite order, in 6,230 deposited comparisons; the exact min-entropy tie of the two four-vertex unicyclic graphs excludes the infinite order and is retained as a control.

202. **Tree one-vertex extension law (Conjecture 202).** Pendant attachment and edge subdivision raise the normalized distance signless Laplacian entropy of a tree at every order, in 168,168 deposited comparisons with minimum margin 4.44×10^{-2}, the first entropy law posed for this spectrum.

203. **Low-order extension law (Conjecture 203).** For arbitrary connected graphs the same two extensions raise the metric entropy on the window of orders up to two, in 10,884 deposited and 7,672 independent comparisons, with explicit high-order violations at orders five and ten retained as controls.

204. **Star maximality (Conjecture 204).** The star uniquely maximizes the distance signless Laplacian entropy among trees at every order, reversing the Laplacian programme's path law, in 8,775 deposited and 3,031 independent comparisons with smallest margin 4.72×10^{-4}.

205. **Bipartite path minimality at the Shannon order (Conjecture 205, resolved false).** An interval-certified fourteen-vertex tree refutes the proposed path minimum with gap 3.15×10^{-5}. Exhaustive enumeration of all 3,159 unlabelled trees of that order finds a sharper witness with gap 9.10×10^{-5}.

206. **Balanced-biclique window maximality (Conjecture 206).** The balanced complete bipartite graph maximizes the metric entropy on the window from the Shannon order to order two, in 5,322 deposited and 195 independent comparisons with smallest margin 1.60×10^{-3}.

207. **Unicyclic cycle maximality above the Shannon order (Conjecture 207).** Among unicyclic graphs the cycle uniquely maximizes the metric entropy at every order at least

one, in 2,245 deposited comparisons, while a four-vertex noncycle graph wins at order 0.1 by 6.62×10^{-4}; the boundary control is retained.

208. **Cross-prime carry central limit law (Conjecture 208).** The carry counts of $n+n$ at finitely many distinct odd primes are asymptotically independent Gaussians, with single-base mean and variance fixed by the proved carry-chain law, marginals and cross-correlations below 0.08 tracked at two million integers; a limit law, flagged, with attribution risk beside the q-additive different-base theorem.

209. **Bounded cross-prime covariance (Conjecture 209).** The covariance of carry counts at two distinct primes stays bounded while each variance grows logarithmically, deposited covariances below 0.24; the part's crispest analytic target.

210. **Cross-prime local limit factorization (Conjecture 210).** Joint point probabilities of the carry vector factor into products of Gaussian lattice masses at one-cell resolution; a limit law beyond any finite check, flagged as such.

211. **Modular equidistribution of carry vectors (Conjecture 211).** Carry counts at distinct primes jointly equidistribute over arbitrary fixed moduli, deposited discrepancies below 10^{-3} at moduli two and three; the Fourier face of cross-base mixing.

212. **Additive large deviations (Conjecture 212).** The vector of carry densities satisfies a large-deviation principle whose rate is the sum of the proved single-base rates, asserted on the interior only, away from the Diophantine boundary where the Graham entanglement lives; a limit law, flagged.

213. **Bounded mixed cumulants (Conjecture 213).** Every mixed cumulant of total order at least three across at least two primes stays bounded while marginal cumulants grow, giving a connected-correlation route to the full decoupling.

214. **Fixed-multiplier carry vectors (Conjecture 214).** The decoupling survives the richer carry automaton of $\binom{an}{n}$ for every fixed multiplier, deposited correlations below 0.11 and independent ones below 0.04 at $a = 3$.

215. **Balanced factorial-ratio vectors (Conjecture 215).** Valuation vectors of every balanced integer factorial ratio decouple across primes, the Landau ratio at $p = 7, 11$ showing the predicted logarithmic variances with cross-correlation -0.016.

216. **Multibase transducer principle (Conjecture 216).** For bounded finite-state transducers in jointly multiplicatively independent bases, automaton mixing plus the absence of a common periodic factor is asserted to be the complete obstruction list for joint Gaussian decoupling; the part's most exposed statement, flagged as a programme conjecture.

Part XII: twenty conjectures on subdivision spectra, spanning-tree correlation, cyclic entropy, and graphon dimension, in statement order.

217. **Critical moment exponent for barycentric vertex Laplacians (Conjecture 217).** The spectral α-moments of the refined 1-skeleton Laplacian have exponential rate $\max\{0, \alpha \log d! - \log(d+1)!\}$, with linear growth at the critical order $\log(d+1)!/\log d!$, the exact divergence threshold that weak universality cannot see.

218. **Hodge incidence-tail hierarchy (Conjecture 218).** Across Hodge degree k, the critical moment exponents follow the codimension factorials $(d-k+1)!$, with degrees zero and one sharing the tail $d!$ by supersymmetry and the top degree carrying no exponential tail; the three-dimensional probe pattern is $6, 6, 2, 1$.

219. **All-degree inclusion-uniform universality (Conjecture 219).** Every inclusion-uniform subdivision rule has universal weak Hodge limits in every degree, the common generalization of Märte's top-degree theorem and Knill's barycentric all-degree theorem; flagged as an attribution risk for that interpolation.

220. **Spectral-tail large deviations from simplex age (Conjecture 220).** Eigenvalues at scale $B_{d,k}^{\theta m}$ occur at exponential frequency $-\theta \log(d+1)!$, the full tail profile whose Legendre data are the critical exponents; the part's most exposed statement, with no direct tail support at reachable depth.

221. **Finite-state renormalization for barycentric Hodge limits (Conjecture 221).** A finite vector of boundary Green functions evolves under refinement by a rational map whose attracting fixed point carries all limiting Hodge laws, generalizing the known one-dimensional quadratic decimation; falsifiable by unbounded growth of the minimal boundary closure rank, on which the statement stands or falls.

222. **Complete-split extremality (Conjecture 222).** The spanning-tree correlation of a connected graph is maximized by a complete split graph with mesoscopic core of order root of n over log n, and the maximum is n minus root of n log n to leading order.

223. **Triangle-free bipartite extremality (Conjecture 223).** Under triangle-freeness the maximizer is a complete bipartite graph whose small side migrates at the same mesoscopic scale; the retained control is the crossover from two to three at seventeen vertices that killed the predecessor claim.

224. **Cycle minimum for bridgeless dependence (Conjecture 224).** Every bridgeless graph has spanning-tree correlation at least that of the cycle of the same order, with equality only for the cycle.

225. **Core-periphery stability at the ceiling (Conjecture 225).** Near-extremal families for the correlation ceiling must contain a mesoscopic set joined to almost the whole graph over a near-independent complement, with no structure asserted inside the core, a freedom the order-one split-bipartite gap shows is genuine.

226. **Complete-multipartite reduction (Conjecture 226).** Among complete multipartite graphs the correlation maximizer is always one macroscopic part plus singletons, collapsing the partition optimization to the split family; verified exhaustively through order forty-five.

227. **Localization of near-minimizers on prime cycles (Conjecture 227).** Mesoscopic sequences attaining the proved half-log-two doubling floor asymptotically must, after affine automorphisms, concentrate on cyclic intervals of length little-o of p: the inverse theorem to the 2026 prime-field entropy-power inequality.

228. **Gaussian rigidity after localization (Conjecture 228).** Under localization and a uniform moment bound, near-minimizing lifts obey a central limit theorem with entropic

convergence; the moment hypothesis is load-bearing, since unrestricted continuous rigidity is known to fail.

229. **Heat-kernel wrap crossover (Conjecture 229, resolved false).** The claimed universal heat-kernel wrap limit is refuted by a mesoscopic jump family satisfying every stated hypothesis whose limit carries an extra compound-Poisson factor; a Lindeberg condition at the modular scale is the recorded repair.
230. **Fixed-m entropy-power law on prime cycles (Conjecture 230).** For every fixed number of summands the mesoscopic entropy gain is at least half the logarithm of the count, with Gaussian rigidity at equality; flagged as an attribution risk, since the integer-lattice analogue is a recent theorem and only the modular content is new.
231. **Uniform pre-wrap entropy monotonicity (Conjecture 231).** Log-concave cyclic laws obey the entropic central-limit increment law uniformly in the number of summands all the way to the wrap scale, with the deposited failure onset near wrap ratio one percent as the retained boundary control.
232. **Exact-dimensional entropy identity (Conjecture 232).** For random-free graphons whose row measure is exact-dimensional with local doubling, sampled-graph entropy per $n \log n$ converges exactly to the dimension.
233. **Multifractal average-dimension law (Conjecture 233).** Under an explicit L^1 convergence of local dimensions, the entropy coefficient is the mean pointwise information dimension, neither a Minkowski dimension nor an essential supremum.
234. **Second-order graph entropy constant (Conjecture 234).** Smooth row manifolds admit a finite linear-order entropy constant, which the threshold-graphon control proves cannot be the quantization constant of the row space.
235. **Equipartition for random-free samples (Conjecture 235).** The sample information content concentrates at the leading scale: almost every sampled graph lies in a typical set of the predicted exponential size, a Shannon–McMillan law for graphon sampling.
236. **Sparse-observation dimension law (Conjecture 236).** Under Bernoulli edge-masking with a growing probe count, conditional entropy scales as n times the logarithm of the probe count times the mean dimension, quantifying information decay under missing data.

Part XIII: nineteen conjectures on flux torsion, quantum merge monotonicity, and effective string geometry, in statement order.

237. **Odd-degree discrete saturation (Conjecture 237).** Sign-valued random r-forms attain the generic rational cohomology profile with exponentially small failure probability for every odd degree, lifting the three-form law of Part X to the full family; sixty deposited and twenty-four independent degree-five trials all saturate.
238. **Odd coverage-scale threshold (Conjecture 238).** Sparse random odd flux saturates exactly at the isolated-vertex scale of random r-uniform hypergraphs, with the suspension dimensions excluded by the essentiality hypothesis; the subcritical direction is now proved and the supercritical one is flagged.

239. **Torsion shape law for the complete flux (Conjecture 239).** The degreewise prime-torsion multiplicities of the complete flux are strictly log-concave in the interior, with the shifted duality and the interval support, now identified as the full range $2p+1$ to $n-2p+2$, both proved; log-concavity is verified for every prime up to twenty-three through dimension one hundred fifty.

240. **Post-onset torsion persistence (Conjecture 240, resolved true).** Once p-torsion appears at dimension $4p-1$ its total strictly grows forever, now proved for every prime from the normal form; the two-primary totals run $1, 2, 10, 20, 66, 132$.

241. **Normalized mirror-map contraction (Conjecture 241).** In the scale-free normalization by the discriminant scale, every coefficient of the mirror map contracts under equation merges from dimension three on, with the dimension-three order-two equalities as the exact boundary; five hundred eighty-eight independent comparisons pass.

242. **Gopakumar–Vafa amplification (Conjecture 242).** Across every merge edge of the five one-parameter threefolds, the ratio of genus-zero curve counts grows strictly with the degree, an ordering of geometries by quantum growth refined degree by degree.

243. **Monotone conifold approach (Conjecture 243).** The genus-zero invariants of each of the five threefolds are strictly log-convex from degree one, with ratios climbing monotonically to the conifold growth constant.

244. **Merge monotonicity of the conifold constant (Conjecture 244).** The conifold point in the flat coordinate strictly decreases under every merge, so consolidation strictly raises the quantum growth constant, the transcendental image of the proved algebraic ordering.

245. **Conifold-normalized Yukawa monotonicity (Conjecture 245).** At equal fractions of the respective conifold radii, the Yukawa coupling normalized by the classical intersection number strictly increases under merges, with the raw instanton comparison failing everywhere as the retained control.

246. **Quadratic wall complexity (Conjecture 246).** The number of numerical walls meeting a compact region grows at most quadratically in the charge norm. A support-property count is only polynomial and the quadratic exponent requires extra Picard-rank-one geometry.

247. **Sharp Gamma for Brill–Noether stability (Conjecture 247).** The Gamma-constant produced by the Brill–Noether construction of stability conditions is exactly optimal on its ray, its topological floor computed by the proved Chern calibration as five sixths down to one third across the five threefold configurations.

248. **Tame wall atlas (Conjecture 248).** In a fixed effective analytic-exponential presentation, the aggregate wall arrangement has polynomial formula format and a polynomial-size compatible stratification up to every charge cutoff.

249. **Polynomial conditioning of Ricci-flat metrics (Conjecture 249).** All curvature derivatives, the diameter and the regularity scale of Ricci-flat metrics are polynomially controlled in a power-equivalence class of discriminant gauges.

250. **Discriminant-effective balanced convergence (Conjecture 250).** Balanced metrics converge at the classical rate uniformly in the family once the level exceeds a power of the inverse discriminant gauge.

251. **Discriminant-effective Hermitian–Yang–Mills convergence (Conjecture 251).** Balanced-bundle approximations carry a polynomial condition number in the minimum of a discriminant gauge and a quantitative stability margin.

252. **Polynomial matter spectral gap (Conjecture 252).** The first positive bundle-valued Dolbeault eigenvalue is bounded below by a power of the pair gauge wherever the matter cohomology is constant.

253. **Certified heterotic Yukawa convergence (Conjecture 253).** Numerically computed physical Yukawa couplings admit a-priori error certificates polynomial in the pair and coupling-zero gauge classes.

254. **Saturation-index special-Lagrangian realization (Conjecture 254).** At infinite-distance degenerations, every extremal light charge has a calibrated representative after multiplication by a divisor of the saturation exponent of the monodromy-orbit lattice.

255. **LMHS-controlled Laplace towers (Conjecture 255).** The vanishing Ricci-flat Laplace spectrum has finitely many exhaustive towers. Their rational exponents come from the integral limiting mixed Hodge structure and skeleton combinatorics, while the limiting operators may retain metric data.

256. **Coordinate monotonicity of normalized complexity (Conjecture 256).** The degree-normalized primitive middle Betti number of a smooth complete intersection strictly increases in every defining degree, growth beyond the trivial multiplicative degree scale.

257. **Strict discrete convexity in each degree (Conjecture 257).** Each one-variable restriction of the normalized complexity to the degree lattice is strictly convex, so marginal topological growth accelerates.

258. **Additive supermodularity across degrees (Conjecture 258).** Raising one degree strictly increases the absolute marginal value of raising another: distinct defining equations reinforce.

259. **Strict log-submodularity (Conjecture 259).** Proportional marginal gains shrink where absolute gains grow, so the logarithm of the normalized complexity is strictly submodular against the supermodularity of the complexity itself.

260. **Schur-convexity at fixed degree sum (Conjecture 260).** At fixed dimension, codimension, and adjunction coefficient, majorization of multidegrees raises normalized complexity: balanced degrees minimize it.

261. **Adjunction-preserving splits decrease complexity (Conjecture 261).** Splitting one degree d into a pair summing to $d+1$ strictly lowers normalized complexity; the odd-dimensional Calabi–Yau slice is proved via the merge order of Part VII.

262. **Strict log-concavity of the nonlinear row (Conjecture 262).** The $q = 2$ row of every $\mathbf{P}^1 \times \mathbf{P}^1$ Betti table is strictly log-concave across its support, beyond the published unimodality conjecture; flagged for its thirteen-table corpus.

263. **Linear row log-concave through its mode (Conjecture 263).** The $q = 1$ row keeps strict log-concavity on its rising side and at its mode, every observed failure lying in the descending tail.

264. **At most one defect in the linear row (Conjecture 264).** The linear row admits at most one log-concavity failure, a nearly totally positive shape with one controlled exception.

265. **Defects occur only inside the nonlinear overlap (Conjecture 265).** A linear-row defect can occur only where the nonlinear row is nonzero, so vanishing of the nonlinear syzygies forces local log-concavity of the linear ones.

266. **Defect-mode locking (Conjecture 266).** The unique linear-row defect sits at the nonlinear row's mode or one step beyond it, in all eight defect tables; the part's boldest and highest-risk law.

267. **Likelihood-ratio flattening in the degree (Conjecture 267).** Outer primitive Hodge pieces of a smooth hypersurface gain relative weight against inner ones at every degree step, a monotone likelihood-ratio comparison across varieties.

268. **Central-window escape (Conjecture 268).** Every central window of the Hodge diamond loses normalized mass monotonically as the degree grows.

269. **Convex-order dispersion (Conjecture 269).** Each degree step is a mean-preserving spread of the normalized Hodge profile, so every convex dispersion functional grows.

270. **Monotone total-variation convergence (Conjecture 270).** The normalized profiles approach the Eulerian descent law monotonically in total variation, with a single equality at the cubic–quartic threefold step.

Abstract

We state two hundred seventy conjectures in fourteen parts. The twenty-five conjectures of Part I are each derived from the local–global random model of the primes: Cramér's model corrected by Hardy–Littlewood singular series, with Bateman–Horn as the organizing framework and Borel–Cantelli accounting for sparse events. Each statement is calibrated to the strongest form the heuristic supports. Every computable constant is computed from its definition, with local admissibility asserted programmatically; every statement was tested against exact counts (probable-prime counts, so labelled, where the integers involved exceed the deterministic certification range), with success measured by the *shape* of agreement (predicted constant, square-root residuals, no drift) rather than by the size of the range searched. The claimed content includes: a prime-power contamination calculus for pattern races—twin races modulo 5 and 8 driven by the pairs $(q^2 - 2, q^2)$, cousin, sexy-pair, and triplet predictions derived from the calculus before testing, a Goldbach lane race with a verified internal null lane, a Stern-representation lane race whose null classes are provable by direct congruence, and an algebraic null control; the measured sub-diffusivity of balanced races (universal negative step correlations tied to the Lemke Oliver–Soundararajan repulsion); canonical ordering deficits for least primes in progressions and for least Goldbach summands; derived covariance kernels for moving-window residual fields, with the diagonal deficit reduced to a pinned singular-series average whose growth already exceeds the single-prime Montgomery–Soundararajan term; distribution laws for the constants of the quadratic-pair and cubic-shift families, with exact mean-value lemmas; an entanglement-aware local-global law for $n^2 + 2^n$ with an exact-rational CRT factorization; a boundary trichotomy for polynomial ladders; a multibase subtorus law for Fermat quotients; and singular-series waiting-time refinements for first prime gaps.

Every statement was checked against the literature by an independent search, with novelty labelled conservatively: statements already known in essence are attributed and kept *inside* the conjecture whose claimed content they calibrate, since those benchmarks are what make the derived constants credible. Every falsifiable-by-instance statement was then stress-tested computationally well beyond its original range, and every constant was recomputed independently from its definition. Part II adds twenty-five structural conjectures in five programmes—connected prime-pattern fields, arithmetic first-arrival fields and class groups, finite logarithms and algebraic tori, arboreal arithmetic dynamics, and adelic factorization processes—each introducing a canonical operator, process, invariant, classification, or phase boundary, with its mechanism, nearest literature boundary, first decisive theorem, and failure mode. Part III adds six conjectures across fields: a relative pretentious inverse principle for polynomial arguments, a threshold purification law for monotone vector paths, an entropy–dimension law for random-free graphons, a Hessian principle for microcanonical graph phases, a sharp stability law for entropy idempotence on finite abelian groups, and a convolution-entropy rigidity for stationary processes. Part IV (Conjectures 57–76) adds twenty conjectures on a single further object, Rossmann's class-counting polynomial of graphical Lie algebras: rigidity and reconstruction laws for the polynomial and its binary specialization, recovery of the subgraph-component polynomial and of domination invariants, a strict log-concavity ladder with proven failure boundaries, and extremal laws for trees, each tested exhaustively at small order, on adversarial random holdouts, and at every known collision class of the polynomial. Two of the twenty, the pseudoforest one-field rigidity claim and the interval-support claim, have since been refuted, by an explicit seven-vertex pair and a connected eight-vertex graph, and are recorded as resolved false with their counterexamples. Part V (Conjectures 77–96) adds a second algebraic family, the nilpotent pattern Lie algebras of finite posets, and studies two rank-weight enumerators of the bracket tensor, the adjoint and the Kirillov enumerators: determination of structural and coadjoint invariants by the adjoint enumerator, characteristic and field-monotonicity behaviour, a positive unimodal centre-one incidence quotient, and reverse determination, saturation, and a breadth bound for Hasse-forest posets, fifteen of them supported throughout the tested corpus, three stated in greater generality than the computations reach, and two since refuted by explicit seven-point posets and recorded as resolved false, all reverified on an independently generated poset corpus over a proved calibration identity. Part VI (Conjectures 97–116) returns to the local–global aesthetic of Part I on new objects: prime compasses at polynomial centres matched to local-sieve hazards, order patterns of the radical ratio with every fixed conditional density now proved to exist and nonuniformity open beyond length four, order constants of abundancy and totient ratios, unbounded prime-factor shocks in combinatorial sequences resting on seven proved propositions, and a local–global Euler sum for the prime-factor anatomy of prime-gap boundaries, one of the twenty since resolved false by a theorem of G. Martin, the three two-sided shock laws and the sign-balance law since proved, and six stated as measured bands rather than derived constants. Part VII (Conjectures 117–136) crosses into algebraic topology on four canonical geometric families: Euler indices of Calabi–Yau complete intersections ending in an exact dimension-wise gcd law, middle Betti numbers of Brieskorn–Pham links under degree caps and common scaling, Hochster strands of flag moment-angle complexes with interval, log-concavity, and Hurwitz-stability shape laws, and torus mapping tori with signed monodromy over proved signed-orbit and normalised-merge calibrations, two of the twenty asserting limits beyond the tested range and two flagged for elevated attribution risk, the quadric ratio law and normalised merge law since proved in full, the compression-exponent law since resolved false with true exponent two, and the common-scale law since proved by an Ehrhart model. Part VIII (Conjectures 137–156) continues into enumerative topology with five shape-law programmes: Euler indices of anticanonical hypersurfaces in products of projective spaces under dominance and refinement, Betti profiles of Hilbert schemes over a proved pair of exact moment identities, twisted-sector age histograms of cyclic Calabi–Yau quotients, exterior torsion-entropy profiles of hyperbolic toral automorphisms, and bigraded Hochster tables of simple binary matroids, with two limit clauses and three random-model statements flagged and one attribution risk marked, two statements since proved by convexity, and four since refuted by exact graphic certificates. Part IX (Conjectures 157–176) passes to spectra and filtrations of combinatorial geometries: real-rooted h-polynomials and Mahler orders for graph associahedra, one-crest laws for age-filtered McKay complexes, temperature and shape laws for lens-space sine-torsion sectors over a proved pair of log-sine moment identities, and pseudodeterminant entropy laws for simplicial spheres, with one iterated-limit and one random-model statement flagged and five attribution risks marked. Part X (Conjectures 177–196) enters string-geometric index theory: saturation, threshold, and torsion-staircase laws for the cohomology of integral three-form flux on tori over a proved coefficient-and-parity calibration, and endpoint, monotonicity, real-rootedness, Mahler, ratio, and gcd laws for the signed χ_y-profiles and signatures of Calabi–Yau complete intersections along the merge order, with four random-model statements, one statement beyond the computed range, three numerical-evidence flags, and three attribution risks marked, the quadric ratio law and the central-binomial law since proved in full, and the prime-torsion staircase proved for every prime but its elementary-abelian clause resolved false by a summand $\mathbb{Z}/4$ at dimension fifteen. Part XI (Conjectures 197–216) pairs a spectral programme with an arithmetic one: all-order Rényi entropy monotonicity and extremal laws for the Laplacian density matrices of trees and unicyclic graphs and for the normalized distance signless Laplacian, over a proved degree-and-majorization calibration, and complete cross-prime decoupling laws, Gaussian, local, modular, large-deviation, and cumulant, for the carry fields of central binomial coefficients, over a proved single-base carry-chain law, with the spectral scans flagged as floating-point checks, the nine carry statements flagged as limit laws, two attribution risks marked, and the boundary controls of the entropy windows retained in place, one of the twenty, the bipartite path minimum, since resolved false by an explicit fourteen-vertex tree. Part XII (Conjectures 217–236) runs four programmes at the interface of spectra and information: factorial critical-moment exponents and tail large deviations for iterated barycentric subdivision beyond weak universality, the extremal theory of a new spanning-tree total-correlation invariant with a mesoscopic core-periphery shape principle, the inverse theory of the prime-field entropy-doubling floor through localization, rigidity, and a heat-kernel wrap crossover, and the identification of random-free graphon entropy with the information dimension of the latent row space, over a proved block-additivity and cycle law and a proved threshold-graphon entropy bound, with all scans flagged as floating-point checks, two attribution risks and one evidence gap marked, and four boundary controls retained in place, one of the twenty, the heat-kernel wrap crossover, since resolved false by an explicit jump family. Part XIII (Conjectures 237–255) extends the string-geometric programmes in five directions: odd-degree flux saturation and thresholds with complete torsion shape laws over the staircase of Part X, strict quantum monotonicity of mirror maps, Gopakumar–Vafa invariants, conifold constants, and normalized Yukawa couplings along the merge order, wall-counting, sharp-constant, and tame-atlas laws for Bridgeland stability, polynomial condition numbers for Ricci-flat metrics, balanced and Hermitian–Yang–Mills convergence, spectral gaps, and certified Yukawa couplings in an algebraic discriminant gauge, and special-Lagrangian realization with LMHS-indexed Laplace towers at infinite distance, over a proved merge-and-Chern calibration, with seven statements beyond any finite check flagged as such, three attribution risks marked including one uncounted remark, the raw-instanton reversal retained as an adversarial control, and the post-onset persistence law since proved in full. Part XIV (Conjectures 256–270) closes with three programmes on classical projective invariants: a discrete-convexity package for the degree-normalized primitive middle cohomology of complete intersections, controlled-defect laws for the two Betti rows of Segre–Veronese embeddings of $\mathbf{P}^1 \times \mathbf{P}^1$, and four monotone-dispersion laws carrying hypersurface Hodge profiles toward the Eulerian descent law, over three proved calibrations, the curve, antidiagonal, and surface propositions, with the syzygy laws flagged for their thirteen-table corpus and the dispersion laws flagged as beyond any finite check.

Contents

1 Introduction

This collection states two hundred seventy conjectures in fourteen parts. Part I (Conjectures 1–25) is a single sustained experiment: the primes, beyond their local structure, are modelled as a random set of density $1/\log n$, and the model is pushed to its sharpest consequences. Part II (Conjectures 26–50) leaves the single model for five structural programmes—connected prime-pattern fields, arithmetic first-arrival fields and class groups, finite logarithms and algebraic tori, arboreal dynamics, and adelic factorization—each contributing one canonical object and its laws. Part III (Conjectures 51–56) leaves number theory itself, carrying the same discipline into multiplicative number theory's pretentious frontier, discrepancy theory, graph limits, additive information theory, and ergodic theory. Part IV (Conjectures 57–76) turns to a single object on the algebraic side, the class-counting polynomial of graphical Lie algebras, and states for it a rigidity and reconstruction programme, a log-concavity programme, and a tree extremal programme, with the proved layer, an exact rank formula and two lower bounds for trees, separated from the twenty conjectural laws. Part V (Conjectures 77–96) turns to a second algebraic family, the nilpotent pattern Lie algebras of finite posets, whose adjoint and Kirillov rank enumerators carry a determination programme, a characteristic-and-field programme, a centre-one positivity programme, and a Hasse-forest programme over a proved calibration identity, three of the twenty being stated in greater generality than the computations reach and flagged as such, and two since resolved false with explicit counterexamples. Part VI (Conjectures 97–116) returns to the local–global aesthetic of Part I, reading it off prime compasses at polynomial centres, order patterns of the radical ratio, order constants of multiplicative ratios, prime-factor shocks in combinatorial sequences, and the arithmetic anatomy of prime-gap boundaries, over a proved layer of factorial-ratio jump identities, short-window exchangeability, existence of every fixed radical-pattern density, exact triangular–hexagonal finite-sieve equivalence, and the $k = 2$ dyadic multinomial slope, one of the twenty since resolved false by a theorem of G. Martin and the shock and sign-balance laws since proved. Part VII (Conjectures 117–136) crosses into algebraic topology, reading exact classical calibrations, the Chern-class formula, the Milnor–Orlik formula, Hochster's formula, and the Wang sequence, across four canonical families of spaces, over proved signed-orbit and normalised-merge calibrations, with Conjectures 117, 118, and 125 since proved and Conjecture 135 since resolved false. Part VIII (Conjectures 137–156) continues the topological turn with shape laws, order and concavity statements for the exact Euler, Betti, age, entropy, and Hochster arrays of five classical families, over a proved pair of profile moment identities, with Conjectures 151 and 152 since proved and Conjectures 153 to 156 since refuted by exact graphic certificates. Part IX (Conjectures 157–176) passes to spectra and filtrations, reading root location, connectivity, convexity, and determinant laws across four combinatorial geometries, graph associahedra, age-filtered McKay complexes, sine-torsion sectors, and simplicial spheres, over a proved pair of log-sine moment identities. Part X (Conjectures 177–196) enters string-geometric index theory, reading saturation and torsion laws off flux complexes on tori and extremal, monotonicity, and divisibility laws off the protected χ_y-profiles and signatures of Calabi–Yau complete intersections, over a proved coefficient-and-parity calibration, its quadric ratio law since proved in full. Part XI (Conjectures 197–216) pairs spectral entropy with carry arithmetic, reading all-order Rényi monotonicity and extremal laws off the Laplacian and distance signless Laplacian density matrices of graphs, over a proved degree-and-majorization calibration, and cross-prime decoupling laws off the carry fields of central binomial coefficients at distinct primes, over a proved single-base carry-chain law. Part XII (Conjectures 217–236) runs four programmes at the interface of spectra and information, reading factorial moment thresholds off barycentric subdivision tails, extremal laws off a new spanning-tree total-correlation invariant, an inverse localization-and-rigidity theory off the prime-field entropy-doubling floor, and dimension identities off random-free graphon entropy, over a proved block-additivity and cycle law and a proved threshold-graphon entropy bound. Part XIII (Conjectures 237–255) extends the string-geometric programmes, reading odd-degree saturation and torsion shape laws off the flux complexes of Part X, strict quantum monotonicity off the mirror data of the five one-parameter threefolds along the merge order, wall and sharp-constant laws off Bridgeland stability, and effective convergence, spectral-gap, and tower laws off the metric geometry of Calabi–Yau families, over a proved merge-and-Chern calibration. Part XIV (Conjectures 256–270) closes on three classical projective invariants, reading convexity laws off the normalized middle cohomology of complete intersections, controlled-defect laws off the Betti rows of $\mathbf{P}^1 \times \mathbf{P}^1$, and monotone-dispersion laws off hypersurface Hodge profiles, over proved curve, antidiagonal, and surface calibrations.

One standard governs all fourteen parts. A good conjecture is not merely a statement that has resisted counterexample: it is a determinate proposition about a canonical object, with a stated mechanism, and with enough exposed structure to be refuted in pieces. Concretely, every statement here is asked to (i) survive every congruence and size obstruction; (ii) come with a quantitative form whose constants are derived, not fitted; (iii) demand exactly the fluctuations its mechanism earns and no fewer (the Mertens conjecture died of demanding fewer); (iv) sit inside a known hierarchy where one exists (for Part I, Hardy–Littlewood k-tuples $\subset$ Schinzel's Hypothesis H $\subset$ Bateman–Horn), so that failure would propagate; and (v) expose instance-falsifiable content—congruences, null lanes, exact identities, derived constants—so that computation can refute the derivation even where the asymptotic itself lies beyond finite refutation. Numerical range is the least important column in the ledger: most deep phenomena drift at the rate $\log\log x$, and $\log\log 10^{18} \approx 3.7$, so verification "to 10^{18}" is weak evidence by itself. What persuades is the *shape* of the agreement—a count that tracks a derived constant through several decades with residuals that look like noise.

Part I is presented in three sections: the principal conjectures (numbers 1–9), family laws and second-order refinements (numbers 10–17), and instances and structural companions (numbers 18–25). The material they draw on—Bateman–Horn systems, barely-divergent sparse sequences, representation problems with Borel–Cantelli accounting, and statistical laws of the prime sequence—runs through all three. Section 60 reports the stress tests and the independent recomputation of every constant.

One structural lesson organizes several of the statements below. A probability model integrates over density, and is therefore blind to algebraic families of density zero; the admissibility clause, checked at every prime for every system, is what repairs that blindness. The cubes inside the representation problem $n = p + k^3$ (Theorem 1) are the sharpest instance: they obstruct the problem outright while contributing nothing to any Borel–Cantelli sum taken over all n. This is the classical lesson of admissibility, and it is why the statements below separate what is provable by congruence or factorization from what a probabilistic accounting supplies. Parts II to V inherit the same separation in a different key: each statement names its canonical object and averaging law, identifies its source, states a first decisive theorem where one exists, and records an honest failure mode.

Three checks stand behind every statement, and §60 reports their outcomes for Part I. First, an independent literature search at neighbourhood depth—defining objects, derived sequences, OEIS comments where relevant, and the abstracts and bodies of near-neighbour papers, read rather than skimmed from search summaries—establishes what is already known. Where a statement exists in essence it is attributed and labelled: some verbatim (Dubner's twin-sum conjecture, the Stern list for $p+2k^2$, Caldwell–Gallot's $e^{\gamma} \log N$ laws), one with an exact constant catalogued as an OEIS entry (A188596), and the rest attributed accordingly. No such statement stands as a conjecture of its own; each is retained *inside* the conjecture whose claimed content it calibrates, restated in our uniform framework and re-verified at our bounds, and where a statement strengthens a named open problem the containment is stated as its headline. Second, every falsifiable-by-instance statement was verified against exact counts and then re-tested well past its original bound, the computation being designed as an attempt at refutation rather than at confirmation. Third, constants and counts were recomputed from the definitions alone by independent implementations sharing no code with the primary computation. Our novelty policy throughout is conservative: where priority is uncertain we attribute rather than claim.

Taxonomy: claimed content and its calibration layer

Within Part I, one criterion governs the taxonomy. An unrecorded specialization of Bateman–Horn is analogous to evaluating a classical special function at a new argument—worthwhile as data, but not a new conjecture in the conceptual sense; no conjecture's claimed content is a bare specialization, and the taxonomy is two-layered *within* each conjecture. The *calibration layer*—the previously-known statements this paper begins from (Dubner's conjectures, the Stern list, the Caldwell–Gallot laws, Martin's refined Goldbach, Kourbatov's record framework, the classical single instances of each family)—appears as attributed remarks inside the conjectures, each with its derived constant re-verified at our bounds; membership in the standard hierarchy of conjectures is itself a virtue, and the calibration layer is what makes the novel layer's constants credible. The *claimed layer* is graded as follows: mechanism-level content (the contamination calculus 1(v) with its instances 18, 19, 14, 12, 1, 4; the ordering deficits 5, 3; race sub-diffusivity 9; the entanglement law 6); derived second-order and distributional laws (10, 15, 2, 8(ii), 11(iii), 13, 7(iv), 16(ii)); and structural classifications (23, 25, 24, 20(i), 21(i), 22)—with framework attributions stated where an instance lives inside someone else's theory (Kowalski's for 15(ii), Montgomery–Soundararajan's for 2 and 13). Every statement was searched against the literature at neighbourhood depth and verified at the scales reported in §60. We regard the prime-power contamination calculus (Conjecture 1(v)) and its mechanism family—1 and 4, the fresh cousin-race predictions it generated, and the negative control 14—together with Conjecture 3(ii) and the derived covariance kernels of 8(ii) and 16(ii) as the paper's core, joined by 13 and 23; Conjectures 20, 6 and 11 carry open modelling questions flagged in place.

One tagging convention runs through every statement below: clauses marked [theorem] or "elementary" are proved; clauses marked "conditional proposition" follow from the hypotheses stated with them; the remaining clauses are conjectures; and open questions are labelled as such and are not counted among the two hundred seventy conjectures. A title tagged "resolved true" or "resolved false" marks a statement settled since its deposit, each carrying a *Resolution* paragraph with the proof or the counterexample.

Computational-check standards

All counts below are exact (sieves; deterministic Miller–Rabin for $n < 3.3 \times 10^{24}$; Baillie–PSW beyond, flagged as such). Every comparison reports the Poisson-normalized residual $z = (\text{obs} - \text{pred})/\sqrt{\text{pred}}$. Every quoted z-score is such a residual and is used *descriptively*: where the counts compared are correlated, overlapping, or examined after selection, these residuals are not calibrated tail probabilities, and no significance claim in this paper rests on them. For sparse counts the cumulative comparison is contaminated at small x by the phantom mass of the main-term integral near its lower limit; in those cases we also report per-decade increments, which are clean. Constants are computed by truncated Euler products with the truncation wobble (difference between cutoffs) reported as an error bar; for conditionally convergent quadratic and cubic products this wobble, not the last digit, is the effective precision.

Primality certification is stratified, and every count inherits the stratum of the integers it involves. Primality of every integer below 3.3×10^{24} is decided *deterministically*, by fixed-base Miller–Rabin over a witness set verified for that range. For larger integers the classification is made by the Baillie–PSW test, and is a *probable-prime* classification, not a proof: no Baillie–PSW pseudoprime is known—and none exists below 2^{64}, where the test is therefore deterministic—but their nonexistence is unproved. Consequently every count involving such integers is a probable-prime count, and is labelled as one wherever it appears. The computations affected are the factorial scans of Conjectures 21 and 2 ($n! \pm 1$ for n beyond about 26), the $n^2 + 2^n$ scan of Conjecture 6 beyond $n \approx 80$, the quartic values of the cyclotomic chain of Conjecture 24 beyond $p \approx 2 \times 10^6$, and the Fibonacci–Lucas values of Conjecture 20 at all but the smallest indices. Every other count below—sieve censuses, race counts, window fields, and the inputs to every constant—lies entirely inside the deterministic range.

Data availability. The accompanying repository (`conjecture-engine`) contains the primary verification programs and the independent reimplementations used for recomputation, the machine-readable results files and exception lists, the Euler-product truncation points used for every constant, and the window-selection rules and random seeds of every sampled computation; `run_all.py` regenerates every number in this paper. Probable-prime classifications are recorded as such in the results files, separately from the deterministically certified ones, so that each count can be read at its own certification stratum.

2 Notation

p, q denote primes; γ is Euler's constant, $e^{\gamma} = 1.781072\ldots$; φ is Euler's totient; $\phi = (1 + \sqrt{5})/2$; F_n the Fibonacci numbers; $p\#$ the primorial. For irreducible $f_1, \ldots, f_k \in \mathbb{Z}[x]$ with positive leading coefficients, the Bateman–Horn singular series is

$$C(f_1,\dots,f_k) \;=\; \prod_p \frac{1-\omega(p)/p}{(1-1/p)^k}, \qquad \omega(p) = \#\{n \bmod p:\ p \mid f_1(n)\cdots f_k(n)\},$$

and the corresponding main term is

$$I(N) \;=\; \int_2^N \frac{dt}{\prod_i \log f_i(t)},$$

which absorbs the usual $1/\prod_i \deg f_i$. The Bateman–Horn conjecture [2] asserts $\#\{n \le N :$ all $f_i(n)$ prime$\} \sim C \cdot I(N)$ whenever $C \ne 0$; each Bateman–Horn conjecture below is an instance with its constant evaluated. For nonlinear systems the product converges only conditionally; throughout, it is taken in the canonical ordering of increasing primes, in which convergence follows from the equidistribution of $\omega(p)$ about its Chebotarev mean. Our numerical truncation wobble is an error bar for this canonically ordered product. We write $\mathrm{Li}_k(x) = \int_2^x (\log t)^{-k}\,dt$. The twin singular series is $\mathfrak{S}(d) = 2C_2 \prod_{p\mid d,\, p>2} \frac{p-1}{p-2}$ for even d, with $2C_2 = 1.3203236\ldots$; the Goldbach series $\mathfrak{S}(n)$ is the same product over odd $p \mid n$.

Eight of the conjectures below are built over the Bateman–Horn move—choose an admissible system, compute its singular series, state the count—but none is a bare instance of it: 10 and 15 are family laws with derived constant distributions, 23 and 25 carry structural classifications, 24 is a searched chain, and 18, 19, 14 are contamination-calculus predictions (two fresh races and the negative control) whose mechanism is the contamination calculus of Conjecture 1(v). Each conjecture carries its classical calibration instance as an attributed remark. The heuristic is uniform, so we give it once. Model the primality of the values $f_i(n)$ as independent events of probability $1/\log f_i(n)$, corrected by the factor $(1-\omega(p)/p)/(1-1/p)^k$ at each prime p to account for the joint local behaviour; multiplying the corrections and integrating gives $C \cdot I(N)$.

For locally integrable F we write $\mathcal{M}_x(F) = \frac{1}{\log x}\int_2^x F(t)\,\frac{dt}{t}$ for the logarithmic mean; all race conjectures below state their drift and null clauses through $\mathcal{M}_x$, so that "no persistent leader" and "systematic surplus" are claims about a defined functional rather than about pointwise behaviour. Drift clauses are stated at the *drift scale*—for a pair race, $\mathcal{M}_x(D(t)\log^2 t/\sqrt{t})$ converging to an explicit constant—never as $\mathcal{M}_x((D-T)/\sqrt{\pi}) \to 0$, which is vacuous for the drift coefficient (both $T/\sqrt{\pi}$ and its multiples have logarithmic mean $\to 0$). Convergence of $\mathcal{M}_x$ at drift scale is itself part of each such conjecture: a pure random-walk fluctuation model would make these means divergent in variance, so their convergence encodes the Rubinstein–Sarnak-type almost-periodicity that the zero-oscillation hypothesis names.

Admissibility ($\omega(p) < p$ for all p) is asserted programmatically during every constant computation. The natural-looking pair $\{n^2+n+1,\ n^2+n+3\}$ has $\omega(3) = 3$ and is rejected; the $k=4$ branch of Conjecture 25 and the naive iterated chain behind Conjecture 24 were both rejected by the same machinery, and both rejections became part of the statements.

Novelty labels, from the independent literature search: **(a)** previously stated in essence; **(b)** classical family, quantified instance apparently unstated.

How to read the two hundred seventy. The count is a numbering convention, not a claim of two hundred seventy independent mechanisms. The contamination calculus is one operator, stated with the twin races and projected onto the Goldbach, Stern, triplet, and sexy-pair patterns, together with one provably null pattern as its control, and those six statements stand or fall largely together. Several conjectures pair a proved classification with a Bateman–Horn lane that is standard once stated, and several keep attributed benchmarks inside them as calibration; the tagging convention above separates these layers, and the novelty labels mark what is claimed as new. Part II is five programmes, each contributing one central object and its laws. Part IV is one polynomial invariant carrying three programmes, and its twenty statements stand or fall in blocks rather than singly, with two, Conjectures 63 and 71, already fallen and recorded as resolved false. Part V is two rank enumerators of one bracket tensor, fifteen conjectures supported throughout the tested corpus, three stated in greater generality than the computations reach, and two resolved false with explicit counterexamples, flagged in place. Part VI is five local-structure programmes in the aesthetic of Part I over a proved layer of jump identities and short-window exchangeability, six of its twenty stating a bias as a measured band, two extrapolating in a parameter rather than the index, one since resolved false, and four since proved. The corresponding per-part registers for Parts VII through XIII, their programme counts, evidence flags, retained controls, and resolutions, are carried once by the abstract and the summary sheet rather than repeated here. Depth and independence therefore vary by design across the two hundred seventy, and the summary sheet's ordering, not the raw count, carries the priority claim.

Part I
Conjectures from a local–global random model

Added to the deposit: 28 July 2026.

3 Principal conjectures

The twin-race statement is split into its provable algebraic core and its conjectural clauses, stated through the logarithmic-mean functional $\mathcal{M}_x$ of the notation section.

Lemma 1 (Orientation elimination and class assignment)**.** *In $\psi_2(x) = \sum_{n\le x} \Lambda(n)\Lambda(n+2)$, the total contribution of terms in which n or $n+2$ is a proper prime power is, apart from $O(x^{1/3}\log^2 x)$ from cubes and higher powers, carried entirely by the patterns $(q^2-2,\, q^2)$ with q prime and q^2-2 prime: the mirror pattern $(q^2,\, q^2+2)$ is annihilated by $3 \mid q^2+2$ for every prime $q > 3$. Moreover, for every prime $q \neq 5$, $q^2 - 2 \equiv 2$ or $4 \pmod 5$ (never 1), and for every odd q, $q^2 - 2 \equiv 7 \pmod 8$. (Two bounded exceptions: at $q = 5$ the term $(23, 25)$ lands outside the twin-start classes altogether, and at $q = 3$ the mirror term $(9, 11)$ survives, the annihilation beginning only at $q > 3$; each is a single term, absorbed in the bounded error.)*

The proof is elementary (squares mod 3, 5, 8; the prime-power count). The lemma orients the race: prime-square contamination can enter only specific residue classes, and the model converts that into a drift prediction for the *unweighted* twin counts.

Conjecture 1 (Twin races modulo 5 and 8, and the prime-power contamination calculus; (b), apparently new)**.** *Twin starts $p > 5$ lie in classes $p \equiv 1, 2, 4 \pmod 5$; write $\pi_t(x; m, a)$ for the* *number of twin starts at most x in the class a modulo m. Let $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}\big(\pi_t(x; 5, 2) + \pi_t(x; 5, 4)\big)$ and*

$$T(x) \;=\; \frac{1}{2\log^2 x} \sum_{\substack{q \le \sqrt{x} \\ q^2-2 \text{ prime}}} \log q \, \log(q^2 - 2).$$

Then: (i) *[drift law, at drift-scale normalization] the clause asserts two sub-claims, labelled separately because they can fail independently.* (i-a) *[mechanism] class 1 carries the systematic surplus T: one has the decomposition $D_1(t) = T(t) + R(t)$ in which the normalized remainder $R(t)\log^2 t/\sqrt{t}$ possesses a limiting logarithmic distribution with mean zero, so that the entire drift-scale deterministic content of D_1 is the contamination term and its coefficient is thereby identified.* (i-b) *[averaging] at the drift-scale normalization the remainder averages away:*

$$\mathcal{M}_x\!\left(\frac{R(t)\,\log^2 t}{\sqrt{t}}\right) \longrightarrow 0, \qquad \textit{equivalently} \qquad \mathcal{M}_x\!\left(\frac{D_1(t)\,\log^2 t}{\sqrt{t}}\right) \;\longrightarrow\; c_T,$$

where

$$c_T \;:=\; \lim_{x\to\infty} \frac{T(x)\log^2 x}{\sqrt{x}} \;=\; \tfrac{1}{2}\, C(x,\, x^2 - 2)$$

is the half of the Bateman–Horn constant of the pair $(q,\, q^2{-}2)$ (the weighted census sum is $\sim C\sqrt{x}$, so the limit exists and is explicit). A normalization at the noise scale, $\mathcal{M}_x((D_1 - T)/\sqrt{\pi_t}) \to 0$, would be vacuous *for the coefficient of (i-a): since $T/\sqrt{\pi_t} \asymp 1/\log t$, one has $\mathcal{M}_x(T/\sqrt{\pi_t}) \asymp$ $\log\log x/\log x \to 0$, so the condition holds with T replaced by 0, $2T$, or any fixed multiple, and does not express "exactly the removed mass." The statement therefore normalizes at the* drift *scale, where the claimed limit is a nonzero constant. Convergence at this normalization is a Rubinstein–Sarnak-type regularity assertion: under a pure random-walk model the logarithmic mean would not converge (its variance grows like $\log x$), so (i-b) asserts almost-periodic structure for the twin race and is the precise content of the zero-oscillation hypothesis, while (i-a) is what fixes the limit at c_T rather than 0 or $2c_T$, and is the calculus. The convergence asserted in (i-b) selects the rigid branch of the dichotomy registered at Conjecture 9(iii); the two statements stand or fall together at drift scale;* (ii) *[zero deterministic drift in the symmetric differences] the symmetric classes carry no deterministic term* at the same normalization*: $\mathcal{M}_x\big((\pi_t(\cdot; 5, 2) - \pi_t(\cdot; 5, 4))\log^2 t/\sqrt{t}\big) \to 0$, and likewise for each of the three pairwise differences among classes $\{1, 3, 5\}$ mod 8 in clause (iv)—now a statement distinct from (i), since at drift scale "zero" and "c_T" are different limits;* (iii) *[sign density, conditional] conditional on the Gaussian fluctuation model for the normalized remainder in (i)–(ii), the logarithmic density of $\{D_1 > 0\}$ exists and equals $\frac{1}{2}$, approached from above with a finite-x excess of order $\log\log x/\log x$: the pointwise drift-to-noise ratio is $T/\sqrt{\pi_t} \asymp 1/\log t$, and its* logarithmic average *$\mathcal{M}_x(1/\log t) \sim \log\log x/\log x$ carries an extra* $\log\log$ *(the scale is not* $1/\log x$*). In contrast to Chebyshev's races, the twin race has no persistent leader;* (iv) *[mod-8 companion] since $q^2 - 2 \equiv 7 \pmod 8$ always (Lemma 1), the* entire *deterministic term falls on the single class $7 \pmod 8$: with $D_7(x) = \frac{1}{3}\sum_{a\in\{1,3,5\}} \pi_t(x; 8, a) - \pi_t(x; 8, 7)$,*

$$\mathcal{M}_x\!\left(\frac{D_7(t)\,\log^2 t}{\sqrt{t}}\right) \longrightarrow 2c_T$$

(twice the mod-5 constant, undiluted—at this normalization the factor 2 is a falsifiable prediction, the mechanism and averaging claims separating here exactly as in (i-a)–(i-b)), while classes $1, 3, 5$ are mutually symmetric. The mod-5 and mod-8 races are two projections of one mechanism, so

their joint behaviour is a family test in the sense of the shape-of-agreement criterion; (v) *[contamination calculus for balanced races] for a pattern* $(n, n+d)$ *and modulus* m*, call the race* balanced *if the prime-prime singular series is identical across the competing residue classes (as CRT gives for the classes compared in (i)–(iv)) and no mechanism other than prime powers contributes at the* $\sqrt{x}$ *scale under the zero-oscillation hypothesis. For balanced races the deterministic drift vector is computed by one operator. Define the* contamination operator

$$\mathcal{C}_{d,m}(a;x) \; = \sum_{\textit{orientations } o \in \{(q^2-d, q^2),\,(q^2, q^2+d)\}} \; \sum_{\substack{q \le \sqrt{x},\ o \textit{ prime-compatible} \\ \mathrm{start}(o) \equiv a\ (m)}} \Lambda(q^2)\,\Lambda(\textit{prime member of } o),$$

where prime-compatible means the non-square member is prime and no fixed prime annihilates the orientation. The provable layer is the orientation census and class assignment (which o survive, and where their starts land—Lemma 1 and its analogues); the conjectural layer is the transfer: the unweighted class counts are depressed by $\mathcal{C}_{d,m}(a;x)/\log^2 x$ *relative to the clean classes, in the drift-scale limiting-log-mean sense of (i), with the mechanism and averaging claims separated as in (i-a)–(i-b)—for each class* a*, the deficit obeys* $\mathcal{M}_x(\mathrm{deficit}_a(t)\log^2 t/\sqrt{t}) \to c(a) := \lim \mathcal{C}_{d,m}(a;x)/\sqrt{x}$*, an explicit Bateman–Horn constant per class, so every coefficient in the vector is identified—i.e. "exactly the removed mass," which requires the weighted identity, partial summation uniform in classes, higher-power bounds (exponents* $r \ge 3$ *contribute* $O(x^{1/3+\varepsilon})$ *against the operator's* $\asymp \sqrt{x}$ *and are negligible—the one easy layer of the transfer), and the zero-oscillation hypothesis, and is the calculus's single analytic conjecture rather than a rule. The drift is* $\asymp \sqrt{x}/\log^2 x$ *against a count fluctuation* $\asymp \sqrt{x}/\log x$*, i.e. smaller by* $1/\log x$ *in standard-deviation units at every height—so the calculus predicts occupation biases under long logarithmic aggregation, never a fixed nonzero standardized mean, and every verification in this family is designed around that fact. (Races with unequal primary constants, or with interfering orientations beyond those enumerated, are outside the calculus until those terms are separated.) In particular, for the cousin pattern* $d = 4$ *the calculus makes two predictions with no free choices: the orientation* (q^2-4, q^2) *is dead apart from a single term (*$q^2 - 4 = (q-2)(q+2)$ *is composite for* $q > 3$*, while at* $q = 3$ *the value* 5 *is prime: one bounded term, whose start* 5 *lies outside the mod-*5 *cousin classes and contributes one bounded term to class* 5 *mod* 8*), the orientation* (q^2, q^2+4) *survives (no prime annihilates it),* q^2+4 *prime forces* $q \equiv \pm 2 \pmod 5$ *for* $q \ne 5$ *(at* $q = 5$ *the value* 29 *is prime, but the start* $25 \equiv 0 \pmod 5$ *lies outside the cousin start classes* $\{2,3,4\}$*), so the contaminated cousin-start class is* $4 \pmod 5$ *(of* $\{2,3,4\}$*) and—since* $q^2 \equiv 1 \pmod 8$*—the class* $1 \pmod 8$ *(of* $\{1,3,5,7\}$*): both deficits converge, in the drift-scale limiting-log-mean sense of (i)–(ii), to* $c^{\mathrm{c}} = \lim_{x\to\infty} T^{\mathrm{c}}(x)\log^2 x/\sqrt{x}$*, the Bateman–Horn constant of the pair* $(q,\, q^2+4)$*, where* $T^{\mathrm{c}}(x) = \frac{1}{\log^2 x}\sum_{q \le \sqrt{x},\ q \ne 5,\ q^2+4 \text{ prime}} \log q\, \log(q^2+4)$.

Clause (v) answers the structural question of what general principle the mod-5 and mod-8 races instantiate, and the cousin predictions it yields were derived first and tested second. At $x = 10^9$ the predicted drift is far inside the noise ($T^{\mathrm{c}}/\sqrt{\pi_{\mathrm{c}}} \approx 0.09$, as the $1/\log x$ law requires), so the sharp statistic is leadership under logarithmic averaging, calibrated against the null of Conjecture 14(ii) (log-occupation $\frac{1}{2}$ with Gaussian spread $\sqrt{\log 2/\log x} \approx 0.18$ at 10^9, at the occupation-fraction constant): the class-4 deficit race led on the predicted side with log-density 0.99 (+2.7 null standard deviations), the class-1 mod-8 race with 0.92 (+2.3), both signs agreeing with the prediction, while the control differences among uncontaminated classes stayed within one noise unit of zero in log-mean. This is moderately strong directional evidence—two independent projections each between two and three null standard deviations on the predicted side—though the races share underlying primes and the two statistics are positively correlated, so we do not call it sharp confirmation; the family now has five projections of one mechanism (twin mod 5, twin mod 8, cousin mod 5, cousin mod 8, and the Goldbach lanes of Conjecture 4) plus two algebraic nulls (Conjecture 14 and the dead cousin orientation), which is the shape-of-agreement standard applied to the calculus itself.

A limiting sign density strictly between $\frac{1}{2}$ and 1 would be inconsistent with the scale analysis: when drift/noise $\to 0$, a Gaussian model forces the sign density to $\frac{1}{2}$, and a nontrivial limiting density would require a persistent normalized bias that no mechanism here produces. The fluctuation in (iv) is not written as $O^*(\sqrt{\pi_t})$, since O-notation asserts a provable bound where only a conjectural fluctuation statement is meant. Two model assumptions remain open and are stated as such: the passage from the Λ-weighted count to unweighted twin counts is a partial-summation argument whose error terms need writing out (higher prime powers are disposed of by Lemma 1); and the *zero-oscillation caveat*—the drift law treats the oscillatory explicit-formula contributions to class differences as having vanishing logarithmic mean, the exact analogue of the Rubinstein–Sarnak GRH-plus-linear-independence framework [6], and a conspiracy among hypothetical zeros could in principle mimic or cancel part of the drift; clause (i) is conjectured on the standard side of that hypothesis.

The mechanism is the twin analogue of the classical explanation of Chebyshev's bias: prime-square terms contaminate the classes that can contain them. (Biases of twin primes in residue classes have been studied before—Sahoo [27] for the mod-4 race, and [28] for prime pairs versus isolated primes—but we could find no prior statement of the mod-5 race, the (q^2-2, q^2) mechanism, or the quantified surplus below.) Mod 3 kills the pattern (q^2, q^2+2) outright, a local accident that halves the mechanism and (as a mod-3 computation shows, overturning the natural first guess of a quadratic-residue bias against class 4) concentrates the surplus entirely on class 1. Data to 4×10^9: class counts (3980017, 3982505, 3981922), D_1 within one noise unit of T throughout, no persistent leader, and the control race $\pi_{t,2}-\pi_{t,4}$ wandering like a fair coin. Mod-8 data to 10^9: class counts (856684, 855807, 856046, 855967) for $a = 1, 3, 5, 7$; $D_7 = +212$ at 10^9 against predicted $T = +254$ with noise $\sim$1068, positive through most of the range (log-density 0.775), and the three pairwise $\{1, 3, 5\}$ controls all within one noise unit of zero—the two projections of the mechanism behave coherently, none of it provable at present.

Conjecture 2 (The pair-level Montgomery–Soundararajan reduction; apparently new)**.** *For the window field of Conjecture 8(ii), the diagonal variance of the twin-pair count reduces to a* pinned *singular-series average. With $R(h) = C_4(0, 2, h, h+2)/\mathfrak{S}(2)^2$ for $h \geq 1$, C_4 the Hardy–Littlewood constant of the displayed quadruple, (zero when the offset set is degenerate or inadmissible—all odd h, and $h = 2$ since $\{0, 2, 4\}$ dies at 3; consecutive twin pairs cannot overlap beyond $(3, 5, 7)$, so no overlap term exists) and*

$$G(H) \;=\; \sum_{1\leq h\leq H}\Big(1-\frac{h}{H}\Big)\big(R(h)-1\big),$$

the derived identity (clause (i)*) is*

$$\frac{\operatorname{Var}_t\,\pi_2(t;H)}{\mathbb{E}_t\,\pi_2(t;H)} \;=\; 1+\frac{\mathfrak{S}(2)\,\big(2G(H)-1\big)}{\log^2 x}+o\Big(\frac{1}{\log^2 x}\Big):$$

with $p = \mathfrak{S}(2)/\log^2 x$ the per-site intensity, the variance is $Hp(1-p) + 2p^2HG(H)$, so $\operatorname{Var}/\mathbb{E} = 1 - p + 2pG(H)$—the Bernoulli diagonal term $-p = -\mathfrak{S}(2)/\log^2 x$ is of exactly the order of *the claimed error and cannot be dropped (it is the same diagonal that, in the single-prime case, merges into Montgomery–Soundararajan's constant $\gamma + \log 2\pi - 1$). With that term retained, the entire deficit left open at Conjecture 8(ii) is the single computable function G plus the explicit Bernoulli correction—the pair analogue of Montgomery–Soundararajan's reduction of the prime-count variance to $\sum_{h \le H} \mathfrak{S}(h)(H-h)$ [19]. Clause (i) is a* conditional proposition*, not an independent conjecture: given quantitative Hardy–Littlewood for 4-tuples, the reduction follows. The conjectural clause is (ii): $-G(H)/\log H$ diverges—the pinned average grows strictly faster than the single-prime secondary term $\frac{1}{2}(\log H + \gamma + \log 2\pi - 1)$, so pair counts in windows are more sub-Poisson than prime counts—with the caveat that data to $H = 3000$ cannot distinguish divergence from a pure* log *with a large constant (nor $c \log H \log\log H$ from $(\log H)^{\alpha}$): what the computation* establishes *is only that the local slope exceeds the single-prime value sevenfold across the computable range; the divergence and its exact form are registered, not evidenced beyond that.*

The pinning leaves the local densities in exact balance. At every odd prime p the relative local factor of $R(h)$ takes the value $p/(p-2)$ for $h \equiv 0$, the value $p(p-3)/(p-2)^2$ for $h \equiv \pm 2$, and the value $p(p-4)/(p-2)^2$ otherwise, and their exact average over h mod p equals 1, since $(p-2) + 2(p-3) + (p-3)(p-4) = (p-2)^2$. Any persistent negativity of $G(H)$ is therefore not a first-order local-density effect but a global correlation effect across primes, which is what makes its growth rate delicate.

G was computed exactly to $H = 3000$ (4-tuple constants by Euler products to 2×10^4): $G(3000) = -20.9$, with local logarithmic slope increasing from 3.4 to 3.7 across the range—seven times the single-prime slope $\frac{1}{2}$, consistent with (ii) over the computable range while leaving the limiting form undetermined (a quadratic-in-log fit is unstable on subranges). The window measurement brackets the extrapolation: at $H = 10^5$, $x \in [10^8, 2 \times 10^8]$ the pure-log form predicts $\mathrm{Var}/\mathbb{E} = 0.97$, the $\log^2$ form 0.72, and the observed value is 0.815—intermediate growth, exactly where the computable range points. (Caldwell–Gallot's factorial-prime law $\#\{n \le N : n!+1 \text{ prime}\} \sim e^{\gamma} \log N$ [4], the divergent single-sided companion of Conjecture 21, is included as an attributed calibration benchmark: 15 primes to $n = 700$ against 11.7, $z = +0.98$—a probable-prime count, $n! + 1$ leaving the deterministic range at $n \approx 26$.)

Computational checks. The pinned singular-series computation and window comparison are reported at the statement; its factorial benchmark stands at 15 primes with $n \le 700$ against 11.7 ($z = +0.98$; a probable-prime count for n beyond about 26), the divergent single-sided companion of Conjecture 21's convergent side.

Conjecture 3 (Least primes in progressions; (a) for part (i) [20, 21, 30, 29], (b) for part (ii))**.** *Let $p(a,q)$ be the least prime $\equiv a \pmod q$ and $U(a,q) = \mathrm{Li}(p(a,q))/\varphi(q)$. Then:* (i) $U \Rightarrow \mathrm{Exp}(1)$ *marginally as $q \to \infty$; and, under the additional hypothesis that the class processes are asymptotically jointly independent in the extreme-value sense (which marginal convergence alone does not supply), $\max_a U - H_{\varphi(q)}$ converges to a Gumbel law ($H_n = \sum_{k \le n} 1/k$), the order-statistics form of Wagstaff's $\max_a p(a,q) \sim \varphi(q) \log^2 q$;* (ii) *[canonical deficit, dyadic-average form] define, with no reference to any control model, $\Theta(q) = \big(1 - \mathbb{E}_a[U(a,q)]\big) \log q$. The conjecture is stated for dyadic averages over prime moduli—pointwise convergence along every prime q may be too strong, since arithmetic fluctuations in q could persist:*

$$\frac{1}{\#\{q \in [Q,2Q] \text{ prime}\}} \sum_{\substack{Q \le q \le 2Q \\ q \text{ prime}}} \Theta(q) \;\longrightarrow\; \Theta \;>\; 0 \qquad (Q \to \infty).$$

(The formal claim is model-free: existence and strict positivity of the dyadic-average limit. The

stronger reading—that Θ exceeds the discreteness level any parity-aware control produces—is the diagnostic layer below, not part of the display, since a control constant has no place in a model-free statement.) The limiting distribution *of $\Theta(q)$ over the dyadic block is left open, and in particular we do* not *assert* $\mathrm{sd}(\Theta(q)) \to 0$*: fluctuations driven by the arithmetic of q and of $q-1$ (through the singular-series pair terms $\mathfrak{S}(kq)$ and the class structure mod q) may well persist at bounded size, so $\Theta(q)$ need not concentrate around its dyadic mean.*

The decomposition $\Theta(q) = \theta_{\mathrm{disc}}(q) + \theta_{\mathrm{corr}}(q)$ is a *diagnostic*, not a canonical invariant: its two parts move if the benchmark moves, and clause (ii) is therefore stated through the model-free $\Theta(q)$ alone. The named benchmark (the parity-aware Cramér–Bernoulli control: candidates a, $a+q$, $a+2q, \ldots$ independently "prime" with hazard $h_i = (q/\varphi(q)) \cdot s_i / \log(a+iq)$, $s_i = 2$ for odd candidates and 0 for even, matching density and parity) defines $\theta_{\mathrm{disc}}(q) = (1 - \mathbb{E}_a[U^{\mathrm{model}}]) \log q$ exactly for each q, and $\theta_{\mathrm{corr}} := \Theta - \theta_{\mathrm{disc}}$ is the part of the measured deficit that the control cannot produce; it is this diagnostic that localizes the anomaly in the *ordering* of the primes (Remark 1). Measured on prime moduli $q \in [1500, 6000]$: $\Theta = 1.671 \pm 0.009$, of which the control accounts for 0.847.

One further deterministic baseline must be subtracted before any of the residue is called an anomaly: for prime q, the primes $p < q$ occupy *distinct* classes mod q—an injective initial phase that no exchangeable allocation reproduces. Its size has a closed form in index time (where $\mathrm{Li}(p_i) \approx i$, so U is the first-hit index over φ): forcing the first $\pi(q)$ arrivals to be collision-free and leaving the rest exchangeable gives $\mathbb{E}[U] = 1 - \pi(q)^2/2\varphi^2 + O(\pi(q)/\varphi^2)$, hence

$$\Theta_{\mathrm{inj}}(q) \;=\; \frac{\pi(q)^2}{2\varphi(q)^2}\, \log q\; (1 + o(1)) \;=\; \frac{1 + o(1)}{2 \log q},$$

closed form $0.092,\ 0.082,\ 0.074$ at $q = 1499,\ 3001,\ 5987$, with Monte Carlo agreeing within sampling error and the exchangeable control at 1.000. The injective phase is therefore real but an order of magnitude too small to be the explanation: it accounts for $\approx$0.08 of the measured $\theta_{\mathrm{corr}} = 0.824 \pm 0.009$, and carries a $1/\log q$ decay that the flat measured values do not show. (One might expect the effect to sit at the full deficit scale, since the *affected-class fraction* is $\pi(q)/\varphi \asymp 1/\log q$; but the deficit is the affected fraction times the collision probability an exchangeable model would have suffered in that phase, $\asymp \pi(q)/2\varphi$, giving $1/2\log^2 q$ in $\mathbb{E}[U]$, not $1/\log q$.) The injective phase is in any case the $y < q$ head of the pair-correlation expansion already registered below—collisions require $q \mid p_2 - p_1$, impossible before $y = q$—so subtracting it is the first term of the stated programme, not a rival explanation. The residue $\theta_{\mathrm{corr}} - \Theta_{\mathrm{inj}} \approx 0.74$ remains the unexplained ordering anomaly.

A derivation route for the deficit, which we adopt as the conjecture's stated programme, runs as follows. Let $V_q(y)$ be the number of reduced classes mod q not yet visited by a prime $\le y$; then $\mathbb{E}_a[U] = \int_0^\infty \overline{V_q(y_\tau)}/\varphi(q)\, d\tau$ with the Li-time change y_τ defined by $\mathrm{Li}(y_\tau) = \tau\varphi(q)$, and inclusion–exclusion expands $V_q(y)$ in prime-tuple correlations *within* a class:

$$\mathbb{E}\binom{N_a}{2} \text{ summed over } a \;=\; \#\{p_1 < p_2 \le y :\; q \mid p_2 - p_1\} \;\approx\; \sum_{1 \le k \le y/q} \mathfrak{S}(kq) \int_2^{y-kq} \frac{dt}{\log t\, \log(t + kq)},$$

so the first correction to the exchangeable occupancy model is carried by the Hardy–Littlewood singular series of gaps divisible by q. The truncation is structural, not a convenience: a gap kq requires $p_1 \le y - kq$, so the k-sum stops at y/q and the endpoint-shortened integrals supply the triangular weights, while the untruncated sum $\sum_{k \ge 1} \mathfrak{S}(kq)$ diverges (the singular series has mean one on average over its argument) and the formally factorized expression $\mathrm{Li}_2(y)\sum_k \mathfrak{S}(kq)$ is meaningless. Evaluating this expansion to second order, and comparing it with the measured Θ, is the concrete open computation that would either derive the ordering anomaly from pair correlations or prove it lies deeper; either outcome retires Question 1's mystery one level.

The stratified experiment behind these numbers separates moduli by factorization type in the matched range $q \in [1500, 6000]$: over 40 *prime* moduli, $\theta_{\mathrm{disc}} = 0.847$ (essentially constant across q) and $\theta_{\mathrm{corr}} = 0.824 \pm 0.009$; over 40 7-*smooth* moduli in the same range, $\theta_{\mathrm{disc}} = 4.32 \pm 0.22$ (the large $q/\varphi(q)$ hazards make discreteness dominant) while $\theta_{\mathrm{corr}} = -2.32 \pm 0.22$ reverses sign. The comparison with Leung [29] is thereby given its answer-shaped experiment: Leung derives the exponential law under a uniform Hardy–Littlewood hypothesis and reports discrepancies for *smooth* moduli, and our smooth stratum indeed behaves anomalously—but the positive ordering term survives, with small error bars, exactly where smooth-modulus effects cannot reach. Whether θ_{corr} tends to a constant, and the exact functional form in $1/\log q$, remain open: data at $q \leq 6000$ cannot distinguish $1/\log q$ from $(\log\log q)/\log q$-type refinements—the aggregate $\Theta \approx 1.5$–1.9 over $q \in [10^2, 6 \times 10^3]$, with $\mathbb{E}_a[U]$ bottoming near 0.737 around $q \approx 200$ and recovering through 0.776 by $q \approx 6000$, is calibration, not a determination of the form. The averaging family is fixed as: all moduli of the stated type in dyadic ranges of q, per stratum.

Remark 1. *The deficit is reproducible and is* not *an artefact of the time-change: a Cramér pseudo-prime control (independent Bernoulli "primes" with the correct densities and parity) shows roughly half of it—the discreteness of large per-candidate hazards—while the other half vanishes under any reshuffling of the actual prime residues (iid labels, or a random permutation of the true residue sequence, both restore* $\mathbb{E}[U] = 1.00$*). The ordered sequence of primes fills residue classes measurably faster than any exchangeable model, and the stratified experiment above shows the effect is not a smooth-modulus artefact: it is largest and cleanest on prime moduli. We measure* θ_{corr}*; we cannot yet derive it.*

Conjecture 4 (The Goldbach lane race; apparently new)**.** *For* $n \equiv 2 \pmod 4$ *let* $R_1(n)$, $R_3(n)$ *be the ordered Goldbach representation counts in the lanes* $p \equiv q \equiv 1$ *and* $p \equiv q \equiv 3 \pmod 4$ *(the lane convention of the Martin benchmark recorded at Conjecture 5), and* $D(n) = R_3(n) - R_1(n)$*. Prime-square terms* $n = q^2 + p$ *(*q *odd prime) land* exclusively *in the* $(1,1)$ *lane, since* $q^2 \equiv 1 \pmod 4$ *and then* $n - q^2 \equiv 1 \pmod 4$ *automatically. Consequently the* $(3,3)$ *lane leads on average. Define*

$$D_{\mathrm{sys}}(n) \;=\; \frac{2}{\bar{\ell}(n)} \sum_{\substack{q \text{ odd prime},\ q \leq \sqrt{n} \\ n - q^2 \text{ prime}}} \log q \, \log(n - q^2), \qquad \bar{\ell}(n) \;=\; \frac{n-6}{\displaystyle\int_3^{n-3} \frac{dt}{\log t \, \log(n-t)}},$$

(the analytic mean of $\log p \log(n-p)$ *under the lane profile; the empirical lane mean is its estimator). Write* $\mathbb{E}_x$ *for the average over* n *drawn from* $\{n \equiv 2\ (4),\ n \leq x\}$ *with weight proportional to* $1/n$ *(the logarithmic sampling measure, now explicit). Then:* (i) *[drift law]* $\mathbb{E}_x[D - D_{\mathrm{sys}}] = o(\mathbb{E}_x[D_{\mathrm{sys}}])$ *as* $x \to \infty$*: the* $(3,3)$ *lane leads on average by exactly the square contamination;* (ii) *[weighted mean and internal null lane]* $\mathbb{E}_x[D_{\mathrm{sys}}]$ *is given by the Hardy–Littlewood (Conjecture H) local model for* $n - q^2$*, whose local factors vary strongly with* n*—most drastically at* $p = 3$*: for* $n \equiv 1 \pmod 3$*,* $3 \mid n - q^2$ *for* every *prime* $q > 3$*, so the contamination collapses to a single boundary family, the* $q = 3$ *term (when* $n - 9$ *is prime): the complementary possibility* $n - q^2 = 3$ *is empty on this ensemble, since* $n \equiv 2 \pmod 4$ *forces* $n - 3 \equiv 3 \pmod 4$*, never a square. The family is negligible in logarithmic mean, so the race carries no drift at*

the systematic scale on the subprogression $n \equiv 10 \pmod{12}$*: with the positive comparison scale* $A_G(x) := \mathbb{E}_x^{(\not\equiv 1\,(3))}[D_{\mathrm{sys}}]$*, the logarithmic-ensemble mean of the systematic term over the contaminated lane* $n \not\equiv 1 \pmod 3$ *(a comparison against this subprogression's own systematic term, which vanishes, would be empty), the claim is* $\mathbb{E}_x[D] = o(A_G(x))$ *on* $n \equiv 10 \pmod{12}$*, while the drift concentrates on* $n \not\equiv 1 \pmod 3$*: an internal null lane, predicted by the same mechanism that predicts the lead;* (iii) *[sign density] the per-*n *drift-to-noise ratio* $\kappa(n) = D_{\mathrm{sys}}(n)/\sqrt{\mathfrak{S}(n)J(n)}$ *(where* $J(n) = \int_3^{n-3} dt/(\log t \log(n-t))$*, so that* $\mathfrak{S}(n)J(n)$ *is the total ordered representation count) satisfies* $\kappa(n) \asymp c(n)/\log n \to 0$*; under the Gaussian fluctuation model the logarithmic density of* $\{D > 0\}$ *is therefore* $\frac{1}{2}$ *in the limit, with the finite-*x *sign fraction predicted to be* $\mathbb{E}_x[\Phi(\kappa)]$ *(*Φ *the standard normal distribution function)—decaying, but far from* $\frac{1}{2}$ *at any accessible height.*

Derivation status: the identity behind D_{sys} is the Λ-weighted lane symmetry minus its prime-power part; the transfer to unweighted counts is a partial-summation argument whose error terms (higher prime powers, which enter only at the $n^{1/3}$ scale, and the variation of the weight across the lane) remain to be written out—the same open transfer flagged at Conjecture 1—and clause (iii) inherits the zero-oscillation caveat stated there. One further disclosure: the census D_{sys} is computed from the primes in the same sample it is compared against, so the level agreement is in-sample; the out-of-sample content of the verification is the null lane and the class assignment, which the mechanism fixes in advance.

This is the Goldbach-partition analogue of Chebyshev's bias, produced by the same explicit-formula mechanism as Conjecture 1 but through $q^2 + p$ patterns rather than twin patterns; the single-sign prediction (every square contamination falls into one lane) makes it cleaner than the classical race. We could find no prior statement. The neighbouring phenomenon for consecutive primes is the Lemke Oliver–Soundararajan mod-3 law, whose verification here—$\hat{c}$ drifting $0.400 \to 0.370$ over $10^7 \to 4 \times 10^9$ with $(1,1)/(2,2)$ symmetry 0.99989—serves as calibration for the present race.

Computational checks. Over 500 log-spaced samples $n \le 10^8$: mean $D = 76.5$ against mean empirical $D_{\mathrm{sys}} = 64.5$, the $(3,3)$-lane lead significant at 5.2 standard errors of the mean. Clause (ii)'s weighted model, computed in presieve-exact form (trial division of each $n - q^2$ to 10^3 with the exact Mertens normalization), reproduces the empirical contamination with no fitted parameter: model mean 66.3 against empirical 67.8 (ratio 0.98) on a 400-sample run. The internal null lane behaves as predicted: on $n \equiv 1$ (3) (123 samples) the model and empirical D_{sys} both vanish and the measured lead drops to 31 ± 23 (consistent with zero), while on $n \not\equiv 1$ (3) (277 samples) the lead is 88 ± 21 against predicted contamination 96—the drift lives exactly where the mechanism puts it. Clause (iii): computed $\mathbb{E}_x[\Phi(\kappa)] = 0.572$ (κ ranging 0–0.64 with mean 0.18) against observed sign fraction 0.588; weak, and quantitatively the weakness the model requires.

Conjecture 5 (The least Goldbach summand: exponential law and ordering deficit; the time-changed law and deficit constant apparently unstated)**.** *Let* $s(n)$ *be the least prime* $p \nmid n$ *with* $n - p$ *prime (the exclusion* $p \nmid n$ *removes a trivial obstruction:* $p \mid n$ *forces* $p \mid n - p$*, so* $n - p$ *is composite except in the diagonal case* $n = 2p$*, which would otherwise distort the hazard at one atypical point), and time-change by the expected-arrivals clock* $U(n) = \mathfrak{S}(n) \sum_{p \le s(n),\, p\nmid n} 1/\log(n-p)$*. Here* $s(n) = +\infty$ *when no admissible prime exists, which among even* n *happens only at* $n = 4$ *and* $n = 6$ *and conjecturally at no larger even* n*, so the law below runs over the even* n *with* $s(n)$ *finite. The ensemble is fixed first, since* $U(n)$ *is deterministic at each* n *and an expectation* *therefore needs a declared measure: for a scale X,*

$$\mathbb{E}_X U = \frac{\sum_{X<n\le 2X,\ 2|n} U(n)/n}{\sum_{X<n\le 2X,\ 2|n} 1/n}, \qquad \Theta_G(X) = (1-\mathbb{E}_X U)\log X.$$

Then: (i) *[limit law] under $\mathbb{E}_X$-sampling, $U \Rightarrow \mathrm{Exp}(1)$ as $X\to\infty$;* (ii) *[canonical ordering deficit] $\Theta_G(X)\to\Theta_G>0$—the Goldbach sibling of the least-prime deficit of Conjecture 3(ii), produced by the same occupancy mechanism: the ordered primes fill the available representations faster than an exchangeable model, so the first arrival comes* early *and $\mathbb{E}_X[U]<1$.*

The extremal theory of $s(n)$ is well developed—Granville, van de Lune and te Riele conjectured $s(n) = O(\log^2 n \log\log n)$, and Oliveira e Silva–Herzog–Pardi [12] tabulate first occurrences of minimal Goldbach summands to 4×10^{18} against prime k-tuple predictions—but the time-changed distributional law (i) and the deficit constant (ii) appear unstated (we could not rule out that the untransformed version of (i) is implicit in the comparisons of [12]). Measured on 2,000 log-sampled $n \le 10^8$: $\mathbb{E}[U] = 0.796$, with $\Theta_G = 3.12\pm 0.35$ on $[10^6, 10^7)$ and 3.42 ± 0.45 on $[10^7, 10^8)$—positive, stable, and roughly *twice* the least-prime-in-progressions deficit $\Theta = 1.67$ of Conjecture 3, a comparison the occupancy expansion there should eventually explain; the Kolmogorov–Smirnov distance to Exp(1) (0.088 at this size) is dominated by the deficit displacement itself, as (ii) requires. The derivation route for Θ_G, parallel to the occupancy expansion at Conjecture 3: expand the survival probability $\Pr(s(n)>y) = 1-\sum_{p\le y} h_p + \sum_{p_1<p_2\le y} h_{p_1,p_2} - \cdots$, where the pair term requires $n-p_1$ and $n-p_2$ simultaneously prime and so carries the Hardy–Littlewood singular series of the shift p_2-p_1; inserting the k-tuple predictions and integrating against the time change isolates the $1/\log X$ coefficient—deriving Θ_G rather than measuring it is this conjecture's stated programme. $\mathfrak{S}(n)$ alone does not encode the between-candidate dependence that the pair term carries. (The $(3,3)$-lane refined Goldbach statement—every $n\equiv 2\ (4)$, $n\ge 6$, is $p+q$ with $p\equiv q\equiv 3\ (4)$, with the halved singular-series count law, formulated by K. Martin [13]—is included as an attributed calibration benchmark: exhaustively verified to 10^9, count formula matching to 0.2–1.3%; its comparative lane statement is Conjecture 4.)

For sequences whose nth term has size e^{cn} the expected number of primes is $\sum_n \kappa_n/(cn)$: convergence or divergence of this sum is the entire question, and when it diverges it diverges logarithmically, so the right conjecture has counting function $\alpha\log x$ with a derived α.

Conjecture 6 (n^2+2^n; sequence is OEIS A064539)**.** *Every prime of the form n^2+2^n with $n>1$ has $n\equiv 3 \pmod 6$ (elementary), and:* (i) *infinitely many such primes exist, with counting function $\asymp \log N$;* (ii) *[candidate law, entanglement-aware form] for a finite set S of primes $p\ge 5$ (the primes 2 and 3 are exact in the lane reduction, $\delta_2=\delta_3=0$, and $\mathrm{ord}_2 2$ is undefined) let D_S be the* exact *joint density, within the lane $n\equiv 3\ (6)$, of n with $p\nmid n^2+2^n$ for all $p\in S$, computed over the full joint period $\mathrm{lcm}_{p\in S}\,\mathrm{lcm}(6,\, p\cdot\mathrm{ord}_p 2)$, and set $\kappa_S = 3\,D_S/\prod_{p\in S}(1-1/p)$: the factor $3 = 2\cdot\frac{3}{2}$ is the exact lane bonus at the primes 2 and 3 (within the lane $\delta_2=\delta_3=0$, so each contributes its full $(1-1/p)^{-1}$), included in the definition so that κ_S is the complete local constant of the count in (ii-c). The conjecture is stated along the* canonical exhaustion *$S_z = \{p \text{ prime} : 5\le p\le z\}$, and its three layers are logically separate and are asserted separately:* (ii-a) *the sequence κ_{S_z} converges as $z\to\infty$ (a general net over arbitrary finite S carries an order-of-exhaustion ambiguity we do not need), with limit κ;* (ii-b) *whether the limit* factors*—i.e. whether the exact finite-level factorization observed below persists, so that entanglement is asymptotically absent—is an* open question*, expressly not counted among this paper's conjectural assertions, on which (ii-a) takes no position; and* (ii-c) *the count is $\sim \sum_{n\le N,\, n\equiv 3\,(6)} \kappa/\log(n^2+2^n)$—a genuine further step, since* *a convergent local density does not automatically govern the global count. The working value $\kappa^* = 4.2734\ldots$ is a* hybrid*, not a computation of any single κ_{S_z}: the lane factor 3 times the CRT-exact core S_{19} times independent per-prime factors for $19 < p \le 300$—so $\kappa^* = \kappa_{S_{300}}$ exactly if and only if the observed exact factorization persists to 300.*

This is not a Bateman–Horn problem: the local conditions at different primes are tied through the orders $\operatorname{ord}_p 2$, so the per-prime product $\prod_p (1-\delta_p)/(1-1/p)$ would tacitly assume their independence. The law is therefore stated through the CRT-exact quantities κ_S, which make no independence assumption. The computation then returned a small surprise in the other direction: through $p \le 19$ (joint period 116,396,280) the joint survivor fraction equals the product of the per-prime fractions as an *exact rational identity*, verified in exact arithmetic at every level $S = \{5 \le p \le P\}$, $P \in \{5, 7, 11, 13, 17, 19\}$. Entanglement is thus absent at all computed levels; whether exact factorization persists for all p—equivalently, whether the survival events are exactly independent over every joint period, which would itself be a nontrivial equidistribution statement about the orbits of 2—is part of what clause (ii) asks. We separate the robust claim (i) from the candidate law (ii), and note that the eight hits below 6000 are far too few to validate a four-digit constant—the verification below checks consistency, not the constant. The OEIS records further candidate indices beyond our completed range: 29355, 34653, 57285, 99069, and the probable-prime candidate 1933695. We could not independently re-verify these, but all five satisfy the lane constraint $n \equiv 3$ (6), and taking them at face value the κ^*-model predicts 2.9 hits in $(6 \times 10^3, 10^5]$ against 4 reported and 5.9 in $(6 \times 10^3, 1.94 \times 10^6]$ against 5 ($z = +0.65$ and -0.38): the records extend the consistency check by two decades.

The congruence claim is elementary: n odd is forced by parity, and for odd n with $3 \nmid n$ we have $n^2 + 2^n \equiv 1 + 2 \equiv 0 \pmod 3$. What appears to be a harsh obstruction is, inside the surviving lane, a local bonus: $\delta_2 = \delta_3 = 0$ contribute the factors $2 \cdot \frac{3}{2} = 3$ carried explicitly in the definition of κ_S.

Computational checks (probable-prime count beyond $n \approx 80$). Hits $\{3, 9, 15, 21, 33, 2007, 2127, 3759\}$ up to 6000: observed 8, model 8.55, $z = -0.19$ (over the shorter range $n \le 4200$: 8 vs 8.19, $z = -0.07$); the lane constraint was verified exhaustively (no prime off $n \equiv 3$ (6) to $n = 3000$, independently reconfirmed), and the CRT-exact factorization of clause (ii) was verified through $p \le 19$ as described at the statement.

Conjecture 7 (Fermat quotients: the multibase subtorus law and the Wieferich ledger; single-base heuristic is (a) [22], the vertical multibase law apparently new)**.** *With $q_p(a) = (a^{p-1}-1)/p \bmod p$ (defined for primes $p \nmid a$; all clauses below range over such p only) and $q_p = q_p(2)$, clauses* (0)–(v)*, stated multibase group first:* (0) *[structure, classical] $q_p(ab) \equiv q_p(a) + q_p(b) \pmod p$ (the Eisenstein–Lerch homomorphism; verified exactly on every tested prime), so for multiplicatively* dependent *bases the vector $(q_p(a_1), \ldots, q_p(a_r))/p$ is confined to the rational subtorus cut out by the relations;* (iv) *[multibase equidistribution, the claimed law] for multiplicatively* independent *bases the vector equidistributes on the full torus as p varies—the* vertical *joint law, complementing the fixed-p, varying-base statistics of Ostafe–Shparlinski and Cobeli–Zaharescu—with correlations vanishing and the same LIL calibration as (i) below;* (v) *[simultaneous Wieferich] the expected count of $p \le x$ with $q_p(2) = q_p(3) = 0$ has convergent sum $\sum 1/p^2$: at most finitely many simultaneous Wieferich primes exist (the single-base folklore of this accounting is Conrad's), and the empirical list is empty to 10^7. Three further clauses on the base-2 quotient alone:* (i) *[global equidistribution] q_p/p equidistributes on $[0, 1)$ with discrepancy at exactly the calibrated* *random-model scale: in its sharp form,*

$$\limsup_{x\to\infty} \frac{\sqrt{\pi(x)}\, D_{\mathrm{KS}}(x)}{\sqrt{\log\log \pi(x)}} \;=\; \frac{1}{\sqrt{2}},$$

the Chung–Smirnov law-of-the-iterated-logarithm constant for an i.i.d. uniform sequence—demanding both no more (a bound $O(\pi(x)^{-1/2})$ would be stronger than the random model supports*, the same over-demand that killed the Mertens conjecture) and no less than the model earns: the upper bound alone would be consistent with hidden rigidity, and the matching lower bound asserts the quotients fluctuate like genuine noise;* (ii) *[shrinking targets] for every fixed $K \geq 1$, and uniformly for $K = K(x) \leq \log x$, $\#\{p \leq x :\ q_p < K\} \sim \sum_{p\leq x} \min(1, K/p)$—the fixed-$K$ case down to $K = 1$ is what governs the zero event and does NOT follow from (i), since $\{q_p = 0\}$ is a target of shrinking measure $1/p$;* (iii) *[Wieferich count, fluctuation-calibrated, with the probability space declared] $W(x) = \#\{$ Wieferich $p \leq x\}$ is a deterministic function, so its fluctuation clauses are stated in the two legitimate forms, an unqualified CLT for a fixed sequence having no meaning: the* deterministic envelope conjecture

$$\limsup_{x\to\infty} \frac{W(x) - \log\log x}{\sqrt{2\log\log x\, \log\log\log\log x}} \;=\; 1$$

(the iterated-logarithm scale at variance $V(x) \sim \log\log x$; a bound $O_\varepsilon(V^{1/2+\varepsilon})$ would be weaker than the model's exact prediction), and the almost-sure central limit law, at the correct logarithmic weighting*: with $u = \log\log x$ the variance-time of the count, sample u with density du/u on $[\mathrm{e}, L]$, $L = \log\log X$ (equivalently: $\log u = \log\log\log x$ uniform); then the sampled distribution of $(W(x) - u)/\sqrt{u}$ converges to $N(0, 1)$ as $X \to \infty$. The weighting is what makes the clause true, and the two natural alternatives both fail. A block-sampled version on $[X, X^A]$ fails because only* $\log A = O(1)$ *new events enter the block. A* uniform*-in-u version on $[1, L]$ fails too, by Brownian scaling: with $W(\mathrm{e}^{\mathrm{e}^u}) - u \approx B(u)$ and $u = Ls$, the self-similarity $B(Ls)/\sqrt{Ls} \overset{d}{=} \widetilde{B}(s)/\sqrt{s}$ means the uniform-u empirical distribution converges not to Φ but to the* random *occupation functional $\int_0^1 \mathbf{1}\{\widetilde{B}(s)/\sqrt{s} \leq z\}\, ds$—values at proportional times stay correlated, so uniform sampling never averages over enough independent scales. The du/u weighting is the classical almost-sure-CLT weighting (the continuous analogue of the $1/k$ weights), for which the model does predict almost-sure convergence of the sampled law to Φ. Both clauses are conjectured on the Crandall–Dilcher–Pomerance side of the model choice that clause (iv)'s discussion records.*

Two things do not follow. Clause (iii) is not a consequence of (i): a square-root-size error in the aggregate distribution is astronomically larger than $\log\log x$, and only the shrinking-target clause (ii) at bounded K speaks to the zero event. And an $O(1)$ fluctuation claim would violate this paper's calibration principle of demanding exactly the fluctuations the model earns: with event probabilities $1/p$ the count has standard deviation $\sim \sqrt{\log\log x}$. Nor is any location predicted for a third Wieferich prime: the expected count reaching 3 "at doubly exponential heights" is an expected-count statement whose additive constant is uncalibrated, not a forecast. Consistency data: $\{1093, 3511\}$ below 10^8 against expected ≈ 2.7; $\sqrt{n}\,\mathrm{KS} = 0.82,\ 0.82,\ 0.52,\ 0.63$ at $x = 10^5, \ldots, 10^8$ (bounded, no drift); census at $K = 100$: observed 169 against model 161.2.

The multibase clauses were verified at 10^7 (664,577 primes): the homomorphism (0) held *exactly* on every spot-checked prime; for the independent pair $(2, 3)$, correlation $+0.0023 \pm 0.0012$, a 20×20 joint occupancy χ^2 of 441 on 399 degrees of freedom ($+1.5\sigma$), joint small-quotient census 46 against model 41.2 ($z = +0.75$), and no simultaneous Wieferich prime. Two positions are taken here: the vertical multibase law (iv) sides with the Crandall–Dilcher–Pomerance accounting against Gras's competing probabilistic model (which predicts $q_p(a) = 0$ *rarer* than $1/p$, hence possibly finitely many Wieferich primes even for a single base)—the disagreement is what gives clauses (iii)–(iv) content; and the horizontal literature (fixed p, base varying: Ostafe–Shparlinski's joint distributions and the Fermat-quotient-matrix statistics of Cobeli–Zaharescu and coauthors) is the attributed neighbour whose vertical counterpart is what this conjecture claims.

Conjecture 8 (Uniform quantitative de Polignac; (a)-core)**.** (i) $\pi_d(x) = \#\{p \le x - d : p,\ p+d \text{ prime}\} = \mathfrak{S}(d)\ \mathrm{Li}_2(x)\,(1+o(1))$ *uniformly for even* $d \le (\log x)^A$. (ii) *[residual field, with derived covariance kernel] The same primes participate in many pair counts, and the covariance of two of them is carried by triple configurations sharing one prime. For even $d \neq d'$ define the overlap constant*

$$K(d,d') = C_3(0,d,d') + C_3(0,d',d+d') + C_3(0,d,d+d') + C_3(0,\,|d-d'|,\,\max(d,d')),$$

where C_3 denotes the Hardy–Littlewood triple constant of the offset set (degenerate or inadmissible triples contributing 0*). The field is randomized canonically—covariances of the deterministic cumulative counts are otherwise undefined: for t uniform on $[x, 2x]$ and a window length H, let $\pi_d(t;H) = \#\{t < n \le t+H :\ n,\, n+d \text{ prime}\}$ and let $Z_d(t;H)$ be its studentization. The mean count per window is $\asymp H/\log^2 x$, and the regime is set by that ratio (a phase condition: a bare $H \to \infty$ does not fix the regime):*

$$\mathrm{Cov}_t(\pi_d, \pi_{d'}) \sim \frac{K(d,d')\,H}{\log^3 x}, \qquad \rho(d,d') \sim \frac{K(d,d')}{\sqrt{\mathfrak{S}(d)\mathfrak{S}(d')}\,\log x},$$

valid for $\max(d,d') = o(H)$*, which is the standing restriction of this clause and holds throughout the window class fixed below. In general each overlap orientation o—the in-window displacements $h_o \in \{0,\ d,\ -d',\ d-d'\}$ and their reflections—enters with multiplicity $(H - |h_o|)_+$ rather than H, so that the shared-prime term reads*

$$\frac{1}{\log^3 x}\sum_o (H - |h_o|)_+\, K_3^{(o)}(d,d'), \qquad \sum_o K_3^{(o)}(d,d') = K(d,d'),$$

and an orientation with $|h_o| > H$ is absent from the window altogether; under $\max(d,d') = o(H)$ *one has $(H - |h_o|)_+ = H(1+o(1))$ for every orientation and the displayed form is recovered, so the uniform-in-d statement carries that restriction throughout. Here the shared-prime (triple) term is the* first *covariance contribution but not the whole of it: the full expansion is*

$$\mathrm{Cov}_t(\pi_d, \pi_{d'}) = \frac{K(d,d')\,H}{\log^3 x} + \frac{1}{\log^4 x}\sum_{|h| \le H} (H - |h|)\,K_4(d,d';h) + \textit{error},$$

with K_4 the centred singular series of the off-index four-point sets $\{0, d, h, h+d'\}$—the cross analogue of the pinned diagonal average of Conjecture 2—where degenerate offset relations are handled inside the kernels: whenever the four-point set collapses (repeated entries, as at $h = 0,\ h = d,\ h = -d'$, or $d' - d = \pm h$) the configuration is a triple or pair and its contribution belongs to the K-term (or to the mean), so K_4 is defined as the centred singular series of genuinely four-element *sets, with special pairs such as $d' = 2d$ acquiring their extra overlap orientations in $K(d,d')$'s census rather than double-counted across kernels. The triple term dominates only where the pinned sum satisfies $\sum(1 - |h|/H)K_4 = o(\log x)$; the dominance range* *is stated as an explicit function class: for $H = \lambda(\log x)^\alpha$, any fixed $\lambda > 0$ and any $\alpha \ge 1$ with $\alpha > A$—the inequality on the exponents that makes $\max(d, d') = o(H)$ hold uniformly for all $d, d' \le (\log x)^A$—the per-shift centred average grows only polylogarithmically in H (the Conjecture-2 scale), so $\sum(1 - |h|/H)K_4/H = O((\log\log x)^{2+\varepsilon}) = o(\log x)$ and the triple term dominates throughout the polylogarithmic window class; the conjecture asserts the two-term form on exactly that class and takes no position outside it. When $H/\log^2 x \to \lambda$ the window counts converge jointly to Poisson laws with these vanishing covariances, and when $H/\log^2 x \to \infty$ every fixed finite collection $\{Z_{d_1}, \dots, Z_{d_r}\}$ is asymptotically jointly normal with* independent *limiting coordinates: both kernels are finite-x corrections to independence (test below), not nondegenerate limiting kernels. Both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances, namely uniform Hardy–Littlewood tuple estimates on the window class fixed above, strong enough to control all higher joint cumulants of the window counts: every joint cumulant of order at least three vanishes in the respective normalizations. The diagonal correction to* $\mathrm{Var}_t(\pi_d)$ *is Conjecture 2's pinned average, stated there.*

The kernel is verified twice over. Direct window-field measurement ($x = 10^8$, $H = 10^5$, 2000 windows, all even $d \le 40$, 190 pairs): the empirical correlation matrix matches the predicted kernel entrywise with correlation 0.86, mean off-diagonal 0.33 observed against 0.37 predicted, the small deficit having the sign of the omitted diagonal corrections, which are also directly visible as $\mathrm{Var}/\mathrm{mean} = 0.94$ on average across d. Second, the deterministic cumulative profile $z_d(x) = (\pi_d(x) - \mathfrak{S}(d)\,\mathrm{Li}_2(x))/\sqrt{\mathfrak{S}(d)\,\mathrm{Li}_2(x)}$—the estimator the uniformity verification in (i) uses—shows the same structure at cumulative scale (there $\rho \sim K\,\mathrm{Li}_3/\sqrt{\mathfrak{S}\mathfrak{S}'}\,\mathrm{Li}_2$): mean off-diagonal $\rho = 0.43$ over all even $d, d' \le 60$ at $x = 10^8$, hence predicted profile spread $\sigma^2 \le 1 - 0.41 = 0.59$ before the diagonal correction, against observed $\approx$0.27, implying a diagonal factor $\approx$0.5 of Montgomery–Soundararajan sign and order.

The force of the statement is its uniformity: the singular series takes a thousand different values over $d \le 2000$, spanning a factor of about three between $d = 2$ and the primorial-rich d, and the counts must reproduce the entire profile simultaneously with square-root errors. A conspiracy would need to bend a thousand-dimensional vector, not one number. At $x = 10^8$: correlation 0.999999, slope 0.99966, $\max_d |z_d| = 1.80$; stressed to $d \le 6000$, $\max |z_d| = 2.01$ over three thousand values.

(A further Hardy–Littlewood calibration, on a prime quintuplet constellation, gives observed 15,236 against predicted 15,230.3 at 4×10^9, ratio 1.0004, $z = +0.05$.)

Conjecture 9 (The sub-diffusivity of balanced prime races; apparently new)**.** *Model a balanced race (Conjecture 1(ii)'s symmetric class differences) as the walk of its class-assignment steps, the process defined at event index: enumerate the pattern occurrences $n_1 < n_2 < \cdots$ in the two competing classes, set $\xi_i = \pm 1$ by class membership of n_i, and let $\rho_k(x)$ denote the lag-k sample autocorrelation of (ξ_i) over the events with $n_i \le x$. Then the walk is measurably* sub-diffusive*:* (i) *[decaying-repulsion law] the step autocorrelations are negative at small lags—the race analogue of the Lemke Oliver–Soundararajan consecutive-pattern repulsion [24], which supplies the mechanism—and, since that repulsion is scale-dependent rather than fixed, the conjectured form is a decay law*

$$\rho_k(x) \;=\; -\,\frac{c_k \log\log x + d_k}{\log x}\,(1 + o(1)),$$

with constants c_k, d_k in principle derivable from the LOS correlation series (underived here; and the measured values $\rho_1 \approx -0.037$, $\rho_2 \approx -0.009$ at 10^9*—universal across twin and cousin races* *mod* 5 *and* 8 *to within* 0.004*, each* $\sim$50 *standard errors from zero—are observations at one height, consistent with* $c_1 \approx 0.25$ *there, not asymptotic constants);* (ii) *[long memory, measured] short lags do not exhaust the deficit: the running maxima* $M(x) = \max_{t \le x} |D(t)|/\sqrt{count}$ *of the four races land at quantiles* 0.02–0.17 *of the simulated independent-step null (median* 2.34 *at this span), and still only* 0.03–0.23 *after correcting for* $\rho_{1,2}$*: the negative correlation persists across lags, so the cumulative diffusivity* $\sigma^2(x) = 1 + 2\sum_k \rho_k(x)$ *sits well below* 1 *at this height. The model layer is declared:* (ξ_i) *is a deterministic, nonstationary family and the* $\rho_k(x)$ *are empirical statistics of it, so the Green–Kubo identity* $\sigma^2 = 1 + 2\sum_k \rho_k$ *is invoked under an explicit* local-stationarity model hypothesis*—the step process restricted to a window* $[x, 2x]$ *is modelled as a stationary sequence whose autocorrelations are the measured* $\rho_k(x)$*—and the variance statement can equivalently be taken as a direct* definition *of* $\sigma^2(x)$ *through block averages of* $(S_{n+m} - S_n)^2/m$ *over the events up to* x*, a form in which no stationarity is assumed at all. The Green–Kubo sum requires more than the fixed-lag law of (i): the per-lag decay controls each* $\rho_k(x)$ *but not the sum unless the decay is* uniform in k with summable tail*, since the number of relevant lags may grow with* x*. We therefore state the needed hypothesis as part of the clause:* $\rho_k(x)$ *obeys the law of (i) uniformly for* $k \le K(x)$ *with* $\sum_{k>K(x)} |\rho_k(x)| = o(\log\log x/\log x)$ *for some* $K(x) \to \infty$, and*—since the tail bound alone leaves the growing head sum uncontrolled—the repulsion constants are summable,* $\sum_k c_k < \infty$ *and* $\sum_k |d_k| < \infty$*, so that* $\sigma^2(x) = 1 - 2(\sum_k c_k \log\log x + \sum_k d_k)/\log x\,(1 + o(1))$ *is a genuine consequence rather than an unjustified interchange of limits. The measured correlation length at* 10^9 *(a few tens of lags, with* $|\rho_k|$ *decreasing geometrically in the observable range) is consistent with both hypotheses. Under them,* $\sigma^2(x)$ *returns to* 1 *asymptotically, making sub-diffusivity a finite-height phenomenon of* $\log\log x/\log x$ *size, like every other second-order law in this paper;* (iii) *[registered dichotomy, open] asymptotically the two possibilities are* mutually exclusive*, and which of them holds is the open question. Either the fixed-lag expansion of (i) extends uniformly in k with summable constants—in which case the hypotheses of (ii) hold, (ii) applies, and the race is asymptotically diffusive with* $\sigma^2(x) \to 1$ *(running maxima* $\sim \sqrt{2\log\log x}$*, Darling–Erdős class)—or that expansion fails at long lags, and only in that case can the race inherit the almost-periodic rigidity that Rubinstein–Sarnak-type structure [6] forces on classical races (maxima* $\asymp (\log\log\log x)^A$*, the Montgomery–Ng class). Rigidity therefore requires a failure of the hypotheses of (ii) at long lags, and cannot coexist with them. No explicit formula is known for twin patterns, the two branches differ only beyond any computable height, and we register the dichotomy without choosing* in this conjecture*. What is conjectured here is therefore the fixed-lag law (i) outright, and the variance law of (ii)* conditionally on *its stated uniformity and summability hypotheses, which belong to the finite-height model layer and are not asserted asymptotically: asserting them would silently select the diffusive branch of (iii). The pattern-specific selection is made elsewhere—Conjecture 1 asserts the rigid branch for its races, through (i-b) for the contaminated combinations and through the drift-scale nulls of its clause (ii) for the symmetric differences—and consistency between the two conjectures then requires exactly that the hypotheses of (ii) fail at long lags for those races.*

The repulsion constants ρ_1, ρ_2 are pattern-independent across our four races—a universality the LOS correlation series should predict quantitatively, and the conjecture's stated derivation programme—and the suppressed maxima are the first (to our knowledge) direct measurement of race sub-diffusivity, an effect invisible to endpoint statistics. (The twin-gap record benchmark—$G_t(x) \asymp \log^3 x$ with working constant $1/(2C_2)$, inside Kourbatov's extreme-value framework [15]—is included as an attributed calibration benchmark: records to 4×10^9 wander in $[0.41, 0.56] \log^3 x$, approaching from below on a $\log\log$ clock, largest observed twin-to-twin gap 5292 after the twin at 2,466,641,069; the Cramér–Granville fragility of such constants [17] is why the constant was always flagged as first-order.)

4 Family laws and second-order refinements

Conjecture 10 (Uniform quadratic de Polignac; family law apparently unstated, the $d = 2$ instance is (a), cf. OEIS A080149, [32]). *For even $d \geq 2$ let $\pi^{\mathrm{q}}_d(N) = \#\{n \leq N : \ n^2+1, \ n^2+1+d$ both prime$\}$ and $C(d) = C(x^2+1, x^2+1+d)$, with $I_d(N)$ the integral $I(N)$ of this pair. Then:* (i) *[uniform family law] for every fixed $B > 0$, $\pi^{\mathrm{q}}_d(N) = C(d)\, I_d(N)(1 + o(1))$ uniformly over even $d \leq (\log N)^B$;* (ii) *[registered rate, strictly stronger] there is $\eta_B > 0$ with*

$$\sup_{\substack{2 \leq d \leq (\log N)^B \\ 2|d}} \Big| \frac{\pi^{\mathrm{q}}_d(N)}{C(d)\, I_d(N)} - 1 \Big| \ll_B (\log N)^{-\eta_B}$$

—an error clause that no version of Bateman–Horn supplies, registered separately from (i) since its analytic content is independent; (iii) *[family statistic: the law of the constants] each local factor of $C(d)$ depends only on $d \bmod p$, and the factors at any* finite *set of primes are exactly independent as d varies (CRT), with the passage to the infinite product controlled by the convergent tail-variance sum—which supplies the uniform integrability (via L^2-boundedness of the normalized partial products) that the moment identities require, since almost-sure convergence alone would not carry means through the limit; so the moments of $C(d)$ over even $d \leq D$ converge, as $D \to \infty$, to* derived *Euler products; in particular*

$$\frac{1}{\#\{2 \mid d \leq D\}} \sum_{2|d \leq D} C(d) \to \bar{C} = 2.7456\ldots, \qquad \mathrm{sd}\big(C(d)\big) \to 1.6840\ldots,$$

and the empirical distribution of $C(d)$ converges to the law of the random product $\prod_p f_p(U_p)$ with independent uniform residues U_p. In particular ($d = 2$, the classical instance): $\#\{n \leq N : n^2+1, \ n^2+3$ both prime$\} \sim C(2)\, I(N)$, $C(2) = 2.954014\ldots$

This is the quadratic analogue of the uniform de Polignac law of Conjecture 8: not one count but the whole profile of constants must be matched at once, and the family, unlike its linear parent, has no known prior statement. Every even shift is admissible (for $p \geq 5$, $\omega(p) \leq 4 < p$; the classes mod 3 and 5 vary with d, which is what makes the $C(d)$-profile nonconstant, spanning a factor of about twenty over $d \leq 300$, from 0.47 at $d = 298$ to 9.06 at $d = 162$). At $d = 2$ the local analysis reads: n even, $n \not\equiv 0 \pmod 3$, $\omega(p) = 2 + (\frac{-1}{p}) + (\frac{-3}{p})$ for $p \geq 5$, and the conjecture implies infinitely many twin pairs $(m+1, m+3)$ with m a perfect square. Profile verification at $N = 10^6$ over all 150 even shifts $d \leq 300$: correlation 0.99976 between observed and predicted counts, regression slope 1.0022, residual mean $z = +0.24$ with spread 0.84 and $\max_d |z| = 2.28$—the shape-of-agreement standard applied across the family. Clause (iii) was verified by computing both sides independently: the derived Euler-product moments give mean 2.7456 and standard deviation 1.6840, against 2.7434 and 1.6726 measured over the 150 constants with $d \leq 300$ (ratios 0.9992 and 0.9932)—a family-level statistic whose constants are derived, not fitted. Clause (iii), unlike (i) and (ii), may prove accessible unconditionally by the exact-CRT and martingale route recorded at the cubic family below; the hard content of the conjecture is the uniform law and the rate clause. One question the family raises is recorded as open rather than conjectured: the maximal range $D(N)$ for which the uniform law can hold (the analogue of the Elliott–Halberstam question for this family—whether there is a sharp transition at some power scale N^δ rather than a logarithmic one).

Computational checks (the classical single instance). At $d = 2$ and bound 10^7: observed 32,898 against predicted 32,862.7, ratio 1.0011, $z = +0.19$; the family-profile verification across the 150 shifts $d \le 300$ is reported above.

Conjecture 11 (First occurrences of prime gaps, liminf form; (b))**.** *Let $\mathcal{G}$ be the set of even g that occur as gaps between consecutive primes (all even g, under Polignac's conjecture), and for $g \in \mathcal{G}$ let $p(g)$ be the prime starting the first such gap. Then:* (i) $\log p(g)/\sqrt{g}$ *is bounded between positive constants on $\mathcal{G}$;* (ii) *[liminf law, stated as the dual of the maximal-gap limsup,* conditional on the realization hypothesis below*]*

$$\liminf_{\substack{g\to\infty \\ g\in\mathcal{G}}} \frac{\log p(g)}{\sqrt{g}} \;=\; \sqrt{\mathrm{e}^{\gamma}/2} \;=\; 0.943682\ldots$$

The inversion of Granville's $\limsup G(X)/\log^2 X = 2\mathrm{e}^{-\gamma}$ *into first-occurrence coordinates is* not *automatic: the limsup controls how large maximal gaps get, not which gap* sizes *are realized as first occurrences near the envelope. Clause (ii) is therefore conditional on the* realization hypothesis*: infinitely many $g \in \mathcal{G}$ first occur inside a maximal-gap event at the Cramér–Granville envelope scale, i.e. with $g = (2\mathrm{e}^{-\gamma} + o(1))\log^2 p(g)$. Under that hypothesis the displayed value is forced by algebra; without it only the lower bound* $\liminf \log p(g)/\sqrt{g} \ge \sqrt{\mathrm{e}^{\gamma}/2}$ *follows from the envelope. The value is registered* as *that dual (against the slope-1 laws), not claimed as an independent phenomenon;* (iii) *[second-order, singular-series dependence] sampling g uniformly from the realized gaps $\mathcal{G} \cap [G, 2G]$ and letting $G \to \infty$,*

$$\log p(g) \;=\; \sqrt{g} + \tfrac{1}{2}\log g - \tfrac{1}{2}\log \mathfrak{S}^*(g) + E_g, \qquad \mathfrak{S}^*(g) = \prod_{\substack{p\mid g \\ p>2}} \frac{p-1}{p-2},$$

where the error E_g converges in distribution: for the sampled g,

$$\Pr\left[E_g \le t\right] \;\longrightarrow\; 1 - \exp(-\mathrm{e}^{2(t-\mu)}) \qquad (G \to \infty)$$

for some centring constant μ—the min-type (reversed) Gumbel law at scale $\frac{1}{2}$*, i.e. the law of $\frac{1}{2}\log W$ for W a unit exponential. The scale is forced by the hazard computation: the cumulative intensity of gap-g events is $\Lambda_g(y) \asymp (\mathfrak{S}(g)/y^2)\exp(y - g/y)$ in $y = \log x$ (the factor* $\mathrm{e}^{-g/y}$ is *the no-intervening-prime probability), so $\frac{d}{dy}\log\Lambda_g = 1 + g/y^2 - 2/y = 2 + o(1)$ at the centring point $y_0 \approx \sqrt{g}$, and the first-passage identity $\Lambda_g(y) = W$ gives $E_g = y - y_0 = \frac{1}{2}\log W + o(1)$. The scale is $\frac{1}{2}$ and not 1 precisely because the slope of the log-intensity at y_0 is 2 (both the y-term and the g/y-term contribute), and the same slope-2 expansion is what produces the $\frac{1}{2}$ coefficients of the centring. (Gumbel-type modelling of gap extremes and first occurrences is Kourbatov–Wolf territory [15, 25]; the claim of this clause is only the $\mathfrak{S}^*$-centring and the scale-$\frac{1}{2}$ first-passage form, not the extreme-value framework. Sampling uniformly over the* realized *gaps in* $[G, 2G]$ *conditions on realization, which the independent first-passage model does not represent; the clause conjectures that this selection does not shift the limiting law, and that assumption is part of what a refutation would refute.) Smooth gaps, whose tuple constant is large, appear* earlier *by exactly half a logarithm of that constant—the waiting-time first-passage computation makes the coefficient $-\frac{1}{2}$, not a free parameter, while μ is left as the one unresolved constant of the clause.*

(The restriction to $\mathcal{G}$ is needed: $p(g)$ is not known to be defined for every even g, and quantifying over all g would smuggle Polignac's conjecture into the statement. The $\mathfrak{S}^*$-clause records that the Hardy–Littlewood factor of g materially shifts waiting times; it does not move the liminf constant, since $\log \mathfrak{S}^*(g) = O(\log\log g)$.)

Clause (iii) was tested on the full first-occurrence table to 10^9: over the 100 even gaps realized in $[60, 500]$, regressing $\log p(g) - \sqrt{g} - \frac{1}{2}\log g$ on $\log \mathfrak{S}^*(g)$ (with a $\sqrt{g}$ trend term absorbing the liminf-versus-typical drift) gives coefficient -0.466 against the predicted $-\frac{1}{2}$, partial correlation -0.24: the singular series of the gap is visible in when the gap first appears, at the predicted strength.

The full limit does not follow by inverting Granville's $\limsup G(x)/\log^2 x = 2\mathrm{e}^{-\gamma}$: that inversion is invalid, since a limsup envelope for maximal gaps controls only where gaps *can first appear at the earliest*, i.e. a lower envelope for $p(g)$—hence a liminf claim, not a limit. Whether $\log p(g)/\sqrt{g}$ converges at all, and if so whether the limit is Shanks's classical 1 [16] (as Wolf's refinement $p(g) \sim \sqrt{g}\,\mathrm{e}^{\sqrt{g}}$ and the Kourbatov–Wolf residue-class laws [25] also assert) or the envelope value, involves at least three distinct phenomena—whether a given g occurs near the extreme scale at all, the distribution of exact gap lengths near records, and the difference between record gaps and non-record first occurrences—and is left open here. The two candidate constants differ by 6%, beneath the resolution of any feasible computation: our regression slope over $g \geq 100$ falls from 1.307 at 10^9 to 1.297 at 4×10^9, far above both and decreasing, as it must be for either.

Conjecture 12 (The Stern lane race; apparently new)**.** *In Stern's representation problem $n = p + 2k^2$ (n odd), race the k-even and k-odd lanes. Prime-square contamination of the Λ-weighted lane counts comes from $n = q^2 + 2k^2$—the norm form $x^2 + 2y^2$ of $\mathbb{Q}(\sqrt{-2})$ with prime x—and since $q^2 \equiv 1 \pmod 8$ while $2k^2 \equiv 0$ or $2 \pmod 8$ by the parity of k:* (i) *[class assignment, direct congruence] contamination sits in the k*-even *lane iff $n \equiv 1 \pmod 8$, in the k*-odd *lane iff $n \equiv 3 \pmod 8$, and for $n \equiv 5, 7 \pmod 8$* no *square contamination exists at all: $q^2 + 2k^2 \equiv 1$ or $3 \pmod 8$ always, a two-line congruence (genus theory enters only for the converse question of which n* are *represented), so the null classes are provable, not conjectured;* (ii) *[drift law, in the ensemble form of Conjecture 4] Fix the ensemble: for a scale X, let $\mathbb{E}_X$ average over odd $n \in (X, 2X]$ of the given residue class mod 8 with weight proportional to $1/n$ (the logarithmic sampling measure, as at Conjectures 5* and *4). Define, per class, with $R_{\mathrm{o}}(n)$ and $R_{\mathrm{e}}(n)$ the numbers of representations $n = p + 2k^2$ with k odd and k even respectively, the* clean-minus-contaminated *difference with explicit sign: $D(n) = R_{\mathrm{o}} - R_{\mathrm{e}}$ for $n \equiv 1 \pmod 8$ (even lane contaminated), $D(n) = R_{\mathrm{e}} - R_{\mathrm{o}}$ for $n \equiv 3 \pmod 8$, and $D(n) = R_{\mathrm{e}} - R_{\mathrm{o}}$ (sign immaterial) on the null classes $n \equiv 5, 7 \pmod 8$. Set $D_{\mathrm{sys}}(n) = \big[\sum_{(q,k):\, q^2+2k^2=n,\ q\ \mathrm{prime},\, k\geq 1} \log q\big]/\lambda_{\mathrm{an}}(n)$—the weight is $\Lambda(q^2) = \log q$, each representation counted once in the $k \geq 1$ convention (the ordered-representation factor 2 of Conjecture 4 has no counterpart here, and doubling would be a normalization error)—where $\lambda_{\mathrm{an}}(n)$ is the* analytic *lane mean of* $\log p$ *over the candidate profile, defined with no reference to which candidates are prime:*

$$\lambda_{\mathrm{an}}(n) \;=\; \frac{\displaystyle\int_0^{\sqrt{(n-2)/2}} du}{\displaystyle\int_0^{\sqrt{(n-2)/2}} \frac{du}{\log(n-2u^2)}},$$

the mean of $\log(n-2u^2)$ *under the density proportional to* $1/\log(n-2u^2)$ *on the candidate interval (the upper limit keeps $n - 2u^2 \geq 2$, the convention of $I(N)$, avoiding the nonintegrable point*

$n - 2u^2 = 1$*), which is the profile a Bateman–Horn heuristic assigns to the prime representations. The definition is out-of-sample: the verification's empirical lane mean over the observed prime representations is an estimator of* $\lambda_{\rm an}$*, used as a consistency check, not part of the definition. The law, coefficient-identifying as at Conjecture 4(i): on the contaminated classes*

$$\mathbb{E}_X[D - D_{\rm sys}] = o(\mathbb{E}_X[D_{\rm sys}]) \qquad (X \to \infty),$$

and on the null classes—where $D_{\rm sys}$ *vanishes identically, so that a comparison against it would be empty—the statement is made against the positive comparison scale*

$$A(X) \;:=\; \mathbb{E}_X^{(1)}[D_{\rm sys}],$$

the logarithmic-ensemble mean of the systematic census taken over the contaminated *class* $n \equiv 1 \pmod 8$*: writing* $D_{\rm null}$ *for the difference* D *on a null class* $n \equiv 5$ *or* $7 \pmod 8$*, the claim is* $\mathbb{E}_X[D_{\rm null}] = o(A(X))$*.*

This is the calculus applied to a *representation* problem with norm-form contamination—the drift weight is carried by the arithmetic of $\mathbb{Q}(\sqrt{-2})$ rather than by a polynomial lane—and it comes with two provable null classes. On depth: an *asymptotic* for the contaminating count $\#\{(q,k) : q^2 + 2k^2 = n,\ q \text{ prime}\}$, averaged over n, is a norm-form-with-prime-argument problem of genuine analytic difficulty (the same species as primes represented with restricted variables); $D_{\rm sys}(n)$ is *computed exactly per sample*, so the verification tests the transfer conjecture against the true census, not against an unproved averaged asymptotic. Verified on 2,500 log-sampled odd $n \le 10^8$ (mean $\sim$330 representations each): contaminated strata pooled $D = +0.27 \pm 0.33$ against predicted $D_{\rm sys} = +0.15$ (at the $\Lambda(q^2)$ weight); null strata pooled -0.22 ± 0.33, consistent with zero (one of the four strata sits at -2σ individually, within a four-test family). (Stern's exceptional list—the ten known odd integers that are not $p + 2k^2$, the largest 5993, with the seven prime members the Stern primes; completeness of the list is conjectural, OEIS A060003—is included as the attributed calibration benchmark anchoring this conjecture: no further exception occurs below 10^9 in our computation.)

Conjecture 13 (Microscopic variance law; (b); the constant is Montgomery–Soundararajan's [19])**.** *Fix* $\lambda > 0$ *and* $c \in (0,1]$*, and let* X *count primes in* $[t, t + \lambda \log x)$ *for* t *uniform on* $[x, (1+c)x]$*. Then, under the Hardy–Littlewood* k*-tuple conjectures,* $X \Rightarrow \mathrm{Poisson}(\lambda)$ *(Gallagher's argument [18], which is conditional on exactly those conjectures), and*—conditional on the quantitative hypothesis *spelled out after the statement (uniform Hardy–Littlewood with* $o(h)$ *control of the averaged singular series), without which the display below is a registered extrapolation rather than a consequence—at finite* x

$$\frac{\mathrm{Var}\,X}{\mathbb{E}\,X} \;=\; 1 - \frac{\log(\lambda \log x) + \gamma + \log 2\pi - 1}{\log x} + o\Big(\frac{1}{\log x}\Big),$$

with $\gamma + \log 2\pi - 1 = 1.41509\ldots$*, the leading correction being independent of the sampling constant* c*. [Interpolation clause] More generally, for windows of length* $H = \lambda(\log x)^\alpha$ *the* same *formula with* $\log H$ *in place of* $\log(\lambda \log x)$*,*

$$\frac{\mathrm{Var}\,X}{\mathbb{E}\,X} \;=\; 1 - \frac{\log H + \gamma + \log 2\pi - 1}{\log x} + o\Big(\frac{1}{\log x}\Big),$$

holds uniformly for $1 \le \alpha \le A$ *(any fixed* A *with* $H \le x^{o(1)}$*): one expression interpolates from Gallagher's Poisson boundary (*$\alpha = 1$*, deficit* $\asymp \log\log x / \log x$*) through the mesoscopic*

Montgomery–Soundararajan regime (deficit $\asymp \alpha \log\log x/\log x$), with no transition other than the slow growth of $\log H$*.*

The deficit arises by inserting Montgomery's singular-series average $\sum_{d\le h}\mathfrak{S}(d)(h-d) = \frac{h^2}{2} - \frac{h}{2}(\log h + \gamma + \log 2\pi - 1) + o(h)$ into the second factorial moment at the microscopic window $h = \lambda \log x$. The constant $\gamma + \log 2\pi - 1$ is Montgomery–Soundararajan's, was identified in their framework long before this paper, and the statement is a *microscopic extrapolation and numerical test* of their second-moment formula at Gallagher's boundary—not a newly discovered second-order law. The hypothesis must also be quantitative: what is required is not the qualitative k-tuple conjectures but uniform Hardy–Littlewood estimates for all shifts up to $O(\log x)$, control of the average singular series at the $o(h)$ level, and bounds for prime powers, endpoint weights, and the variation of $\log t$ across the sampling range. Three further qualifications are built into the statement: Gallagher's Poisson law is conditional, not a theorem about primes; the sampling measure for t is specified, since different weightings shift lower-order terms; and the error is claimed only as $o(1/\log x)$. The naive claim $\mathrm{Var}/\mathbb{E} = 1$ fails at every accessible scale by 18–28%; the corrected law matches with no fitted parameter, and the observed residual $\approx +0.003$ is consistent with (but not evidence for) a genuine $1/\log^2 x$ term. (Numerical precedent: Sanchis-Lozano [26] fitted the Montgomery–Soundararajan variance shape at $x \sim 10^8$ in the mesoscopic regime $h \gg \log N$; the microscopic statement and test appear to be new.)

	$x = 10^9$		$x = 4\times 10^9$	
λ	observed	predicted	observed	predicted
1/2	0.8241	0.8189	—	—
1	0.7910	0.7854	0.7998	0.7960
2	0.7527	0.7520	0.7670	0.7646
4	0.7201	0.7185	—	—

The residual $\approx +0.003$ sits at the scale of a $1/\log^2 x$ term, consistent with the hedged reading at the statement.

Conjecture 14 (The null-mechanism race; apparently new)**.** *For the quadratic twin pairs of Conjecture 10 at $d = 2$, square contamination is* algebraically impossible*: $n^2+1 = q^2$ has no solution with $n \ge 1$, and $n^2+3 = q^2$ only at $(n,q) = (1,2)$. The surviving classes of n mod 5 are $\{0,1,4\}$ (the single escape $n = 2$, the pair $(5,7)$ in the killed class 2, is one bounded term), and the classes 1 and 4 have identical singular series (elementary, via CRT). Let $\pi_a^{\mathrm{q}}(x)$ count solutions with $n \equiv a \pmod 5$, $D(x) = \pi_1^{\mathrm{q}}(x) - \pi_4^{\mathrm{q}}(x)$, and define the normalized race process $Y(x) = D(x)/\sqrt{\pi_1^{\mathrm{q}}(x) + \pi_4^{\mathrm{q}}(x)}$. Then, in contrast to the twin race of Conjecture 1, this race is* driftless*:* (i) *$\mathcal{M}_x(D(t)\log^2 t/\sqrt{t}) \to 0$ at the* drift-scale *normalization of Conjecture 1(i)—the scale at which the contaminated races converge to nonzero constants—so this is the genuine negation of Conjecture 1(i) for this pair (the weaker normalization $\mathcal{M}_x(Y) \to 0$ with $Y = D/\sqrt{\pi^{\mathrm{q}}}$ would not exclude a drift at the contamination scale, hence would not make this race a control; the form stated excludes exactly that);* (ii) *[occupation law, conjectured directly, with a separately falsifiable local hypothesis] the process is defined precisely at event index: let $n_1 < n_2 < \cdots$ enumerate the n with n^2+1 and n^2+3 both prime and $n \equiv 1$ or $4 \pmod 5$, set $\xi_i = +1$ if $n_i \equiv 1$ and -1 if $n_i \equiv 4$, and let $S_N = \xi_1 + \cdots + \xi_N$ (so $D(x) = S_{N(x)}$ with $N(x)$ the number of such events to x). The sequence (ξ_i) is deterministic, so the hypothesis is stated for a randomly sited window rather than for S_N itself. Fix a growth scale N_0 and a window length $N = N(N_0)$ with*

$N = o(N_0)$, and choose the starting index M of the window according to the logarithmic measure on $[1, N_0]$:

$$\Pr[M = m] \;=\; \Big(\sum_{1\le m'\le N_0} \tfrac{1}{m'}\Big)^{-1}\frac{1}{m}, \qquad 1 \le m \le N_0.$$

The hypothesis is an invariance principle of ASIP type, stated in triangular-array form, for the induced random path $(S_{M+j} - S_M)_{0\le j\le N}$ under that sampling law: on a probability space carrying M and a standard Brownian motion B,

$$S_{M+j} - S_M \;=\; B(j) + O(N^{1/2-\delta}) \qquad \textit{uniformly for } 0 \le j \le N,$$

with probability tending to one as $N_0 \to \infty$, for some $\delta > 0$; the sampling law of M changes with N_0, so the coupling is asserted per N_0 with vanishing exceptional probability rather than almost surely on a single space. The logarithmic sampling law is the same clock in which the occupation statement below is read, and randomizing the window start is what supplies the deterministic step sequence with a probability space. The weak functional limit theorem—$S_{\lfloor N\tau\rfloor}/\sqrt{N}$ converging weakly to standard Brownian motion on $\tau \in [0,1]$—is retained as the weaker, separately falsifiable consequence of the windowed ASIP. The occupation law itself, however, is not *a consequence of the windowed hypothesis, and is conjectured directly. The windowed path $S_{M+j} - S_M$ is recentred at the window start, and recentring erases exactly the absolute level that occupation measures: a deterministic bias $b(n) = \sqrt{n}/\log n$ added to S_n shifts the level by $\sqrt{n}/\log n$ while contributing window increments of size $\sim j/(2\sqrt{n}\,\log n) \ll \sqrt{j}$, invisible to the windowed ASIP at every window scale, yet it would drive the occupation fraction to 1. A global coupling $D(t) = B(N(t)) + o(\sqrt{N(t)})$ with $N(t) \asymp t/\log^2 t$ would suffice, but for the deterministic sequence (ξ_i) no randomization is available that makes such a coupling well posed, so we do not hypothesize it. The Brownian computation is instead the motivation that identifies the limit and the constant: for a coupled walk, in* logarithmic *time $u = \log t$ the Lamperti reduction $Z(s) = \mathrm{e}^{-s/2}B(\mathrm{e}^s)$ is a stationary, ergodic Ornstein–Uhlenbeck process, and the logarithmic occupation measure of leadership converges almost surely to $\frac{1}{2}$. The conjectured law is that conclusion itself:*

$$\frac{1}{\log x}\int_2^x \mathbf{1}_{\{D(t)>0\}}\,\frac{dt}{t} \;\longrightarrow\; \frac{1}{2},$$

with fluctuations of the Gaussian scale $\sqrt{\log 2/\log x}$ (the occupation-indicator covariance $\frac{1}{2\pi}\arcsin \mathrm{e}^{-u/2}$ integrates to $\frac{1}{2}\log 2$; note that the integrated covariance for the sign *average $2\mathbf{1} - 1$ is four times this, its standard deviation twice, the occupation fraction itself carrying $\sqrt{\log 2/\log x}$); the classical arcsine spread survives only in the* natural *event clock, i.e. for occupation fractions of $\{B(m) > 0,\ m \le N(x)\}$. A persistent systematic leader—a log-occupation stuck near 0 or 1 beyond the stated Gaussian scale—is the refutation.*

The clock matters here. An arcsine-distributed limit for the logarithmic occupation itself is incompatible with the change of clock: the arcsine law lives in the natural time of the walk, whereas under dt/t averaging the Lamperti transform is ergodic and forces the limit $\frac{1}{2}$. The distinction is what makes the statistic informative: the observed log-occupation 0.21 at 10^7 sits 1.4 null standard deviations below $\frac{1}{2}$ (scale $\sqrt{\log 2/\log x} \approx 0.21$ there)—still consistent with the driftless null—whereas read as an arcsine draw the same number would carry no information at all.

This is the negative control that completes the mechanism family of Conjectures 1 and 4: those races carry a predicted drift *because* prime squares can infiltrate specific residue classes of specific patterns; here the quadratic form blocks squares identically, so the same model predicts *nothing*—a falsifiable contrast. If a persistent drift were found in this race, the mechanism story behind the family would be wrong no matter how well the positive cases fit. The absence of *this* drift source does not by itself prove the race driftless—oscillatory zero terms, higher prime powers, and endpoint corrections are other conceivable sources—so clause (i) is a conjecture under the same zero-oscillation hypothesis as Conjecture 1, not a consequence of the square-blocking algebra alone. What the algebra supplies is the *contrast*: whatever residual model one adopts, it must produce drift in Conjectures 1 and 4 and none here. Computational checks at 10^7: 32,898 pairs, class counts 10,917 against 10,981 ($D = -64$, i.e. -0.43 noise units), log-mean normalized drift -0.46 (consistent with 0 at one noise unit), log-occupation of leadership 0.21—1.4 null standard deviations below $\frac{1}{2}$ on the clock and constant of clause (ii), within the fair race's 95% band. (The quadratic triple $n^2 + \{1, 3, 7\}$, $C = 10.64599 \pm 0.006$, is included as a calibration benchmark—observed 3,963 against predicted 3,998.6 at 10^7, $z = -0.56$, independently recomputed to 3×10^7; the longer pattern $n^2 + \{1, 3, 7, 9, 13\}$ has been recorded in the OEIS since 2000, so the tuple family has prior presence and only its singular-series evaluation is plausibly new.)

Computational checks. The race data are reported at the statement above; independent recomputation of this conjecture's calibration member, the quadratic triple, gave 10.647706, with primes to 10^9—inside our stated wobble—and an independent count to 3×10^7 with final ratio 1.0059.

Conjecture 15 (The cubic shift family: the distribution of its constants; instance-level law apparently unstated)**.** *For non-cube $a \geq 1$ let $C(a) = C(x^3 + a)$. Only $p \equiv 1 \pmod 3$ moves the product, and the local root count there takes* three *values as a varies mod p: $\omega = 3$ with probability $(p-1)/3p$ ($-a$ a nonzero cube), $\omega = 1$ with probability $1/p$ (the case $p \mid a$, easy to overlook and essential to the mean-value identity below), and $\omega = 0$ with probability $2(p-1)/3p$; for $p \equiv 2 \pmod 3$ the cube map is a bijection and $\omega = 1$ identically. Then:* (i) *[mean-value one, elementary lemma] $\mathbb{E}_a[\omega(p)] = 3 \cdot \frac{p-1}{3p} + \frac{1}{p} = 1$ exactly at every p, so the derived mean of $C(a)$ is exactly 1—the cubic analogue of the Korevaar–te Riele mean-value-one theorem for linear prime-pair constants, here a one-line computation at each finite level; passing the mean through the infinite product needs more than almost-sure convergence, and the needed ingredient is on hand: the convergent variance sum makes the partial products an L^2-bounded martingale (each factor has mean 1 and is independent of the earlier ones), so they are uniformly integrable and the limit law has mean exactly 1, not merely limit-of-means 1;* (ii) *[limit law] the empirical law of $\{C(a) : a \leq A,\ a \text{ a non-cube}\}$ converges, as $A \to \infty$, to the law of the random Euler product $\prod_{p \equiv 1(3)} f_p(\xi_p)$ with the three-state local variables ξ_p above—independent at any finite set of primes exactly, by CRT, with the infinite product's tail controlled by the convergent variance sum (unlike the quadratic family of Conjecture 10(iii), no normalization is needed)—with derived standard deviation $0.2762\ldots$ computed from the full three-state law;* (iii) *[uniformity] $\#\{n \leq N : n^3 + a \text{ prime}\} = C(a) I_a(N)(1 + o(1))$ uniformly over non-cube $a \leq (\log N)^B$.*

The framework here is Kowalski's theory of averages of Euler products and singular-series distributions [11], which proves such limit laws for k-tuple singular series; the one-parameter cubic shift family, its exact mean-one lemma, and the short-range uniformity appear unstated, and we present (ii) as an instance-level law inside that framework rather than a new mechanism. Computational checks: derived moments (mean exactly 1, sd 0.2762) against the 294 non-cube shifts $a \leq 300$: empirical mean 1.0215 (+1.5 profile standard errors), sd 0.2481 (finite-family tail correlations account for the sd gap: for $p > a_{\max}$ the residues of a are not yet equidistributed); uniformity profile over 57 shifts at $N = 2 \times 10^5$: mean $z = +0.09$, spread 0.72, $\max |z| = 2.31$.

(The single instance $a = 2$, $C = 1.298435 \pm 0.0003$, verified at 10^7 with $z = +0.32$, is included as the calibration member; no statement of Bunyakovsky type is proven for any cubic, the nearest theorem being Heath-Brown's $x^3 + 2y^3$ [3], which is what keeps the whole family conjectural.)

Computational checks (the classical single instance). At $a = 2$ and bound 10^7: observed 287,956 against predicted 287,784.8, ratio 1.0006, $z = +0.32$; the family-profile verification across the 57 shifts is reported above.

Conjecture 16 (Hardy–Littlewood's Conjecture F as a family; (a), [1], cf. [23])**.** *For odd A let $Q_A(N) = \#\{n \le N :\ n^2 + n + A$ prime$\}$ and $C(A) = C(x^2 + x + A)$. Then:* (i) *for every fixed $B > 0$, $Q_A(N) = C(A)\, I_A(N)\,(1 + o(1))$ uniformly over odd $1 \le A \le (\log N)^B$;* (ii) *[window field] the residual field obeys the analogue of Conjecture 8(ii), kernel derived and probability space specified: for t uniform on $[N, 2N]$ and window length $H \le N^{o(1)}$, let $Q_A(t; H) = \#\{t < n \le t + H :\ n^2 + n + A$ prime$\}$ (mean $\asymp H/2\log N$, which sets the regime). Two members share the index window, so with $C(A, A') = C(x^2 + x + A,\ x^2 + x + A')$,*

$$\mathrm{Cov}_t(Q_A, Q_{A'}) \ \sim\ [C(A, A') - C(A)\, C(A')]\, \frac{H}{4\log^2 N},$$

with $C(A, A') = 0$ (hence negative *correlation) for the locally exclusive pairs. This is the* same-index *contribution only: the full covariance adds the off-index pinned sum*

$$\frac{1}{4\log^2 N} \sum_{0<|h|\le H} (H - |h|)\, [C(f_A(x), f_{A'}(x + h)) - C(A)C(A')],$$

the two-parameter analogue of Conjecture 2's $G(H)$, whose evaluation is this family's open component. (The order of this sum matters: the joint event here is two single prime values *at distinct indices, of probability $\asymp 1/4\log^2 N$, unlike Conjecture 8's pair-of-pairs events at $1/\log^4 x$. At that order the off-index sum, which grows like a pinned G-average over $\asymp H$ shifts against the same-index term's single H, is generically the* larger *component, so the numerical cancellation of the same-index kernel establishes nothing about the full cross-member covariance, and the localization of the observed profile deficit to the diagonal is an open question rather than an inference.) For $H/\log N \to \lambda$ the counts are jointly Poisson with local exclusions, and for $H/\log N \to \infty$ each fixed finite collection of studentized counts is asymptotically jointly normal with independent limiting coordinates, all correlations of size $\asymp 1/\log N$ making both kernels finite-N corrections to independence. As at Conjecture 8(ii), both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances—uniform Hardy–Littlewood tuple estimates over the family, strong enough that every joint cumulant of order at least three of the studentized counts vanishes in the respective normalizations—since the covariance kernels alone do not determine the limit laws. (Uniformity over a range growing with N is the substantive claim; uniformity over any* fixed *finite set of A is logically automatic from the individual asymptotics.) Among odd $A \le 199$, Euler's $A = 41$ has the largest constant, $C(41) = 6.6395463\ldots$ (Cohen's high-precision value [31]).*

Numerical evaluation of the kernel (all odd $A, A' \le 99$ at $N = 10^6$: 1,225 pairs, of which 289 locally exclusive) returned a structurally clean answer: the positive correlations of admissible pairs and the negative correlations of exclusive pairs *cancel*, mean off-diagonal $\rho = -0.003$, so the derived *same-index* cross-member kernel is null. With the off-index sum at its correct order (larger, not smaller, than the same-index term), this cancellation does not license the conclusion that the members are asymptotically independent or that the observed profile variance ≈0.37

must be diagonal; the natural reading remains that each $\mathrm{Var}(Q_A)$ is sub-Poisson through within-sequence pair correlations, the quadratic-family analogue of the Montgomery–Soundararajan deficit (Conjecture 13), but the off-index cross terms are an unevaluated competitor—and part of the spread is truncation noise in the constants themselves (the $\pm 10^{-3}$ wobble moves each z_A by $\sim$0.3). Deriving the diagonal law and the off-index pinned sums is the family's remaining open component, and the more important one; the numerical cancellation of the *same-index* cross-member kernel is a finite computation over $A \leq 99$, which localizes the deficit off the same-index diagonal without proving that the averaged kernel or the off-index contributions vanish—an analytic classification of the sign of $C(A, A') - C(A)C(A')$, and the evaluation of the two-parameter pinned sums, remain open. At $N = 10^6$: mean ratio 1.0006, standard deviation 0.0027, $\max |z| = 1.92$, rank correlation 0.99988; for $A = 41$, observed 261,080 against predicted 261,017.6 ($z = +0.12$, using Cohen's constant). On precision: a truncated Euler product for $C(41)$ at cutoff 2×10^5 reads 6.64092—wrong in the fourth significant decimal—because these conditionally convergent products drift at the 10^{-3} scale over practical cutoffs. Quoted digits for all quadratic-family constants in this paper are limited by the stated truncation wobble, not by the last printed digit. The entire 10^8-bit primality dataset compresses to one product formula per A.

Conjecture 17 (Twin-member Goldbach, orientation-resolved; the basis conjecture and the integral kernel are Dubner's [9]; the orientation decomposition apparently new)**.** *For even n let $R_T(n)$ count ordered (a, b), $a + b = n$, with a and b both members of twin pairs, each ordered role (lower/upper) counted. Each of the four role assignments is a 4-form linear system in $t = a$ with root set $\{0, \pm 2\} \cup \{n, n \pm 2\}$-type mod each p, hence its own n-dependent singular series $\mathfrak{S}_4^{(o)}(n)$, and*

$$R_T(n) \;=\; \Big[\sum_o \mathfrak{S}_4^{(o)}(n)\Big] \int_5^{n-5} \frac{dt}{\log^2 t \, \log^2(n-t)} \;(1 + o(1)),$$

the four constants computed exactly per n from the coincidence pattern of the root sets (which of $n, n\pm 2, n\pm 4$ each small prime divides). The error term is stated as $(1+o(1))$, not $(1+O(1/\log n))$: our own measurements show a second-order deficit whose $1/\log n$ coefficient is not yet pinned down (below). Two precision notes: the asymptotic runs over varying n, *so what is assumed is a Hardy–Littlewood estimate for 4-form systems* uniform in the n-dependent coefficients*—strictly stronger than the fixed-system conjecture—and the number 5, the unique prime that is simultaneously a lower twin member (of $(5, 7)$) and an upper one (of $(3, 5)$), is double-counted consistently by the ordered-role convention, affecting finitely many representations per n and no asymptotic.*

Dubner's paper already contains the integral kernel (his quotient μ_4) and the qualitative basis conjecture; the content claimed here is the orientation-resolved singular series and its n-profile. Two exact identities compress the four orientations: the substitution $t \mapsto n - t$ exchanges the two summand roles, forcing $\mathfrak{S}_4^{(\mathrm{lu})}(n) = \mathfrak{S}_4^{(\mathrm{ul})}(n)$ identically (each of the (ll) and (uu) systems being self-dual under it), and the affine substitution $t \mapsto t+2$ gives $\mathfrak{S}_4^{(\mathrm{uu})}(n) = \mathfrak{S}_4^{(\mathrm{ll})}(n-4)$. So only one function of n per symmetry class is free, and the profile is really two-dimensional; determining the average of $\sum_o \mathfrak{S}_4^{(o)}(n)$ over n in closed form is the conjecture's registered identity-hunting programme. Verified on 150 log-sampled $n \leq 10^8$: the profile *shape* is confirmed at log-log correlation 0.9993 across two decades, while the *level* runs at 0.87 of prediction on $[10^6, 10^7)$ and 0.81 on $[10^7, 10^8)$—a systematic deficit of second-order size ($\approx 3.4/\log n$ at the top range, the same order as the 18–28% finite-height Poisson deficit of Conjecture 13), which we flag as its open component rather than absorb into a fitted constant: either the 4-tuple second-order correction accounts for it, or the orientation model needs repair, and the two are distinguishable at 10^{10}. The *direction* of the trend is itself the anomaly: a genuine $1 - c/\log n$ correction shrinks as n grows, so a ratio moving *away* from 1 across the decades indicates either a different error shape or a normalization problem, and settling which is part of what the open component must resolve. (The attributed benchmark stands inside the statement: every even $n \geq 4210$ is a sum of two twin members, with the conjectured complete list of 35 exceptions (OEIS A007534)—verified to 10^8, re-swept to 10^9, independently re-verified by a different algorithm, the same 35 exceptions each time.)

5 Instances and structural companions

Conjecture 18 (Contamination in prime triplets; apparently new)**.** *The contamination calculus of Conjecture 1(v) extends to k-tuples, and the triplet pattern* $(n,\, n+2,\, n+6)$ *is its first instance beyond pairs. Of the three configurations placing a prime square inside the pattern, two die algebraically for* $q > 3$*—*(q^2, q^2+2, q^2+6) *by* $3 \mid q^2+2$*, and* (q^2-6, q^2-4, q^2) *because* $q^2-4 = (q-2)(q+2)$ *is composite (at* $q = 3$ *the factorization is trivial,* $q^2 - 4 = 5$ *is prime, and the single bounded exceptional configuration* $(3, 5, 9)$ *occurs—one term, harmless to every asymptotic)—leaving the single* doubly-thinned *survivor* $(q^2-2,\ q^2,\ q^2+4)$*, which requires* q^2-2 and q^2+4 *simultaneously prime and forces* $q \equiv \pm 2 \pmod 5$ *for* $q \neq 5$ *(else* $5 \mid q^2+4$*; at* $q = 5$ *all three conditions do hold, the configuration being* $(23, 25, 29)$*, but its start* $23 \equiv 3 \pmod 5$ *lies outside the triplet start classes* $\{1, 2\}$*, so the single term is harmless). Triplet starts lie in classes* $n \equiv 1, 2 \pmod 5$ *(single bounded exception* $(5, 7, 11)$*, start* $\equiv 0$*), and the surviving configuration has start* $q^2 - 2 \equiv 2$*: class* 2 *is contaminated and class* 1 *leads, in the drift-scale form of Conjecture 1(i):*

$$\mathcal{M}_x\left(\frac{(\pi_3(t;5,1) - \pi_3(t;5,2))\log^3 t}{\sqrt{t}}\right) \to c_3 := \lim_{x\to\infty} \frac{T_3(x)\log^3 x}{\sqrt{x}}, \quad T_3(x) = \frac{1}{\log^3 x} \sum_{\substack{q \leq \sqrt{x},\ q \neq 5 \\ q^2-2,\ q^2+4 \text{ prime}}} \log(q^2-2)\log q \log$$

(c_3 *exists and is the Bateman–Horn constant of the triple* $(q,\, q^2-2,\, q^2+4)$*, since the census sum is* $\sim c_3\sqrt{x}$*; convergence of the logarithmic mean at this normalization is the same Rubinstein–Sarnak-type structure hypothesis as at Conjecture 1(i); a normalization at the noise scale,* $\mathcal{M}_x((D - T_3)/\sqrt{\pi_3})$ *with* D *the class difference displayed above and* π_3 *the total triplet count to* t*, would be vacuous for the coefficient, exactly as at the pair races), with drift* $\asymp \sqrt{x}/\log^3 x$*, one logarithm weaker than the pair races, and drift-to-noise* $\asymp \log^{-3/2} x$ *against their* $\log^{-1}$*, a scale the calculus itself predicts and the verification must respect. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).*

This is the calculus applied at tuple level: the configuration census, the double thinning, and the mod-5 class assignment are all forced, and the prediction was derived before the data were taken. At 10^9: classes 189,837 against 189,670 ($D = +167$ on the predicted side; $T_3 = +25$ against noise 616, so the endpoint is uninformative exactly as the $\log^{-3/2}$ law requires), leadership log-density 0.65 (+0.8 null standard deviations at the occupation constant of Conjecture 14(ii)—directionally consistent, sharpness unavailable at this height by the calculus's own accounting). (The Chernick chain $\{p, 2p-1, 3p-2\}$—the population behind the universal-form Carmichael numbers, quantified by Dubner [10]—is included as an attributed calibration benchmark: $C = 2.858249$, observed 125,379 against predicted 125,429.4 at 3×10^8, $z = -0.14$.)

Computational checks. The triplet-race data and the Chernick-chain calibration count are reported at the statement above.

Conjecture 19 (The contamination matrix: sexy pairs with two surviving orientations; apparently new)**.** *For the pattern* $(n, n+6)$*,* both *prime-square orientations survive, on complementary classes of* q *mod* 5*—the first matrix instance of the calculus (every pair race so far had exactly one): orientation* $A = (q^2 - 6,\ q^2)$ *requires* $q^2 - 6$ *prime, forcing* $q \equiv \pm 2 \pmod 5$ *for* $q \neq 5$ *(at* $q = 5$ *the value* 19 *is prime, but its start* $19 \equiv 4 \pmod 5$ *lies outside the sexy start classes* $\{1, 2, 3\}$*), and lands in start class* 3 *(mod* 5*), class* 3 *(mod* 8*); orientation* $B = (q^2,\ q^2 + 6)$ *requires* $q^2 + 6$ *prime, forcing* $q \equiv \pm 1 \pmod 5$ *(the term* $q = 5$ *falls outside the start classes), and lands in class* 1 *(mod* 5*), class* 1 *(mod* 8*). Sexy starts occupy classes* $\{1, 2, 3\}$ *mod* 5 *and* $\{1, 3, 5, 7\}$ *mod* 8*; with* $T_A(x) = \frac{1}{\log^2 x} \sum_{q^2-6 \text{ prime}, q\neq 5} \log q \log(q^2 - 6)$*,* $T_B(x) = \frac{1}{\log^2 x} \sum_{q^2+6 \text{ prime}, q\neq 5} \log q \log(q^2 + 6)$*, and* c_A, c_B *their drift-scale limits (*$c_A = \lim T_A \log^2 x/\sqrt{x}$*, the Bateman–Horn constant of* $(q, q^2 - 6)$*, likewise* c_B *for* $(q, q^2 + 6)$*), the predicted drift vector, in the drift-scale limiting-log-mean sense of Conjecture 1(i) (an* $\mathcal{M}_x$*-over-*$\sqrt{\pi}$ *normalization would be coefficient-blind), is, writing* $\pi(a)$ *for the number of sexy starts up to* t *in class* a *of the modulus in question: mod 5:* $\mathcal{M}_x((\pi(2) - \pi(3)) \log^2 t/\sqrt{t}) \to c_A$*,* $\mathcal{M}_x((\pi(2) - \pi(1)) \log^2 t/\sqrt{t}) \to c_B$ *(class* 2 clean*); mod 8:* $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(3)) \log^2 t/\sqrt{t}) \to c_A$*,* $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(1)) \log^2 t/\sqrt{t}) \to c_B$ *(classes* 5, 7 *clean, mutually symmetric, with* $\mathcal{M}_x((\pi(5) - \pi(7)) \log^2 t/\sqrt{t}) \to 0$*). Two independently computable drift constants, one certified null class per modulus. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).*

The matrix structure—two orientations feeding disjoint classes on complementary q-classes, with an interior null—is what distinguishes this from every single-orientation race, and it tests the calculus beyond its balanced one-mechanism cases; the class assignments were verified independently before the data were taken. At 10^9 the predicted components ($T_A = 191$, $T_B = 110$) sit far inside the noise ($\approx$ 2,100), as the $1/\log x$ law forces; all four measured components and the 5–7 control lie within one noise unit of their predictions (trivially, at this height), and the four leadership log-densities $(0.48, 0.56, 0.53, 0.73)$ sit at $(-0.1, +0.3, +0.1, +1.2)$ null standard deviations (occupation constant 0.18): three at the null, the mod-8 B-component mildly positive on the predicted side. The matrix is *registered and consistent*, with sharpness unavailable below $\sim 10^{14}$ by the calculus's own accounting. (The pair $\{p,\ p^2 - 2\}$, OEIS A062326, $C = 3.383216$, the input that sets Conjecture 1's drift scale, is included as an attributed calibration benchmark: verified at 10^7, $z = -1.31$, independently recomputed to 10^9; a refutation of its infinitude would remove the twin race's predicted mechanism.)

Computational checks. The sexy-pair race data are reported at the statement above; independent recomputation of this conjecture's calibration member, the pair $\{p, p^2 - 2\}$, gave 3.383227, with primes to 10^9—inside our stated wobble—and an independent count to 3×10^7 with final ratio 0.9965.

Conjecture 20 (Fibonacci–Lucas twins: the convergent side stress-tested; the object is OEIS A080327)**.** (i) *[rank-disjointness, elementary, with its exception stated] every* odd *prime factor* r *of* L_p *has rank of apparition exactly* $2p$ *(from* $z(r) \mid 2p$*,* $z(r) \nmid p$*, and* $z(r) = 2$ *being impossible), and every prime factor of* F_p *(*$p \neq 5$ *prime) has rank exactly* p*: the odd divisor pools of the two numbers are disjoint. The prime* 2 *is the unique exception:* 2 *divides both* F_p *and* L_p *precisely when* $p = 3$ *(*$F_3 = 2$*,* $L_3 = 4$*;* $\gcd(F_p, L_p) = 2$ *iff* $3 \mid p$*)—the restriction to odd divisors is essential, the clause being false at* $p = 3$ *without it. Disjoint pools remove shared-*prime *correlation but not all correlation: dependence can still enter through the order structure of the* *recurrence, which is why (iii) matters;* (ii) *[finiteness] only finitely many primes p have F_p and L_p simultaneously prime;* (iii) *[calibration clause, quantitative] the naive joint accounting—probability* $(\mathrm{e}^{\gamma}\log p/(p\log\phi))^2$ *per index—assigns prior mass about* 1.4×10^{-2} *to the whole index range beyond* 10^4*, in which the catalogued index* $p = 148091$ *falls (OEIS A080327;* both F_p *and* L_p *are* probable *primes—numbers of roughly* 30,950 *digits, with Lucas numbers proved prime only up to index* 56003*—so everything recorded here is conditional on both probable-prime classifications, for which no Baillie–PSW pseudoprime is known: no unconditional refutation is claimed from uncertified numbers). That rare-event datum is recorded descriptively, with no significance claim attached to it: the model was examined after the event was known, and no test protocol was fixed in advance. A single surprising event is not a strict refutation, and it does not by itself estimate the* size *of the correction: an underestimated constant, genuine cross-dependence, and a mis-calibrated finite-range tail are competing explanations that one event cannot separate. What the datum does bear on (conditionally as above) is not the model but the list: it is fatal to any Goldilocks completeness list; the conjectural content of (ii) is that even the corrected accounting has convergent sum.*

This conjecture is the collection's designated stress test of the convergent Borel–Cantelli template that Conjecture 21 exemplifies. Our scan to $p \le 10^4$ found exactly the joint indices $\{5, 7, 11, 13, 17, 47\}$ with expected further mass 1.4×10^{-2}, while the catalogued index 148091, at which F_p and L_p are both probable primes, lies beyond that range. We therefore conjecture finiteness *without* a completeness clause: the Goldilocks-maximal list that is defensible for factorial twins is indefensible here. The lesson, in the language of Grantham–Granville [7], is that recurrence-sequence constants need their local corrections *before* tail masses are trusted. The single-sided calibration (probable-prime counts: F_p and L_p leave the deterministic range at very small p): 25 Fibonacci prime indices $\le 10^4$ against the naive screening prediction 29.2 (the finite sum of the declared per-index hazard over primes to 10^4) ($z = -0.8$, a mild deficit, not significant on its own); 29 Lucas prime indices $\le 10^4$; the corrected c_F needs a Granville-type patch, and clause (iii) quantifies the same need at the joint level.

Computational checks. Joint scan and single-sided calibration reported at the statement (Fibonacci prime indices $\{3, 5, 7, \ldots, 9311, 9677\}$, 25 of predicted 29.2, $z = -0.78$; Lucas indices 29; joint $\{5, 7, 11, 13, 17, 47\}$ to 10^4)—probable-prime counts at all but the smallest indices, the Fibonacci and Lucas values leaving the deterministic range almost at once.

Conjecture 21 (Factorial twins; the uniqueness core is (a), OEIS A088054; clauses (i) and (iii) are ours)**.** (i) *[window rigidity, elementary] for $n \ge 4$ and $2 \le |a| \le n$ the number $n! + a$ is composite (any prime factor p of a satisfies $p \le n$, so $p \mid n! + a$, and $n! + a > n \ge p$); the two small anomalies* $2! - 2 = 0$ *and* $3! - 3 = 3$ *are the only exceptions, which is why the clause is restricted to* $n \ge 4$*. Hence among all offsets of bounded size the only candidates for primality near* $n!$ *are* $a = \pm 1$*: the twin question below is the* unique *bounded-offset constellation question at* $n!$*, and every admissible offset set is a subset of* $\{-1, +1\}$*.* (ii) *[uniqueness; (a)] Only finitely many n have $n! - 1$ and $n! + 1$ simultaneously prime, and the complete list is $n = 3$: i.e. $6 = 3! = 3\#$ is the unique factorial lying between twin primes.* (iii) *[joint fluctuation model] the two single-sided counts* $F_{\pm}(N) = \#\{n \le N :\ n! \pm 1 \text{ prime}\}$ *are each* $\sim \mathrm{e}^{\gamma}\log N$ *and,* under the independent-indices model stated below*, asymptotically independent, with* $F_+ - F_-$ *normalized by* $\sqrt{2\mathrm{e}^{\gamma}\log N}$ *asymptotically standard normal. The probability space is declared: the clause is about a random model in which the events* $E_n^{\pm} = \{n! \pm 1 \text{ prime}\}$ *carry the Caldwell–Gallot marginal hazards* e^{γ}/n *with an* unspecified *dependence structure, and the Gaussian limit is asserted under three explicit hypotheses on that structure: the same-index bound* $\sum_{n\le N}|\mathrm{Cov}(\mathbf{1}_{E_n^+}, \mathbf{1}_{E_n^-})| = o(\log N)$*, which* *pins the variance at* $2e^{\gamma}\log N\,(1+o(1))$ *and which the joint law* $\sim(e^{\gamma}/n)^2$ *derived below supplies with a convergent sum—without it the remaining hypotheses leave the second cumulant free, and a model with* $\Pr(E_n^+\cap E_n^-)=e^{\gamma}/2n$ *halves the limiting variance; the cross-index covariance bound* $\sum_{m<n\le N}|\mathrm{Cov}(\mathbf{1}_{E_m},\mathbf{1}_{E_n})|=o(\log N)$ *over the four sign pairs (the variance itself is* $\asymp\log N$*),* and*—since pairwise control alone cannot preclude higher-order dependence—the higher-cumulant condition* $\kappa_r(F_+-F_-)=o((\log N)^{r/2})$ *for each fixed* $r\ge 3$*, the method-of-moments sufficient condition. None of the three is imported from an independence model; whether they hold for the arithmetic dependence across factorial indices is the open component of this clause.*

Each event separately has Caldwell–Gallot probability $\sim e^{\gamma}/n$ [4] and divergent sum (the $n!+1$ side is the classical law retained as calibration at Conjecture 2); the *joint* event has probability $\sim(e^{\gamma}/n)^2$, whose sum converges. This is the suite's exemplar of the convergent side of the Borel–Cantelli dichotomy—the accounting that predicts finitude rather than infinitude. The at-a-common-index independence in (iii) has a derivation: a generic twin pair near $n!$ would carry the Hardy–Littlewood twin factor $2C_2\prod_{2<p\le n}\frac{p-2}{p-1}/\prod_{2<p\le n}(\frac{p-1}{p})^2$-type couplings, but here the screening (coprimality to all $p\le n$) is *deterministic*—both neighbours always pass it—and the twin coupling factor and the reciprocal sieve factors cancel to first order, leaving the joint probability $\sim(e^{\gamma}/n)^2$ with no singular-series correlation; what remains open, and is flagged in (iii), is dependence *across* distinct indices n, whose divisibility structures are nested. That open component has a concrete first computation, which we register as the conjecture's programme: for $m<n$ and a prime $p>n$, the residues $m!\bmod p$ and $n!\bmod p$ are deterministically linked by $n!=m!\cdot(m{+}1)\cdots n$, so the covariance of the events $p\mid m!\pm 1$ and $p\mid n!\pm 1$ is an explicit character-sum average over the multiplier $(m{+}1)\cdots n$; summing it over p either bounds the cross-index dependence at $o(1/n)$—vindicating the independent-indices model—or exhibits genuine coupling, and the computation is finite at each height. Verified for $n\le 700$ ($F_+=15$, $F_-=16$: difference -1 against noise scale ≈ 4.8); the model-expected number of further twin examples beyond 700 is $e^{2\gamma}/700\approx 4.5\times10^{-3}$ (computed tail). The defensible core is finiteness; the exact one-element list is the Goldilocks-maximal form—strictly stronger than the tail estimate can guarantee. On priority: the exact uniqueness conjecture is on the public record, OEIS A088054 stating that 3 is conjecturally the intersection of A002981 and A002982, so clause (ii) is attributed there and the conjecture's claimed content is clauses (i) and (iii).

The primorial analogue—$p\#\pm1$ both prime only for $p\in\{3,5,11\}$—is due to Lillie [5], whose abstract gives both the $O(n^{-2})$ joint probability and the prediction that there are three instances in total; it is included here as an attributed calibration benchmark, independently verified to $p\le 4000$.

Computational checks (probable-prime count for n beyond about 26). Exhaustive to $n\le 700$: the only factorial twin is $n=3$; individually, $n!{+}1$ is prime for $n\in\{2,3,11,27,37,41,73,77,116,154,320,340,399,4$ and $n!-1$ for $n\in\{3,4,6,7,12,14,30,32,33,38,94,166,324,379,469,546\}$—two divergent single-sided laws straddling their shared constant while the joint count stops at one, the convergent accounting in action. (The primorial companion, attributed to Lillie [5]: exhaustive to $p\le 4000$, twins exactly $\{3,5,11\}$, with $p\#+1$ prime for eleven p against the Caldwell–Gallot prediction 12.4, $z=-0.40$; again a probable-prime count above the deterministic range.)

Theorem 1 (Cube obstruction)**.** *For* $k\ge 2$*, the cube* k^3 *is representable as* $p+j^3$ *with* p *prime,* $j\ge 1$*, if and only if* $3k^2-3k+1$ *is prime. Consequently the set of integers not representable as* $p+k^3$ *is infinite: the non-representable* cubes *alone have counting function* $\sim x^{1/3}$*. (That the full exceptional set is* $\sim x^{1/3}$ *follows only when combined with Conjecture 22(iii), and is conjectural.)*

Proof. $k^3 - j^3 = (k-j)(k^2+kj+j^2)$; for $1 \le j \le k-2$ both factors exceed 1, so the difference is composite. The only candidate is $j = k-1$, giving $3k^2-3k+1$. Since $3k^2-3k+1$ is composite for a density-one set of k (all but $O(K/\log K)$ of $k \le K$, by an upper-bound sieve of Brun or Selberg type applied to the quadratic's prime values), all but a vanishing proportion of cubes are unrepresentable. □

The restriction to non-cubes in the next conjecture is therefore essential: the exception set of $n = p + k^3$ over *all* n is infinite, and the cubes are exactly what makes it so. The effect is visible immediately in the data—all 412 exceptions found in $(10^8, 10^9]$ are perfect cubes—and it is invisible to a Borel–Cantelli sum taken over all n, since the cubes have density zero. The cube lane is not lost, only relocated: by Theorem 1 it is governed by the primality of $3k^2-3k+1$, which is a Bateman–Horn family in its own right and appears as such in Conjecture 22.

Conjecture 22 (The boundary trichotomy for polynomial ladders; the divisibility principle is classical and its cubic case is Cunningham's cuban observation (OEIS A002407); the family classification is apparently new)**.** *For $F \in \mathbb{Z}[x]$ with* $\deg F \ge 2$ *and* positive leading coefficient*, and $m > j \ge 1$, $(m-j) \mid F(m)-F(j)$ (textbook), and the divided difference $(F(m)-F(j))/(m-j)$—whose leading form is then positive on the cone $m > j \ge 1$—exceeds 1 outside an effectively bounded region; so $F(m)-F(j)$ prime forces $m-j=1$ apart from finitely many exceptional pairs (for the family below the exceptional set is provably* empty*). Both hypotheses are necessary: for* $\deg F = 1$ *the cofactor can be identically 1 and the collapse fails, and for negative leading coefficient the divided difference tends to $-\infty$ on the cone, so the "exceeds 1" clause would be false as stated (one may equivalently keep general sign and read the claim through $|F(m)-F(j)|$). Every representation problem $F(m) = p + F(j)$ in this range thus collapses to the boundary polynomial $D_F(m) = F(m) - F(m-1)$. The content is the classification of the boundary lanes. For $F = x^3 + cx$, $c \ge 0$:* (i) *[trichotomy, elementary] $D_F = 3m^2-3m+1+c$ and the lane is* dead-parity *for c odd (D_F always even),* dead-3-adic *for even $c \equiv 2 \pmod 3$ ($3 \mid D_F$ always), and admissible exactly for $c \equiv 0, 4 \pmod 6$;* (ii) *[uniform boundary law] over the admissible lanes, $\#\{m \le M : D_F(m) \text{ prime}\} \sim C(D_F)\, I(M)$ uniformly for $c \le (\log M)^B$—a Bateman–Horn family whose members are indexed by the ladder classification;* (iii) *[attributed core] the case $c = 0$ is the cuban ladder: Cunningham's classical observation that a prime difference of cubes forces consecutive arguments, with the non-cube exceptional set of $n = p + k^3$ finite (Hardy–Littlewood's $E_3(X) = O(1)$ [1]).*

The reducible-boundary phenomenon is part of the same classification: $F = x^4$ has $D_F = (2m-1)(2m^2-2m+1)$—the boundary lane itself factors, which is precisely the composite-k obstruction of Conjecture 23 seen from the ladder side, and $F = x^3 + x^2$ gives $D_F = m(3m-1)$, dead by reducibility. Verified: the collapse is algebraically empty for $x^3 + cx$ ($c \ge 0$), the trichotomy checked to $m = 5000$ on all $c \le 12$, and the two smallest admissible new lanes counted at $M = 10^6$: $c = 4$ ($C = 2.12956$, $z = +0.27$) and $c = 6$ ($C = 2.68954$, $z = -0.92$). (Empirically the non-cube exception census of (iii) by decade is 4, 27, 168, 763, 2011, 2808, 1181, 88 up to 10^8, peaking at the sixth decade and collapsing thereafter, largest found 78,526,384.)

Conjecture 23 (The power-obstruction ladder: an elementary structural proposition with a Bateman–Horn corollary; apparently new as a family)**.** *For $k \ge 2$ and $m \ge 2$, write $D_k(m) = m^k - (m-1)^k$, and call m^k representable if $m^k = p + j^k$ for some prime p and $j \ge 1$. Then:* (i) *[theorem] for* composite k*, no k-th power is representable: if r is a proper divisor of k with $r > 1$ (such r exists precisely because k is composite; the excluded value $r = 1$ would allow* *only the useless factor $m - j = 1$ at $j = m - 1$), then for every $1 \le j < m$ the difference $m^k - j^k$ has the proper factor $m^r - j^r > 1$ with cofactor > 1;* (ii) *[elementary] for* prime k *(including $k = 2$), m^k is representable if and only if $D_k(m)$ is prime; and D_k is irreducible for prime k (a root α has $(\alpha/(\alpha-1))^k = 1$ with $\alpha/(\alpha-1) \ne 1$, so $\zeta = \alpha/(\alpha-1)$ is a primitive k-th root of unity and $\mathbb{Q}(\alpha) = \mathbb{Q}(\zeta)$ has degree $k - 1 = \deg D_k$, forcing irreducibility; explicitly $D_k(x) = (x-1)^{k-1}\,\Phi_k(x/(x-1))$. The forcing $j = m - 1$ needs only $k \ge 2$);* (iii) *[conjecture] for each prime $k \ge 3$ the representable lane follows Bateman–Horn ($k = 2$ is a theorem, the prime number theorem on the lane $2m - 1$): $\#\{m \le M :\ D_k(m)\ \text{prime}\} \sim C_k \int_2^M dt/\log D_k(t)$.*

The dividing line is prime versus composite k, not odd versus even: for *odd* composite $k = rs$ the factorization of clause (i) applies verbatim, so that $D_9(m)$, for instance, is *never* prime. This was checked directly for $k = 9, 15, 21, 25, 27, 33$—no prime value of D_k occurs up to $m = 2000$, and the predicted divisor $D_r(m) \mid D_k(m)$ is present identically—and it is the same algebraic mechanism, seen from the ladder side, that Theorem 1 exhibits for cubes. The family generalizes the cubic case $k = 3$ of Conjecture 22.

Computational checks. The even-rung theorem was checked numerically to 10^8 (no fourth power is $p + j^4$); the three verifiable Bateman–Horn lanes $k = 2, 3, 5$ (constants 2 exactly, 3.36181, and 3.67770 ± 0.007, the quartic product computed by brute-force root counts to 10^5) all sit within $|z| < 0.6$: at bound 10^7 the lane $k = 2$ gives observed 1,270,606 against predicted 1,270,902.8 (ratio 0.9998, $z = -0.26$); at 10^6 the lane $k = 3$ gives 126,826 against 126,641.6 (ratio 1.0015, $z = +0.52$); and at 8×10^5 the lane $k = 5$ gives 56,925 against 57,021.3 (ratio 0.9983, $z = -0.40$).

Conjecture 24 (The alternating cyclotomic chain; apparently new)**.** *There are infinitely many primes p such that, writing $u = \Phi_3(p) = p^2 + p + 1$, the three integers*

$$p, \qquad u = p^2 + p + 1, \qquad \Phi_6(u) = u^2 - u + 1 = p^4 + 2p^3 + 2p^2 + p + 1$$

are all prime, with $\#\{p \le x\} \sim C_6^{\mathrm{ch}}\, I(x)$ for the Bateman–Horn system $\{x,\ x^2 + x + 1,\ x^4 + 2x^3 + 2x^2 + x + 1\}$, $C_6^{\mathrm{ch}} = 3.6143 \pm 0.011$. The first chains are $(2, 7, 43)$ *and* $(3, 13, 157)$.

This is the repunit analogue of a Cunningham chain: a prime base p, its length-3 repunit $u = (p^3 - 1)/(p - 1)$, then $(u^3 + 1)/(u + 1)$. The chain must *alternate* between Φ_3 and Φ_6: the naive iterate $\{p,\ \Phi_3(p),\ \Phi_3(\Phi_3(p))\}$ is *inadmissible*—if $p \equiv 2 \pmod 3$ then $u \equiv 1 \pmod 3$, so $3 \mid \Phi_3(u)$ always, while $p \equiv 1 \pmod 3$ forces $3 \mid u$, so the admissibility check rejects the naive chain outright. The uniqueness claim is made over a specified construction space: among length-3 chains $\{x,\ \Phi_j(x),\ \Phi_k(\Phi_j(x))\}$ with $j, k \in \{3, 6\}$ *extending the repunit sub-chain* (i.e. $j = 3$), the mod-3 arithmetic above forces $k = 6$: alternation is the unique admissible continuation. (Dropping the repunit anchor admits one sibling, the pure-Φ_6 chain $\{x, \Phi_6(x), \Phi_6(\Phi_6(x))\}$ on the complementary lane $x \equiv 1 \pmod 3$; since $\Phi_6(x) = \Phi_3(x - 1)$, its second element is again a repunit value, and we regard the classification of all admissible words in the two-letter alphabet $\{\Phi_3, \Phi_6\}$—which words are obstruction-free at every prime, and whether they form a regular language—as a natural open programme.) The length-2 sub-chain $\{p,\ p^2 + p + 1\}$ is the classical cyclotomic Sophie Germain statement, included here as an attributed calibration benchmark ($C = 1.521661 \pm 0.0005$, catalogue value $1.5217315\ldots$, OEIS A188596; verified at $z = -1.06$ at 10^7): infinitely many prime bases have prime repunits of length 3, the base-independent home of the phenomenon whose base-2 shadow is the Mersenne sequence. The length-3 chain stated here is, as far as we can find, unstated, and its quartic layer makes it the highest-degree system in the suite.

Computational checks. At bound 10^7: observed 1,362 against predicted 1,357.2, ratio 1.0035, $z = +0.13$. The chain census by decade is $4,\ 11,\ 41,\ 224,\ 1362$ at $N = 10^3, \ldots, 10^7$ (a probable-prime count for the quartic layer beyond $p \approx 2 \times 10^6$; z from $+0.40$ to $+0.13$, no drift), the quartic layer's singular series computed by brute-force root counts to 10^5; the retained sub-chain benchmark $\{p,\, p^2 + p + 1\}$ stands at 33,661 against 33,855.5 at 10^7 ($z = -1.06$). Independent recomputation of that calibration member, the Sophie Germain sub-chain, gave 1.521730, with primes to 10^9—inside our stated wobble—and an independent count to 3×10^7 with final ratio 0.9962.

Conjecture 25 (Twin cyclotomic bases: the complete quadratic family; no prior trace found)**.** *For the three cyclotomic polynomials of degree* 2 *(indices k with $\varphi(k) = 2$, i.e. $k \in \{3, 4, 6\}$), consider $\Phi_k(n)$ and $\Phi_k(n+1)$ simultaneously prime. Then:* (i) *[parity obstruction, elementary] for $k = 4$ one member of the pair $(n^2+1,\, n^2+2n+2)$ is even for* every n*, so the only prime pair is $n = 1$, giving $(2, 5)$—the same obstruction that forbids consecutive n^2+1 primes;* (ii) *[collapse, elementary] $\Phi_6(x) = \Phi_3(x-1)$ identically, so the $k = 6$ pair at index n* is *the $k = 3$ pair at index $n - 1$: the family contains exactly one nontrivial statement;* (iii) *[conjecture, the live instance] there are infinitely many n for which the length-3 repunits in bases n and $n+1$ are simultaneously prime: $\#\{n \le N :\ \Phi_3(n) = n^2 + n + 1$ and $\Phi_3(n+1) = n^2 + 3n + 3$ both prime$\} \sim C_5\, I(N)$, with $C_5 = C(x^2 + x + 1,\, x^2 + 3x + 3)$.*

Both forms in (iii) are odd for every n (each is Φ_3 of an integer), and mod 3 each vanishes on exactly one class ($n \equiv 1$ and $n \equiv 0$ respectively), so the pair is admissible with $\omega(3) = 2$. Unlike an arbitrary shifted pair, the companion here is structurally forced: it is the same cyclotomic value at the consecutive base, making this the natural "twin" statement inside the cyclotomic-chain family of Conjecture 24. The family analysis (i)–(ii) is what a family *lift* of the statement uncovers, and both clauses were found by the admissibility computation itself: the $k = 4$ branch was rejected as inadmissible at $p = 2$ at singular-series time (both-odd count to 10^6: zero, prime pairs: $n = 1$ only), and the $k = 6$ run returned counts *identical* to $k = 3$, which is the numerical shadow of the polynomial identity in (ii).

Computational checks. $C_5 = 2.964239 \pm 0.002$; at bound 10^7, observed 33,274 against predicted 32,976.7, ratio 1.0090, $z = +1.64$.

Part II

Structural conjectures in five programmes

Added to the deposit: 3 August 2026.

The twenty-five conjectures of this part are organized by mechanism rather than by the contamination calculus of Part I. Each names a canonical object and averaging law, identifies an arithmetic, geometric, spectral, or adelic source, and states a first decisive theorem and a failure mode.

6 Programme I: connected prime-pattern fields

Let $H \subset \mathbb{Z}$ be finite and admissible, and write

$$\Lambda_H(n) = \prod_{h \in H} \Lambda(n+h), \qquad Y_H(x,L) = \sum_n w\Big(\frac{n-x}{L}\Big)\,(\Lambda_H(n) - \mathfrak{S}(H)),$$

where w is a fixed smooth compactly supported function and $\mathfrak{S}(H)$ is the Hardy–Littlewood singular series $\prod_p (1-\nu_H(p)/p)(1-1/p)^{-|H|}$, with $\nu_H(p)$ the number of residue classes occupied by H modulo p. The window position x is sampled from $[X, 2X]$ with density $dx/(x \log 2)$ and $X^\varepsilon \le L \le X^{1-\varepsilon}$. For translated motifs $H_i + t_i$, their *exact overlap type* records every equality among shifted prime constraints. The connected singular series $\mathfrak{S}^{\mathrm{c}}(K_1, \dots, K_r)$ is obtained by Möbius inversion over set partitions of the indexed tuple $(K_1, \dots, K_r)$.

Conjecture 26 (Connected motif generating functional)**.** *Fix admissible motifs $H_1, \dots, H_m$ and put*

$$\mathcal{Z}_{X,L}(\mathbf{z}) = \mathbb{E}_x \exp\Bigg(\sum_{i=1}^m z_i Y_{H_i}(x,L)\Bigg).$$

For every fixed total degree R, the degree-$\le R$ Taylor polynomial of $\log \mathcal{Z}_{X,L}$ has a uniform asymptotic expansion indexed by connected exact-overlap diagrams of at most R translated motifs. A diagram whose coincidence hypergraph connects the r translated sets $H_{i_1} + t_1, \dots, H_{i_r} + t_r$ contributes, at the leading order of its stratum,

$$W_\tau(L;\mathbf{t})\,\mathfrak{S}\Big(\bigcup_{j=1}^r (H_{i_j} + t_j)\Big) \prod_v (\log X)^{m_v - 1},$$

where $W_\tau(L;\mathbf{t}) = L\int_{\mathbb{R}} \prod_{j=1}^r w(u + t_j/L)\,du$ is the archimedean window-overlap weight of the diagram and m_v is the multiplicity of the prime constraint at v: any set partition that separates a coincidence loses at least one factor of $\log X$, so the Möbius alternation leaves the single-block moment as the leading term of a connected diagram. The same alternation cancels the leading terms of disconnected diagrams, whose contribution is carried by the connected singular series of their disjoint blocks at strictly smaller logarithmic order, reducing for the fully disjoint type to $\mathfrak{S}^{\mathrm{c}}(H_{i_1} + t_1, \dots, H_{i_r} + t_r)$ with no logarithmic enhancement. The log powers are therefore partition dependent and are never factored across the alternation, and every prime-power counterterm is included in the local weight attached to its multiplicity. Ordering terms lexicographically by powers of L and $\log X$, the remainder after any fixed truncation is smaller than the last retained scale, uniformly in the mesoscopic range.

Significance. This is a multitype connected field theory for prime constellations. It simultaneously organizes covariance, odd moments, mixed motifs, overlap singularities, and lower-order prime-power effects. Montgomery–Soundararajan and Kuperberg provide the closest one-field moment frameworks, while constrained singular-series sums provide the nearest local input [19, 34, 35]. The new content is the complete connected multitype functional. The first decisive theorem is the full third-cumulant formula for two distinct pair motifs. A failure would reveal a term of $\log \mathcal{Z}_{X,L}$ at a scale indexed by no connected exact-overlap diagram, a correlation source among the motif fields $Y_{H_i}(x,L)$ invisible to the Möbius alternation over set partitions.

Conjecture 27 (Complete overlap-renormalization filtration)**.** *Fix a finite motif family $H_1, \dots, H_m$. Apply the degree-two specialization of Conjecture 26 and order all of its connected covariance strata by their asymptotic scales $s_1 \succ s_2 \succ \cdots$ (lexicographically in powers of L, $\log X$, and the renormalized disjoint scales). In the mesoscopic range scales sharing the same total power of L and of logarithms differ only by powers of $\theta = \log L/\log X$ and merge into a single stratum whose limiting form is polynomial in θ. The filtration, the graded convergence, and the eigenvalue asymptotics below are asserted for every fixed $\theta \in (\varepsilon, 1-\varepsilon)$. Recursively subtract the earlier strata and let $A_\nu = A_\nu(\theta)$ be the limiting covariance form at scale s_ν, and put*

$$\mathcal{W}_\nu = \bigcap_{\mu<\nu} \ker A_\mu.$$

Each A_ν is asserted to induce a nondegenerate form on the graded quotient $\mathcal{W}_\nu/\mathcal{W}_{\nu+1}$ for every such θ, so that $\mathcal{W}_{\nu+1} = \mathcal{W}_\nu \cap \ker A_\nu$ and the filtration does not collapse. For every fixed truncation of this ordered diagram expansion, the spectral projections of the covariance matrix converge to the associated graded spaces $\mathcal{W}_\nu/\mathcal{W}_{\nu+1}$, and every eigenvalue whose first nonzero term occurs within the truncation is asymptotic to s_ν times a positive eigenvalue of the induced form A_ν. Vectors annihilating all retained forms pass canonically to the next connected diagram scales of Conjecture 26.

Consequently every asymptotic variance scale of a finite motif statistic is generated by a connected overlap or renormalized disjoint diagram from Conjecture 26, and no scale external to that diagrammatic expansion occurs. The filtration is invariant under replacing the motif family by one with isomorphic exact-overlap incidence algebra and identical local Euler weights.

Significance. The claim classifies all mechanisms by which a finite linear combination of prime motifs can become rigid. It predicts a canonical renormalization filtration, not merely one variance formula, and gives an algorithm for constructing observables that expose deeper arithmetic strata. A counterexample would identify a genuinely new source of rigidity outside overlap and disjoint correlation. The first theorem is a three-motif example in which one overlap form has a rational kernel and the next form is nondegenerate on it.

Conjecture 28 (Regularized mesoscopic local–spectral trace formula)**.** *Fix motifs H_i, H_j, a Schwartz function ϕ with $\phi(0) = 0$, an even cutoff $\eta \in C_c^\infty(\mathbb{R})$ with $\eta = 1$ near the origin, and an even mollifier $\kappa \in C_c^\infty(\mathbb{R})$ of integral one. Let $Q_{ij}^{\mathrm{loc}}(\phi; L)$ be the exact-overlap-renormalized singular-series form*

$$\sum_{t\in\mathbb{Z}} \left[L \int_{\mathbb{R}} w(u)w(u + t/L)\, du \right] \phi(t/L)\, \mathfrak{S}^{\mathrm{c}}(H_i, H_j + t).$$

Let $\Lambda^\sharp_{T,\eta}$ be the Weil explicit-formula distribution in the logarithmic variable, with each nontrivial zero of ordinate γ multiplied by $\eta(\gamma/T)$ and with the pole, trivial-zero, archimedean, and prime-power terms kept as separately named summands. For $\delta > 0$, put

$$\Lambda^\sharp_{T,\eta,\delta} = \Lambda^\sharp_{T,\eta} *_{\log} \kappa_\delta, \qquad \kappa_\delta(u) = \delta^{-1}\kappa(u/\delta).$$

This is a smooth function, so products of its shifted copies are unambiguous. Define

$$Q_{ij}^{\mathrm{spec}}[\eta, \kappa] = Q_{ij}^{\mathrm{spec}}(\phi; X, L; T, \delta, \eta, \kappa)$$

by inserting these mollified copies into the smoothed motif covariance, applying the same exact-overlap Möbius projection as on the local side, and subtracting the explicitly specified pole, diagonal, trivial-zero, archimedean, and prime-power counterterms before the mollifier is removed. In normal form: expand each inserted copy into its separately named summands together with the mollified zero sum and expand the covariance multilinearly; Q_{ij}^{spec} *is then the zero–zero block of that expansion with its diagonal term removed, the subtracted counterterms being exactly the cross terms containing a pole, trivial-zero, archimedean, or prime-power summand, together with the diagonal term of the zero–zero block, so that two readers form the same object.*

For every sufficiently large fixed A, B*, take*

$$T = (X/L)(\log X)^{A}, \qquad \delta = (L/X)(\log X)^{-B}.$$

Then

$$Q_{ij}^{\mathrm{spec}}(\phi; X, L; T, \delta, \eta, \kappa) - Q_{ij}^{\mathrm{loc}}(\phi; L) = o\big(V_{ij}(X, L)\big)$$

uniformly for $X^{\varepsilon} \le L \le X^{1-\varepsilon}$ *and for* ϕ *in bounded Schwartz sets, where* V_{ij} *is the first nonzero disjoint covariance scale. The same limit holds jointly for every fixed matrix of motifs and tests. Changing* $A, B, \eta,$ *or* κ *within the stated admissible class changes the renormalized spectral form by* $o(V_{ij})$*, so the limiting trace functional is canonical.*

Significance. The spectral side is a finite smooth expression before any products are taken, and regularization independence is part of the conjecture rather than an implicit convention. The statement extends the Goldston–Montgomery and Chan prime/zero equivalence from one prime-counting field to products of shifted von Mangoldt functions [55, 33]. Failure can occur in two mathematically different ways: a missing mesoscopic covariance functional, or a genuine regularization anomaly. The first decisive theorem is the pair-motif covariance with one factor unshifted and two independent admissible mollifiers.

Conjecture 29 (Anchored arithmetic polymer expansion for prime motifs)**.** *Fix an admissible motif* H *of size* k*. A finite set* $A \subset \mathbb{Z}$ *of occurrence starts is a polymer when* $0 \in A$*, the overlap graph on* A *is connected, and* $U_A = \bigcup_{t \in A}(H + t)$ *is admissible. Define its* anchored codimension

$$d_H(A) = |U_A| - |H|;$$

this is the number of new prime constraints beyond the occurrence anchored at 0*. In the regularized spectral model of Conjecture 28, let* Z *be the limiting low-mode vector (the joint limit, asserted as part of this conjecture, of the mode integrals of the Conjecture 28 functional against frequencies below a fixed cutoff* K *on the* L*-scale, the Conjecture 28 limits taken first and* $K \to \infty$ *last), let* $\lambda(t \mid Z)$ *be the conditional one-point occurrence intensity, and set*

$$\lambda_H^0 = \frac{\mathfrak{S}(H)}{(\log X)^k}, \qquad \widetilde{\lambda}(t \mid Z) = \lambda(t \mid Z)/\lambda_H^0.$$

For every fixed polymer A*, uniformly for bounded diameter,*

$$\frac{(\log X)^{d_H(A)} \rho^{(|A|)}(A \mid Z)}{\lambda(0 \mid Z) \prod_{t \in A \setminus \{0\}} \widetilde{\lambda}(t \mid Z)} \longrightarrow \frac{\mathfrak{S}(U_A)}{\mathfrak{S}(H)}$$

in probability in the successive limits defining Z*. Thus the conditional environment supplies only the one-point intensity, while every connected Palm activity is graded by* $(\log X)^{-d_H(A)}$ *and weighted by the complete-union singular series.*

If admissible overlapping translates exist, put

$$\delta(H) = \min_{\substack{t\neq 0\\ H\cup(H+t)\ \text{admissible}}} (|H\cup(H+t)|-|H|), \qquad \mathcal{T}_H = \{t \neq 0 :\ H\cup(H+t) \text{ admissible},\ |H\cup(H+t)|-|H| = \delta(H)\}$$

Then the cluster-start fraction has the first nontrivial expansion

$$\theta_X(H) = 1 - \frac{c_H}{(\log X)^{\delta(H)}} + o\big((\log X)^{-\delta(H)}\big), \qquad c_H = \sum_{\substack{t\in\mathcal{T}_H\\ t<0}} \frac{\mathfrak{S}(H\cup(H+t))}{\mathfrak{S}(H)}.$$

The set $\mathcal{T}_H$ is symmetric under $t \mapsto -t$, and the cluster-start fraction counts, for each occurrence, only whether it has an overlapping predecessor, so the sum is restricted to $t < 0$ and equals half the symmetric sum over $\mathcal{T}_H$. If no admissible overlap exists, $\theta_X(H) = 1 + o((\log X)^{-M})$ for every fixed M. Higher corrections are the connected Ursell sums (the signed connected sums of cluster-expansion theory) of anchored polymers ordered by $d_H(A)$. These finite-X Gibbs expansions are projectively consistent and subcritical, and after division by the low-zero intensity the cluster-start process is Poisson to each fixed order of this triangular expansion.

Significance. The relevant rarity is the number of new constraints after one occurrence is given, not the enhancement relative to a product of all one-point intensities. For two translates sharing s of the k prime positions, the Palm scale is $(\log X)^{-(k-s)}$. A universal $1/\log X$ correction therefore holds only for special motifs, such as pair motifs that admit an overlapping translate. The law keeps polymers of every size, supplies a canonical extremal-index exponent $\delta(H)$, and links Hardy–Littlewood constants, low-zero environments, Palm theory, and extreme motif gaps. For $H = \{0, 6\}$ one has $\delta(H) = 1$ and the pair-motif coefficient is recovered. The first decisive theorem is the two-element polymer law for one pair motif, the Palm limit at $A = \{0, t\}$ with $t \in \mathcal{T}_H$ together with the expansion of $\theta_X(H)$ at order $(\log X)^{-\delta(H)}$. A failure would show that the low-mode environment Z supplies more than the one-point intensity, so connected Palm activities would not be graded by $(\log X)^{-d_H(A)}$ and weighted by $\mathfrak{S}(U_A)$ alone.

Conjecture 30 (Topological expansion of non-Gaussian prime cumulants)**.** *For every connected exact-overlap diagram in Conjecture 26, expand its connected singular series over the linking primes and form the linkage graph of each term, whose vertices are the motif occurrences and whose edges join two occurrences sharing a linking prime in that term. Let g be the first Betti number of the linkage graph. Once all larger coincidence strata have been subtracted, the total contribution of the terms of Betti number g is smaller than that of the spanning-tree terms with the same external motifs by at least $(\log L)^{-g}$. The complete leading coefficient at each logarithmic order is the sum of the linked-term weights with the corresponding Betti number. In particular, the first nonzero odd cumulant of a balanced motif statistic, a combination Y_c whose coefficient vector annihilates every coincidence-stratum form of Conjecture 27, is carried by incidence trees. For the single-prime field the fully spread connected part $\kappa^{\mathrm{sp}}_{2m+1}$, defined by subtracting every coincidence stratum as in Conjecture 27, obeys*

$$\kappa^{\mathrm{sp}}_{2m+1} = (c_m + o(1))L^m(\log L)^{m+1},$$

where c_m is the total connected tree weight. The raw cumulant adds the explicit coincidence diagrams of Conjecture 26, which carry powers of $\log X$ and are computable term by term, so the tree law is a statement about the spread diagrams, exactly the regime of Kuperberg's singular-series sums.

Significance. This predicts a topological genus expansion for prime statistics: every independent cycle costs a logarithmic order. Kuperberg's odd-moment conjecture is the one-field calibration, and the incidence-homology filtration with its motif-dependent coefficients is the new mechanism [34]. The first theorem is the case $m = 1$ for the single-prime field, the spread third-cumulant tree law $\kappa_3^{\mathrm{sp}} = (c_1 + o(1))L(\log L)^2$ with its one-cycle terms suppressed by $(\log L)^{-1}$. A failure would show that logarithmic suppression is controlled by an arithmetic invariant not visible in diagram topology.

7 Programme II: arithmetic first-arrival fields and class groups

For a prime modulus q and $a \in \mathbb{F}_q^\times$, let $p(q, a)$ (argument order reversed from Part I's $p(a, q)$) be the least prime in the class a and put

$$T_q(a) = \frac{\mathrm{Li}(p(q, a))}{q - 1}, \qquad \mathcal{N}_{q,a} = \sum_{p \equiv a\,(q)} \delta_{\mathrm{Li}(p)/(q-1)}.$$

The probability space chooses q uniformly among primes in $[Q, 2Q]$ and then chooses a uniformly from $\mathbb{F}_q^\times$.

Conjecture 31 (Connected first-arrival functional)**.** *Let $w^{\mathrm{c}}_{r,Q}$ be the connected factorial-cumulant measure of the random point process $\mathcal{N}_{q,a}$. For every fixed $r \geq 2$ there is a signed measure K_r, locally finite on the off-diagonal region $\{u \in \mathbb{R}_+^r : u_i \text{ distinct}\}$, such that, on that region (the diagonal strata carry strictly smaller normalizations, as in the stratification of Conjecture 26, and are excluded here),*

$$(\log Q)^{r-1} w^{\mathrm{c}}_{r,Q} \Longrightarrow K_r$$

vaguely, and K_1 denotes the limiting intensity measure, which in this clock is Lebesgue measure on $\mathbb{R}_+$. The measure K_r is obtained by averaging, over the modulus aspect, the connected Hardy–Littlewood densities of r primes constrained to one residue class. For every nonnegative compactly supported f in a fixed uniform ball on which the Laplace functional is analytic,

$$\log \mathbb{E} \exp(-\langle f, \mathcal{N}_{q,a} \rangle) = \sum_{r \geq 1} \frac{(-1)^r}{r!} \int \prod_{j=1}^{r} (1 - e^{-f(u_j)})\, dw^{\mathrm{c}}_{r,Q}(\mathbf{u}),$$

and the asymptotic expansion obtained by inserting the limits K_r is uniform to every fixed connected order. In particular the entire survival function, not merely its first correction, is determined by the hierarchy $(K_r)_{r \geq 2}$.

Significance. This is an arithmetic point-process theory of least primes. The coupon-collector model of Li–Pratt–Shakan supplies the nearest extremal analogy, but not a connected Laplace functional [36]. Truth would determine waiting times, occupancy covariance, cover times, and terminal clustering from one hierarchy. The first theorem is the existence and explicit evaluation of K_2 on compact off-diagonal sets. A failure would show that least primes in residue classes cluster through a mechanism not captured by modulus-averaged connected Hardy–Littlewood densities, a connected correlation of $\mathcal{N}_{q,a}$ living outside the hierarchy $(K_r)_{r \geq 2}$.

Conjecture 32 (Tested Gauss-polyspectral reciprocity for the least-prime field)**.** *Set $T_q(0) =$ $\mathrm{Li}(q)/(q-1)$ and centre the resulting field f_q on all of $\mathbb{F}_q$. Let $\mathbf{A}_q$ and $\mathbf{M}_q$ be its complete additive* *and multiplicative Fourier coefficient vectors, including the explicit rank-one coordinate needed to record the value at* 0*. There is then an explicit invertible Gauss matrix* U_q *with*

$$\mathbf{M}_q = U_q \mathbf{A}_q.$$

Choose q *uniformly among primes in* $[Q, 2Q]$*. For every finite family of degrees* $r_1, \ldots, r_m$ *and uniformly bounded fibrewise test tensors* $\Phi_{j,q}$*, define the scalar additive observables*

$$X_{j,Q}(q) = \langle \Phi_{j,q}, \mathbf{A}_q^{\otimes r_j} \rangle$$

and the transported multiplicative observables

$$Y_{j,Q}(q) = \left\langle (U_q^{-1})^{*\otimes r_j} \Phi_{j,q}, \mathbf{M}_q^{\otimes r_j} \right\rangle.$$

For every modulus, $X_{j,Q}(q) = Y_{j,Q}(q)$*, so all joint cumulants under the random-modulus law satisfy the exact finite identity*

$$\mathrm{Cum}_Q(X_{1,Q}, \ldots, X_{m,Q}) = \mathrm{Cum}_Q(Y_{1,Q}, \ldots, Y_{m,Q}).$$

This is the probability-space-correct form of Gauss transport. It makes no reference to a cumulant of a deterministic coefficient vector.

The conjectural content is that, after normalization by the first nonzero connected scale, tests supported on additive zero-sum tensors converge to contractions of the Fourier transforms of the arrival kernels K_r *from Conjecture 31. Transporting those tests fibrewise by the exact Gauss matrices and only then taking the modulus average gives the same limits as the corresponding multiple explicit-formula correlations of Dirichlet* L*-function zeros. The two descriptions agree test by test: every uniformly bounded fibrewise family of test tensors yields the same normalized limits along both routes, so the two limiting multilinear functionals coincide on the tested class itself, no completion being invoked.*

Significance. The elementary input is the finite Gauss matrix. The new claim is a complete higher-order tomography principle for the random-modulus least-prime field. Every scalar statistic is compared on a genuine probability space, while the varying finite fields are handled fibrewise before averaging. Failure would isolate a tensor class on which local same-class prime correlations and character-zero correlations disagree even though the underlying finite transforms are exact. The first theorem is the quadratic identity for tests localized to one additive lag and its transported character kernel.

Conjecture 33 (Rubinstein–Sarnak terminal chaos dichotomy)**.** *Assume GRH and a* q*-aspect linear-independence/random-phase hypothesis for the low zeros of primitive Dirichlet* L*-functions. Let* x_q *satisfy* $\mathrm{Li}(x_q)/(q-1) = \log(q-1)$ *and let* $G_{q,A}(a)$ *be the centred, variance-normalized low-zero field obtained from characters* $\chi \neq \chi_0$ *and zeros* $|\gamma| \leq (\log q)^A$*, with the standard compensator* $-\frac{1}{2}\operatorname{Var} G_{q,A}(a)$*. Define the random empirical chaos measure*

$$\mathcal{M}_{q,A}(dg) = \frac{1}{q-1} \sum_{a \in \mathbb{F}_q^\times} \exp\Big(G_{q,A}(a) - \tfrac{1}{2}\operatorname{Var} G_{q,A}(a)\Big)\, \delta_{G_{q,A}(a)}(dg).$$

Let b_Q *be the smallest level with* $\mathbb{E}_q \#\{a : T_q(a) > b_Q\} \leq 1$*. In the successive limits* $Q \to \infty$ *and then* $A \to \infty$*, the pair consisting of* $\mathcal{M}_{q,A}$ *and*

$$\sum_a \delta_{(G_{q,A}(a),\, T_q(a) - b_Q)}$$

converges jointly. Conditionally on the limiting chaos measure $\mathcal{M}$, the terminal cluster-start process is Poisson with directing measure

$$e^{-x}\,\mathcal{M}(dg)\,dx.$$

At finite modulus the cluster-start thinning enters through the terminal extremal index $\theta_{\mathrm{occ}}(q)$, the analogue for the last classes of the cluster-start fraction $\theta_X(H)$ of Conjecture 29, together with the terminal connected decorations supplied by the hierarchy of Conjecture 31. Both corrections enter at the first nonzero connected scale and vanish in the successive limit, so $\theta_{\mathrm{occ}}(q) \to 1$ and the limiting directing measure carries no cluster factor.

The nature of the directing measure is governed by the covariance-energy statistic

$$\mathfrak{E}_{q,A} = \frac{1}{(q-1)^2} \sum_{a,b\in\mathbb{F}_q^\times} \mathrm{Cov}(G_{q,A}(a), G_{q,A}(b))^2.$$

If $\mathfrak{E}_{q,A} \to 0$ along the successive limit, $\mathcal{M}$ self-averages to the deterministic exponentially tilted Gaussian law and the terminal process is an ordinary decorated Poisson process. If $\liminf \mathfrak{E}_{q,A} > 0$ and the Gaussian-chaos second moments are uniformly integrable, $\mathcal{M}$ is nondegenerate and the limit is a genuine Cox process. No nondegeneracy is asserted without this criterion.

Significance. Rubinstein–Sarnak theory supplies the fixed-modulus prime-race environment [6]. The new object is its terminal first-arrival chaos measure in the modulus aspect. The statement now treats self-averaging as a serious competing model rather than assuming a Cox limit by terminology. It predicts the complete last class process, its prime-race marks, and an explicit spectral criterion separating a deterministic environment from persistent random mixing. The first theorem is the self-averaging branch at one fixed cutoff A, where $\mathfrak{E}_{q,A} \to 0$ forces the deterministic exponentially tilted Gaussian limit of $\mathcal{M}_{q,A}$ and an ordinary decorated Poisson law for the terminal cluster-start process.

Conjecture 34 (Nonlinear spectral response calculus for arithmetic first arrivals)**.** *Let a sequence of modulus ensembles have a small explicit-formula perturbation of the arrival intensity*

$$V_Q(a,t) = \sum_{\chi\in\mathcal{C}_Q} v_{Q,\chi}(t)\chi(a), \qquad \|V_Q\|_T \to 0,$$

where $\mathcal{C}_Q$ has uniformly bounded cardinality and, on every fixed interval $[0,T]$, the canonical norm is $\|V\|_T = \sup_{a,0\le t\le T} |V(a,t)|$. Assume that the connected hierarchy of Conjecture 31 satisfies an exponential cluster bound there, so its Laplace functional is analytic in a fixed $\|\cdot\|_T$-ball. If $S_Q(a,t)$ and $S_Q^{(0)}(t)$ are the perturbed and ordinary survival functions, then for every fixed R

$$\log \frac{S_Q(a,t)}{S_Q^{(0)}(t)} = \sum_{m=1}^{R} \mathcal{R}_m[V_Q^{\otimes m}](a,t) + o(\|V_Q\|_T^R)$$

uniformly on compact t-ranges. The causal Volterra operator $\mathcal{R}_m$ is the universal linked-cluster contraction of the full hierarchy $(K_r)_{r\ge1}$ with m marked perturbation insertions, and truncating the connected hierarchy at order J gives a convergent approximation as $J \to \infty$.

If the input character support is $\mathcal{C}_Q$, the mth response has Fourier support contained in the products of at most m input characters, with the reverse inclusion possibly failing through cancellation or the trivial character. In particular the linear response preserves character rank, while any new harmonics first appear at quadratic order with coefficients determined by connected three-point arrival correlations.

Significance. This is a nonlinear transfer calculus from zeros to first-hit statistics. A Siegel zero, a cluster of low zeros, and an ordinary prime-race bias become different inputs to the same response operators. The exceptional-zero rank-one statement is a special case, while the second-order harmonics provide a direct falsification test. The first theorem is the Fréchet derivative $\mathcal{R}_1$ from the two-point occupancy kernel.

Conjecture 35 (Local-information capacity of odd class groups)**.** *Fix an odd prime ℓ. Sample negative fundamental discriminants $D \in [-2Y, -Y]$, conditioned on a fixed coarse two-primary invariant. For a set of odd primes S_Y, put $M_Y = \prod_{p \in S_Y} p$, let $\Sigma_Y(D) = ((D/p))_{p \in S_Y}$ be the attainable sign vector, and let H_Y be the logarithm of the number of attainable cells. Let $\delta_Y(\sigma)$ be the exact CRT density of the sign cell after imposing quadratic reciprocity and the fixed two-primary data (for a single odd unramified prime the density of each sign is $p/(2(p+1))$, and an unconditioned cell density is the corresponding product), and let $\mathcal{D}_Y$ be the ambient discriminant family. Write*

$$\Delta_Y(S_Y) = \sup_{\sigma} \left| \frac{\#\{D \in \mathcal{D}_Y : \Sigma_Y(D) = \sigma\}}{|\mathcal{D}_Y|\, \delta_Y(\sigma)} - 1 \right|$$

over attainable cells with $|\mathcal{D}_Y| \delta_Y(\sigma) \geq \log Y$.

There is a sequence $\varepsilon_Y \to 0$, independent of S_Y in the subcritical range, such that uniformly over those cells

$$d_{\mathrm{TV}}(\mathcal{L}(\mathrm{Cl}(D)[\ell^\infty] \mid \Sigma_Y = \sigma, \textit{2-data}), \mathcal{L}_{\mathrm{CL},\ell}) \leq \varepsilon_Y + O(\Delta_Y(S_Y)),$$

and the analogous bound holds for every fixed Cohen–Lenstra moment. In particular, for every $\varepsilon > 0$, the conclusion is uniform whenever

$$H_Y \leq (1-\varepsilon)\log Y, \qquad \log M_Y \leq (\tfrac{1}{2} - \varepsilon)\log Y, \qquad \Delta_Y(S_Y) = o(1).$$

Since every odd prime contributes at most $\log 3 \leq \log p$ of cell entropy, $H_Y \leq \log M_Y$ throughout, so the conductor constraint is the binding one in the displayed range, while the entropy governs the impossibility clause below. At $H_Y \geq (1+\varepsilon)\log Y$ no uniform statement is possible. Up to the conductor scale at which Δ_Y becomes macroscopic, every first-order failure of local independence factors through this discriminant-cell discrepancy, and there is no earlier odd-class-group obstruction.

Significance. Wood's framework treats fixed local conditions [37]. The new statement proposes the full information capacity of an odd class group under a growing local sigma-algebra, with the entropy governing impossibility and the conductor governing equidistribution. It would transfer directly to Selmer groups, ray class groups, and unramified extension statistics. The first theorem is uniformity for $|S_Y| \to \infty$ with $M_Y = Y^{o(1)}$. The discrepancy half is not the obstruction: for cells of macroscopic mass the equidistribution of discriminants is elementary for $M_Y \leq Y^{1/2-\varepsilon}$, and squarefree-in-progressions technology carries it to $Y^{2/3-\varepsilon}$, so the conjectural content lies entirely in the class-group law over the growing cell family.

8 Programme III: finite logarithms, regulators, and algebraic tori

For $p \nmid a$, write

$$q_p(a) = \frac{a^{p-1} - 1}{p} \pmod{p}.$$

For a generating tuple $g = (g_1, \dots, g_s)$, let $R(g)$ be its multiplicative relation lattice and let $\mathbb{T}_g \subset (\mathbb{R}/\mathbb{Z})^s$ be the identity component of its annihilator. Its dimension is the saturated multiplicative rank. Write $e(x) = \mathrm{e}^{2\pi i x}$.

Conjecture 36 (Kummer–Haar law on relation tori)**.** *Let $\Gamma \le \mathbb{Q}^\times$ be finitely generated and torsion-free, saturated modulo torsion (if $x^n \in \Gamma \cdot \{\pm 1\}$ for some $n \ge 1$ then $x \in \Gamma \cdot \{\pm 1\}$), and let g be any ordered generating tuple. As p varies in every fixed compatible cyclotomic–Kummer Chebotarev class, the vectors*

$$\frac{(q_p(g_1), \dots, q_p(g_s))}{p} \in (\mathbb{R}/\mathbb{Z})^s$$

equidistribute with Haar measure on $\mathbb{T}_g$. More strongly, for every finite collection of compatible Chebotarev restrictions, the conditional empirical measures converge to the Haar disintegration on the corresponding components, and the support is contained in no proper closed coset of $\mathbb{T}_g$. The law is invariant under changing generators and depends on Γ only through its saturation and the fixed Kummer conditioning data.

Significance. Katz formulated horizontal Wieferich equidistribution principles for algebraic groups, and Shparlinski proved estimates in different horizontal and averaged regimes [38, 39]. The new content here is the exact saturated relation torus, full generator invariance, and disintegration under fixed Kummer classes. It turns multiplicative rank into a geometric support theorem rather than a heuristic count of coordinates. The first decisive theorem is the Haar law on $\mathbb{T}_g$ for one rank-one saturated Γ presented by two dependent generators, restricted to a single fixed compatible cyclotomic–Kummer Chebotarev class. A failure would show that the vectors $q_p(g_i)/p$ remember more of Γ than its saturation, concentrating on a proper closed coset of $\mathbb{T}_g$ that no relation in $R(g)$ predicts.

Conjecture 37 (Rank-dimensional large sieve for finite logarithms)**.** *Choose a Smith-normal-form basis of the character lattice $X^*(\mathbb{T}_g) = \mathbb{Z}^s/R(g)^{\mathrm{sat}}$. There is a constant $\delta_\Gamma > 0$ such that, for every $\varepsilon > 0$, every $M \le X^{\delta_\Gamma}$, and all complex weights a_p supported on $X < p \le 2X$,*

$$\sum_{\substack{m \in X^*(\mathbb{T}_g) \\ \|m\|_\infty \le M}} \left| \sum_{X < p \le 2X} a_p e\Big(\frac{\langle m, q_p(g)\rangle}{p}\Big) \right|^2 \ll_{\Gamma,\varepsilon} (M^{\operatorname{rank}\Gamma} + \pi(X)) X^\varepsilon \sum_p |a_p|^2.$$

The same inequality holds after every fixed compatible cyclotomic–Kummer restriction, and for matrix-valued weights with the Hilbert–Schmidt norm. The exponent δ_Γ is invariant under changing the generating tuple.

Significance. This is the quantitative mechanism behind Conjecture 36. It predicts that saturated rank is not only the support dimension but the exact dual dimension in a prime-varying large sieve. A proof for any positive power range would control genuine shrinking targets and would be far beyond currently available fixed-frequency tests. Failure would reveal an unrecognized spacing or energy obstruction among finite-logarithm vectors.

Conjecture 38 (Mesoscopic-to-lattice shrinking-target transition)**.** *Let $\Gamma \le \mathbb{Q}^\times$ be torsion-free of rank r and saturated modulo torsion, and choose a basis of Γ, identifying its relation torus with $(\mathbb{R}/\mathbb{Z})^r$. For integers w_p with $1 \le w_p = o(p)$, let $E_p(w)$ be the event that every coordinate of the Fermat-quotient vector has a representative of absolute value at most w_p.*

Mesoscopic universality. *If $w_p \to \infty$, then in every range in which the Haar mass*

$$\lambda(X) = \sum_{p \le X} \left(\frac{2w_p + 1}{p} \right)^r$$

tends to infinity, the count of $E_p(w)$ is asymptotic to $\lambda(X)$. On prime blocks with bounded Haar mass, the randomly translated event process converges to a Poisson process with that mass.

Lattice transition. *For every fixed finite target set $B \subset \mathbb{Z}^r$, there is an arithmetic regulator intensity $\kappa_{\Gamma,B} \ge 0$ such that*

$$\#\{p \le X : q_p|_\Gamma \in B\} = (\kappa_{\Gamma,B} + o(1)) \sum_{p \le X} p^{-r}$$

whenever the sum diverges, while only finitely many such primes occur when $r \ge 2$. As a separately falsifiable genericity clause, for Kummer-generic Γ (the Kummer fields $\mathbb{Q}(\zeta_m, \Gamma^{1/m})$ have the maximal degree $\varphi(m)m^r$ for every m) and $B = \{0\}$ one has $\kappa_{\Gamma,B} = 1$. The transition from Haar universality to the arithmetic intensity occurs only when w_p remains bounded.

Significance. This separates two universality classes that a single undifferentiated model conflates. Growing targets are controlled by Haar geometry, while exact Wieferich targets can carry a genuinely arithmetic regulator intensity. Gras's much rarer fixed-base model and the Haar model therefore disagree precisely at the lattice boundary [40, 41]. The first theorem is mesoscopic universality for $w_p = p^\eta$ with one rank-one base.

For a p-adic unit a, write $\langle a \rangle_p$ for the Teichmüller lift of its residue, and for a prime $\mathfrak{p} \mid p$ of a number field write $\tau_\mathfrak{p}$ for the Teichmüller lift in the completion at $\mathfrak{p}$. Define $q_{p,k}(a) = p^{-1} \log(a\langle a \rangle_p^{-1}) \pmod{p^k}$. The matrix entries below are the $\mathfrak{p}$-adic analogues of $q_{p,k}$, and on $\mathbb{Q}$ with $k = 1$ one has $q_{p,1}(a) \equiv -q_p(a) \pmod p$, the Fermat quotient up to sign.

Conjecture 39 (Global-lattice horizontal p-adic regulator-matrix law)**.** *Let $K/\mathbb{Q}$ be a fixed Galois number field with group G, and let $\Gamma \le \mathcal{O}_K^\times$ be torsion-free and saturated modulo $\mu(K)$, with ordered basis $\epsilon_1, \dots, \epsilon_s$. Define the fixed global relation lattice*

$$\mathcal{R}_\Gamma = \left\{ (c_{\sigma j}) \in \mathbb{Z}^{G \times s} : \prod_{\sigma \in G} \prod_{j=1}^{s} \sigma(\epsilon_j)^{c_{\sigma j}} \in \mu(K) \right\}.$$

Let $\mathbb{T}_\Gamma \subset (\mathbb{R}/\mathbb{Z})^{G \times s}$ be its annihilator. This lattice contains the norm and Galois relations and is independent of the chosen basis up to the natural integral change of coordinates.

For a rational prime p splitting completely in K, label the primes above p by G, use the Teichmüller lift in each completion, and form the depth-k matrix

$$L_{p,k}(\Gamma) = \left(p^{-1} \log_\mathfrak{p}(\epsilon_j \tau_\mathfrak{p}(\bar{\epsilon}_j)^{-1}) \bmod p^k \right)_{\mathfrak{p}|p,\, 1 \le j \le s}.$$

It lies in the annihilator of $\mathcal{R}_\Gamma \otimes \mathbb{Z}/p^k\mathbb{Z}$. As p varies through any fixed compatible Frobenius class, $L_{p,1}/p$ equidistributes with Haar measure on $\mathbb{T}_\Gamma$, and the depth-k matrices are Haar at every fixed depth k, with the depth-$(k+1)$ law pushing forward to the depth-k law under reduction (the moduli vary with p, so this is a per-depth statement with compatible marginals, not Haar on one fixed group).

Let $\mathcal{D}$ be a basis-invariant determinantal or Smith stratum in this projective module, and let $\mu_{p,k}(\mathcal{D})$ be its exact local Haar mass. The primes for which $L_{p,k} \in \mathcal{D}$ have counting law governed by $\sum_{p\le X} \mu_{p,k}(\mathcal{D})$. For a determinantal or Smith stratum of codimension c, whose point counts are polynomial in p, $\mu_{p,1}(\mathcal{D}) = p^{-c} + O(p^{-c-1})$, and codimension one has a Poisson point process in $\log\log p$ after uniform random translation, while codimension at least two has a summable large-prime tail together with the exact small-prime atoms.

Significance. The horizontal state space is one fixed global torus rather than a relation module that changes with p. The conjecture turns mod-p Leopoldt defects into a Galois-equivariant random-matrix problem, with exact Smith strata retaining higher depth. Existing mod-p Leopoldt and random-matrix theories provide neighbouring languages [41, 42, 43]. Recent number-field Wieferich results strengthen the arithmetic motivation without supplying this horizontal matrix law [58, 59]. The first theorem is the rank-one split-prime law for a real quadratic field. A failure would show that the matrices $L_{p,k}(\Gamma)$ favour some Smith stratum beyond its exact local Haar mass, so mod-p Leopoldt defects would be governed by an arithmetic invariant outside the fixed global torus $\mathbb{T}_\Gamma$.

Conjecture 40 (Functorial horizontal finite logarithms on algebraic tori)**.** *Let $T/\mathbb{Q}$ be an algebraic torus, let $\mathcal{T}$ be its smooth integral model away from a finite set, and let $\Gamma \le T(\mathbb{Q})$ be finitely generated. Let S be the identity component of the Zariski closure of Γ. At a good unramified prime p, let $\tau_p(\bar{P})$ be the canonical prime-to-p torsion lift of the reduction of $P \in \Gamma$ and define*

$$\ell_p(P) = p^{-1}\log_T(P\tau_p(\bar{P})^{-1}) \pmod p \in \mathrm{Lie}(S)(\mathbb{F}_p).$$

Here $\mathrm{Lie}(S)(\mathbb{F}_p)$ is identified with $(\mathbb{Z}/p)^{\dim S}$ through a fixed integral basis of the Lie algebra of $\mathcal{T}$, with representatives in $[0,p)$; a different basis changes the matrix by a fixed element of $\mathrm{GL}_{\dim S}(\mathbb{Z})$, which preserves the Haar law and the shrinking-target intensities. For any generators $P_1,\dots,P_s$ of Γ, the normalized matrix $(\ell_p(P_i)/p)_i$ equidistributes, in every compatible Frobenius class, with Haar measure on the real relation subtorus cut out jointly by the algebraic subgroup S and the integral relations among the P_i. For every morphism of tori $\phi : T \to T'$,

$$\ell_p(\phi(P)) = d\phi(\ell_p(P)),$$

and the horizontal Haar measures push forward under the induced map. Isogenies preserve the shrinking-target intensities after the explicit finite kernel correction.

Significance. Katz already proposed Wieferich equidistribution beyond $\mathbb{G}_m$ [38]. The additional content is a matrix-valued law for a finitely generated global subgroup, relation-subtorus support, fixed-Frobenius disintegration, and exact functoriality under morphisms and isogenies. It turns horizontal finite logarithms into a functor on the category of tori with arithmetic subgroups. The first decisive theorem is the Haar law for the group generated by a single point of a one-dimensional nonsplit torus T, in a single fixed compatible Frobenius class. A failure would show that the matrices $(\ell_p(P_i)/p)_i$ concentrate on a proper closed subset of the relation subtorus, an arithmetic constraint beyond the algebraic subgroup S and the integral relations among the P_i.

9 Programme IV: arboreal arithmetic dynamics

For a degree-d map f and basepoint a, write $K_n = \mathbb{Q}(f^{-n}(a))$, $G_n = \mathrm{Gal}(K_n/\mathbb{Q})$, and let $\mu_n(C) = |C|/|G_n|$ on conjugacy classes. For primes $p \le X$ unramified in K_n, let $\widehat{\mu}_{n,X}$ be the empirical Frobenius measure.

Conjecture 41 (Entropy–conductor profile of arboreal resolution)**.** *For a degree-d map f and level n, let $G_{n,f}$ be the arboreal quotient and $\mu_{n,f}$ its Haar measure on conjugacy classes. For N independent samples from $\mu_{n,f}$ define the exact sampling obstruction*

$$\mathcal{S}_{n,f}(N) = \mathbb{E}\, d_{\mathrm{TV}}\left(N^{-1}\sum_{j=1}^{N}\delta_{Z_j}, \mu_{n,f}\right).$$

For each nontrivial irreducible Artin representation ρ of $G_{n,f}$, let $q(\rho)$ be its Artin conductor and define the GRH conductor energy

$$\mathcal{A}_{n,f}(X) = \frac{\log X}{\sqrt{X}}\left(\sum_{\rho\neq 1}(\dim\rho)^2\big(\log q(\rho) + \dim(\rho)\log X\big)^2\right)^{1/2}.$$

This is a canonical character-theoretic replacement for a single discriminant proxy. For a parameter family $\mathcal{F}(B)$ put

$$n^*_{\mathcal{F}}(B, X; \eta) = \max\Big\{n : \mathbb{E}_{f\in\mathcal{F}(B)}\big[\mathcal{S}_{n,f}(\pi(X)) + \mathcal{A}_{n,f}(X)\big] \le \eta\Big\}.$$

Assume the generic arboreal image is open, exceptional parameters have density zero, Artin holomorphy and GRH hold at the levels considered, and the family is in the large-sieve range of Conjecture 42 below. Then the actual empirical resolution depth, defined by the same total-variation threshold with prime Frobenius samples, differs from $n^*_{\mathcal{F}}(B, X; \eta)$ by $O_{\mathcal{F},\eta}(1)$.

More precisely, suppose, uniformly for a density-one set of parameters, that the class measures have a limiting complete Rényi profile

$$\psi_{\mathcal{F}}(\alpha) = \lim_{n\to\infty} d^{-n}\log\sum_{C}\mu_{n,f}(C)^{\alpha}$$

uniformly for α in a fixed interval containing $[1/2, 2]$, and the corresponding Artin-conductor energies have finite positive exponential profiles on the same d^n scale. Then

$$n^*_{\mathcal{F}}(B, X; \eta) = \log_d \log X + O_{\mathcal{F},\eta}(1).$$

Two families of the same degree with boundedly different complete Rényi and conductor profiles have resolution depths differing by $O(1)$, and in particular a fixed-index open subgroup changes only the bounded shift.

Significance. The observable depth is defined through the exact nonuniform sampling law and a representation-by-representation conductor energy. The $\log_d \log X$ scale follows from explicit Rényi and conductor profiles on the natural d^n complexity scale. Large arboreal images and effective Chebotarev are inputs [44, 45, 46]. Recent work on fixed-point profiles shows why the full class-measure tail, rather than group order alone, matters [60]. The first theorem is bounded-shift stability for one explicit index-two subgroup of the binary wreath tower. A failure would show that resolution depth responds to a family invariant beyond the complete Rényi profile $\psi_{\mathcal{F}}(\alpha)$ and the conductor energy $\mathcal{A}_{n,f}(X)$, so two families with matching profiles could separate by an unbounded shift.

Conjecture 42 (Arboreal family large sieve and cumulant independence)**.** *Let $\mathcal{F} \to \mathbb{A}^1$ be a generically finite-index arboreal family. For each level n and irreducible representation ρ of the generic quotient G_n, let $\mathcal{V}_{n,\rho}$ be the associated middle-extension sheaf on the good parameter locus (the canonical extension of the corresponding local system across the bad parameters). Define the geometric conductor budgets*

$$\mathfrak{C}_{n,r} = \sum_{\rho_1,\ldots,\rho_r} \left(\prod_{j=1}^{r} \dim \rho_j \right)^2 \prod_{j=1}^{r} \operatorname{cond}\left(\mathcal{V}_{n,\rho_j}\right)^2, \qquad \mathfrak{C}_n = \mathfrak{C}_{n,1},$$

where each conductor includes rank, the number of singular points, and the tame-drop and Swan terms (the tame and wild ramification invariants at those points); for $r \geq 2$ the budget is attached to the product trace function on the CRT parameter set, whose factors live over distinct characteristics, so its complexity is by definition this product of local conductors rather than the conductor of an external tensor product over a single base.

Square-root tier. *For every $\varepsilon > 0$ and all complex coefficients $\alpha_{p,\rho}$,*

$$\sum_{|c| \leq B} \left| \sum_{p \leq X} \sum_{\rho \neq 1} \alpha_{p,\rho} \operatorname{tr} \rho(\mathrm{Frob}_{p,c}) \right|^2 \ll_{\varepsilon,\mathcal{F}} (B + X^2 \mathfrak{C}_n^{1+\varepsilon}) \sum_{p,\rho} |\alpha_{p,\rho}|^2,$$

uniformly when $X^2 \mathfrak{C}_n^{1+\varepsilon} \leq B^{1-\varepsilon}$, after deleting the singular fibres.

Connected tier. *For every fixed $r \geq 3$, centred trace functions at distinct primes have family-averaged connected cumulants with square-root cancellation in the parameter-family count, uniformly in the range $X^r \mathfrak{C}_{n,r}^{1+\varepsilon} \leq B^{1-\varepsilon}$, and equivalently, after normalization, every fixed connected Frobenius cumulant tends to zero. An exponential connected-moment bound makes this convergence uniform in r over the range needed for occupancy functionals.*

Consequently, whenever the analytic Chebotarev error is $o(1)$ but the iid sampling functional of μ_n is nonvanishing, the complete conjugacy-cylinder count vector, averaged over parameters, has the nonuniform multinomial occupancy law of independent samples from μ_n, including its missing-mass, collision, and total-variation profile. The multinomial conclusion is derived from the connected tier, not from the L^2 bound alone.

Significance. This separates the two analytic inputs that a single-tier statement compresses into one claim. The square-root tier is a growing-monodromy large sieve in the spirit of Kowalski [47]. The connected tier is the genuinely stronger cross-prime independence principle required for an occupancy limit. The conductor budget is attached to the actual sheaves and their product trace functions rather than to an undefined scalar conductor of a cover. The first theorem is the connected tier at $r = 3$ for one fixed level n, square-root cancellation of the family-averaged third connected cumulant within the range governed by $\mathfrak{C}_{n,3}$. A failure identifies either geometric complexity missed by the sheaf conductors or arithmetic entanglement surviving all fixed-order trace tests.

Conjecture 43 (Coloured dynatomic Galois-factor process)**.** *For $f_c(x) = x^d + c$, let the critical exact-period polynomial factor as $\Psi_n(c) = \prod_{j=1}^{s_n} F_{n,j}(c)$ over $\mathbb{Z}$. Sample c from a dyadic height interval away from postcritically finite, discriminant, and fixed-resultant loci.*

For every fixed n, condition on any fixed finite local valuation state and normalize the prime factors of each $F_{n,j}(c)$ by that component's residual logarithmic mass. The component-coloured factor processes converge jointly to independent Frobenius-marked Poisson–Dirichlet

PD(1) *processes, with the mark in colour j given by the fixed-point-size-biased conjugacy law in the splitting field of $F_{n,j}$. The total logarithmic mass of ramified factors tends to zero.*

Uniformly for $n = n(H)$ with $\deg \Psi_n = o(\log H)$, every fixed correlation functional whose support lies inside the available level of distribution converges to the corresponding restricted-support correlation of these coloured marked processes. No full-process claim is made at growing degree without level-one distribution.

Significance. Exact-period components are genuine geometric colours, and their Galois marks distinguish dynamical families with similar unmarked factor sizes. The statement deliberately separates the full fixed-degree process from the restricted-support growing-degree law, in accordance with the distribution thresholds in Bharadwaj–Rodgers [51]. The first theorem is a two-colour correlation law at one fixed period. A failure would show that distinct exact-period components are entangled, with the splitting fields of two factors $F_{n,j}$ sharing Frobenius information that survives conditioning on the fixed finite local valuation state.

Conjecture 44 (Complexity-uniform primitive valuation process for dynatomic values)**.** *For the critical exact-period polynomial $\Psi_n(c)$, remove its canonical greatest common divisor in $\mathbb{Q}[c]$ with the earlier critical-orbit product $\prod_{m<n} f_c^m(0)$, in the notation of Conjecture 43, and denote the result by Ψ_n^{new}. Define the logarithmic arithmetic complexity*

$$\mathfrak{H}_n = \deg \Psi_n^{\mathrm{new}} + \log^+ H_{\mathrm{coeff}}(\Psi_n^{\mathrm{new}}) + \log^+ |\operatorname{Disc}(\Psi_n^{\mathrm{new}})| + \sum_{m<n} \log^+ |\operatorname{Res}(\Psi_n^{\mathrm{new}}, f_c^m(0))|,$$

where zero resultants have already been removed by the canonical gcd. For each prime p, let $\nu_{n,p}$ be the exact Haar law on $\mathbb{Z}_p$ of the primitive valuation

$$V_{n,p}(c) = v_p(\Psi_n^{\mathrm{new}}(c))\mathbf{1}_{p \nmid f_c^m(0)\ \forall m<n}.$$

All discriminant, resultant, and collision primes remain inside these local laws.

For every control function $\omega(H) = o(\log H)$, uniformly over levels with $\mathfrak{H}_n \le \omega(H)$, the finite-prime valuation vectors for c in a dyadic height interval converge to the corresponding marginals of $\bigotimes_p \nu_{n,p}$. In addition, for every $\eta > 0$,

$$\lim_{z\to\infty} \limsup_{H\to\infty} \sup_{n:\,\mathfrak{H}_n \le \omega(H)} \mathbb{P}\left(\sum_{p>z} (\log p)(V_{n,p} - 1)_+ > \eta \log |\Psi_n^{\mathrm{new}}(c)| \right) = 0,$$

and the analogous unnormalized tail $\sum_{p>z} \mathbb{P}(V_{n,p} \ge 2)$ tends to zero whenever the local squarefree product is positive. These two clauses give convergence in the topology generated by bounded cylinder functions and the logarithmic squarefull-mass functional.

Consequently

$$\mathbb{P}(\Psi_n^{\mathrm{new}}(c) \text{ is squarefree}) = \prod_p \tilde{\nu}_{n,p}(\{0,1\}) + o(1)$$

uniformly in the stated complexity range, where $\tilde{\nu}_{n,p} = \nu_{n,p}$ except at the finitely many collision primes dividing one of the resultants $\operatorname{Res}(\Psi_n^{\mathrm{new}}, f_c^m(0))$, at which $\tilde{\nu}_{n,p}$ is the law of the raw valuation $v_p(\Psi_n^{\mathrm{new}}(c))$, and the logarithmic squarefull mass converges to the law determined by the complete local product.

Significance. Degree alone does not control a growing polynomial family. The controlling range includes coefficient height, discriminant, and resultants, while the infinite-product conclusion is supported by an explicit uniform squarefull-tail assertion. The local process still retains every bad-prime and collision state and remains level dependent, as primitive hits at a fixed prime occur by bounded time modulo p^k. For the unicritical families $x^d + c$ the resultants at the first levels are units, so the stated complexity range is nonempty there. The first decisive theorem is the primitive valuation law with its uniform squarefull tail at one fixed level of the unicritical family $x^d + c$, giving the squarefree asymptotic for that single Ψ_n^{new}. A failure would show that $\mathfrak{H}_n$ is not the right complexity gauge, with some growing-level family carrying squarefull mass or valuation correlations that degree, coefficient height, discriminant, and resultants together do not see. The nearest literature treats primitive divisors and fixed-polynomial squarefree values, not this complexity-uniform adelic process [49, 48].

Conjecture 45 (Divisor-sensitive classification of dynamical gcd height)**.** *Let $F = (f, g)$ be a split map on $(\mathbb{P}^1)^2$, with f, g disintegrated (in the sense of Medvedev and Scanlon [57], so neither is linearly conjugate to a power map, a Chebyshev map, or a close relative of these) of the same degree, and let $P = (a, b)$ have positive canonical heights. Let D_1, D_2 be the two coordinate-zero divisors and let E be the exceptional divisor of the blow-up at $D_1 \cap D_2$. Then*

$$\limsup_{n\to\infty} \frac{h_E(F^n(P))}{\max\{h(f^n(a)), h(g^n(b))\}} > 0$$

if and only if there exist $m \geq 0$ and an irreducible F-periodic curve Z containing $F^m(P)$, with Z not contained in $D_1 \cup D_2$ (automatic when both coordinates have positive canonical height), such that, on the normalization of Z, the pullbacks of D_1 and D_2 have a common nonzero effective component. In the absence of such eventual entry into a divisor-compatible periodic curve, the normalized gcd height tends to zero.

The converse is unconditional in every setting where the required Vojta inequality for the blow-up is known, and otherwise is explicitly conditional on that inequality. The same criterion extends to split maps on $(\mathbb{P}^1)^r$ by replacing a common component with positive codimension-one intersection multiplicity among the pulled-back divisors.

Significance. The obstruction is not merely a common dynamical quotient. It is a periodic geometric relation carrying the actual divisors measured by the gcd height. Eventual entry is the right notion, since a preperiodic tail can carry the divisor relation before periodicity begins. Recent work resolves several set-theoretic dynamical-GCD problems, while this height-theoretic divisor classification is the additional boundary [50]. The first theorem is the positive-limsup direction, that eventual entry of the orbit into a divisor-compatible periodic curve Z forces $\limsup h_E(F^n(P))/\max\{h(f^n(a)), h(g^n(b))\} > 0$. Failure would point to a new source of macroscopic Diophantine proximity outside invariant geometry.

10 Programme V: adelic factorization and sieve flow

Throughout this programme $P^+(m)$ and $P^-(m)$ denote the largest and smallest prime factors of m, the constant γ is Euler's, $\rho_f(p)$ counts the roots of f modulo p, and every sample $n \in [N, 2N]$ is drawn uniformly.

Conjecture 46 (Frobenius-marked Poisson–Dirichlet process)**.** *Let $f \in \mathbb{Z}[x]$ be irreducible with splitting field L and transitive Galois group G. Sample $n \in [N, 2N]$. Mark each unramified prime factor p of $f(n)$ by its conjugacy class $C \subset G$ and give it mass $u = \log p/\log|f(n)|$. The macroscopic marked process converges to a marked* PD(1) *process with conditional mark law*

$$\mathbb{P}(C \mid u) = \frac{|C|}{|G|}\operatorname{fix}(C), \qquad \operatorname{fix}(C) \text{ the number of fixed roots of an element of } C,$$

independent of u. Given the masses and every fixed finite local valuation state, the marks of finitely many macroscopic factors are asymptotically independent. Under a normal quotient equipped with the induced permutation representation, the marked process pushes forward functorially.

Significance. Chebotarev contributes $|C|/|G|$, divisibility by a polynomial value size-biases by the number of fixed roots, and Burnside's lemma normalizes the result. Bharadwaj–Rodgers supply the unmarked factor-process framework [51]. Simultaneous Galois marking, conditional independence, and functorial pushforward are the new content. It connects Galois theory to probabilistic factorization directly. The first theorem is the mark law for the largest prime factor of one irreducible quadratic f, the two-class case where the size bias $\operatorname{fix}(C)$ already separates the marked law from the plain Chebotarev weight $|C|/|G|$. A failure would show that divisibility size-biases Frobenius by more than the fixed-root count, a coupling between factor masses u and Galois marks that survives conditioning on the finite local valuation state.

Conjecture 47 (Three-scale factorization of polynomial values)**.** *Let $f \in \mathbb{Z}[x]$ be irreducible, sample $n \in [N, 2N]$, choose $y = y(N) \to \infty$ with $\log y = o(\log N)$, and write $R_y = f(n)/S_y(f(n))$, where $S_y(m) = \prod_{p \le y} p^{v_p(m)}$ is the y-smooth part. Jointly with the complete small-prime valuation field $(v_p(f(n)))_{p \le y}$, the residual logarithmic factor process*

$$\sum_{p \mid R_y} v_p(R_y)\, \delta_{(\log p/\log|R_y|,\, \mathrm{Frob}_p)}$$

converges to the following conservation-corrected product object: the small valuations have their polynomial Kubilius law (independent exact local limit laws for the small-prime valuations), and conditional on their consumed logarithmic mass the residual process is a scale-invariant Poisson process of intensity du/u, marked by Conjecture 46, conditioned to have total mass one. The convergence is in total variation for the complete valuation field, the range $\log y = o(\log N)$ being the classical Kubilius range for the field at $f(x) = x$, and in the sense of finite-dimensional distributions for the residual process jointly with the field. The small field and the unconditioned residual Poisson process are asymptotically independent, and all leading dependence after conditioning is the single residual-mass constraint.

For disjoint normalized size bands bounded away from zero, the unconditioned residual process has independent increments. Its mesoscopic restriction and its ranked macroscopic atoms yield respectively the scale-invariant factor process and the Frobenius-marked PD(1) *partition. Thus the local Kubilius field is the boundary variable fixing the available mass, while the mesoscopic and macroscopic laws are two projections of one residual process, and no additional cross-scale coupling survives.*

Significance. Raw small–large independence is false because small factors consume mass. This conjecture identifies the complete conservation-corrected product structure and adds the mesoscopic scale that is absent from a two-scale statement. It would justify, in one limit theorem, the local sieve model, the scale-invariant factor process, and the macroscopic partition. The first decisive theorem is the conservation-corrected joint law at one fixed prime p and one normalized size band, where the only surviving dependence is the residual-mass constraint. A failure would show a cross-scale coupling between the small-prime valuation field and the residual process beyond the single residual-mass constraint, so smooth-part consumption would not exhaust the dependence between the sieve scale and the factor process.

Conjecture 48 (Adelic Gibbs gluing for reducible polynomial values)**.** *Let $f = \prod_{i=1}^{s} g_i$ be squarefree with irreducible components, sample $n \in [N, 2N]$, and put $t = n/N$. For each prime p, let ν_p be the exact Haar law on $\mathbb{Z}_p$ of the valuation vector $(v_p(g_1(x)), \ldots, v_p(g_s(x)))$. For $y \to \infty$ with $\log y = o(\log N)$, consider stable convergence jointly with the archimedean coordinate t and condition on $\mathscr{L}_y = (v_p(g_i(n)))_{p \le y, i \le s}$. After normalizing each component by its own residual logarithmic mass, the coloured Frobenius-marked factor processes converge to conditionally independent copies of the three-scale law of Conjecture 47, with the marks of Conjecture 46.*

Unconditionally, the joint process is the projective Gibbs mixture obtained by integrating those conditional product laws against Lebesgue measure in t and the adelic local law $\bigotimes_p \nu_p$, in the iterated order $N \to \infty$ and then $y \to \infty$. Every bounded finite-dimensional Laplace functional is determined by

$$Z_p(\mathbf{z}) = \int_{\mathbb{Z}_p} \prod_i z_i^{v_p(g_i(x))} \, dx$$

together with the archimedean size profile $(\log |g_i(Nt)|)_i$. No additional colour coupling survives after this adelic–archimedean mixture.

Significance. A scalar Euler factor cannot specify a point process. This statement gives the full local-to-global gluing rule and includes the archimedean variable that can couple component sizes. It is functorial under multiplying, splitting, or regrouping components. The first theorem is the case of two linear components, where $Z_p(\mathbf{z})$ is computable in closed form and the mixture over t and $\bigotimes_p \nu_p$ already yields the full two-colour gluing law. A persistent medium-prime or archimedean coupling after the stated mixture would identify a new nonadelic invariant.

Conjecture 49 (Full multivariate adelic saddle law for joint smoothness)**.** *Let $f_1, \ldots, f_s \in \mathbb{Z}[x]$ be pairwise coprime, primitive, and jointly admissible, meaning that for every prime p some residue x has $p \nmid \prod_i f_i(x)$, sample $n \in [N, 2N]$, and put $t = n/N$. For complex variables $\mathbf{s}$ and smoothness bounds $\mathbf{y}$, define*

$$\mathscr{Z}_{p,\mathbf{y}}(\mathbf{s}) = \int_{\mathbb{Z}_p} p^{-\sum_{i: p \le y_i} (s_i - 1) v_p(f_i(x))} \, dx, \qquad \widehat{\mathscr{Z}}_{p,\mathbf{y}}(\mathbf{s}) = \frac{\mathscr{Z}_{p,\mathbf{y}}(\mathbf{s})}{\mathscr{Z}_{p,\mathbf{y}}(+\infty)}.$$

The denominator is the exact Haar probability that p divides none of the relevant $f_i(x)$. Put

$$\Phi_t(\mathbf{s}) = \sum_p \log \widehat{\mathscr{Z}}_{p,\mathbf{y}}(\mathbf{s}) + \sum_i (s_i - 1) \log |f_i(Nt)|.$$

For $f(x) = x$, this gives the usual truncated zeta factor $(1 - p^{-s})^{-1}$. Define the joint local probability by the Perron-calibrated multivariate inverse Mellin integral

$$\mathscr{P}_t^{\text{joint}} = \frac{1}{(2\pi i)^s} \int_{\boldsymbol{\sigma}_t + i\mathbb{R}^s} e^{\Phi_t(\mathbf{s})} \prod_{i=1}^{s} \frac{ds_i}{s_i},$$

with the standard truncated-contour limit, where $\boldsymbol{\sigma}_t$ is the unique real saddle in the admissible region. The factors s_i^{-1} are part of the canonical cumulative smoothness event, and for one variable they recover the Hildebrand–Tenenbaum Perron normalization [56].

Uniformly when every $u_i(t) = \log|f_i(Nt)|/\log y_i$ lies in a fixed compact saddle-point region,

$$\mathbb{P}(P^+(f_i(n)) \le y_i \ \forall i) \sim \int_1^2 \mathscr{P}_t^{\mathrm{joint}}\, dt.$$

The integral has the full multivariate saddle expansion. Writing $H_t = \nabla^2\Phi_t(\boldsymbol{\sigma}_t)$, its leading term includes the joint action, the amplitude $\prod_i \sigma_{i,t}^{-1}$, and $(\det H_t)^{-1/2}$, with the usual lattice correction when the valuation span is proper.

If $\mathscr{P}_{i,t}^{\mathrm{marg}}$ are the identically calibrated one-polynomial probabilities, then $\mathscr{P}_t^{\mathrm{joint}}/\prod_i \mathscr{P}_{i,t}^{\mathrm{marg}}$ is the complete ratio of joint and marginal actions, Perron amplitudes, Hessian determinants, and lattice corrections. Its Euler-product component is the ratio of the exact p-adic partition functions evaluated at the joint and marginal saddles. None of these factors may be replaced by a scalar Euler correction at the marginal saddle.

Significance. The statement is calibrated to the cumulative smoothness probability. It combines exact higher p-adic valuations, moving archimedean sizes, the joint saddle displacement, Perron amplitudes, mixed curvature, and possible lattice effects. Even one-polynomial smooth-value asymptotics are difficult [52, 54]. A complete multivariate adelic saddle law would be a major new local–global bridge. A failure would show a joint smoothness dependence not produced by the exact p-adic partition functions and the moving archimedean sizes, an arithmetic coupling among the f_i outside the saddle displacement, the Hessian H_t, and the lattice correction.

Conjecture 50 (Buchstab–Bateman–Horn component flow)**.** *Let $f \in \mathbb{Z}[x]$ be primitive, irreducible, and admissible ($\rho_f(p) < p$ for every prime p), and sample $n \in [N, 2N]$. For y with $u_n = \log|f(n)|/\log y$ in a fixed compact subset of $(1,\infty)$, let $\Omega_y(f(n))$ be the number of prime factors of $f(n)$ exceeding y, counted with multiplicity, on the event $P^-(f(n)) > y$. Put*

$$V_f(y) = \prod_{p \le y}\left(1 - \frac{\rho_f(p)}{p}\right).$$

Define the Buchstab component functions by

$$\omega_1(u) = \frac{1}{u}, \qquad \omega_j(u) = \frac{1}{u(j-1)!}\int_{\substack{t_1,\dots,t_{j-1}\ge 1\\ t_1+\cdots+t_{j-1}\le u-1}} \frac{dt_1\cdots dt_{j-1}}{t_1\cdots t_{j-1}} \quad (j \ge 2),$$

so that $\sum_{j\ge1}\omega_j(u) = \omega(u)$. Then for every fixed j,

$$\mathbb{P}(P^-(f(n)) > y,\ \Omega_y(f(n)) = j) = V_f(y)e^{\gamma}\mathbb{E}_{n\in[N,2N]}\omega_j(u_n) + o(V_f(y)),$$

uniformly on compact u-ranges. Conditional on the ordered logarithmic sizes of the j remaining factors, their unramified Frobenius marks are independent with the fixed-point-size-biased law of Conjecture 46. Summing over j gives the rough-value Buchstab law, $j = 1$ gives the Bateman–Horn prime density, and $j = 2$ gives the semiprime transition. Whenever $\max_{N\le n\le 2N}|f(n)| < y^2$, every rough value has exactly one prime factor above y, so the $j = 1$ component exhausts the rough set, while the density formula itself remains conjectural.

Significance. The conjecture resolves a rough polynomial value into its complete finite-u factor count and Galois mark content. Primes and semiprimes are not inserted as separate heuristics. They are components of one universal flow. Generalized Buchstab equations are the random-integer antecedent [61], and the fixed-polynomial, Galois-marked component law with its local factors is the new statement. It is strongly falsifiable even when total roughness follows Buchstab.

Part III
Six conjectures across fields

Added to the deposit: 4 August 2026.

The six conjectures below leave the prime model. They range over multiplicative number theory, discrepancy theory, graph limits, additive information theory, and ergodic theory, and each is held to the same standard as the rest of the collection: a canonical object, a stated mechanism, the nearest literature boundary, a first decisive theorem where one exists, and an honest failure mode.

11 Relative polynomial pretentiousness

Let $P \in \mathbb{Z}[x]$ be primitive (the greatest common divisor of its coefficients is 1) and irreducible, with positive leading coefficient. For a prime p write

$$\rho_P(p) = \#\{a \in \mathbb{Z}/p\mathbb{Z} : P(a) \equiv 0 \pmod{p}\}.$$

The weight $\rho_P(p)/p$ is the first-order frequency with which p divides a polynomial value $P(n)$. A function f on the positive integers is completely multiplicative if $f(mn) = f(m)f(n)$ for all $m, n \geq 1$. For completely multiplicative f, g with values in the closed complex unit disc, define the truncated P-relative pretentious distance

$$\mathbb{D}_P(f, g; X)^2 = \sum_{p \leq X} \frac{\rho_P(p)}{p} \big(1 - \mathrm{Re}\big(f(p)\overline{g(p)}\big)\big), \qquad \mathbb{D}_P(f, g)^2 = \lim_{X \to \infty} \mathbb{D}_P(f, g; X)^2 \in [0, \infty].$$

Every summand is nonnegative, so the truncated distance is nondecreasing in X and the limit exists. Throughout, a primitive Dirichlet character χ of conductor c is regarded as the c-periodic completely multiplicative function on $\mathbb{Z}$ that vanishes on integers not coprime to c, so that expressions such as $\chi(P(a))$ are defined for every residue a modulo any multiple of c. For $t \in \mathbb{R}$, the comparison function $n \mapsto \chi(n)n^{it}$ is completely multiplicative with values in the closed unit disc.

Let $P_1, \ldots, P_m \in \mathbb{Z}[x]$ be pairwise distinct primitive irreducible polynomials with positive leading coefficients and no fixed prime divisor, meaning $\rho_{P_j}(p) < p$ for every j and every prime p. Assume multiplicative independence modulo constants:

$$\prod_{j=1}^{m} P_j(x)^{e_j} \in \mathbb{Q}^\times, \quad e_j \in \mathbb{Z} \quad \Longrightarrow \quad e_1 = \cdots = e_m = 0.$$

Write $\mathbb{S}^1$ for the unit circle in the complex plane.

Conjecture 51 (Relative polynomial pretentiousness inverse principle)**.** *Let $f_1, \ldots, f_m : \{1, 2, 3, \ldots\} \to \mathbb{S}^1$ be completely multiplicative. If*

$$\limsup_{X\to\infty} \left| \frac{1}{\log X} \sum_{\substack{n\le X \\ P_j(n)>0\ \forall j}} \frac{\prod_{j=1}^m f_j(P_j(n))}{n} \right| > 0,$$

then there exist primitive Dirichlet characters $\chi_1, \ldots, \chi_m$ and real numbers $t_1, \ldots, t_m$ such that the following three conditions hold simultaneously:

(i) $\mathbb{D}_{P_j}(f_j, \chi_j(n)n^{it_j}) < \infty$ *for every* $1 \le j \le m$,

(ii) $\sum_{j=1}^{m} t_j \deg P_j = 0$,

(iii) *for some common multiple q of the conductors of $\chi_1, \ldots, \chi_m$,*

$$\frac{1}{q} \sum_{a \bmod q} \prod_{j=1}^{m} \chi_j(P_j(a)) \neq 0,$$

the characters being extended by 0 on nonunits as above.

Significance. The mechanism is that a persistent logarithmic correlation should have exactly two sources: an archimedean resonance carried by the factors n^{it_j} and a finite local resonance carried by the characters χ_j. The balance condition (ii) is forced by the expansion $P_j(n)^{it_j} = c_j^{it_j} n^{it_j \deg P_j}(1 + o(1))$, with c_j the leading coefficient, because logarithmic averaging annihilates any nonzero total power of n, and (iii) asserts that the finite-modulus resonances are jointly compatible, so that the model correlation does not vanish identically for local reasons. Packaging the conclusion as one existential block matters: the characters and archimedean parameters cannot be chosen independently for each j, and it is the joint system (i)–(iii) that reproduces the correlation. The decisive-theorem ledger for the linear case is as follows. For linear polynomials the two-point case is Tao's logarithmically averaged two-point Elliott theorem [75]. The odd-order cases are the theorems of Tao–Teräväinen, and the shape of the conclusion here, one existential block coupling characters, archimedean parameters, and a nonvanishing local average, is the template of their structure theorem for logarithmically averaged correlations [76]. The general even-order linear case is precisely the logarithmically averaged Elliott conjecture. All of the novelty of the present statement therefore sits at $\max_j \deg P_j \ge 2$, where even the logarithmic mean value of $f(n^2 + 1)$ for a single nonpretentious completely multiplicative f is open, and a logarithmic mean value theorem for $f(n^2 + 1)$ with a conclusion of the stated relative-pretentious type would be the first decisive theorem. The failure mode is a system of functions nonpretentious in every relative metric whose logarithmic correlation has positive limsup, which would reveal a correlation mechanism beyond the archimedean and finite local resonances.

Remark 2. *The restriction to completely multiplicative f_j is not cosmetic. For 1-bounded multiplicative functions that are not completely multiplicative, the fixed-parameter form of the conclusion is false: Matomäki–Radziwiłł–Tao [71] constructed a 1-bounded multiplicative function that pretends to be n^{it_k} for different parameters t_k on different scales, refuting the original fixed-parameter Elliott conjecture, and the corrected conjecture must allow the comparison parameter to vary with the scale X. For the unimodular completely multiplicative f_j of the present statement* *the fixed-parameter form survives, and by a rigidity argument rather than by monotonicity alone. Fix a member f_j of the system, a primitive Dirichlet character χ, and scales $X_k \le X_{k+1}$, the scale index k being unrelated to the function index j. If χn^{is_k} and $\chi n^{is_{k+1}}$ are within bounded truncated distance of f_j at the scales X_k and X_{k+1} respectively, monotonicity of the truncated distance in the scale places both within bounded distance of f_j at the common scale X_k, the triangle inequality places them within bounded distance of each other, and the growth in X of $\sum_{p\le X}(1-\cos(\delta\log p))/p$ for fixed $\delta \ne 0$ then forces $|s_{k+1}-s_k| = O(1/\log X_k)$. After passing to a geometrically growing subsequence of the scales, $\log X_{k+1} \ge 2\log X_k$ say, which monotonicity permits without weakening the conclusion, the increments are summable, the parameters converge, and monotonicity converts boundedness along the scales into finiteness of the full distance at the limit parameter. Unimodularity at the primes is consumed at the triangle-inequality step: the passage through the middle function f_j rests on the identity $\chi n^{is_k}\,\overline{\chi n^{is_{k+1}}} = (\chi n^{is_k}\,\bar f_j)(f_j\,\overline{\chi n^{is_{k+1}}})$, valid at the primes exactly when $|f_j(p)| = 1$, and for values of modulus less than one the two comparisons decouple on the primes where the modulus drops, which is exactly the room the scale-switching construction occupies.*

Remark 3. *For each j, the finitely many primes dividing $\operatorname{disc}(P_j)\operatorname{lc}(P_j)$, the discriminant times the leading coefficient, may be weighted arbitrarily: replacing the weights $\rho_{P_j}(p)/p$ at those primes by any bounded values changes $\mathbb{D}_{P_j}(f,g)^2$ by a bounded amount and leaves the finiteness condition (i) unchanged. The relative metric is therefore canonical up to the bad primes, and the conjecture is insensitive to that choice.*

12 Monotone threshold purification of vector paths

Let $d, m \ge 1$. Let $v_1, \dots, v_m \in \mathbb{R}^d$ satisfy $\|v_i\|_2 \le 1$, where $\|\cdot\|_2$ is the Euclidean norm. Let $a_i : [0,1] \to [0,1]$ be continuous nondecreasing functions with $a_i(0) = 0$ and $a_i(1) = 1$, called a monotone activation schedule, and define the fractional path

$$F(t) = \sum_{i=1}^{m} a_i(t) v_i.$$

A threshold rounding of F is a choice of pairwise distinct $\tau_1, \dots, \tau_m \in (0,1]$ together with the pure path

$$G(t) = \sum_{i:\ \tau_i \le t} v_i.$$

Conjecture 52 (Monotone threshold purification of vector paths)**.** *There exists a universal constant $C < \infty$ such that for every $d, m \ge 1$, every $(v_i)_{i\le m}$, and every monotone activation schedule $(a_i)_{i\le m}$ as above, some threshold rounding satisfies*

$$\sup_{0\le t\le 1} \|G(t) - F(t)\|_2 \le C\sqrt{d}.$$

Significance. The headline attribution comes first. The special case $a_i(t) = t$ for all i with $\sum_i v_i = 0$ is exactly the open prefix-sum form of the Euclidean Steinitz problem at the scale $O(\sqrt{d})$: there $F \equiv 0$, and ordering the vectors by their thresholds turns the display into the assertion that some permutation π has all prefix sums bounded, $\max_{k\le m} \|\sum_{i\le k} v_{\pi(i)}\|_2 \le C\sqrt{d}$. The distinctness of the thresholds is what makes this reduction exact: allowing coincident thresholds would admit the rounding with every $\tau_i = 1$, whose pure path is identically zero in this special case and exposes no prefix sum. Grinberg–Sevast'yanov [66] proved the bound d in every norm, the best known Euclidean bound is Banaszczyk's [62, 63] $O(\sqrt{d + \log m})$, now matched constructively in the high-dimensional regime $d \geq \mathrm{polylog}(m)$ [172], and so already in this special case the conjecture asserts the removal of the logarithmic term uniformly in m. For the general threshold form, even the existence of any bound $C(d)$ depending on d alone and uniform in m appears to be open. The mechanism is that F is a monotone trajectory in the zonotope generated by the v_i, the set of sums $\sum_i \lambda_i v_i$ with $\lambda_i \in [0,1]$, and the conjecture asserts that every monotone zonotope path admits an integral shadow at the optimal Euclidean discrepancy scale. The order $\sqrt{d}$ is verified sharp: take $m = d$, $v_i = e_i$ the standard basis, and $a_i(t) = t$, so that at $t = 1/2$ every coordinate of $G(1/2) - F(1/2)$ equals $\pm 1/2$ whatever the thresholds, giving $\|G(1/2) - F(1/2)\|_2 = \sqrt{d}/2$. No decisive theorem exists: the first would be any m-uniform bound $C(d)$ for the threshold form, even with a d-dependence far worse than $\sqrt{d}$. A falsification honesty note: since even the existence of a finite bound at each fixed d is open, the conjecture can fail in two ways, at fixed dimension, through paths with d fixed and $m \to \infty$ whose best threshold roundings have unbounded discrepancy, or asymptotically, through an infinite family of monotone paths with $d \to \infty$ every threshold rounding of which has discrepancy $\omega(\sqrt{d})$. Small-d computer searches at bounded m cannot decide either failure mode.

13 Entropy dimension of random-free graphons

A graphon is a symmetric measurable function $W : [0,1]^2 \to [0,1]$, and W is random-free if it takes values in $\{0,1\}$ almost everywhere. Define the row pseudometric

$$d_W(x,y) = \int_0^1 |W(x,z) - W(y,z)| \,\mathrm{d}z.$$

Identifying points at zero distance and completing yields a metric-measure space (Ω_W, d_W, ρ_W), the row space, where ρ_W is the pushforward of Lebesgue measure. The row space is Ahlfors s-regular if it is compact and there exist $c, C, r_0 > 0$ such that

$$cr^s \leq \rho_W(B(x,r)) \leq Cr^s$$

for every $x \in \mathrm{supp}\,\rho_W$ and every $0 < r < r_0$. Let $G(n,W)$ be the random labelled graph on $[n] = \{1,\dots,n\}$ obtained from independent uniform $U_1,\dots,U_n$ on $[0,1]$ by declaring ij an edge exactly when $W(U_i,U_j) = 1$, and let $H(G(n,W))$ be its Shannon entropy with natural logarithms.

Conjecture 53 (Entropy dimension of random-free graphons)**.**

(i) [nonvacuity] *For every $s \in (0,2]$ there exists a random-free graphon whose row space is Ahlfors s-regular.*

(ii) [refined entropy-dimension law] *If the row space of a random-free graphon W is Ahlfors s-regular for some $s > 0$, then*

$$H(G(n,W)) = s\,n\log n + O_W(n).$$

Proposition 1 (Entropy-dimension limit law)**.** If the row space of a random-free graphon W is Ahlfors s-regular for some $s > 0$, then

$$H(G(n,W)) = s\,n\log n + O_W(n\log\log n),$$

and in particular $\lim_{n\to\infty} H(G(n,W))/(n\log n) = s$.

Proof. Write $d = d_W$ and $\rho = \rho_W$, and use natural logarithms. For the lower bound, set $k = \lfloor n/\log n \rfloor$, fix the labelling, and designate the last k vertices as tests, with latent points $Z = (Z_1, \ldots, Z_k)$ and remaining latent points $X_1, \ldots, X_{n-k}$. The cross matrix $M = (W(X_i, Z_j))_{i,j}$ is a function of the labelled graph, and conditioning cannot increase entropy, so $H(G(n, W)) \geq H(M) \geq H(M \mid Z)$. Conditionally on Z the signature rows $S_Z(X_i) = (W(X_i, Z_1), \ldots, W(X_i, Z_k))$ are independent with a common law, whence $H(M \mid Z) = (n-k)\, \mathbb{E}_Z H(S_Z(X))$ with X an independent uniform point. Let C_Z be the collision probability of the signature law. Shannon entropy dominates the Rényi entropy of order two, so $H(S_Z(X)) \geq -\log C_Z$, and convexity of $-\log$ gives $\mathbb{E}_Z H(S_Z(X)) \geq -\log \mathbb{E}_Z C_Z$. For independent uniform X and Y the tests are independent uniform points, two rows agree at a test with probability $1 - d(X, Y)$, and Fubini gives the exact identity $\mathbb{E}_Z C_Z = \mathbb{E}_{X,Y}(1 - d(X, Y))^k$. Since $(1-u)^k \leq \mathrm{e}^{-ku}$ and the upper mass bound gives $\rho(B(x, r)) \leq Cr^s$, a dyadic shell decomposition yields, uniformly in x,

$$\int \mathrm{e}^{-k\, d(x,y)}\, d\rho(y) \leq \rho(B(x, 1/k)) + \sum_{m \geq 0} \mathrm{e}^{-2^m} \rho\big(B(x, 2^{m+1}/k)\big) = O_W(k^{-s}).$$

Hence $\mathbb{E}_Z C_Z = O_W(k^{-s})$, so $\mathbb{E}_Z H(S_Z(X)) \geq s \log k - O_W(1)$ and $H(G(n, W)) \geq (n-k)(s \log k - O_W(1)) = s\, n \log n - O_W(n \log \log n)$.

For the upper bound, let $\delta \in (0, \frac{1}{2})$. The lower mass bound makes the balls of radius $\delta/2$ around a maximal δ-separated set disjoint, each of mass at least $c(\delta/2)^s$, so the row space admits a Borel partition into $N_\delta = O_W(\delta^{-s})$ cells of diameter at most 2δ. Let Q_i be the cell of the ith latent row. Subadditivity and conditioning give $H(G(n, W)) \leq n \log N_\delta + \binom{n}{2} \mathbb{E}_{(a,b)} h(p_{ab})$, where (a, b) is the cell pair of two independent rows, p_{ab} is the conditional edge probability, and h is the binary entropy. For a $\{0, 1\}$-valued W, draw independent latent points X, X' with rows in the cell a and independent latent points Y, Y' with rows in the cell b, so that $W(X, Y)$ and $W(X', Y')$ are independent indicators of mean p_{ab} and $p_{ab}(1 - p_{ab}) = \frac{1}{2}\mathbb{P}(W(X, Y) \neq W(X', Y'))$, while the union bound gives $\mathbb{P}(W(X, Y) \neq W(X', Y')) \leq \mathbb{P}(W(X, Y) \neq W(X', Y)) + \mathbb{P}(W(X', Y) \neq W(X', Y'))$. Averaging each term over the opposite cell with its mass removes that cell restriction, so $\mathbb{E}_{(a,b)}[p_{ab}(1 - p_{ab})] \leq \frac{1}{2}\big(\mathbb{E}_a \mathbb{E}_{X,X' \in a}\, d(X, X') + \mathbb{E}_b \mathbb{E}_{Y,Y' \in b}\, d(Y, Y')\big) \leq 2\delta$. The bound $h(p) \leq \varphi(p(1-p))$ with the concave increasing $\varphi(u) = 2u(1 + \log(1/u))$ and Jensen give $\mathbb{E}_{(a,b)} h(p_{ab}) \leq \varphi(2\delta) = O(\delta \log(1/\delta))$. Therefore $H(G(n, W)) \leq s\, n \log(1/\delta) + O_W(n) + O(n^2 \delta \log(1/\delta))$, and the choice $\delta = (n \log^2 n)^{-1}$ gives $H(G(n, W)) \leq s\, n \log n + O_W(n \log \log n)$. □

Remark 4. *The conjecture is the refinement of the error term of proposition 1 to $O_W(n)$, which lies beyond both bounds of the proof: locating the mispredicted edges of a resolution-$1/n$ code costs order $n \log n$ if done naively, and the two-scale correction that codes coarse types first stops at $O(n \log \log n)$.*

Significance. The mechanism is metric distinguishability of rows: two row types at d_W-distance ε disagree on an ε-fraction of potential neighbours and become statistically distinguishable against n sampled vertices at a scale comparable to $1/n$, an Ahlfors s-regular row space has about n^s distinguishable types at that scale, and the latent type of each of the n labelled vertices then carries $s \log n + O(1)$ nats. Clause (i) certifies that the hypothesis class is nonvacuous at every fractional s: it is expected to be provable by Cantor-type threshold constructions, in which W is the indicator of a threshold event read along a self-similar Cantor set, and it is included so that clause (ii) quantifies over a genuinely rich family. The nearest literature boundary is Hatami–Norine [141]: a graphon is random-free exactly when $H(G(n, W)) = o(n^2)$, and every subquadratic growth order occurs for suitable random-free graphons, so no hypothesis short of a metric-measure condition on the row space can pin the coefficient of $n \log n$, and Ahlfors regularity is exactly the condition that does. The decisive theorem half-exists: the limit law is provable from regularity by the collision-entropy and quantization bounds of proposition 1, so the first theorem for the conjecture proper is the linear error term for one explicit self-similar threshold graphon of noninteger s, together with a proof of clause (i). The failure mode is a genuinely growing correction: an Ahlfors regular random-free graphon whose entropy exceeds $s\,n \log n + C\,n$ for every C would show that the resolution-$1/n$ coding loss is intrinsic rather than an artefact of the two-scale method.

14 A Hessian principle for isolated microcanonical graph phases

Fix a partition

$$[0,1] = I_1 \sqcup \cdots \sqcup I_q, \qquad |I_a| = \alpha_a > 0.$$

A coloured graphon is a symmetric measurable $U : [0,1]^2 \to [0,1]$ considered together with this fixed colour partition. Let $p(U) \in [0,1]^D$ with $D = q(q+1)/2$ denote the block-average vector

$$p_{ab}(U) = \frac{1}{|I_a||I_b|} \int_{I_a \times I_b} U(x,y)\, \mathrm{d}x\, \mathrm{d}y, \qquad 1 \le a \le b \le q.$$

A coloured finite graph is a finite simple graph F with a colour $c(v) \in \{1, \dots, q\}$ attached to each vertex, and its coloured homomorphism density in U is

$$t(F,U) = \int_{[0,1]^{V(F)}} \prod_{v \in V(F)} \alpha_{c(v)}^{-1} \mathbf{1}_{I_{c(v)}}(x_v) \prod_{uv \in E(F)} U(x_u, x_v) \prod_{v \in V(F)} \mathrm{d}x_v.$$

Fix coloured finite graphs $F_1, \dots, F_k$ and put $\mathbf{t}(U) = (t(F_1,U), \dots, t(F_k,U))$. Define the graphon entropy

$$\mathcal{S}(U) = \frac{1}{2} \int_{[0,1]^2} \big[-U \log U - (1-U)\log(1-U)\big]\, \mathrm{d}x\, \mathrm{d}y,$$

with $0 \log 0 = 0$, and for a feasible constraint vector $\mathbf{a}$ the contracted entropy

$$\Psi_{\mathbf{a}}(p) = \sup\{\mathcal{S}(U) : \mathbf{t}(U) = \mathbf{a},\ p(U) = p\}.$$

The finite graphs are coloured with pinned classes. For each n set $n_a = \lfloor \alpha_a n \rfloor$ and $N_n = \sum_a n_a$, and fix a partition of the vertex set $[N_n]$ into colour classes $V_1, \dots, V_q$ with $|V_a| = \lfloor \alpha_a n \rfloor$ exactly, so that $N_n = n + O(q)$. For a graph G on $[N_n]$ with these colour classes, partition each I_a into n_a consecutive intervals of length $|I_a|/n_a$ assigned to the vertices of V_a, let U_G be the $\{0,1\}$-valued coloured graphon equal to 1 on a product of two vertex intervals exactly when the corresponding pair is an edge, and put $\mathbf{t}(G) = \mathbf{t}(U_G)$ and $\widehat{p}(G) = p(U_G)$.

Fix $0 < \eta < 1$ and assume the following regularity hypotheses.

(H1) The supremum of $\mathcal{S}(U)$ over $\mathbf{t}(U) = \mathbf{a}$ is attained at a unique coloured graphon U_*, up to measure-preserving maps preserving each colour class.

(H2) Writing $p_* = p(U_*)$, the feasible block-average set $\{p(U) : \mathbf{t}(U) = \mathbf{a}\}$ is, near p_*, a C^3 embedded manifold $M \subset \mathbb{R}^D$.

(H3) The restriction $\Psi_{\mathbf{a}}|_M$ is C^3 near p_* and has there a unique local maximum.

(H4) The quadratic form $A = -D^2_M \Psi_{\mathbf{a}}(p_*)$ is positive definite on the tangent space $T = T_{p_*} M$.

(H5) [transversally Lipschitz constraint correspondence] There exist $\delta_0 > 0$ and $L < \infty$ such that every coloured graphon U with $\|p(U) - p_*\| \le \delta_0$ and $\|\mathbf{t}(U) - \mathbf{a}\|_\infty = \delta \le \delta_0$ satisfies $\operatorname{dist}(p(U), M) \le L\delta$. In particular the conditioning window below confines $\widehat{p}_n$, on the event $\|\widehat{p}_n - p_*\| \le \delta_0$, to an $o(n^{-1})$ thickening of M.

(H6) [nonemptiness] For every sufficiently large n the conditioning set $\{G : \|\mathbf{t}(G) - \mathbf{a}\|_\infty \le n^{-1-\eta}\}$ over graphs G on $[N_n]$ with the pinned colour classes is nonempty.

Let G_n be uniform on the conditioning set of (H6), let $\widehat{p}_n = \widehat{p}(G_n)$, and let Π_T be orthogonal projection of $\mathbb{R}^D$ onto T.

Conjecture 54 (Hessian principle for an isolated microcanonical phase)**.** *Under hypotheses* (H1)–(H6)*,*

$$n\,\Pi_T(\widehat{p}_n - \mathbb{E}\,\widehat{p}_n) \xrightarrow{d} N_T(0, A^{-1}),$$

where $N_T(0, A^{-1})$ *denotes the centred Gaussian law on* T *whose precision form is* A*.*

Significance. The mechanism is a Laplace expansion at the dense-graph entropy exponent n^2: on the feasible manifold, $\Psi_{\mathbf{a}}(p_* + u) = \Psi_{\mathbf{a}}(p_*) - \frac{1}{2}\langle Au, u\rangle + O(\|u\|^3)$, so the fluctuation scale is $u \asymp n^{-1}$ and the predicted covariance on the tangent space is A^{-1}, while (H5) suppresses the transversal directions below the fluctuation scale, which is exactly why the conclusion is stated only after projection by Π_T and centring at $\mathbb{E}\,\widehat{p}_n$. The nearest literature boundary is the Chatterjee–Varadhan [64] large deviation principle, which supplies the n^2 normalization and the variational description of conditioned dense graphs, together with the multipodal structure theory of Kenyon–Radin–Ren–Sadun [67] and Neeman–Radin–Sadun [72] (constrained entropy maximizers are step-function graphons with finitely many blocks), which makes hypotheses (H1)–(H4) verifiable in concrete constrained ensembles. A scale check disposes of one suspect: a single edge toggle moves each coordinate of $\mathbf{t}(G)$ by $O(n^{-2})$, so the lattice of attainable constraint values has spacing of order n^{-2}, far below the fluctuation scale n^{-1}, and the genuine risk is therefore Laplace-ratio control, namely uniform comparison of microcanonical counts with their Laplace approximations across windows of width $o(n^{-1})$, not arithmetic oscillation. Nonemptiness (H6) is a genuine hypothesis and may fail along subsequences of n for arithmetic reasons, in which case the statement is asserted only along the subsequence where it holds. No decisive theorem exists at this generality: the natural first one is the case $q = 1$ with the edge-triangle constraint pair in a region where the entropy maximizer is bipodal, a two-block step function, and (H1)–(H5) have been verified. The failure mode is an isolated nondegenerate phase whose tangent fluctuations are non-Gaussian or live at a different scale, which would necessarily come from a failure of Laplace-ratio control. The conjectural content is exactly that (H1)–(H6) already force this local regularity: a subexponential prefactor in the microcanonical counts varying across the conditioning window would leave every hypothesis intact while tilting the limit away from $N_T(0, A^{-1})$, and the conjecture asserts that isolated nondegenerate phases exclude it.

15 Sharp stability of entropy idempotence on finite abelian groups

Let G be a finite abelian group and μ a probability measure on G. Write $H(\mu)$ for Shannon entropy with natural logarithms, $\mu * \mu$ for the convolution, the law of $X+Y$ with X, Y independent of law μ, and

$$\Delta(\mu) = H(\mu * \mu) - H(\mu)$$

for the convolution entropy defect. If $H \le G$ is a subgroup and $a \in G$, let u_{a+H} be the uniform measure on the coset $a + H$. Total variation distance is

$$\|\mu - \nu\|_{\mathrm{TV}} = \frac{1}{2} \sum_{x \in G} |\mu(x) - \nu(x)|.$$

Conjecture 55 (Sharp stability of entropy idempotence)**.** *For every finite abelian group G and every probability measure μ on G, with entropies in natural logarithms,*

$$\inf_{\substack{H \le G \\ a \in G}} \|\mu - u_{a+H}\|_{\mathrm{TV}}^2 \le \frac{2}{\log 2}\big(H(\mu * \mu) - H(\mu)\big),$$

and the constant $2/\log 2$ is sharp.

Remark 5. *The equality case is verified. With X, Y independent of law μ,*

$$\Delta(\mu) = H(X+Y) - H(X) = H(X+Y) - H(X+Y \mid Y) = I(X+Y; Y) \ge 0,$$

where I denotes mutual information. If $\Delta(\mu) = 0$ then $X + Y$ is independent of Y, which forces the translates of μ by the elements of its support to coincide, and hence forces μ to be uniform on a coset of a subgroup. The right-hand side therefore vanishes exactly on the proposed extremizers.

Significance. The mechanism is that the defect is the mutual information created by one self-convolution, so the conjecture is a dimension-free quantitative classification of near-idempotent measures with cosets as the exact equality case. The sharpness discussion identifies the extremal geometry. Coset perturbations $\mu = (1 + \varepsilon f)u_H$ with $\sum_{x \in H} f(x) = 0$ give, as $\varepsilon \to 0$, the ratio

$$\frac{\|\mu - u_H\|_{\mathrm{TV}}^2}{\Delta(\mu)} \longrightarrow \frac{(\mathbb{E}|f|)^2}{2\,\mathbb{E}f^2} \le \frac{1}{2}$$

by Cauchy–Schwarz, so perturbative examples certify at most the constant $1/2$. The near-extremizers are instead discretized Gaussians at intermediate scale on $\mathbb{Z}/p\mathbb{Z}$: for $1 \ll \sigma(p) \ll p$, the discretization μ_p of a centred Gaussian of standard deviation $\sigma(p)$ has $\Delta(\mu_p) \to \frac{1}{2}\log 2$, the entropy-power constant recording the entropy gain of doubling a Gaussian variance, while its total variation distance to every u_{a+H} tends to 1, since the only subgroups of $\mathbb{Z}/p\mathbb{Z}$ are trivial and full and μ_p is asymptotically singular to every point mass and to the uniform measure. Hence no constant below $2/\log 2$ can serve, the sharpness half of the statement. The positioning against the literature is as follows: the entropic inverse theorems of Tao's sumset entropy theory [74] and the entropic polynomial Freiman–Ruzsa theorem of Gowers–Green–Manners–Tao [65] conclude in Ruzsa distance, an entropic closeness of X to a subgroup-uniform variable, not in total variation; the subgroup-form inverse theorem of Green–Manners–Tao [87], supplying for small defect a finite subgroup within entropic Ruzsa distance a bounded multiple of $\Delta(\mu)$, upgrades to total variation by a Pinsker argument, and with the trivial bound in the large-defect regime this settles the existence of some finite universal constant, so the sharp value is what the statement adds. On $\mathbb{Z}/p\mathbb{Z}$ there are no proper nontrivial subgroups, so the conjecture specializes to a discrete entropy-power-type lower bound for spread measures: any μ far in total variation from every point mass and from the uniform measure must create a definite amount of entropy under self-convolution. The natural first decisive theorem is that $\mathbb{Z}/p\mathbb{Z}$ specialization with the sharp constant. The defect also has the exact form $H(\mu * \mu) - H(\mu) = \mathbb{E}_Y D_{\mathrm{KL}}(\tau_Y \mu \,\|\, \mu * \mu)$, with

τ_y translation by y, so a small defect makes typical support-translates of μ close to $\mu * \mu$ in relative entropy, and Pinsker's inequality converts that closeness to total variation: the route to a universal constant is to upgrade approximate invariance under many translations to proximity to a coset uniform. The failure mode is a sequence μ_n with $\Delta(\mu_n) \to 0$ whose coset distance is much larger than $\sqrt{\Delta(\mu_n)}$, and the natural hunting ground is measures straddling several subgroup scales inside groups with rich subgroup lattices, a family for which computational search through $\mathbb{Z}/2^8\mathbb{Z}$ finds no ratio $\|\mu - u_{a+H}\|_{\mathrm{TV}}^2/\Delta(\mu)$ above the discretized-Gaussian value $2/\log 2$. Of the original assertion's three layers, the existence of some universal constant is settled by the inverse theorem as noted, and the conjecture is the remaining two: the sharp value $2/\log 2$ and the classification of near-extremizers as the discretized Gaussians at intermediate scale.

16 Stationary convolution-entropy rigidity

Let K be a finite abelian group and $\Omega = K^{\mathbb{Z}}$ with coordinatewise addition and the left shift σ. Let μ be a σ-invariant ergodic Borel probability measure on Ω, and write $h_\sigma(\mu)$ for its Kolmogorov–Sinai entropy. Define the translational stabilizer

$$H_\mu = \{h \in K^{\mathbb{Z}} : (\tau_h)_*\mu = \mu\}, \qquad \tau_h(x) = x + h,$$

a closed shift-invariant subgroup of $K^{\mathbb{Z}}$, and let $\pi_\mu : K^{\mathbb{Z}} \to K^{\mathbb{Z}}/H_\mu$ be the quotient homomorphism. Let $\mu * \mu$ be the law of $X + Y$ for independent $X, Y \sim \mu$.

Conjecture 56 (Stationary convolution-entropy rigidity)**.** *For every shift-ergodic μ,*

$$h_\sigma(\mu * \mu) = h_\sigma(\mu) \quad \iff \quad h_\sigma((\pi_\mu)_*\mu) = 0.$$

Remark 6. *The binary specialization deserves prominence. Take $K = \mathbb{Z}/2\mathbb{Z}$. By Kitchens' structure theory of group shifts [68], every closed shift-invariant subgroup of $(\mathbb{Z}/2\mathbb{Z})^{\mathbb{Z}}$ is a group shift of finite type, and every proper one is the solution set of a nontrivial linear recurrence and hence finite. If H_μ is finite the quotient map has finite fibres and $h_\sigma((\pi_\mu)_*\mu) = h_\sigma(\mu)$, while if H_μ is everything then μ is the Haar measure, the uniform Bernoulli process, with entropy* $\log 2$. *Since $h_\sigma(\mu * \mu) \geq h_\sigma(\mu)$ always, by conditioning on one summand, the conjecture therefore asserts: every ergodic binary process with* $0 < h_\sigma(\mu) < \log 2$ *gains entropy under independent XOR, the coordinatewise sum modulo* 2 *with an independent copy.*

Proposition 2 (Easy direction, conditional)**.** Suppose $h_\sigma((\pi_\mu)_*\mu) = 0$. Assume the following two ingredients.

(a) The Abramov–Rokhlin entropy addition formula holds for the Haar-fibred compact group extension $\pi_\mu : (\Omega, \mu, \sigma) \to (\Omega/H_\mu, (\pi_\mu)_*\mu, \sigma)$ and for the corresponding extension of $\mu * \mu$, so that each total entropy splits as the quotient entropy plus the common Haar fibre entropy, the conditional measures of μ and of $\mu * \mu$ over the quotient being Haar measures on cosets of H_μ.

(b) The entropy of a factor of an independent joining is bounded by the sum of the entropies of the factors, so that $(\pi_\mu)_*(\mu * \mu) = ((\pi_\mu)_*\mu) * ((\pi_\mu)_*\mu)$ has entropy at most $2h_\sigma((\pi_\mu)_*\mu) = 0$.

Then $h_\sigma(\mu * \mu) = h_\sigma(\mu)$.

Significance. The mechanism is the stationary analogue of entropy idempotence on a finite group: equality of entropy under independent self-convolution should force all positive entropy to be Haar-type randomness along a translation-invariant subgroup process, modulo a quotient that may retain arbitrarily rich zero-entropy dynamics such as rotations, substitutions, or Toeplitz systems. The easy direction is recorded above as a conditional proposition with its two ingredients named, the Abramov–Rokhlin formula for Haar-fibred compact group extensions and the factor-of-joining entropy bound, and the difficult direction is the rigidity claim that no other equality mechanism exists. The nearest literature boundary consists of Kułaga-Przymus–Lemańczyk [69] on the entropy of products of independent stationary processes and Lindenstrauss–Meiri–Peres [70] on the entropy of convolutions, neither of which decides the binary question of remark 6. No decisive theorem appears to be on record even for Markov processes, and strict XOR entropy gain for mixing two-state Markov chains would be the first. The failure-mode honesty note is this: the natural counterexample hunting ground is the Ornstein–Weiss [73] bilaterally deterministic processes, positive entropy processes whose distant past and distant future jointly determine the whole trajectory, together with T, T^{-1}-type constructions (skew products driven by a random walk, in the style of Kalikow), and no search has yet probed either family, so the negation of the conjecture has an unexplored natural habitat. The gap the rigidity must close is quantitative: equality of entropy rates asserts only that the block mutual information $I(X_{1:n} + Y_{1:n}; Y_{1:n})$ is $o(n)$, not that it is bounded, so sublinear information spread through arbitrarily long-range structure is the resource a counterexample would exploit. The natural staircase of evidence runs from independent letters through Markov and mixing processes to completely positive entropy.

Part IV

The class-counting polynomial of graphical Lie algebras

Added to the deposit: 5 August 2026.

The twenty conjectures below concern a single canonical object, the class-counting polynomial of a finite graph, approached through the two-step nilpotent Lie algebras that realize it as a rank generating function. They form three programmes. A rigidity and reconstruction programme (Conjectures 57 to 66) asserts that coarse specializations of the polynomial already determine it and that the polynomial in turn determines classical invariants. A log-concavity programme (Conjectures 67 to 71) locates the exact strength of positivity that the coefficient arrays support, with the boundary calibrated by explicit counterexamples one level higher. A tree extremal programme (Conjectures 72 to 76) places the path and the star at the two ends of every ordering the polynomial induces on trees. The proved layer is separated from the conjectural one: the rank formula (Proposition 3) and the two lower bounds on the number of rank values in a tree (Lemma 2) are theorems with full proofs, joined by the attainment of those bounds by the path and the star (Proposition 4) and the reduction of the star majorization to a single anti-concentration bound (Proposition 5), and every conjecture is stated against that floor. Each statement is held to the standard of the rest of the collection: a canonical object, a stated mechanism, the nearest literature boundary, a first decisive theorem where one exists, and an honest failure mode. Two statements, Conjectures 63 and 71, have since been resolved false by explicit counterexamples recorded with them, a seven-vertex unicyclic pair and a connected eight-vertex graph respectively, and both are retained with their resolutions rather than renumbered.

For a finite simple graph G on vertex set $V(G)$ with edge set $E(G)$ and a field F, the *graphical*

Lie algebra $L_G(F)$ is the two-step nilpotent Lie algebra over F with basis $\{v_i : i \in V(G)\}$ together with the central basis $\{z_e : e \in E(G)\}$, the bracket being $[v_a, v_b] = z_{ab}$ for each edge $\{a, b\}$ with $a < b$, $[v_b, v_a] = -z_{ab}$, and all other brackets of basis vectors zero. For $S \subseteq V(G)$ write $N_G[S]$ for the closed neighbourhood of S, the set of vertices lying in S or adjacent to it, and $c(G[S])$ for the number of connected components of the induced subgraph $G[S]$, and set

$$\rho_G(S) = \big|N_G[S]\big| - c(G[S]), \qquad \rho_G(\emptyset) = 0.$$

Proposition 3 (Rank formula)**.** Let G be a graph on n vertices, let F be a field, and let $x \in L_G(F)$ have vertex part supported on S, all coordinates x_i with $i \in S$ being nonzero. Then

$$\operatorname{rank}(\operatorname{ad}_x) = \rho_G(S),$$

independently of the field and of the nonzero values.

Proof. The map ad_x kills the centre and sends v_j to $\sum_{i\in S,\, i\sim j} \pm x_i z_{ij}$, so $\operatorname{rank}(\operatorname{ad}_x)$ is the rank of the linear map $y \mapsto [x, y]$ on the vertex space, whose coefficient at z_{ab} is $x_a y_b - x_b y_a$. A vector y lies in the kernel if and only if $x_a y_b = x_b y_a$ for every edge $\{a, b\}$. If both endpoints lie in S this forces the ratio y_i/x_i to be constant along edges of $G[S]$, one free parameter per connected component of $G[S]$. If exactly one endpoint a lies in S it forces $y_b = 0$, for every $b \in N(S) \setminus S$. Edges with neither endpoint in S impose nothing, so y_j is free for every j outside $N[S]$. Hence $\dim\ker = (n - |N[S]|) + c(G[S])$, and the rank is $|N[S]| - c(G[S])$. □

The formula was verified before it was proved: on eighteen thousand random pairs of a graph and an element, the rank of ad_x computed by direct linear algebra over $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ agreed with $\rho_G(S)$ in every case, and the proof above was written only afterwards.

The *class-counting polynomial* of G is

$$C_G(X, Y) = \sum_{S\subseteq V(G)} (X-1)^{|S|}\, Y^{\rho_G(S)}.$$

The polynomial is Rossmann's [77]: he introduced it, proved that its specializations enumerate the conjugacy classes of graphical groups over $\mathbb{F}_q$ by size, proved the multiplicativity $C_{G\sqcup H} = C_G\, C_H$ over disjoint unions, and showed that the cardinalities of the connected dominating sets of G can be read off from it. Set

$$H_G(U, Y) = C_G(1 + U, Y), \qquad F_G(X, Y) = X^{|E(G)|}\, C_G(X, X^{-1}Y), \qquad f_G(X) = F_G(X, 1),$$

so that the coefficient of $U^k Y^r$ in H_G counts the sets S with $|S| = k$ and $\rho_G(S) = r$. At $X = q$ the polynomial records the distribution of $\operatorname{rank}(\operatorname{ad}_x)$ over $L_G(\mathbb{F}_q)$, by Proposition 3. The *subgraph-component polynomial* of Tittmann–Averbouch–Makowsky [78] is

$$Q_G(U, W) = \sum_{S\subseteq V(G)} U^{|S|}\, W^{c(G[S])}.$$

For a tree T on n vertices let $s(T)$ be the number of distinct values taken by ρ_T, let $\ell(T)$ be the number of leaves, and let $\operatorname{diam}(T)$ be the diameter.

Lemma 2 (Two lower bounds on the number of rank values)**.** *For every tree T on $n \geq 2$ vertices,*

$$s(T) \geq n - \ell(T) + 2 \qquad \textit{and} \qquad s(T) \geq \operatorname{diam}(T) + 1.$$

Proof. For the first bound, when $n = 2$ the values 0 and 1 occur. For $n \geq 3$ the internal vertices, those that are not leaves, induce a subtree with $m = n - \ell(T) \geq 1$ vertices. Order them $v_1, \ldots, v_m$ by breadth-first search from an internal root, so that every prefix $S_j = \{v_1, \ldots, v_j\}$ is connected and $\rho(S_j) = |N[S_j]| - 1$. Each v_{j+1} has degree at least two in T and its parent lies in the prefix, so it has a neighbour c on the far side, whose only neighbour inside $N[S_j]$ could be v_{j+1} itself, which is not in S_j. Hence c lies in $N[S_{j+1}]$ but not in $N[S_j]$, the sizes strictly increase, and the chain realizes m distinct values, all at least $\deg(v_1) \geq 2$. Together with $\rho(\emptyset) = 0$ and the value 1 at a single leaf this gives at least $m + 2 = n - \ell(T) + 2$ values. For the second bound take a diametral path $v_0, v_1, \ldots, v_d$ with $d = \operatorname{diam}(T)$, whose endpoint v_0 is a leaf. For the prefixes $S_j = \{v_0, \ldots, v_{j-1}\}$ with $1 \leq j \leq d$, the vertex v_{j+1} witnesses strict growth of $|N[S_j]|$ as long as it exists, since a tree has no edge from v_{j+1} back into $\{v_0, \ldots, v_{j-1}\}$. The chain realizes d distinct values starting from $\rho(S_1) = 1$, and $\rho(\emptyset) = 0$ gives $d + 1$ in total. □

Proposition 4 (Attainment of the bounds: path and star)**.** The path P_n has $s(P_n) = n$ for every $n \geq 2$, and the star $K_{1,n-1}$ has $s(K_{1,n-1}) = 3$ for every $n \geq 3$. Both trees attain equality in each bound of Lemma 2.

Proof. For a connected graph on n vertices and a nonempty subset S one has $1 \leq \rho(S) \leq n - 1$, the lower bound because a single vertex gives $\rho(\{v\}) = \deg(v) \geq 1$ and the upper because $|N[S]| \leq n$ and $c(G[S]) \geq 1$. With $\rho(\emptyset) = 0$ this leaves at most n distinct values, so $s(T) \leq n$ for every tree T. The path has $\operatorname{diam}(P_n) = n - 1$, so Lemma 2 gives $s(P_n) \geq n$, forcing $s(P_n) = n$, and with $\ell(P_n) = 2$ both $n - \ell(P_n) + 2$ and $\operatorname{diam}(P_n) + 1$ equal n. For the star with $n \geq 3$, Proposition 3 gives attained ranks $\{0, 1, n - 1\}$, so $s(K_{1,n-1}) = 3$, and with $\ell(K_{1,n-1}) = n - 1$ and $\operatorname{diam}(K_{1,n-1}) = 2$ both $n - \ell + 2$ and $\operatorname{diam} + 1$ equal 3. □

17 Rigidity and reconstruction

Conjecture 57 (Irreducibility and connectivity)**.** *For every graph G with at least one vertex, $H_G(U, Y)$ is irreducible in $\mathbb{Z}[U, Y]$ if and only if G is connected.*

Significance. Rossmann's multiplicativity [77] makes the disconnected direction immediate, each factor being nonconstant since $H_{K_1} = 1 + U$, so the content is the connected direction. The mechanism is that a factorization of H_G would express the joint array of $(|S|, \rho_G(S))$ over all vertex subsets as a product of two smaller arrays, which is exactly what a separation of G into two vertex-disjoint parts produces, and the conjecture asserts that no other source of such a product structure exists: connectivity of the graph should be readable as algebraic indecomposability of the polynomial. The nearest literature boundary is the multiplicativity theorem itself, which supplies one direction and gives the statement its shape. The verification is exact rather than probabilistic: all 996 connected graphs on at most seven vertices have irreducible H_G, by exact factorization over $\mathbb{Z}$. A first decisive theorem would be irreducibility of H_{P_n} for the paths, a one-parameter family at the bottom of the connectivity hierarchy. The failure mode is a connected graph whose polynomial factors, which would reveal a product structure in the rank array not induced by any decomposition of the graph and would make the polynomial blind to connectivity.

Conjecture 58 (Recovery of the subgraph-component polynomial)**.** *For all graphs G and H, $C_G = C_H$ implies $Q_G = Q_H$.*

Significance. The two polynomials record transverse marginals of the induced-subgraph structure: C_G records the joint distribution of $(|S|,\ |N_G[S]| - c(G[S]))$ over all vertex subsets and Q_G the joint distribution of $(|S|,\ c(G[S]))$, and no functional relation between the two arrays is apparent, so the mechanism must be rigidity of the pairs of arrays that actually arise from graphs. The nearest literature boundary pairs the two sources: the class-counting polynomial and its enumerative theorems are Rossmann's [77], the subgraph-component polynomial is Tittmann–Averbouch–Makowsky's [78], and no bridge between them is on record. The polynomial C_G does not determine the graph, and every known collision class was tested: the 119 collision classes among the 1252 graphs on at most seven vertices, a randomized eight-vertex corpus, the unique collision class among all 3159 trees on fourteen vertices (there is none through thirteen, and the class was refound independently by exhaustive search), and the first unicyclic collision at ten vertices. In every class Q_G agrees, as do the domination invariants of Conjectures 65 and 66. A first decisive theorem would be a derivation of Q_T from C_T for trees. The failure mode is a C-collision pair with distinct Q, which would show that the closed-neighbourhood array and the component array are genuinely independent invariants and would break the entire recovery programme at its first joint.

Conjecture 59 (Two-field rigidity)**.** *If $C_G(2,Y) = C_H(2,Y)$ and $C_G(3,Y) = C_H(3,Y)$, then $C_G = C_H$. Equivalently, the adjoint-rank distributions of the graphical Lie algebras over $\mathbb{F}_2$ and $\mathbb{F}_3$ determine them over every finite field.*

Significance. The two hypotheses are the unweighted and the $2^{|S|}$-weighted rank statistics, $C_G(2,Y) = \sum_S Y^{\rho_G(S)}$ and $C_G(3,Y) = \sum_S 2^{|S|} Y^{\rho_G(S)}$, and recovering a nonnegative integer array from these two evaluations is impossible for general arrays, so the mechanism is rigidity of the arrays that graphs realize rather than linear algebra. The equivalence with the finite-field form is Proposition 3, and the nearest literature boundary is Rossmann's enumeration theorem [77], through which the conjecture reads: two conjugacy-class censuses, over $\mathbb{F}_2$ and $\mathbb{F}_3$, determine the census over every finite field. The verification is exhaustive on at most seven vertices and covers six thousand random eight-vertex graphs. A first decisive theorem would be two-field rigidity for trees. The failure mode is a pair of graphs agreeing over $\mathbb{F}_2$ and $\mathbb{F}_3$ yet separated over some larger field, which would reveal genuinely field-dependent information in the adjoint-rank distribution and would locate it at a specific weight $(q-1)^{|S|}$.

Conjecture 60 (Hybrid rigidity)**.** *If $C_G(2,Y) = C_H(2,Y)$ and $Q_G = Q_H$, then $C_G = C_H$.*

Significance. The mechanism is complementarity of marginals: the binary specialization carries the rank distribution and Q_G carries the joint size-and-component array, and the conjecture asserts that these two projections of the induced-subgraph data jointly pin the full class-counting polynomial. The pairing is calibrated against a cheaper substitute, since $C_G(2,Y)$ together with the degree sequence does not determine C_G, so the component information in Q_G is doing work that no degree data can do. The nearest literature boundary is again the pair Rossmann [77] and Tittmann–Averbouch–Makowsky [78], whose polynomials the hypothesis mixes. The verification is exhaustive on at most seven vertices. A first decisive theorem would be the tree case. The failure mode is a pair agreeing in both hypotheses but not in C, which would show that the joint array holds correlation information invisible to this pair of its natural projections.

Conjecture 61 (Binary edge-deck reconstruction)**.** *For graphs G and H with at least four edges each, if the multisets $\{C_{G-e}(2,Y) : e \in E(G)\}$ and $\{C_{H-e}(2,Y) : e \in E(H)\}$ coincide, then G and H are isomorphic.*

Significance. This is the high-risk member of the part, in the orbit of the edge reconstruction conjecture, for which Bondy–Hemminger [79] is the survey. Each card here retains only the binary rank distribution of an edge-deleted subgraph rather than its isomorphism type, so the deck is a drastic coarsening of the classical edge deck, and the mechanism is that rank distributions vary so tightly under single-edge deletion that the multiset of shadows should already separate graphs. The nearest literature boundary is the reconstruction literature surveyed in [79], against which the present statement trades weaker cards for the same conclusion. Since each binary card is computed from the corresponding ordinary card, the ordinary edge deck determines the binary deck, so the statement implies the edge reconstruction conjecture on this range and is strictly the bolder claim. The verification is exhaustive on at most seven vertices among graphs with at least four edges. A first decisive theorem would be reconstruction of trees from the binary deck. The failure mode is a nonisomorphic pair with equal binary decks, which would measure exactly how much the rank distribution forgets under edge deletion, and a counterexample here would be the least surprising in the part, which is why the statement is flagged as its stress point.

Conjecture 62 (Bipartite binary rigidity)**.** *For bipartite graphs G and H, $C_G(2,Y) = C_H(2,Y)$ implies $C_G = C_H$.*

Significance. A single specialization cannot determine the polynomial in general, since the implication fails already for connected unicyclic graphs on seven vertices (the resolution of Conjecture 63), so the mechanism is that a hereditary class restriction removes the room in which the unweighted rank distribution and the size-weighted array can decouple. The nearest literature boundary is Rossmann's specialization theorem [77], which gives the hypothesis its meaning as the $\mathbb{F}_2$ census. The verification is a six-thousand-graph random holdout of bipartite graphs on nine to twelve vertices, spanning 2747 distinct binary specializations, with no violation. A first decisive theorem would be the tree case, shared with the surviving tree layer of Conjecture 63. The failure mode is a bipartite pair with equal binary specialization and distinct C, which would locate the boundary of one-field rigidity strictly inside the bipartite world and would redirect the class-restriction programme towards sparser families.

Conjecture 63 (Connected-pseudoforest binary rigidity, resolved false)**.** *For G and H each a tree or a connected unicyclic graph, $C_G(2,Y) = C_H(2,Y)$ implies $C_G = C_H$.*

Remark 7 (Resolution)**.** *The conjecture is false as stated. The connected unicyclic graphs on vertex set $\{0,\dots,6\}$ with edge sets $\{\{0,3\},\{0,4\},\{1,2\},\{1,3\},\{1,4\},\{3,6\},\{4,5\}\}$ and $\{\{0,1\},\{0,3\},\{0,4\},\{1,2\},\{2$ a four-cycle carrying three leaves against a triangle carrying a three-vertex path with two further leaves, share the binary specialization $C(2,Y) = 1 + 3Y + 7Y^2 + 29Y^3 + 32Y^4 + 40Y^5 + 16Y^6$ while their class-counting polynomials differ already in the singleton stratum, whose coefficients record the degree multisets $\{1,1,1,2,3,3,3\}$ and $\{1,1,2,2,2,3,3\}$. An exhaustive scan shows the pair is the unique violating class among all trees and connected unicyclic graphs through nine vertices. One member is bipartite and not chordal and the other chordal and not bipartite, and the ternary specializations differ, so Conjectures 62, 64, and 59 are untouched. The tree case survives exhaustively through fourteen vertices and remains open. Connectedness was in any case not the failure boundary, since the implication also fails for disconnected pseudoforests.*

Significance. The statement is retained with its resolution because the failure is itself informative. The refuting pair shows that on unicyclic graphs the binary rank distribution can forget the degree multiset, the coarsest datum beyond the vertex and edge counts, so one cycle already leaves enough freedom to decouple the unweighted rank census from the size-weighted array. The failure calibrates the neighbouring statements rather than undermining them: the pair straddles the classes of Conjectures 62 and 64, one member bipartite and one chordal, so the bipartite and chordal rigidity claims stand on graphs the counterexample cannot reach, and the two-field claim of Conjecture 59 absorbs it, since the ternary specializations differ. The surviving positive content is the tree case, exhaustively unrefuted through fourteen vertices, which is exactly the first decisive theorem the original statement named.

Conjecture 64 (Chordal binary rigidity)**.** *For chordal graphs G and H, $C_G(2,Y) = C_H(2,Y)$ implies $C_G = C_H$.*

Significance. Chordal graphs, those in which every cycle of length at least four has a chord, form the natural dense counterpart to the sparse classes of Conjectures 62 and 63, and the mechanism is the same class rigidity tested against a family with unbounded clique structure rather than bounded cycle structure. The nearest literature boundary is once more the specialization theorem [77] that interprets the hypothesis over $\mathbb{F}_2$. The verification is a six-thousand-graph random holdout of chordal graphs on nine to twelve vertices, spanning 5598 distinct binary specializations, with no violation. A first decisive theorem would be the case of trees, which are chordal, followed by block graphs. The failure mode is a chordal pair separating the binary specialization from the polynomial. The resolution of Conjecture 63 sharpens rather than settles the question here, since the refuting pair there has exactly one chordal member, so chordal rigidity survives it and its status is genuinely open.

Conjecture 65 (Recovery of the domination number)**.** *For all graphs G and H, $C_G = C_H$ implies $\gamma(G) = \gamma(H)$, where γ is the domination number.*

Significance. A set is dominating exactly when its closed neighbourhood is the whole vertex set, and the array behind C_G records $|N_G[S]|$ only through the difference $\rho_G(S) = |N_G[S]| - c(G[S])$, so the dominating sets are entangled with component counts and the mechanism is that the entanglement can be undone at the extreme where γ lives. The conjecture sits strictly between a theorem and a false analogue, which is the nearest literature boundary: the cardinalities of connected dominating sets are readable from the coefficients, by Rossmann [77], while the total domination number is not determined by C_G. Every known collision class of C, the corpus recorded at Conjecture 58, has agreeing domination number. A first decisive theorem would be the tree case, where the unique fourteen-vertex collision class is the only known test. The failure mode is a collision class with distinct domination numbers, which would place γ on the undetermined side of the boundary alongside total domination and would show that the connected-domination theorem is the exact limit of what the polynomial sees.

Conjecture 66 (Recovery of the independent domination number)**.** *For all graphs G and H, $C_G = C_H$ implies $i(G) = i(H)$, where i is the independent domination number, the least cardinality of a set that is simultaneously independent and dominating.*

Significance. Independent dominating sets impose the two conditions the array treats most asymmetrically, domination through $|N_G[S]|$ and independence through $c(G[S]) = |S|$, and the mechanism is that their interaction leaves a trace in the joint distribution of $(|S|, \rho_G(S))$ even though neither condition is separately readable. The nearest literature boundary is the same theorem-versus-false-analogue frame as Conjecture 65: connected domination is readable [77], total domination is not determined, and i is conjectured to fall on the determined side. Every known collision class of C, the corpus recorded at Conjecture 58, has agreeing independent domination number. A first decisive theorem would be recovery of i for trees. The failure mode is a collision class with distinct i, which would split the domination invariants into determined and undetermined families along a line that no present structural explanation predicts.

18 Log-concavity and support

Conjecture 67 (Global double log-concavity)**.** *For every graph G the coefficient sequence a of $f_G(1+Z)$ is log-concave, and so is its transform $L(a)$ defined by $L(a)_i = a_i^2 - a_{i-1}a_{i+1}$, out-of-range terms being zero.*

Significance. The expansion point is canonical: at $X = 1 + Z$ the polynomial is expanded about the degenerate point $X = 1$, where the weights $(X-1)^{|S|}$ vanish for every nonempty set, and f_G folds the rank variable into the size variable at the same point, so the sequence a is the most compressed image of the whole array. The statement is calibrated to the strongest form the data support: log-concavity is asserted, its once-iterated form is asserted, and the calibration is exact because the third iterate of L fails for a thirteen-vertex graph. The nearest literature boundary is Rossmann's paper [77], whose theorems about this polynomial are enumerative, the specialization, multiplicativity, and dominating-set results, so the positivity layer opens a new face of the object. The verification covers the full atlas of graphs on at most seven vertices and twenty-five hundred random graphs on eight to ten vertices. A first decisive theorem would be log-concavity of $f_T(1+Z)$ for trees. The failure mode is graded: a failure of the second level would compress the true boundary into the narrow band between one and three iterates, while a failure of bare log-concavity on some large graph would show that the seven-vertex atlas sits below the threshold where the array's geometry turns.

Conjecture 68 (Strict log-concavity for connected graphs)**.** *For connected G with at least one edge, $a_i^2 > a_{i-1}a_{i+1}$ at every interior position of the support of a, where a is the coefficient sequence of $f_G(1+Z)$.*

Significance. The mechanism is that equality cases of log-concavity should be confined to the degenerate stratum, and in this part the degenerate stratum is disconnectedness, exactly as in Conjecture 57: multiplicativity [77] builds the polynomials of disconnected graphs as products, and products are where interior coincidences are manufactured. The nearest literature boundary is that same multiplicativity theorem, which marks the hypothesis as necessary in spirit. The verification is the corpus of Conjecture 67, the full seven-vertex atlas and twenty-five hundred random graphs on eight to ten vertices, with strictness observed throughout the connected stratum. A first decisive theorem would be strictness for paths and stars, the two ends of the tree order of Conjecture 72. The failure mode is a connected graph with an interior equality, an exact coincidence $a_i^2 = a_{i-1}a_{i+1}$, which would demand an algebraic explanation and would show that connectivity does not exhaust the sources of degeneracy.

Conjecture 69 (Strict second-order log-concavity)**.** *For connected G with at least one edge, the transform $L(a)$ of Conjecture 67 is strictly log-concave at every interior position of its support.*

Significance. The statement pushes the strictness mechanism of Conjecture 68 one level up the L-hierarchy, and its position is the sharpest the evidence permits, since the third iterate of L fails already for a thirteen-vertex graph and the second-order slice strengthening fails for an eight-vertex graph (Conjecture 70), so the conjecture threads a boundary with counterexamples visible on two sides. The nearest literature boundary is as in Conjecture 67: the polynomial's literature [77] is enumerative and the positivity hierarchy is uncharted. The verification is the same corpus, the full seven-vertex atlas and twenty-five hundred random graphs on eight to ten vertices. A first decisive theorem would be second-order strictness for paths. The failure mode is a connected graph whose $L(a)$ has an interior equality or reversal, which by the two flanking counterexamples would pin the exact rung of the hierarchy at which graphs stop being log-concave.

Conjecture 70 (Rank-slice strict log-concavity)**.** *For connected G and every attainable rank r, the slice $A_{G,r}(Z) = [Y^r]\, F_G(1 + Z, Y)$, when it is not a monomial, has strictly log-concave coefficients at every interior position of its support.*

Significance. The mechanism refines Conjecture 68 from the folded sequence to each rank slice separately: the sets of a fixed rank r are counted by size, and strict log-concavity asserts that each of these level sets is unimodal in the strong multiplicative sense, so that positivity holds fibrewise and not only in aggregate. The calibration is exact here too, since the second-order strengthening of this slice statement fails for an eight-vertex graph, a counterexample reconfirmed by the independent implementation, so the slice hierarchy stops one level below the global hierarchy of Conjecture 69. The nearest literature boundary is Rossmann's [77] reading of the slices as class data over finite fields. The verification is the full seven-vertex atlas and twenty-five hundred random graphs on eight to ten vertices. A first decisive theorem would be strict log-concavity of every slice for stars. The failure mode is a connected graph with a non-log-concave slice, which would show that aggregate log-concavity, if it survives in Conjecture 67, is an artefact of summation rather than a fibrewise phenomenon.

Conjecture 71 (Interval support at fixed rank, resolved false)**.** *For every graph G and every attainable rank r, the set $K_r(G) = \{|S| : \rho_G(S) = r\}$ is an interval of integers.*

Remark 8 (Resolution)**.** *The conjecture is false as stated. The connected graph on vertex set $\{0, \dots, 7\}$ with edge set $\{\{0, 3\}, \{0, 6\}, \{0, 7\}, \{1, 4\}, \{1, 6\}, \{1, 7\}, \{2, 5\}, \{2, 6\}, \{3, 6\}, \{3, 7\}, \{4, 6\}, \{4, 7\}, \{5, 7\}\}$ attains adjoint rank five at support sizes one and three but at no size two: the degree-five vertices 6 and 7 give rank five at size one, the independent set $\{0, 1, 2\}$ has closed neighbourhood all eight vertices and so gives $8-3 = 5$ at size three, while every two-vertex set has rank in $\{3, 4, 6\}$. Hence $K_5(G) = \{1, 3, 4, 5, 6\}$ has an interior gap at size two. An exhaustive scan of all 12,346 graphs on eight vertices finds exactly two violations, this graph and one companion, both connected, and none on seven or fewer vertices.*

Significance. The statement is retained with its resolution because the failure is informative and sharply located. A connected eight-vertex graph attains rank five at the support sizes $\{1, 3, 4, 5, 6\}$, skipping size two, so the size direction of the array is not gapless in general, and since the transposed statement was already false, neither direction carries interval structure. The failure does not reach the slice positivity beneath which the support claim was placed: on the two refuting graphs every rank slice $[Y^r]F_G(1+Z,Y)$ of Conjecture 70 stays strictly log-concave, since each slice weights the raw size counts by a full binomial row that closes the gap, so gapless raw support is not after all necessary for the slice statement. The nearest literature boundary is unchanged, Rossmann's [77] reading of one extreme stratum of the array. The surviving positive content is the interval property for all graphs through seven vertices and, conjecturally, for trees, which the eight-vertex counterexample leaves open.

19 Tree extremality and rigidity

Conjecture 72 (Coefficientwise tree extremality)**.** *For every tree T on n vertices,*

$$f_{P_n}(1+Z) \;\leq\; f_T(1+Z) \;\leq\; f_{K_{1,n-1}}(1+Z)$$

coefficientwise, with equality only for the path and the star respectively.

Significance. The mechanism is visible in Lemma 2: the rank landscape of a tree is driven by its internal vertices and its diameter, and the path and the star are the two trees that extremize both statistics simultaneously, the path with two leaves and maximal diameter, the star with maximal leaf count and diameter two. The conjecture asserts that this extremality is not merely a matter of the range of ρ_T but holds coefficient by coefficient in the shifted polynomial, the strongest ordering the object admits. The nearest literature boundary is Rossmann's enumeration [77], which interprets each coefficientwise inequality as a family of inequalities between class censuses over every finite field at once. The verification is exhaustive over all trees through fourteen vertices. A first decisive theorem would be either inequality for one nontrivial family, say the star bound for caterpillars. The failure mode is a tree escaping the sandwich at a single coefficient, which would show that the coefficientwise order on trees is not anchored at the degree extremes and would force the extremal programme to work slice by slice instead.

Conjecture 73 (Star rank-majorization)**.** *For every $n \geq 5$ and every prime power q, the decreasingly ordered adjoint-rank distribution of $L_{K_{1,n-1}}(\mathbb{F}_q)$ majorizes that of $L_T(\mathbb{F}_q)$ for every n-vertex tree T.*

Remark 9. *The hypothesis $n \geq 5$ cannot be dropped: at $n = 4$ and $q = 2$ the star fails to majorize the path. The dual assertion, that the path is majorization-minimal among n-vertex trees, is false. The failure at $(n,q) = (4,2)$ was found in the verification reported below.*

Proposition 5 (Reduction of the star majorization)**.** Let T be a tree on $n \geq 3$ vertices and q a prime power, and write $\pi_{T,q}(r) = q^{-n}\sum_{\rho_T(S)=r}(q-1)^{|S|}$ for the adjoint-rank distribution of a uniformly random element of $L_T(\mathbb{F}_q)$. The decreasingly ordered distribution of the star $K_{1,n-1}$ majorizes that of T if and only if

$$\max_r \pi_{T,q}(r) \leq 1 - \tfrac{1}{q}.$$

Proof. By Proposition 3 the star attains ranks 0, 1, and $n-1$, from the empty support, the nonempty sets of leaves, and the supports meeting the centre, with masses $1-q^{-1}$, $q^{-1}-q^{-n}$, and q^{-n} in decreasing order for every $q \geq 2$. For any tree $\rho_T(S) = 0$ holds only at $S = \emptyset$, so

$\pi_{T,q}(0) = q^{-n}$, and this is the least atom, every nonempty rank carrying mass a sum of terms $q^{-n}(q-1)^{|S|} \geq q^{-n}$. A tree on $n \geq 3$ vertices has diam ≥ 2, so Lemma 2 gives at least three attained ranks. Majorization of the three-atom star vector over the padded tree vector is the family of partial-sum inequalities on decreasingly ordered atoms, both sides totalling 1. The first reads $1 - q^{-1} \geq \max_r \pi_{T,q}(r)$, the displayed bound. The second reads $1 - q^{-n} \geq \pi^{(1)} + \pi^{(2)}$ for the two largest tree atoms, and holds automatically, since removing them leaves at least one atom of mass at least q^{-n}. Every later star partial sum equals 1. Hence the whole family holds if and only if its first member does. $\square$

Significance. Majorization compares two probability vectors through all partial sums of their decreasingly ordered entries, so the conjecture asserts that the star's adjoint-rank distribution is the most concentrated among trees uniformly across the whole partial-sum hierarchy and across every finite field at once. The mechanism is concentration: by Proposition 3 the rank of a random element is a statistic of its vertex support, and the star compresses the possible values harder than any other tree. The remark shows the inequality system is genuinely tight, which is the honest register for a majorization claim: at $(n, q) = (4, 2)$ one partial sum reverses, and the dual minimality of the path fails outright, so the surviving statement is exactly the one the data leave standing. The nearest literature boundary is Rossmann's [77] reading of these distributions as class data of graphical groups. The verification is exhaustive for $q \in \{2, 3, 4, 5\}$ over all trees through twelve vertices, for $q \in \{2, 3\}$ at thirteen, and for $q = 2$ at fourteen. Proposition 5 reduces the whole majorization, across every partial sum and every field, to the single anti-concentration bound $\max_r \pi_{T,q}(r) \leq 1 - q^{-1}$, whose extremal value is the star's own top atom, so the failure at $(n, q) = (4, 2)$ is exactly a violation of that bound. A first decisive theorem would be the reduced bound for all $n \geq 5$ trees, of which the comparison of star against path is the tightest case. The failure mode is a tree and a prime power breaking one partial sum at some $n \geq 5$, which would show that the small-n anomaly is the visible end of a persistent phenomenon rather than a boundary effect.

Conjecture 74 (Tree rank-variance extremality)**.** *For every $n \geq 4$ and every prime power q, among n-vertex trees the variance of* $\operatorname{rank}(\mathrm{ad}_x)$ *over uniformly random $x \in L_T(\mathbb{F}_q)$ is uniquely minimized by the path and uniquely maximized by the star.*

Significance. The two tree orderings of this programme are not redundant: majorization compares the ordered probabilities while the variance measures the spread of the rank values themselves, and the star sits at the top of both scales, most concentrated in the sense of Conjecture 73 and most spread in the sense of variance, with the path now appearing as the genuine minimum. The mechanism is again Proposition 3, which makes the variance a purely combinatorial functional of the array $(|S|, \rho_T(S))$ weighted by $(q-1)^{|S|}$. The nearest literature boundary is the enumerative frame of Rossmann [77], in which the variance is a second moment of the class census. The verification matches Conjecture 73: exhaustive for $q \in \{2, 3, 4, 5\}$ through twelve vertices, for $q \in \{2, 3\}$ at thirteen, and for $q = 2$ at fourteen. A first decisive theorem would be the uniqueness of the star maximum at $q = 2$. The failure mode is a tree overtaking an extreme at some (n, q), and since the statement begins at $n = 4$ where the majorization statement already fails, a violation would separate the variance order from the majorization order and show that the two concentration scales genuinely diverge on trees.

Conjecture 75 (Leaf-count rigidity)**.** *For a tree T on $n \geq 2$ vertices, equality holds in the bound $s(T) \geq n - \ell(T) + 2$ of Lemma 2 exactly when T is a path or a star.*

Significance. The lemma's proof realizes $n - \ell(T) + 2$ rank values along a breadth-first chain of internal vertices, and equality demands that every subset of vertices lands on one of the chain's values, so the mechanism is rigidity of the equality case: only the two extreme trees should leave no room for excess values off the chain. The nearest literature boundary is internal, the lemma itself, whose bound the conjecture upgrades to a characterization, in the same relation as an isoperimetric equality case stands to its inequality. The verification is exhaustive over all trees through fourteen vertices. The forward half, that the path and the star do achieve equality, is proved in Proposition 4, leaving uniqueness, that no third tree is equally tight, as the open content. The failure mode is a third tree achieving equality, which would show that the leaf count fails to see some genuinely different extremal shape and would refute the rigidity reading of Lemma 2 while leaving the lemma intact.

Conjecture 76 (Diameter rigidity). *For a tree T on $n \geq 2$ vertices, equality holds in the bound $s(T) \geq \operatorname{diam}(T) + 1$ of Lemma 2 exactly when T is a path or a star.*

Significance. The diametral chain of the lemma realizes $\operatorname{diam}(T) + 1$ values, and equality requires the whole subset lattice to produce nothing beyond the chain, so the mechanism parallels Conjecture 75 with the diameter replacing the internal vertex count, and the two rigidity statements pin the same extremal pair from two independent directions, each serving as a cross-check on the other. The nearest literature boundary is again Lemma 2 itself. The verification is exhaustive over all trees through fourteen vertices. The forward half, that the path and the star do achieve equality, is proved in Proposition 4, so only uniqueness remains open. The failure mode is a tree of small diameter and few rank values outside the pair, which would show the diameter bound to be loose in a structured way and would decouple the two rigidity statements that the exhaustive search through fourteen vertices keeps locked together.

Part V

Pattern Lie algebras of finite posets

Added to the deposit: 6 August 2026.

The twenty conjectures below concern the pattern Lie algebras of finite posets, studied through two rank-weight enumerators that record how the bracket contracts against an element and against a functional. They form four programmes. A determination programme (Conjectures 77 to 82) asserts that the adjoint enumerator alone determines the coadjoint enumerator and a list of classical invariants. A characteristic-and-field programme (Conjectures 83 to 87) identifies what in these distributions is independent of the ground field and how the field affects them when it does. A centre-one programme (Conjectures 88 to 90) studies the incidence quotient that measures the difference between the two enumerators when the centre is a line. A Hasse-forest and unitriangular programme (Conjectures 91 to 96) relates coadjoint structure to the shape of the cover graph and treats the unitriangular chain as a separate extremal case. The proved results are kept separate from the conjectural ones: the incidence identity (Proposition 6) is a theorem with a full proof, the standard adjoint–coadjoint duality of this family, included for calibration and not claimed as new, and every conjecture is stated relative to it. Fifteen of the twenty are supported throughout the tested corpus. Three more, Conjectures 79, 81, and 85, are stated in greater generality than the computations reach, and each carries a remark giving the restricted case its evidence covers. Two, Conjectures 87 and 90, have since been refuted by explicit seven-point posets and are recorded as resolved false with their counterexamples.

For a finite poset P on a set of points and a prime power q, the *pattern Lie algebra* over $\mathbb{F}_q$ is

$$L_P(\mathbb{F}_q) = \operatorname{span}\{e_{ij} : i <_P j\}$$

inside the strictly upper triangular matrices under a fixed linear extension of P, with the matrix-unit bracket

$$[e_{ij}, e_{kl}] = \delta_{jk}\, e_{il} - \delta_{li}\, e_{kj}.$$

It is nilpotent, and $\dim L_P$ is the number of strict order relations of P. Write $\operatorname{ad}_x(y) = [x, y]$ for the adjoint map of x, and for a functional $f \in L_P(\mathbb{F}_q)^*$ let $B_f(u, v) = f([u, v])$ be the alternating Kirillov form. The two natural contractions of the bracket tensor give two rank-weight enumerators over a single finite field,

$$A_{P,q}(T) = \sum_{x \in L_P(\mathbb{F}_q)} T^{\operatorname{rank}(\operatorname{ad}_x)}, \qquad K_{P,q}(T) = \sum_{f \in L_P(\mathbb{F}_q)^*} T^{\operatorname{rank}(B_f)},$$

the *adjoint enumerator* and the *Kirillov enumerator*, each a distribution over one finite field. Write $b_A(P, q) = \max_x \operatorname{rank}(\operatorname{ad}_x)$ for the *adjoint breadth* and $b_K(P, q) = \max_f \operatorname{rank}(B_f)$ for the *maximal Kirillov rank*, the largest coadjoint-orbit rank. Write $Z(L_P)$, $\operatorname{Cent}(L_P)$, and $\operatorname{Der}(L_P)$ for the centre, the centroid (the linear maps commuting with every ad_x), and the derivation algebra, and $\gamma_i(L_P)$ for the lower central series. Say P is a *Hasse forest* if the undirected Hasse cover graph of P is a forest. The chain on n points gives $L = \mathrm{ut}_n(\mathbb{F}_q)$, the strictly upper triangular Lie algebra.

Proposition 6 (Incidence identity and calibration)**.** Let $L = L_P(\mathbb{F}_q)$ and let

$$I_L = \{(x, f) \in L \times L^* : f([x, L]) = 0\}$$

be the incidence variety. Then

$$|I_L| = q^{\dim L} A_{L,q}(q^{-1}) = q^{\dim L} K_{L,q}(q^{-1}).$$

Consequently $A_{L,q}(1) = K_{L,q}(1) = q^{\dim L}$ and $A_{L,q}(q^{-1}) = K_{L,q}(q^{-1})$, so $(1 - T)(1 - qT)$ divides $A_{L,q}(T) - K_{L,q}(T)$ in $\mathbb{Z}[T]$.

Proof. Count I_L two ways. For fixed x the condition $f([x, L]) = 0$ says f vanishes on the image $[x, L] = \operatorname{im}(\operatorname{ad}_x)$, a subspace of dimension $\operatorname{rank}(\operatorname{ad}_x)$, so the number of admissible f is $q^{\dim L - \operatorname{rank}(\operatorname{ad}_x)}$. Summing over x gives

$$|I_L| = q^{\dim L} \sum_x q^{-\operatorname{rank}(\operatorname{ad}_x)} = q^{\dim L} A_{L,q}(q^{-1}).$$

For fixed f the condition says x lies in the radical of the alternating form B_f, of dimension $\dim L - \operatorname{rank}(B_f)$, giving $q^{\dim L - \operatorname{rank}(B_f)}$ admissible x, so $|I_L| = q^{\dim L} K_{L,q}(q^{-1})$. Equating the two counts gives $A_{L,q}(q^{-1}) = K_{L,q}(q^{-1})$. Taking every element and every functional, the rank-0 terms included, gives $A_{L,q}(1) = K_{L,q}(1) = q^{\dim L}$. The two evaluations exhibit $T = 1$ and $T = 1/q$ as common roots of $A_{L,q} - K_{L,q}$, and since $q > 1$ these are distinct, so $(1 - T)(1 - qT)$ divides $A_{L,q} - K_{L,q}$. $\square$

The identity is the standard adjoint–coadjoint duality of this family, a Knuth-type contraction of one bracket tensor, and is recorded here as calibration rather than as a new result. For the centre-one conjectures define

$$R_{P,q}(T) = \frac{K_{P,q}(T) - A_{P,q}(T)}{(1-T)(1-qT)},$$

a polynomial in $\mathbb{Z}[T]$ by Proposition 6.

20 Determination by the adjoint enumerator

Conjecture 77 (Adjoint–Kirillov determination)**.** *For all finite posets P and Q and every prime power q, if $A_{P,q}(T) = A_{Q,q}(T)$ then $K_{P,q}(T) = K_{Q,q}(T)$.*

Significance. The adjoint and Kirillov enumerators are the two contractions of the same bracket, one in the element variable and one in the functional variable, and Proposition 6 forces them to agree at $T = 1$ and $T = 1/q$. The conjecture asserts that the adjoint enumerator determines the Kirillov enumerator at every value of T, not only at those two points. The nearest related result is Delsarte's [85] analysis of the rank distributions of alternating bilinear forms over a finite field, which describes the Kirillov side alone but does not connect it to the adjoint side. A tractable special case is the chain $L = \mathrm{ut}_n(\mathbb{F}_q)$, where the divisibility of Proposition 6 already relates the two enumerators. In tests on the corpus described in the stress-test section, the adjoint enumerator determined the Kirillov enumerator across 64 adjoint-collision classes with no exception. A counterexample would be an adjoint-collision pair with distinct Kirillov enumerators, which would show that K is not determined by A beyond the two points where the proposition equates them.

Conjecture 78 (Lower-central recovery)**.** *For every finite poset P and every prime power q, the enumerator $A_{P,q}(T)$ determines the vector of lower-central factor dimensions $\left(\dim \gamma_i(L_P)/\gamma_{i+1}(L_P)\right)_i$.*

Significance. The rank of ad_x is the dimension of the image $[x, L_P]$, and the lower central series is built from these images as x ranges over the algebra. The conjecture asserts that the distribution of adjoint ranks determines the graded dimensions $\dim \gamma_i/\gamma_{i+1}$. The nearest related result is the breadth theory of Cameron–Coll–Mayers–Russoniello [80], in which the maximal adjoint rank b_A is one value of the distribution that the full enumerator records. A tractable special case is the chain, where ut_n has a known lower central series matched to its adjoint spectrum. In tests on the corpus described in the stress-test section, the lower-central factor vector was constant on every adjoint-collision class. A counterexample would be an adjoint-collision class with two distinct factor vectors, which would show that the lower-central factors are not determined by A.

Conjecture 79 (Ordinary Poincaré recovery)**.** *For every finite poset P and every prime power q, the enumerator $A_{P,q}(T)$ determines the ordinary cohomological Poincaré polynomial $\sum_i \dim H^i(L_P, \mathbb{F}_q)\, U^i$.*

Remark 10. *The evidence tests only the equality of $\dim H^2(L_P, \mathbb{F}_q)$ and $\dim H^3(L_P, \mathbb{F}_q)$ across adjoint-collision classes, and it rests on the deposited cohomology computation rather than on an independent recomputation of Lie-algebra cohomology. The universal statement over all degrees is more tentative than the other conjectures of this part, the remaining cases being untested.*

Significance. The Chevalley–Eilenberg differential of L_P is built from the same structure constants that the adjoint maps record, so the adjoint-rank distribution constrains the cohomological Betti numbers through the ranks of those differentials. The nearest related result is the poset Lie cohomology of Lampret–Vavpetič [82], the closest computation of $H^i(L_P, \mathbb{F}_q)$ for this family. No decisive special case is proposed, and the tested range is the two degrees recorded in the remark. A counterexample would be an adjoint-collision class with distinct ordinary Betti numbers in some degree, which would show that cohomology is not determined by the adjoint enumerator.

Conjecture 80 (Derivation recovery)**.** *For every finite poset P and every prime power q, the enumerator $A_{P,q}(T)$ determines* $\dim \mathrm{Der}(L_P)$*.*

Significance. A derivation is constrained at each point by its interaction with the adjoint action, so the derivation algebra is determined by the adjoint representation, of which the adjoint-rank distribution is a coarse record. The conjecture asserts that this distribution determines the dimension of the space of derivations. The nearest related result is again the breadth theory of Cameron–Coll–Mayers–Russoniello [80], which reads structural invariants of L_P off the same adjoint data. A tractable special case is the chain, where $\dim \mathrm{Der}(\mathrm{ut}_n)$ is classical. In tests on the corpus described in the stress-test section, $\dim \mathrm{Der}(L_P)$ was constant on every adjoint-collision class. A counterexample would be an adjoint-collision class with distinct derivation dimensions, which would show that $\dim \mathrm{Der}(L_P)$ is not determined by the adjoint distribution.

Conjecture 81 (Adjoint Poincaré recovery)**.** *For every finite poset P and every prime power q, the enumerator $A_{P,q}(T)$ determines the adjoint-cohomology dimensions* $\dim H^i(L_P, L_P)$ *for every i.*

Remark 11. *The evidence tests only the equality of* $\dim H^1(L_P, L_P)$*,* $\dim H^2(L_P, L_P)$*, and* $\dim H^3(L_P, L_P)$ *across adjoint-collision classes, and it rests on the deposited cohomology computation rather than on an independent recomputation. The universal statement over all degrees is more tentative than the other conjectures of this part, the remaining cases being untested.*

Significance. Cohomology with coefficients in the adjoint module governs the deformation theory of L_P, and its differentials are built from the same brackets whose ranks the adjoint maps record, so the situation parallels Conjecture 79 with the algebra acting on itself. The nearest related result is the poset Lie cohomology of Lampret–Vavpetič [82]. No decisive special case is proposed, and the tested range is the three degrees recorded in the remark. A counterexample would be an adjoint-collision class with distinct adjoint Betti numbers in some degree, which would show that the deformation theory is not determined by the adjoint enumerator.

Conjecture 82 (Centroid-dimension recovery)**.** *For every finite poset P and every prime power q, the enumerator $A_{P,q}(T)$ determines* $\dim \mathrm{Cent}(L_P)$*.*

Significance. The centroid is by definition the space of linear maps commuting with every ad_x, so it is an invariant of the adjoint representation, of which the adjoint-rank distribution is a coarse record. The conjecture asserts that this distribution determines its dimension. The nearest related result is again the breadth theory of Cameron–Coll–Mayers–Russoniello [80], whose invariants are defined on the same adjoint data. A tractable special case is the chain, where $\dim \mathrm{Cent}(\mathrm{ut}_n) = n$ for $n \geq 3$. In tests on the corpus described in the stress-test section, $\dim \mathrm{Cent}(L_P)$ was constant on every adjoint-collision class. A counterexample would be an adjoint-collision class with distinct centroid dimensions, which would show that the centroid dimension is not determined by the adjoint-rank distribution.

21 Characteristic and field behaviour

Conjecture 83 (Adjoint spectrum characteristic independence)**.** *For every finite poset P, the set $\{\mathrm{rank}(\mathrm{ad}_x) : x \in L_P(\mathbb{F}_q)\}$ of attained adjoint ranks is independent of q.*

Significance. The attained adjoint ranks are governed by which support patterns of x can produce a given image dimension, a combinatorial condition on P. The conjecture asserts that the set of attained values depends only on the poset, while the multiplicities depend on the field. The nearest related result is Halasi–Pálfy [83], whose theorem that pattern-group class counts are not polynomial in q shows that the counts depend on the field. The invariance of the support is a stronger statement about a smaller object. A tractable special case is the chain, where the attained ranks of ut_n form an interval determined by n. In tests on the corpus described in the stress-test section, the attained-rank sets agreed across $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ in every case. A counterexample would be a poset with an adjoint rank attained over one field but not another, which would show that the support depends on the characteristic.

Conjecture 84 (Kirillov spectrum characteristic independence)**.** *For every finite poset P, the set $\{\mathrm{rank}(B_f) : f \in L_P(\mathbb{F}_q)^*\}$ of attained Kirillov ranks is independent of q.*

Significance. The attained Kirillov ranks are the even numbers realized as ranks of the alternating forms B_f, and which of these occur is a combinatorial condition on the coadjoint action of P. The conjecture is the coadjoint analogue of Conjecture 83. The nearest related results are Delsarte [85] for the rank distributions of alternating forms and Halasi–Pálfy [83] for the field dependence of the associated class counts, against which the invariance of the support is the finer statement. A tractable special case is the chain. In tests on the corpus described in the stress-test section, the attained Kirillov-rank sets agreed across $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ in every case. A counterexample would be a poset with a Kirillov rank attained over one field but not another, which would show field dependence in the coadjoint support.

Conjecture 85 (Two-field arithmetic rigidity)**.** *For all finite posets P and Q, if $A_{P,2} = A_{Q,2}$ and $A_{P,3} = A_{Q,3}$ then $A_{P,q} = A_{Q,q}$ for every prime power q.*

Remark 12. *Among the posets computed over all three fields, the only field beyond $q = 2$ and $q = 3$ at which equality was actually tested is the single holdout $q = 5$. The universal statement over every prime power is more tentative than the other conjectures of this part, the remaining cases being untested.*

Significance. The two adjoint enumerators over the smallest fields are two weighted counts drawn from the poset's rank array. The conjecture asserts that the arrays realized by posets are rigid enough that no third field carries information absent from the first two. The nearest related result is Halasi–Pálfy [83], whose non-polynomiality theorem shows that larger fields can behave differently in general, against which the two-field claim asserts that this family is nonetheless determined. No decisive special case is proposed, and the tested range is the single holdout recorded in the remark. Equality of A at $q = 2$ and $q = 3$ forced equality at $q = 5$ across

16 three-field collision classes. A counterexample would be a pair agreeing over $\mathbb{F}_2$ and $\mathbb{F}_3$ but differing over some larger field.

Conjecture 86 (Adjoint stochastic field monotonicity). *For every finite poset P and prime powers $q < q'$, the distribution of $\mathrm{rank}(\mathrm{ad}_x)$ for uniformly random $x \in L_P(\mathbb{F}_{q'})$ first-order stochastically dominates its distribution over $\mathbb{F}_q$.*

Significance. First-order stochastic dominance compares two distributions across all of their tail probabilities at once, so the conjecture asserts that raising the field can only move the adjoint rank upward in this sense. The reason is that a larger field provides more elements of high rank at each support pattern. The nearest related result is the breadth theory of Cameron–Coll–Mayers–Russoniello [80], in which the top rank b_A is stable while the rest of the distribution shifts. The statement is calibrated to the strongest form that holds. The monotone-likelihood-ratio strengthening is false, failing three times already from $q = 2$ to $q = 3$, so first-order dominance is the appropriate form. In tests on the corpus described in the stress-test section, first-order dominance held in all 221 field-pair tests, while the likelihood-ratio strengthening failed as recorded. A counterexample would be a poset and a field pair reversing one tail probability, which would show that first-order dominance is too strong and that the field can move rank downward.

Conjecture 87 (Kirillov likelihood-ratio field monotonicity, resolved false). *For every finite poset P and prime powers $q < q'$, the normalized even-rank distribution defined by $K_{P,q'}$ dominates that defined by $K_{P,q}$ in monotone likelihood-ratio order.*

Remark 13 (Resolution). *The conjecture is false as stated. The seven-point poset with cover relations $1 \prec 2$, $2 \prec 3$, $2 \prec 4$, $1 \prec 5$, $5 \prec 7$, $1 \prec 6$, $6 \prec 7$ gives a ten-dimensional pattern algebra whose Kirillov enumerator has even-rank coefficients $128, 384, 128, 384$ at ranks $0, 2, 4, 6$ over $\mathbb{F}_2$ and $2187, 17496, 4374, 34992$ over $\mathbb{F}_3$. The likelihood ratio $[T^{2j}]K_{P,3}/[T^{2j}]K_{P,2}$ then takes the values 17.1, 45.6, 34.2, 91.1 across the four even ranks, decreasing from rank 2 to rank 4, so the monotone likelihood-ratio order already fails at the pair $q = 2$, $q' = 3$. The first-order dominance of Conjecture 86 is untouched, being the weaker order the data do support.*

Significance. The statement is retained with its resolution because the failure locates the boundary precisely. Monotone likelihood-ratio order is strictly stronger than first-order dominance and compares the two distributions ratio by ratio across the even ranks, and the refuting poset shows that the coadjoint side does not satisfy this strengthening any more than the adjoint side does in Conjecture 86. The nearest related result is Delsarte [85], whose rank distributions of alternating forms are the model for the even-rank sequence. The surviving positive content is the first-order Kirillov dominance that the same data support, the coadjoint analogue of Conjecture 86, which the counterexample leaves standing.

22 Centre-one incidence quotient

Conjecture 88 (Centre-one reverse determination). *For every finite poset P with $\dim Z(L_P) = 1$ and every prime power q, the enumerator $K_{P,q}(T)$ determines $A_{P,q}(T)$.*

Significance. Conjecture 77 states the implication from the adjoint side to the coadjoint side, and this is the reverse implication under the hypothesis that the centre is a line. The reason is that a one-dimensional centre makes the incidence quotient of Proposition 6 rigid enough to invert. The nearest related result is Delsarte [85], whose alternating-form rank distributions are what $K_{P,q}$ records. A tractable special case is the chain ut_n, whose centre is the single corner entry. In tests on the corpus described in the stress-test section, the Kirillov enumerator determined the adjoint enumerator on all 43 centre-one records. A counterexample would be a centre-one pair with equal K and distinct A, which would show that A is not determined by K even when the centre is a line.

Conjecture 89 (Centre-one Laplace positivity)**.** *For every finite poset P with $\dim Z(L_P) = 1$ and every prime power q, the polynomial $R_{P,q}(T)$ has nonnegative integer coefficients.*

Significance. The quotient $R_{P,q}$ measures the difference between the coadjoint and adjoint enumerators after the two common roots of Proposition 6 are divided out. The conjecture asserts that under a line centre this difference is a counting polynomial, so its coefficients are nonnegative integers. The nearest related result is Delsarte [85], against whose alternating-form model the positivity of $R_{P,q}$ constrains how far K can exceed A. A tractable special case is the chain, where R is computable in closed form. In tests on the corpus described in the stress-test section, $R_{P,q}$ had nonnegative integer coefficients on all 43 centre-one records. A counterexample would be a centre-one poset with a negative coefficient in $R_{P,q}$, which would show that $R_{P,q}$ is not a counting polynomial and would remove the basis for the shape statement of Conjecture 90.

Conjecture 90 (Centre-one quotient shape, resolved false)**.** *For every finite poset P with $\dim Z(L_P) = 1$ and every prime power q, the coefficient sequence of $R_{P,q}(T)$ has interval support and is unimodal.*

Remark 14 (Resolution)**.** *The conjecture is false as stated. The seven-point poset with cover relations $1 \prec 2$, $2 \prec 3$, $2 \prec 4$, $3 \prec 6$, $4 \prec 5$, $5 \prec 6$, $6 \prec 7$ has a one-dimensional centre and a nineteen-dimensional pattern algebra over $\mathbb{F}_2$, and its incidence quotient $R_{P,2}$ has nonnegative integer coefficients with interval support but is not unimodal: the coefficients at degrees* 8*,* 9*, and* 10 *are* 206336*,* 198656*,* 225280*, a strict interior valley between two higher values. The positivity of Conjecture 89 and the interval support both hold on this poset, so the unimodality alone fails.*

Significance. The statement is retained with its resolution because the failure sharpens the neighbouring positivity result. Given the positivity of Conjecture 89, the natural next question was the shape of the coefficient sequence, and the refuting poset shows that positivity does not force a single peak, since the quotient can carry a strict interior valley while keeping nonnegative coefficients and gapless support. The nearest related results are Delsarte [85] for the alternating-form arithmetic and Coll–Mayers [81] for the coadjoint-orbit reading of K. The surviving positive content is exactly the positivity and interval support of Conjecture 89, which the counterexample leaves intact, so the regularity of $R_{P,q}$ under a line centre stops at nonnegativity and gapless support and does not extend to unimodality.

23 Hasse-forest and unitriangular structure

Conjecture 91 (Hasse-forest reverse determination)**.** *For every finite poset P whose Hasse cover graph is a forest and every prime power q, the enumerator $K_{P,q}(T)$ determines $A_{P,q}(T)$.*

Significance. This is the reverse determination of Conjecture 77 under a structural hypothesis complementary to the centre-one hypothesis of Conjecture 88. The reason is that a forest cover graph builds the coadjoint form from independent covers, which makes the incidence quotient rigid enough to invert. The nearest related result is Coll–Mayers [81], whose index computations for Lie poset algebras are sharpest on tree-shaped posets. A tractable special case is the chain, a forest on one branch. In tests on the corpus described in the stress-test section, the Kirillov enumerator determined the adjoint enumerator across 44 forest-collision classes. A counterexample would be a forest pair with equal K and distinct A, which would show that the forest hypothesis is not the right setting for reverse determination.

Conjecture 92 (Hasse-forest coadjoint saturation)**.** *For every finite poset P whose Hasse cover graph is a forest and every prime power q, with $b_K = b_K(P, q)$ the maximal Kirillov rank,*

$$\operatorname{supp} K_{P,q} = \{0, 2, 4, \ldots, b_K\},$$

so every even rank up to b_K is attained.

Significance. An alternating form has even rank, and the conjecture asserts that on a forest every even rank up to the top is realized by some functional. The reason is that the independent covers of a forest allow the coadjoint rank to be adjusted one pair at a time. The nearest related results are Coll–Mayers [81], whose index is the codimension complementary to b_K, and Delsarte [85], whose alternating-form model predicts which even ranks can occur. A tractable special case is the chain, where K_{ut_n} fills its even range. In tests on the corpus described in the stress-test section, the forest Kirillov support filled every even rank up to b_K in each case. A counterexample would be a forest with a gap in its Kirillov support, which would show that the coadjoint ranks of tree-shaped posets can skip values and would remove the support hypothesis used in Conjecture 93.

Conjecture 93 (Hasse-forest Kirillov unimodality)**.** *For every finite poset P whose Hasse cover graph is a forest and every prime power q, the even-rank sequence $\big([T^{2j}]K_{P,q}(T)\big)_j$ is unimodal.*

Significance. On the gapless support of Conjecture 92 the next question is the shape of the even-rank counts. The conjecture asserts that the coadjoint rank distribution of a forest rises to a single peak and falls, with no interior dip. The nearest related results are Delsarte [85] for the alternating-form counts and Coll–Mayers [81] for the coadjoint reading. The statement is calibrated to the strongest form that holds. Strict log-concavity is false, so unimodality is the appropriate form. In tests on the corpus described in the stress-test section, the forest even-rank sequence was unimodal throughout. A counterexample would be a forest whose even-rank sequence has two peaks, which would show that the coadjoint distribution of tree-shaped posets need not be concentrated and would confine the shape statement to the unitriangular chain of Conjecture 95.

Conjecture 94 (Hasse-forest breadth–orbit bound)**.** *For every finite poset P whose Hasse cover graph is a forest and every prime power q,*

$$b_K(P, q) \le 2\, b_A(P, q).$$

Significance. The bound relates the maximal coadjoint-orbit rank to the adjoint breadth. The reason is that on a forest the largest alternating form cannot exceed twice the largest adjoint image, since both come from the same covers. The nearest related results are Cameron–Coll–Mayers–Russoniello [80] for the breadth b_A and Coll–Mayers [81] for the index that controls b_K. The forest hypothesis is essential. The unrestricted inequality is false, with many counterexamples off the forest class. In tests on the corpus described in the stress-test section, the forest bound held everywhere, and the unrestricted bound failed fourteen times off the forest class. A counterexample would be a forest violating $b_K \leq 2b_A$, which would show that a tree-shaped poset can carry a coadjoint orbit large relative to its adjoint breadth.

Conjecture 95 (Unitriangular Kirillov strict log-concavity)**.** *For $L = \mathrm{ut}_n(\mathbb{F}_q)$ and every prime power q, the even-rank sequence $\big([T^{2j}]K_{L,q}(T)\big)_j$ is strictly log-concave at every nontrivial interior index.*

Significance. The chain is the extremal forest, and here the unimodality of Conjecture 93 strengthens to strict log-concavity. The conjecture asserts that the coadjoint rank distribution of ut_n is as concentrated as the family allows. The nearest related result is Marberg [84], whose character and orbit enumeration for the unitriangular group is the closest account of the coadjoint data of ut_n. A tractable special case would be strict log-concavity for a single infinite family of indices, such as the central ones. In tests on the corpus described in the stress-test section, strict log-concavity was confirmed for ut_4, ut_5, and ut_6 over $\mathbb{F}_2$, for ut_4 and ut_5 over $\mathbb{F}_3$, and for ut_4 over $\mathbb{F}_5$. A counterexample would be an interior index of some ut_n where the strict inequality fails, which would show that even the chain falls short of strict concentration.

Conjecture 96 (Adjoint–coadjoint incidence anticorrelation)**.** *For every nonabelian pattern Lie algebra $L = L_P$ and every prime power q, sampling (x, f) uniformly on the incidence variety $I_L = \{(x, f) : f([x, L]) = 0\}$ gives*

$$\mathrm{Cov}\big(\mathrm{rank}(\mathrm{ad}_x), \mathrm{rank}(B_f)\big) < 0,$$

with covariance zero exactly when L_P is abelian.

Significance. The incidence variety of Proposition 6 pairs an element with a functional annihilating its adjoint image. The conjecture asserts that a high adjoint rank leaves little room for the coadjoint form to be large, so the two ranks are negatively correlated on I_L. The nearest related result is Coll–Mayers [81], whose coadjoint-orbit analysis of Lie poset algebras is the setting in which $\mathrm{rank}(B_f)$ is an orbit dimension. A tractable special case is the abelian case, where both ranks are zero and the covariance vanishes, giving the equality clause. In tests on the corpus described in the stress-test section, the covariance was negative on every nonabelian poset tested and exactly zero on all sixteen abelian ones. The statement is calibrated to the strongest form that holds. The stronger conditional stochastic-order and conditional-mean forms are false, so the sign of the covariance is the surviving claim. A counterexample would be a nonabelian poset with nonnegative incidence covariance, which would decouple the two ranks on the variety that defines their duality.

Part VI

Local structure of primes and arithmetic functions

Added to the deposit: 10 August 2026.

The twenty conjectures below return to the local–global aesthetic of Part I, but read it off objects other than residue classes of consecutive primes. Each takes a familiar arithmetic quantity, looks at it through a local lens that ought to have looked symmetric, and takes the residual asymmetry seriously. They form five programmes. A compass programme (Conjectures 97 to 102) measures the direction of the nearest prime to a polynomial centre and matches it to a local-sieve hazard. An order-pattern programme (Conjectures 103 and 104) studies the order patterns of $\mathrm{rad}(n)/n$ on short windows. An order programme (Conjectures 105 to 109) measures how often abundancy and totient ratios move monotonically. A shock programme (Conjectures 110 to 115) tracks the prime-factor count of combinatorial sequences that drop many prime factors in a single step, a *shock*. A boundary programme (Conjecture 116) reads the prime-factor anatomy of the four integers bracketing a prime gap. The proved layer is kept separate. Proposition 7 gives the factorial-ratio jump identities, Proposition 8 gives short-window exchangeability, Proposition 9 proves that every fixed radical-pattern density exists, Proposition 10 proves exact finite-sieve equivalence for triangular and hexagonal centres, Proposition 13 proves the $k = 2$ case of the dyadic multinomial slope, Proposition 11 settles the three shock laws, Conjectures 110 to 112, by an elementary forcing construction, and Proposition 12 evaluates the prime-truncated lag-one covariance of the central-binomial slice at exactly $-\frac{5}{3}$. Six of the twenty state a bias as a numerical band rather than a derived constant, and are flagged as such where they occur, the compass and order constants of Conjectures 98, 99, 100, 105, 106, and 108. Two, Conjectures 102 and 113, extrapolate in a parameter rather than in the index and are the most exposed.

For a composite integer a let $p^-(a) < a < p^+(a)$ be its neighbouring primes and set $L(a) = a - p^-(a)$ and $U(a) = p^+(a) - a$. The *prime compass* of a points right when $U(a) < L(a)$, left when $L(a) < U(a)$, and is tied when the two distances agree. For a sequence $A_1 < A_2 < \cdots$ of composite centres write

$$\theta_A(x) = \frac{\#\{n : A_n \le x,\ U(A_n) < L(A_n)\}}{\#\{n : A_n \le x,\ U(A_n) \ne L(A_n)\}}$$

for the right-closer frequency among the untied centres, and $\theta_A = \lim_{x\to\infty} \theta_A(x)$ when it exists; we call θ_A the *compass constant* of A. The polygonal numbers are $P_s(n) = \frac{1}{2}\big((s-2)n^2 - (s-4)n\big)$ for $s \ge 3$, the pronic numbers are $Q(n) = n(n+1)$, and the squares are n^2. For the radical $\mathrm{rad}(n) = \prod_{p|n} p$ write $R(n) = \mathrm{rad}(n)/n$, and for a window of length d call the permutation $\pi_0 \cdots \pi_{d-1}$ of $\{0, \dots, d-1\}$ the *order pattern*—the ordinal pattern of the window—at n when $R(n+\pi_0) < \cdots < R(n+\pi_{d-1})$, counting only windows with no ties. Write δ_π for the density of the order pattern π among the untied windows, the limiting ratio of the two window counts, so that $\sum_\pi \delta_\pi = 1$; ties have positive density, adjacent squarefree integers already tying at $R = 1$, so an unconditional normalization would be deficient. For the abundancy index $\sigma(n)/n$ and the totient ratio $\phi(n)/n$ call an adjacent triple *monotone* when it is strictly increasing or strictly decreasing, and for a stride W let $\mu(W)$ be the density of n for which $\sigma(n)/n, \sigma(n+W)/(n+W), \sigma(n+2W)/(n+2W)$ is monotone. Write $\Omega(m)$ for the number of prime factors of m with multiplicity, and set the binomial slice $A_{k,n} = \binom{kn}{n}$, the central multinomial $M_{k,n} = (kn)!/(n!)^k$, and the Fuss–Catalan number $F_{r,n} = \frac{1}{rn+1}\binom{(r+1)n}{n}$, with single-step jump $J_{k,n} = \Omega(A_{k,n+1}) - \Omega(A_{k,n})$. Finally, for consecutive odd primes $p < q$ set

$$D(p,q) = \Omega(p+1) + \Omega(q-1) - \Omega(p-1) - \Omega(q+1),$$

the inward-minus-outward prime-factor balance of the four composites bracketing the gap.

Proposition 7 (Factorial-ratio jump identities)**.** For all $k \geq 2$ and $n \geq 1$,

$$J_{k,n} = \sum_{j=1}^{k} \Omega(kn+j) - \Omega(n+1) - \sum_{j=1}^{k-1} \Omega((k-1)n+j).$$

In particular $\Omega\binom{2n+2}{n+1} - \Omega\binom{2n}{n} = 1 + \Omega(2n+1) - \Omega(n+1)$, and for the Catalan numbers $C_n = \frac{1}{n+1}\binom{2n}{n}$, $\Omega(C_{n+1}) - \Omega(C_n) = 1 + \Omega(2n+1) - \Omega(n+2)$. Consequently the central binomial jump $J_{2,n}$, the Catalan jump $\Omega(C_{n+1}) - \Omega(C_n)$, and the central multinomial jump $\Omega(M_{k,n+1}) - \Omega(M_{k,n})$ for every $k \geq 2$ are unbounded below, while the binomial slice $J_{k,n}$ for $k \geq 3$ is settled by the forcing construction of Proposition 11.

Proof. The ratio $A_{k,n+1}/A_{k,n} = \binom{k(n+1)}{n+1}/\binom{kn}{n}$ is the rational number $\prod_{j=1}^{k}(kn+j)/((n+1)\prod_{j=1}^{k-1}((k-1)n+j))$, an exact identity of factorials in which every factor is a positive integer, so Ω is additive across it and the displayed formula follows, the two specializations being the cases $k = 2$ for the central binomial coefficient and its Catalan quotient. For the unboundedness we use the elementary bound that an odd integer m has $\Omega(m) \leq \log_3 m$, since each of its prime factors is at least 3. Taking $k = 2$ and $n + 1 = 2^t$ gives $J_{2,n} = 1 + \Omega(2^{t+1} - 1) - t$ with $2^{t+1} - 1$ odd, so $J_{2,n} < 1 + (t+1)\log_3 2 - t \to -\infty$; the Catalan jump $1 + \Omega(2n+1) - \Omega(n+2)$ at $n + 2 = 2^t$ equals $1 + \Omega(2^{t+1} - 3) - t \to -\infty$ by the same bound. For the central multinomial $M_{k,n+1}/M_{k,n} = \prod_{j=1}^{k}(kn+j)/(n+1)^k$, so $\Omega(M_{k,n+1}) - \Omega(M_{k,n}) = \sum_{j=1}^{k} \Omega(kn+j) - k\,\Omega(n+1)$; at $n+1 = 2^t$ the top factor $k \cdot 2^t$ contributes $t + \Omega(k)$, while each of the $k-1$ remaining factors $k \cdot 2^t - i$ with $1 \leq i \leq k-1$ has $v_2(k \cdot 2^t - i) = v_2(i) \leq \log_2 k$ and odd part below $k \cdot 2^t$, so $\Omega(k \cdot 2^t - i) \leq \log_2 k + t\log_3 2 + O(1)$ and the jump is at most $-(1 - \log_3 2)(k-1)\,t + O(1) \to -\infty$. For the binomial slice with $k \geq 3$ the extra block $-\sum_{j=1}^{k-1} \Omega((k-1)n+j)$ need not be large, so this argument does not force $J_{k,n} \to -\infty$; that case needs the forcing construction of Proposition 11. □

Proposition 8 (Short-window radical exchangeability)**.** For every window length $d \leq 4$ the strict order-pattern densities of $R(n) = \mathrm{rad}(n)/n$ exist and are uniform, each equal to $1/d!$.

Proof. The value $R(n) = \mathrm{rad}(n)/n$ is the reciprocal of the powerful part of n, determined by the prime squares dividing n: $1/R(n) = \prod_{p^2|n} p^{v_p(n)-1}$. In a window $n, n+1, \ldots, n+d-1$ of length $d \leq 4$ no prime square p^2 can divide two distinct entries, since consecutive multiples of p^2 differ by $p^2 \geq 4 \geq d$, with the single boundary case $p = 2$, $d = 4$ excluded because 4 divides only one of any four consecutive integers. Hence the powerful parts of the d entries are supported on disjoint sets of primes, and by the Chinese remainder theorem the vector $(R(n), \ldots, R(n+d-1))$ is asymptotically exchangeable, so every strict order is equally likely. □

Proposition 9 (Existence of radical-pattern densities)**.** For every fixed window length d, each unconditional strict order-pattern count for $R(n) = \mathrm{rad}(n)/n$ has a natural density. The untied windows have positive density. Consequently every conditional density δ_π exists.

Proof. Fix d. For $Y \geq 2$ and $K \geq 3$, truncate $R(m)$ by retaining only primes $p \leq Y$ and replacing each valuation by $\min(v_p(m), K-1)$. The resulting value is periodic modulo $\prod_{p \leq Y} p^K$. The truncated vector on $m, m+1, \ldots, m+d-1$ can differ from the true vector only if $p^2 \mid m+i$ for some $p > Y$, or if $p^K \mid m+i$ for some $p \leq Y$. The upper density of this exceptional set is at most

$$d \sum_{p>Y} \frac{1}{p^2} + d \sum_{p \leq Y} \frac{1}{p^K}.$$

It tends to zero by first taking Y and then K large. Every strict-pattern indicator for the truncated vector is periodic. Its densities form a Cauchy net under refinement, so every unconditional strict-pattern density exists.

It remains to show that their sum is positive. For each prime $p \leq d$, choose an exponent a_p with $p^{a_p} > d$ and a residue class modulo p^{a_p} on which none of the d entries is divisible by p^{a_p}. This fixes every small-prime valuation in the window. Next choose distinct primes $q_0, \ldots, q_{d-1} > d$ and exponents $e_i \geq 2$. The Chinese remainder theorem permits

$$m + i \equiv q_i^{e_i} \pmod{q_i^{e_i+1}}$$

for every i. No q_i divides another entry of the window. The exponents can be chosen so that the resulting d powerful parts, including the fixed small-prime contributions, are pairwise distinct. In this progression, require every remaining prime p to divide no entry to order two. For $p > d$ this excludes at most $d < p^2$ residue classes modulo p^2. The elementary squarefree sieve therefore gives the positive density

$$\prod_{p \nmid Q} \left(1 - \frac{\nu(p)}{p^2}\right) > 0,$$

where Q is the fixed modulus and $0 \leq \nu(p) \leq d$. On this positive-density set the powerful parts, and hence the values of R, are pairwise distinct. The density of untied windows is therefore positive, so division by it gives every δ_π. □

Proposition 10 (Triangular–hexagonal finite-sieve equivalence)**.** For every modulus $m \geq 1$ and residue $c \bmod m$,

$$\frac{1}{2m}\#\{n \bmod 2m : P_3(n) \equiv c \pmod m\} = \frac{1}{m}\#\{k \bmod m : P_6(k) \equiv c \pmod m\}.$$

Consequently the triangular and hexagonal local compass frequencies are exactly equal for every finite set of sieving primes.

Proof. The involution $n \mapsto -n-1$ on $\mathbb{Z}/2m\mathbb{Z}$ reverses parity and preserves $P_3(n) = n(n+1)/2$ modulo m. It therefore pairs every even index with an odd index having the same triangular residue. The odd indices are $n = 2k-1$ with $k \bmod m$, and $P_3(2k-1) = P_6(k)$. This proves the displayed equality. At any finite sieve depth the first surviving offsets on both sides of a centre depend only on its residue modulo the product of the sieving primes, so the equality of residue laws gives equality of the local compass frequencies. □

24 Prime compasses at polynomial centres

Conjecture 97 (Local-sieve limit for polygonal compasses)**.** *For every fixed $s \geq 3$ the limit $\theta_s = \theta_{P_s}$ exists. Moreover, if the nearest-prime race is replaced by the first offset on each side surviving every prime at most y, the resulting local compass frequency tends to θ_s as $y \to \infty$.*

Significance. The direction of the nearest prime to a centre a is decided by two competing waiting times, and for a polynomial centre the admissible offsets on each side are filtered by the residue pattern of P_s modulo the small primes, so the compass should be governed by a local-sieve hazard rather than by the size of P_s. The conjecture asserts both that the frequency settles and that the deep-sieve local model is its limit, which is the content beyond a mere claim of bias. The nearest related result is the Cramér–Granville model of gaps between primes [17], in which such local hazards are the leading correction to the exponential law. A first decisive theorem would be the existence of θ_3 for the triangular numbers. In tests on the corpus described in the stress-test section, the sieve using only the primes through 47 predicted the order-level bias with correlation 0.896 across $s \leq 100$. A counterexample would be a polygonal order whose frequency fails to settle, or whose deep local model does not approach the observed value.

Conjecture 98 (Pentagonal and pronic compasses)**.** *The pentagonal compass satisfies* $0.56 < \theta_{P_5} < 0.62$*, while the pronic compass satisfies* $0.42 < \theta_Q < 0.48$.

Significance. The pentagonal and pronic sequences are the two simplest quadratic centres of opposite local character, and the conjecture records that their compasses point in opposite directions, one right-biased and one left-biased, which rules out any explanation by the size of the centres alone. The bands are stated as numerical intervals rather than derived constants, the honest register for a measured bias whose local model of Conjecture 97 explains its sign but not yet its value. The nearest related result is again the local hazard reading of prime gaps [17]. In tests on the corpus described in the stress-test section, the frequencies were 0.583 over 51,630 pentagonal centres and 0.450 over 63,236 pronic centres, with the signs replicated on a holdout above 10^9. A counterexample would be either frequency settling outside its band, or the two sequences sharing a direction.

Conjecture 99 (Opposite polygonal orders)**.** *The* 60*-gonal compass satisfies* $\theta_{60} > 0.58$ *and the* 69*-gonal compass satisfies* $\theta_{69} < 0.49$.

Significance. Two nearby polygonal orders can carry opposite biases because the coefficients of P_s traverse different residue patterns as s varies, so the compass is a function of the order and not a monotone drift in it. The conjecture fixes the two most extreme observed orders as a right-biased and a left-biased witness, calibrating the spectrum of Conjecture 102. The nearest related result is the local-sieve reading of Conjecture 97, of which this is a two-point instance. In tests on the corpus described in the stress-test section, the frequencies were 0.609 and 0.468, each replicated above 10^9. A counterexample would be either order settling on the wrong side of its stated cut.

Conjecture 100 (Triangular and hexagonal equality)**.** *The triangular and hexagonal compass constants are equal,* $\theta_{P_3} = \theta_{P_6} \in (0.52, 0.56)$.

Significance. The identity $P_6(n) = P_3(2n-1)$ realizes the hexagonal numbers as the odd-index triangular subsequence, and the conjecture asserts that this subsequence carries the same compass limit as the whole triangular sequence, so the bias is insensitive to the arithmetic-progression thinning of the index. Proposition 10 now proves the exact finite-sieve content. The two centre sequences have the same residue law for every modulus and hence the same compass frequency at every finite sieve depth. What remains open is the transfer from those local models to the actual nearest primes in Conjecture 97, together with the common numerical band. In tests on the corpus described in the stress-test section, the two constants agreed to three places, 0.53595 against 0.53572. A counterexample would be a persistent gap between the two frequencies at large centres.

Conjecture 101 (Square compass by root class)**.** *For each residue* $r \bmod 6$ *the conditional square compass* $\theta_{\square,r}$ *over roots* $n \equiv r$ *exists, with* $\theta_{\square,r} < 0.45$ *for* $r \in \{1,5\}$ *and* $\theta_{\square,r} > 0.54$ *for* $r \in \{0,2,3,4\}$.

Significance. The overall imbalance of the nearest prime to a square is a coarse average, and the conjecture is the finer claim that the bias splits sharply and stably by the residue of the root modulo six, a fingerprint that survives conditioning where the pooled rate does not. The mechanism is that $n \bmod 6$ fixes the residues of $n^2 \pm 1$ and $n^2 \pm 2$ at the primes 2 and 3, forcing which side carries the small forced factors. The nearest related result is the local hazard model of Conjecture 97 specialized to s with a square centre. In tests on the corpus described in the stress-test section, the six classes gave frequencies from 0.419 to 0.572 over about 10,540 roots each, with the same signs below 10^9. A counterexample would be a root class settling on the wrong side of 0.5 from the stated sign.

Conjecture 102 (Two-sided order spectrum)**.** *The polygonal compass spectrum straddles one half in the limit of many sides,*

$$\liminf_{s\to\infty} \theta_s < 0.49 < 0.58 < \limsup_{s\to\infty} \theta_s,$$

so infinitely many polygonal orders are left-biased and infinitely many are right-biased.

Significance. Where Conjectures 98 to 101 extrapolate in the index n at fixed order, this statement extrapolates in the order s itself, asserting that the coefficients of P_s keep producing both directions as s grows, and it is flagged as the most exposed of the compass programme for exactly that reason. The mechanism is that the residue pattern of P_s modulo a fixed prime set runs through a full cycle as s varies, so no eventual sign can set in. The nearest related result is the local hazard model of Conjecture 97, which makes the two-sidedness plausible without proving it. In tests on the corpus described in the stress-test section, the observed orders $3 \le s \le 100$ ranged from 0.468 at $s = 69$ to 0.609 at $s = 60$. A counterexample would be an eventual one-sided bias, an order beyond which every θ_s lies on one side of one half.

25 Order patterns of the radical ratio

Conjecture 103 (Permanent nonuniformity from length five)**.** *For every fixed window length* $d \ge 5$ *the strict order-pattern densities of* $R(n) = \mathrm{rad}(n)/n$ *exist, and their distribution is not uniform.*

Significance. The first coupling event, $4 \mid n$ and $4 \mid n+4$, biases the endpoints of the length-five window. Proposition 8 makes the law exactly uniform through length four, while Proposition 9 now proves that every fixed-length conditional law exists by periodic approximation and a positive-density squarefree sieve. The open content is nonuniformity at each length $d \ge 5$. Nonuniformity at length five alone does not settle a longer length because the longer law conditions on the whole longer window being untied. The nearest related result is the distribution theory of powerful numbers [86], which governs the density of the coupling events. In tests on the corpus described in the stress-test section, the chi-square against uniformity jumped from 0.45 at length four to about 2.1×10^7 at length five through 10^9. A counterexample would be a length $d \geq 5$ at which every strict order pattern has density $1/d!$.

Conjecture 104 (Thirteen-fold pattern separation)**.** *At window length five the strict densities satisfy* $12 < \delta_{30241}/\delta_{13240} < 15$*, with* 30241 *in the dominant class and* 13240 *in the suppressed class.*

Significance. The length-five order-pattern law is not merely nonuniform but sharply structured, and the conjecture pins the ratio of its most common to one of its least common patterns as a bounded constant strictly above one, so the asymmetry is a definite feature and not a vanishing edge effect. The mechanism is the same endpoint coupling of Proposition 8, which favours patterns placing the coupled endpoints coherently and suppresses those placing them against the coupling. The nearest related result is again the powerful-number density [86] that sets the coupling strength. In tests on the corpus described in the stress-test section, the ratio fell from 14.06 at 2×10^7 to 13.31 at 10^9 over 23,154,595 strict windows, with the length-four and length-five cell counts reproduced by an independent implementation. A counterexample would be the ratio settling outside the band, or the two patterns exchanging dominance.

26 Order laws for multiplicative ratios

Conjecture 105 (Abundancy order constant)**.** *The density of n for which $\sigma(n)/n, \sigma(n+1)/(n+1), \sigma(n+2)/(n+2)$ is monotone exists and lies in* $(0.094, 0.098)$.

Significance. For three independent continuous samples the monotone density is $2/3! = 1/3$, and the conjecture records that adjacent abundancy triples are far from independent, monotone only about a tenth of the time, with the middle value an extremum of the triple more than two thirds of the time. The band is stated numerically, the honest register for a measured order constant whose value the mechanism does not yet supply. The nearest related result is the classical distribution function of $\sigma(n)/n$ [86], a one-point law that says nothing about the joint order of a triple. In tests on the corpus described in the stress-test section, the pooled rate was 0.09584 over 19,999,991 strict triples, stable across seven disjoint blocks. A counterexample would be the density settling outside the band or failing to converge.

Conjecture 106 (Totient order constant)**.** *The density of n for which $\phi(n)/n, \phi(n+1)/(n+1), \phi(n+2)/(n+2)$ is monotone exists and lies in* $(0.019, 0.022)$.

Significance. The totient ratio is even more strongly anti-monotone than the abundancy index, its adjacent triples monotone only about one part in fifty, and the conjecture records this second measured order constant as a companion to Conjecture 105 on the reciprocal-flavoured multiplicative function. The mechanism is that $\phi(n)/n = \prod_{p|n}(1 - 1/p)$ and $\sigma(n)/n$ move oppositely with the small prime divisors of n, so the two order constants probe the same local dependence from two sides. The nearest related result is the classical distribution function of $\phi(n)/n$ [86]. In tests on the corpus described in the stress-test section, the pooled rate was

0.02025 over 19,999,990 strict triples. A counterexample would be the density settling outside the band.

Conjecture 107 (Five-term totient barrier, resolved false). *No five consecutive values of $\phi(n)/n$ are strictly increasing, and no five are strictly decreasing.*

Remark 15 (Resolution). *The conjecture is false. A theorem of Martin [53] states that for every k and every prescribed strict ordering of $\phi(n+1), \dots, \phi(n+k)$ with each successive ratio exceeding any fixed constant C, the set of n realizing it has positive lower density. Apply it with $k = 5$. Since $\phi(m+1)/(m+1) > \phi(m)/m$ exactly when $\phi(m+1)/\phi(m) > 1 + 1/m$, prescribing $\phi(n+i+1)/\phi(n+i) > 2$ for $i = 1, \dots, 4$ forces $\phi(n+i)/(n+i)$ strictly increasing across the five consecutive integers; prescribing $\phi(n+i)/\phi(n+i+1) > 2$ forces a strictly decreasing run in the same way. Hence strictly monotone runs of $\phi(n)/n$ of length five—indeed of every length, in both directions—occur on a set of positive lower density. That density is astronomically small, which is why no run appears below 4×10^9, but it is positive, so the barrier does not hold.*

Significance. The statement is retained with its resolution because the contrast it draws survives the refutation. Long strictly monotone runs of $\phi(n)/n$ do exist, by Martin's theorem, but only on a set of astronomically small positive density, with none below 4×10^9, whereas the abundancy analogue already fails at a findable height: a strictly increasing five-term run of $\sigma(n)/n$ begins at $n = 36{,}721{,}681$. The two functions therefore differ not in whether long monotone runs occur, since both admit them, but in how far one must search to meet one, a quantitative separation the original barrier misread as an absolute prohibition for the totient. The surviving content is this gap and its mechanism, that $\phi(m+1)/\phi(m)$ must clear $1 + 1/m$ to advance the ratio, which is exactly the simultaneous-inequality construction of [53] pushed to a fixed length.

Conjecture 108 (Primorial-stride order constant). *Along primorial strides $W_y = \prod_{p \le y} p$ the abundancy order density $\mu(W_y)$ has a limit as $y \to \infty$, and $\lim_{y\to\infty} \mu(W_y) \in (0.16, 0.19)$.*

Significance. Widening the stride to a primorial makes the three sampled arguments share their small prime structure in a controlled way, and the conjecture records that the monotone density then rises from the adjacent value of Conjecture 105 toward a distinct limit, so the order constant is a function of the stride's prime support. The band is again a measured interval. The nearest related result is the distribution function of $\sigma(n)/n$ [86], here conditioned along an arithmetic progression of common modulus. In tests on the corpus described in the stress-test section, the rate fell from 0.202 at $W = 30$ to 0.1768 at $W = 30030$, the later blocks decreasing toward a limit below the pooled value. A counterexample would be $\mu(W_y)$ failing to converge or landing outside the band.

Conjecture 109 (Dyadic completion overshoot). *For every even squarefree W divisible by six the limit $\lim_{e\to\infty} \mu(2^e W)$ exists and exceeds $1/3$, so $\mu(2^e W) > 1/3$ for all large e.*

Significance. Raising only the dyadic exponent of a fixed squarefree stride, rather than its prime support, is the sharp control, and the conjecture records that this raises the monotone density past the independent value $1/3$ and holds it there, the opposite direction from the primorial depression of Conjecture 108. The mechanism is isolated by two controls: replacing $\sigma(n)/n$ by the squarefree abundancy or by $n/\phi(n)$ removes most of the effect, so it is carried by the exponents, especially v_2, and not by the set of prime divisors. The nearest related result is the distribution theory of $\sigma(n)/n$ [86]. In tests on the corpus described in the stress-test section, three stride ladders rose monotonically toward values above $1/3$, and an adversarial scan of 73 further squarefree supports had every final level above $1/3$, the smallest being 0.3341. A counterexample would be a stride whose completed density stayed at or below $1/3$.

27 Prime-factor shocks in combinatorial sequences

Conjecture 110 (Binomial-slice negative shocks, resolved true). *For every fixed $k \geq 3$ the jump $J_{k,n} = \Omega\binom{k(n+1)}{n+1} - \Omega\binom{kn}{n}$ is unbounded below.*

Significance. The binomial slice grows super-exponentially while its prime-factor count can fall, because a step can replace many small prime powers by fewer large primes, and the conjecture asserts arbitrarily deep single-step losses for every slice past the central one. The mechanism is the exact identity of Proposition 7, in which the negative term $-\Omega(n+1)$ can be made large by a power-of-two argument. The nearest related result is Proposition 7 itself, which proves the case at hand for $k = 2$; the general k is not reduced to it, since the slice carries a second negative block that the power-of-two argument does not control, and stays open. A first decisive theorem is the $k = 2$ case, already proved. In tests on the corpus described in the stress-test section, the minima reached -21 through -23 for $k = 3$ to 8 over millions of steps each. A counterexample would be a slice with a finite floor on its downward jumps. *Resolution.* The conjecture is true: Proposition 11 forces arbitrarily deep negative jumps for every $k \geq 3$ by a Chinese-remainder construction that loads the denominator forms $(k-1)n + j$ with many prime factors while a first-moment bound keeps the numerator forms typical.

Conjecture 111 (Binomial-slice positive shocks, resolved true). *For every fixed $k \geq 2$ the jump $J_{k,n}$ is unbounded above.*

Significance. The upward companion of Conjecture 110 asserts that a single step can also gain arbitrarily many prime factors, so the jump sequence is two-sided and not merely floored, and the two together make the slice's prime-factor count a genuinely oscillating additive process. The mechanism is again the identity of Proposition 7, now with the positive block $\sum_j \Omega(kn + j)$ made large by choosing n so that the numerator forms carry high prime powers. The nearest related result is the theory of the normal and large values of Ω on linear forms [86]. In tests on the corpus described in the stress-test section, the maxima reached 13 through 26 across $k = 2$ to 8. A counterexample would be a slice whose upward jumps are bounded. *Resolution.* The conjecture is true: Proposition 11 applies the same forcing with the two form families exchanged, loading the numerator forms $kn + j$; the second negative block that resisted the power-of-two argument is controlled on average rather than pointwise.

Conjecture 112 (Fuss–Catalan two-sided shocks, resolved true). *For every fixed $r \geq 1$ the jump $\Omega(F_{r,n+1}) - \Omega(F_{r,n})$ is unbounded both below and above.*

Significance. The Fuss–Catalan numbers are the natural one-parameter generalization of the Catalan sequence, and the conjecture extends the two-sided shock phenomenon to them, so the effect is a feature of factorial-ratio families rather than of the binomial coefficient alone. The mechanism is the exact consecutive-ratio identity for $F_{r,n}$, derived like Proposition 7 without factoring the enormous integer itself. The nearest related result is Proposition 7, whose Catalan case is the instance $r = 1$. In tests on the corpus described in the stress-test section, orders 1 to 8 gave minima from -20 to -25 and maxima from 13 to 24. A counterexample would be an order with a one-sided bound on its jumps. *Resolution.* The conjecture is true in both directions for every $r \geq 1$: Proposition 11 applies to the Fuss–Catalan ratio's two families of forms, which are pairwise nonproportional since $(r+1)j' = rj$ has no admissible solution.

Proposition 11 (Two-sided shock forcing)**.** For every fixed $k \geq 2$ the jump $J_{k,n}$ is unbounded below and unbounded above, and for every fixed $r \geq 1$ the Fuss–Catalan jump $\Omega(F_{r,n+1}) - \Omega(F_{r,n})$ is unbounded below and unbounded above. In particular Conjectures 110, 111, and 112 are true.

Proof. Proposition 7 and $\Omega(kn+k) = \Omega(k) + \Omega(n+1)$ give $J_{k,n} = \Omega(k) + \sum_{j=1}^{k-1}(\Omega(kn+j) - \Omega((k-1)n+j))$, and cancelling $rn+1$ against the denominator block and $(r+1)(n+1)$ against $n+1$ in the exact ratio $F_{r,n+1}/F_{r,n}$ gives $\Omega(F_{r,n+1}) - \Omega(F_{r,n}) = \Omega(r+1) + \sum_{j=1}^{r}\Omega((r+1)n+j) - \sum_{j=2}^{r+1}\Omega(rn+j)$. Each claim is therefore an instance of the following: for affine forms $L_i(n) = a_i n + b_i$, $1 \leq i \leq s$, and $L'_j(n) = a'_j n + b'_j$, $1 \leq j \leq s$, with positive integer coefficients and no L_i proportional to any L'_j, the difference $\sum_i \Omega(L_i(n)) - \sum_j \Omega(L'_j(n))$ is unbounded above. Nonproportionality holds in all four applications: $kj' = (k-1)j$ forces $k \mid j$ with $1 \leq j \leq k-1$, and $(r+1)j' = rj$ forces $r+1 \mid j$ with $1 \leq j \leq r$, both impossible.

For the claim, extract the contents $\gcd(a'_j, b'_j)$, whose contribution to $\Omega(L'_j)$ is constant, so the L'_j may be taken primitive; let R bound the coefficients and the cross-resultants $|a_i b'_j - a'_j b_i|$, fix m, and choose sm distinct primes exceeding R, one block Q_i of m primes for each L_i. Each congruence $L_i(n) \equiv 0 \pmod{Q_i}$ is a single residue class since the primes exceed a_i, and the classes combine by the Chinese remainder theorem into $n \equiv \rho \pmod{Q}$ with $Q = Q_1 \cdots Q_s$, on which $\Omega(L_i(n)) \geq m$ for every i. No forced prime p divides any $L'_j(n)$: together with $p \mid L_i(n)$ it would divide $a_i L'_j(n) - a'_j L_i(n) = a_i b'_j - a'_j b_i$, a nonzero integer of absolute value below p. Now let n run over $\rho + tQ$ for $1 \leq t \leq Q^3$, so that every value $L'_j(n)$ is at most a constant times Q^4. A prime power $p^v \leq Q^3$ with $p \nmid Q$ divides $L'_j(n)$ along at most $Q^3/p^v + 1$ values of t, primes dividing a'_j divide no value of the primitive form, and summing $1/p^v$ over prime powers gives an average of $\log\log Q + O(1)$ per form, the $+1$ terms totalling $O(Q^3/\log Q)$. Prime powers exceeding Q^3 add $O(1)$ per form on average: a prime above Q^3 divides a value at most once and to the first power, since $(Q^3)^2$ exceeds every value, while prime powers $p^v > Q^3$ with $v \geq 2$ are squarefull, number $O(Q^2)$ below the value bound, and each catches at most two values of t. By Markov some $t \leq Q^3$ has $\sum_j \Omega(L'_j(n)) \leq 2s(\log\log Q + O(1))$, whence $\sum_i \Omega(L_i(n)) - \sum_j \Omega(L'_j(n)) \geq sm - 2s\log\log Q - O(s) \to \infty$ as $m \to \infty$, since $\log\log Q = O(\log(sm\log m))$. Explicit witnesses are cheap: six primes per form already force $J_{3,n} \leq -20$ and $J_{4,n} \leq -30$ at heights near 10^{21} and 10^{35}, and $J_{3,n} \geq 21$ on the positive side, beyond every scanned extreme. □

Conjecture 113 (Negative shock covariance)**.** *In every fixed binomial-slice ($k \geq 2$), central-multinomial ($k \geq 2$), and Fuss–Catalan ($r \geq 1$) family the lag-one covariance of successive jumps has a finite strictly negative limit.*

Significance. Beyond their range, the jumps carry a local temporal structure, and the conjecture asserts that consecutive jumps are negatively associated, a large loss tending to be followed by a partial recovery, which is the signature of a mean-reverting additive process rather than an independent one. The statement is on the covariance and not the correlation, since the slowly growing variance can send the correlation toward zero while a definite local covariance survives, and it is flagged as exposed because it asserts a sign uniformly across families. The nearest related result is the theory of correlations of additive functions on nearby arguments [86]. In tests on the corpus described in the stress-test section, all 22 tested families had negative covariance, from about -1.15 to -59.5. A counterexample would be a family with nonnegative or divergent limiting covariance. The central-binomial slice now carries an exact local constant: Proposition 12 evaluates the prime-truncated covariance at exactly $-\frac{5}{3}$ for every cutoff at least 3, so the limit for that family should equal $-\frac{5}{3}$ outright, the surviving step being that primes beyond any cutoff contribute nothing in the limit; the full covariance stands at -1.47 by 2×10^6 and drifts toward the constant.

Proposition 12 (Exact truncated covariance for the central-binomial slice)**.** For a cutoff $y \geq 2$ write $\Omega_y(m) = \sum_{p \leq y} v_p(m)$ and $J_n^{(y)} = 1 + \Omega_y(2n+1) - \Omega_y(n+1)$. The lag-one covariance of the truncated central-binomial jump over $n \leq N$ converges as $N \to \infty$, to -1 at $y = 2$ and to

$$\lim_{N\to\infty} \operatorname{Cov}\big(J_n^{(y)}, J_{n+1}^{(y)}\big) = -\tfrac{5}{3} \qquad \text{for every } y \geq 3.$$

Proof. Write $A = 2n+1$, $B = n+1$, $C = 2n+3$, $D = n+2$, so that $J_n^{(y)} - 1 = \sum_{p\leq y}\big(v_p(A) - v_p(B)\big)$ and $J_{n+1}^{(y)} - 1 = \sum_{p\leq y}\big(v_p(C) - v_p(D)\big)$. For n uniform on $\{1, \ldots, N\}$ the joint residues modulo any fixed prime powers equidistribute as $N \to \infty$, valuations at distinct primes decouple in the limit, and all limiting expectations exist by the geometric decay of $\Pr[p^r \mid an+b]$, so the limiting covariance is the sum over $p \leq y$ of the local covariance $\operatorname{Cov}\big(v_p(A) - v_p(B),\, v_p(C) - v_p(D)\big)$. For a form $an + b$ with $p \nmid a$ the limiting mean of v_p is $\sum_{r\geq 1} p^{-r} = 1/(p-1)$, and two such forms whose roots differ modulo p are never simultaneously divisible by p, so their valuations have covariance $-1/(p-1)^2$. For $p \geq 5$ all four pairs (A, C), (A, D), (B, C), (B, D) have distinct roots, the cross-resultants being 2, 3, 1, 1, and the four equal covariances cancel in the signed sum. For $p = 2$, A and C are odd while B and D are consecutive, so $v_2(B)v_2(D) = 0$ identically, each mean is 1, and the local contribution is $\operatorname{Cov}(v_2(B), v_2(D)) = -1$. For $p = 3$ the exceptional pair is (A, D), with $A = 2D - 3$: divisibility by 3 is simultaneous, along the single class $n \equiv 1 \pmod 3$, while $9 \mid A$ and $9 \mid D$ are incompatible since $9 \nmid 3$, so $\mathbf{E}[v_3(A)v_3(D)] = \frac{1}{3} + 2\sum_{r\geq 2} 3^{-r} = \frac{2}{3}$ and $\operatorname{Cov}(v_3(A), v_3(D)) = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$; the other three pairs have distinct roots modulo 3 and covariance $-\frac{1}{4}$ each, so the local contribution is $-\frac{1}{4} - \frac{5}{12} + \frac{1}{4} - \frac{1}{4} = -\frac{2}{3}$. Summing the local contributions gives -1 at $y = 2$ and $-1 - \frac{2}{3} = -\frac{5}{3}$ for every $y \geq 3$. □

Conjecture 114 (Dyadic multinomial slope)**.** *For every fixed $k \geq 2$ the dyadic jump of the central multinomial obeys*

$$\frac{\Omega(M_{k,2^m}) - \Omega(M_{k,2^m-1})}{m} \longrightarrow -(k-1).$$

Significance. Proposition 7 gives a negative linear bound for the dyadic multinomial jump, and the conjecture sharpens the constant to $-(k-1)$. At $n = 2^m - 1$ the statement is equivalent to $\Omega(k2^m - c) = o(m)$ for every fixed $1 \leq c < k$. Proposition 13 proves this when $k = 2$ by separating the cyclotomic factors of $2^{m+1} - 1$ into small and large orders. The remaining fixed-k cases are prime-factor questions for the exponential shifts $k2^m - c$. In tests on the corpus described in the stress-test section, the observed jumps at $m = 21$ ran from about -16 for $k = 2$ to -118 for $k = 8$, in the predicted direction. A counterexample would be a fixed k with a nonzero linear contribution from one of the shifted terms.

Proposition 13 (The dyadic multinomial slope for $k = 2$)**.** The conclusion of Conjecture 114 holds for $k = 2$.

Proof. At $n = 2^m - 1$, the jump identity gives

$$\Omega(M_{2,2^m}) - \Omega(M_{2,2^m-1}) = 1 - m + \Omega(2^{m+1} - 1).$$

Put $N = m + 1$ and factor $2^N - 1 = \prod_{d|N} \Phi_d(2)$. Uniformly in d,

$$\log \Phi_d(2) = \varphi(d) \log 2 + O(1).$$

This follows from the Möbius product for $\Phi_d(2)$ and the convergence of $\sum_{r\geq 1} |\log(1 - 2^{-r})|$. A prime factor p of $\Phi_d(2)$ either has multiplicative order d modulo p, in which case $p \equiv 1 \pmod{d}$, or is the largest prime factor of d. In the exceptional case it occurs to the first power, by the standard order lemma and lifting the exponent.

For divisors $d \leq \sqrt{N}$, the total number of prime factors with multiplicity is at most

$$O\left(\sum_{\substack{d|N \\ d\leq\sqrt{N}}} \varphi(d)\right) = N^{1/2+o(1)}.$$

For $d > \sqrt{N}$, every nonexceptional prime factor is larger than d, so these factors contribute at most

$$O\left(\sum_{\substack{d|N \\ d>\sqrt{N}}} \frac{\varphi(d)}{\log d}\right) = O\left(\frac{N}{\log N}\right).$$

The exceptional factors contribute at most $\tau(N) = N^{o(1)}$. Here we used $\sum_{d|N} \varphi(d) = N$ and the standard divisor bound. Hence $\Omega(2^N - 1) = o(N)$. Dividing the displayed jump by m gives the limit -1. □

Conjecture 115 (Eventual sign balance, resolved true)**.** *For every fixed binomial-slice and Fuss–Catalan family the positive-jump and negative-jump densities each tend to* $\frac{1}{2}$*, while the zero-jump density tends to* 0*.*

Significance. The jump has a small positive mean and a broadening variance, and the conjecture asserts that the sign nonetheless balances in the limit, an Erdős–Kac-type prediction that the fluctuations swamp the drift so that up-steps and down-steps become equally frequent. The mechanism is that the jump is a difference of additive functions on linear forms whose variance grows without bound, so the centred sign is asymptotically symmetric. The nearest related result is the Erdős–Kac theorem and its descendants for additive functions [86]. In tests on the corpus described in the stress-test section, the negative fractions stood at 0.356 to 0.435 and rose across prefix blocks as the variance widened. A counterexample would be a family whose sign densities settled away from one half. *Resolution.* The conjecture is true. By the identities of Propositions 7 and 11, each jump is, up to a bounded term, a signed sum of Ω over $2(k-1)$, respectively $2r$, pairwise nonproportional affine forms, whose primitive parts are pairwise coprime linear polynomials. The multivariate Erdős–Kac theorem of El-Baz, Loughran, and Sofos [171], whose one-variable case recovers Tanaka's theorem for pairwise coprime irreducible polynomials, makes the vector of ω-values asymptotically Gaussian with identity covariance, so the signed sum, normalized by $\sqrt{2(k-1)\log\log n}$, is asymptotically standard normal; replacing ω by Ω shifts each coordinate by a quantity of bounded mean, negligible at this scale. The diverging variance swamps the bounded centring, giving sign densities $\frac{1}{2}$, and the zero-jump density vanishes because the limit law assigns every shrinking central window vanishing mass.

28 The arithmetic anatomy of prime-gap boundaries

Conjecture 116 (Boundary Euler sum)**.** *For each prime ℓ let r be uniform on $\mathbb{Z}/\ell^a\mathbb{Z}$ conditioned on $r(r+g) \not\equiv 0 \pmod{\ell}$, and set*

$$m_{\ell,a}(g) = \mathbb{E}\big[v_\ell(r+1) + v_\ell(r+g-1) - v_\ell(r-1) - v_\ell(r+g+1)\big].$$

Then for every fixed even g occurring infinitely often as a consecutive-prime gap,

$$\lim_{x\to\infty} \mathbb{E}\big[D(p,q) \mid p \le x,\ q-p=g\big] = \sum_{\ell} \lim_{a\to\infty} m_{\ell,a}(g).$$

Significance. The prime-factor balance of the four composites bracketing a gap oscillates strongly with the gap, and the conjecture is a local–global law identifying its mean with an explicit Euler sum of ℓ-adic expectations, the random variable being the factor anatomy of the neighbours rather than a residue transition of the prime endpoints. The mechanism is elementary at the small primes: for $g \equiv 2 \pmod{6}$ both inward neighbours are forced multiples of three, for $g \equiv 4 \pmod{6}$ both outward neighbours are, and when $6 \mid g$ the three-adic contribution cancels, with the higher primes superimposing smaller waves. The nearest related result is the Hardy–Littlewood correlation heuristic for tuples inside a gap [1], sharing the calculus's aesthetic of the Lemke Oliver–Soundararajan biases [24] while measuring a different object. A first decisive theorem would be the law for $g = 2$ from a Hardy–Littlewood model with the consecutive-gap exclusion built in. In tests on the corpus described in the stress-test section, truncating the sum at $\ell = 47$ gave correlation 0.998 and root-mean-square error 0.24 across the 39 gaps with at least 500 samples. A counterexample would be a well-sampled gap whose mean disagreed with the truncated Euler sum beyond the truncation error.

Part VII
Topological invariants of geometric families

Added to the deposit: 12 August 2026.

The twenty conjectures below leave arithmetic for algebraic topology while keeping the same discipline. Each takes a canonical family of spaces, computes an exact classical invariant across the whole family, and takes the resulting shape law seriously. They form four programmes. A Calabi–Yau programme (Conjectures 117 to 121) studies the Euler indices of smooth complete intersections in projective space, ending in an exact dimension-wise gcd law. A link programme (Conjectures 122 to 126) studies the middle Betti numbers of Brieskorn–Pham links under degree caps and common scaling. A moment-angle programme (Conjectures 127 to 131) studies the Hochster strands of flag moment-angle complexes, their support, shape, and zero locations. A monodromy programme (Conjectures 132 to 136) studies torus mapping tori with signed-permutation monodromy over the signed-orbit calibration proved below. The calibration layer is classical and exact: the Chern-class formula for complete intersections, the Milnor–Orlik formula for Brieskorn links, Hochster's formula for bigraded Betti numbers, and the Wang sequence for mapping tori. Six adversarial controls are retained in place, each refuting a tempting strengthening. Two of the twenty, the limit clause of Conjecture 117 and Conjecture 135, assert limits beyond any finite check and are flagged as the most exposed. Two more, Conjectures 120 and 124, are flagged for elevated attribution risk, lying close to classical methods. Since deposit, Conjectures 117 and 118 have been resolved true by Propositions 15 and 16, and Conjecture 135 has been resolved false by the Landau floor of Proposition 18, which fixes the compression exponent at 2. Conjecture 125 and the convexity clause of Conjecture 126 have since been proved by the Ehrhart model of Proposition 17.

For $n \geq 2$ and a multidegree $\mathbf{d} = (d_1, \ldots, d_r)$ with every $d_i \geq 2$ and $\sum_i d_i = n + r + 1$, let $X_{\mathbf{d}}^{(n)} \subset \mathbf{CP}^{n+r}$ be a smooth complete intersection of multidegree $\mathbf{d}$, a Calabi–Yau n-fold, with Euler characteristic computed exactly by $c(TX) = (1 + H)^{n+r+1} / \prod_i (1 + d_i H)$ and $\chi = (\prod_i d_i)[H^n]\, c(TX)$; write $E_n(\mathbf{d}) = (-1)^n \chi(X_{\mathbf{d}}^{(n)})$, $D(\mathbf{d}) = \prod_i d_i$, and let the *merge* $M_{ij}\mathbf{d}$ replace d_i, d_j by $d_i + d_j - 1$, preserving both n and the Calabi–Yau condition. Write $Q_n = E_n(2, \ldots, 2)$ for the all-quadric configuration of $n+1$ quadrics. For $\mathbf{a} = (a_1, \ldots, a_N)$ with $a_i \geq 2$, let $L(\mathbf{a})$ be the Brieskorn–Pham link $\{z_1^{a_1} + \cdots + z_N^{a_N} = 0\} \cap S^{2N-1}$, of real dimension $2N - 3$, with middle Betti number $\beta(\mathbf{a}) = b_{N-2}(L(\mathbf{a}); \mathbf{Q})$ given exactly by the Milnor–Orlik formula [88]

$$\beta(\mathbf{a}) = (-1)^N + \sum_{\varnothing \neq J \subseteq [N]} (-1)^{N-|J|} \frac{\prod_{j \in J} a_j}{\operatorname{lcm}_{j \in J} a_j},$$

the *Fano* tuples being those with $\sum_i 1/a_i > 1$, with cap $A = \max_i a_i$ and scale sequence $\beta_{\mathbf{a}}(k) = \beta(ka_1, \ldots, ka_N)$. For a finite simple graph G on m vertices with clique complex $K = \mathrm{Cl}(G)$, define the Hochster strands

$$h_{r,s}(G) = \sum_{\substack{I \subseteq V(G) \\ |I| = s}} \dim_{\mathbf{F}_2} \widetilde{H}_r(K_I; \mathbf{F}_2), \qquad H_{G,r}(z) = \sum_s h_{r,s}(G)\, z^s,$$

the bigraded Betti sectors of the Stanley–Reisner ring and of the moment-angle complex $\mathcal{Z}_K$ by Hochster's formula [89, 90]. Finally, for a signed permutation matrix $A \in SL(d, \mathbb{Z})$ let M_A be the mapping torus of the induced automorphism of the torus T^d, set $f_k(A) = \dim_{\mathbf{Q}} \ker(\wedge^k A - I)$, so that the Wang sequence gives $b_k(M_A; \mathbf{Q}) = f_k(A) + f_{k-1}(A)$ [95, 94], and let $\tau(A) = \sum_k d(\operatorname{Tor} H_k(M_A; \mathbb{Z}))$ count minimal torsion generators. The cap A of the link programme and the matrix A of the monodromy programme are unrelated symbols, each local to its own sections.

Proposition 14 (Signed-orbit calibration)**.** Let A be a signed permutation matrix on $\mathbb{Z}^d$. Then $\wedge^k A$ permutes the wedge basis up to sign, and every orbit of the induced permutation of k-subsets carries a holonomy sign, the product of the signs collected around one period. If p_k and n_k are the numbers of positive and negative orbits at level k, then

$$f_k(A) = p_k, \qquad \operatorname{coker}(\wedge^k A - I) \cong \mathbb{Z}^{p_k} \oplus (\mathbb{Z}/2)^{n_k},$$

and consequently $b_k(M_A; \mathbf{Q}) = p_k + p_{k-1}$, $\operatorname{Tor} H_k(M_A; \mathbb{Z}) \cong (\mathbb{Z}/2)^{n_k}$, and $\tau(A) = \sum_k n_k$, while $\sum_k b_k(M_A; \mathbf{Q}) = 2 \sum_k p_k$.

Proof. On the wedge basis e_S, S a k-subset, $\wedge^k A$ acts by $e_S \mapsto \pm e_{\sigma(S)}$ with σ the induced permutation, so $\wedge^k A - I$ is block-diagonal over the orbits of σ. On an orbit of length L, successive substitutions $e_i \mapsto \pm e_i$ absorb all but one sign into the basis; such changes alter individual signs but preserve their product around the orbit, the holonomy ε, which is therefore invariant. The block becomes $C_\varepsilon - I$ with C_ε the cyclic shift carrying a single sign ε on its closing edge. Eliminating $x_{i+1} = x_i$ along the open edges reduces the block to the single relation $(1-\varepsilon)x = 0$, so its cokernel is $\mathbb{Z}$ and its rational kernel one-dimensional when $\varepsilon = +1$, and its cokernel is $\mathbb{Z}/2$ and its rational kernel zero when $\varepsilon = -1$. Summing over orbits gives $f_k = p_k$ and the displayed cokernel. The Wang sequence of the fibration $T^d \to M_A \to S^1$ reads $0 \to \operatorname{coker}(\wedge^k A - I) \to H_k(M_A;\mathbb{Z}) \to \ker(\wedge^{k-1} A - I) \to 0$, and the kernel is free, so the sequence splits, giving $b_k = p_k + p_{k-1}$ and $\operatorname{Tor} H_k \cong (\mathbb{Z}/2)^{n_k}$. The two sums follow by telescoping. □

29 Euler indices of Calabi–Yau complete intersections

Conjecture 117 (Quadric-ladder ratio law, resolved true)**.** *For every $n \geq 5$,*

$$\frac{Q_{n+1}}{Q_n} < 8, \qquad \frac{Q_{n+2}}{Q_{n+1}} > \frac{Q_{n+1}}{Q_n}, \qquad \textit{and} \qquad \lim_{n\to\infty} \frac{Q_{n+1}}{Q_n} = 8.$$

Significance. The all-quadric configuration is the maximal-codimension endpoint of the family, and the conjecture asserts that its Euler index grows with a definite exponential ratio approached monotonically from below, a rare closed growth law for a topological index across Calabi–Yau dimension. The small-dimensional boundary is genuine: the values run $24, 128, 960, 6912$ at $n = 2$ to 5, the ratio rising to 7.5 and dipping to 7.2 before the monotone regime sets in. The limit clause extrapolates beyond any finite check and is flagged accordingly. The nearest related result is the exact Chern-class evaluation itself, surveyed with the complete-intersection landscape in [93]. A first decisive theorem would be the strict bound $Q_{n+1}/Q_n < 8$ from the coefficient recursion. In tests on the corpus described in the stress-test section, all finite clauses held exactly through $n = 200$, the ratio reaching 7.9337 by $n = 60$. A counterexample would be a ratio at least 8 or a convexity break beyond $n = 5$. Since deposit the conjecture has been resolved true, by proof rather than by computation: Proposition 15 shows the ladder's generating function is algebraic and derives an exact recurrence whose interval induction pins every ratio inside the corridor $(8 - 4/(n+1),\, 8 - 4/(n+2))$ from dimension five onward, proving all three clauses with the previously unstated first correction $8 - Q_{n+1}/Q_n \sim 4/n$; the recurrence also reproduces the deposited ladder through $n = 200$. The limit $8 = 2^3$ is the $d = 2$ member of the equal-degree Euler family d^{d+1} of Proposition 31.

Proposition 15 (Resolution of the quadric-ladder ratio law)**.** Conjecture 117 holds in full, with the sharpening $8 - Q_{n+1}/Q_n \sim 4/n$. The generating function $A(z) = \sum_{n\geq 0} Q_n z^n$, with $Q_0 = 2$ and $Q_1 = 0$, is algebraic,

$$A = \frac{2(1-y)^3}{1-3y}, \qquad z = \frac{y(1-2y)}{2(1-y)^2}, \quad y(0) = 0,$$

and the coefficients satisfy $(n+1)(n+2)Q_{n+2} = (n+1)(7n+9)Q_{n+1} + 4(n+2)(2n+3)Q_n$.

Proof. The Chern-class formula gives $Q_n = [t^n]\varphi(t)^{n+1}$ for $\varphi(t) = 2(1-t)^2/(1-2t)$: indeed $E_n(2,\ldots,2) = (-1)^n 2^{n+1} [H^n](1+H)^{2n+2}(1+2H)^{-n-1}$, and the substitution $t = -H$ converts one extraction into the other. For each small z the equation $y = z\varphi(y)$ has exactly one root

$y(z)$ near the origin, and the residue form of Lagrange inversion, summing the geometric series $\sum_n (z\varphi(t)/t)^n$ on a small contour and picking up the simple pole at $y(z)$, gives

$$A(z) = \sum_{n\geq 0} [t^n]\varphi(t)^{n+1} z^n = \frac{\varphi(y)}{1 - z\varphi'(y)};$$

inverting $y = z\varphi(y)$ gives $z = y(1-2y)/(2(1-y)^2)$, and the residue value simplifies to $2(1-y)^3/(1-3y)$, the stated parametrization. Differentiating along it shows that $(1+z)(1-8z)A'' - 9(1+4z)A' - 24A$ is a rational function of y that vanishes identically, a finite identity checked by exact expansion, and extracting the coefficient of z^n from this differential equation yields the recurrence.

Now the three clauses. Set $q_n = Q_{n+1}/Q_n$, so the recurrence reads $q_{n+1} = \frac{7n+9}{n+2} + \frac{4(2n+3)}{(n+1)q_n}$, strictly decreasing as a function of q_n. Write $\delta_n = 8 - q_n$, and call the two-sided bound $4/(n+2) < \delta_n < 4/(n+1)$ the corridor at index n. The corridor holds at the base $n = 5$: $Q_5 = 6912$ and $Q_6 = 51{,}072$ give $\delta_5 = 11/18 \in (4/7, 4/6)$. For the step, the map carries the corridor at index n into the interval bounded by the images of its two walls, since it is decreasing, and the two inequalities placing that image inside the corridor at index $n+1$ clear denominators to polynomials with positive coefficients in $n-5$; so the corridor propagates to every $n \geq 5$. The corridor gives $q_n < 8$; it gives $q_{n+1} > 8 - 4/(n+2) > q_n$, the strict increase; and both walls are asymptotic to $4/n$, so $q_n \to 8$ with $8 - q_n \sim 4/n$. For $2 \leq n \leq 4$ the ratios are $16/3$, $15/2$, and $36/5$, all below 8. □

Conjecture 118 (Normalized merge monotonicity, resolved true)**.** *For every $n \geq 3$, every multidegree $\mathbf{d}$ with $r \geq 2$, and every pair $i \neq j$,*

$$\frac{E_n(M_{ij}\mathbf{d})}{D(M_{ij}\mathbf{d})} > \frac{E_n(\mathbf{d})}{D(\mathbf{d})}.$$

Significance. A merge lowers the codimension by one at fixed dimension, and the conjecture asserts that the degree-normalized Euler index strictly increases along every merge path from the all-quadric floor to the hypersurface ceiling, so the normalized index is a strict order on the merge poset. The normalization is essential and calibrated: the raw strengthening is false, since $E_4(2,2,4) = 1632 > 1476 = E_4(3,4)$ although $(2,2,4) \mapsto (3,4)$ is a merge, and this counterexample is retained. The nearest related result is the complete-intersection Euler data of [93]. A first decisive theorem would be the two-equation case $r = 2$. In tests on the corpus described in the stress-test section, all 1,568,731 merges through dimension 30 satisfied the strict inequality, reverified independently through dimension 18. A counterexample would be one merge with a nonincreasing normalized index. Since deposit the conjecture has been resolved true. Proposition 16 expresses every normalised merge increment as a coefficient whose factors have nonnegative coefficients and whose target coefficient is strictly positive.

Proposition 16 (Resolution of normalised merge monotonicity)**.** Put $a_\ell = d_\ell - 1$, and let $a = a_i$ and $b = a_j$ for the entries being merged. Then

$$\frac{E_n(M_{ij}\mathbf{d})}{D(M_{ij}\mathbf{d})} - \frac{E_n(\mathbf{d})}{D(\mathbf{d})} = ab[u^{n-2}]\frac{1}{(1+u)^2(1-au)(1-bu)(1-(a+b)u)} \prod_{\ell\neq i,j} \frac{1}{1-a_\ell u} > 0.$$

Consequently Conjecture 118 holds in full. The same inequality also holds in dimension $n = 2$.

Proof. The Calabi–Yau condition gives $\sum_\ell a_\ell = n+1$. The Chern-class formula and the coefficient substitution $t = u/(1+u)$ give

$$\frac{E_n(\mathbf{d})}{D(\mathbf{d})} = [t^n]\frac{(1-t)^{n+r+1}}{\prod_\ell (1-d_\ell t)} = [u^n]\frac{1}{(1+u)^2}\prod_\ell \frac{1}{1-a_\ell u}.$$

The merge replaces a, b by $a+b$. Subtracting the two rational series and using

$$\frac{1}{1-(a+b)u} - \frac{1}{(1-au)(1-bu)} = \frac{abu^2}{(1-au)(1-bu)(1-(a+b)u)}$$

gives the displayed coefficient formula. For every integer $c \geq 1$ and $q \geq 0$,

$$[u^q]\frac{1}{(1+u)(1-cu)} = \frac{c^{q+1}+(-1)^q}{c+1} \geq 0.$$

Pair the two factors $(1+u)^{-1}$ with $(1-au)^{-1}$ and $(1-bu)^{-1}$. Every remaining factor is a geometric series with nonnegative coefficients. The coefficient of degree $n-2$ is strictly positive because $(1-(a+b)u)^{-1}$ supplies the term $(a+b)^{n-2}u^{n-2}$ while all other factors supply their constant terms. Since $ab > 0$, the merge increment is positive. □

Conjecture 119 (Maximal-degree merge monotonicity)**.** *For every $n \geq 3$ and every $\mathbf{d}$ with $r \geq 2$, if $d_i = \max_j d_j$ then $E_n(M_{ik}\mathbf{d}) > E_n(\mathbf{d})$ for every $k \neq i$.*

Significance. Raw merges can lower the Euler index, but the conjecture isolates the direction that survives: merging any degree into a largest degree always raises the unnormalized index, so the failure of raw monotonicity is confined to merges among small degrees. The retained counterexample of Conjecture 118 merges the two smallest degrees, and the conjecture asserts that this is the only kind of failure. The nearest related result is again the Euler census of [93]. A first decisive theorem would be the hypersurface-forming merges, where the target is a single degree. In tests on the corpus described in the stress-test section, all 377,356 maximal-degree merges through dimension 30 increased the index strictly. A counterexample would be a maximal-degree merge that fails to increase E_n.

Conjecture 120 (Extremal multidegrees)**.** *For every $n \geq 3$ and every multidegree $\mathbf{d}$,*

$$E_n(\underbrace{2,\ldots,2}_{n+1}) \leq E_n(\mathbf{d}) \leq E_n(n+2),$$

with equality only at the all-quadric and hypersurface configurations.

Significance. The statement pins the two ends of the family as the unique extremes of the Euler index at every dimension, turning the visually extreme configurations into a precise rigidity claim. It is flagged for elevated attribution risk: the bounds are likely accessible by elementary coefficient inequalities, and the retained content is the equality rigidity together with its calibrating role for Conjectures 118 and 119, whose merge order it bounds. The nearest related result is the complete-intersection census of [93]. A first decisive theorem would be either inequality with its equality case. In tests on the corpus described in the stress-test section, all 35,467 configurations through dimension 30 obeyed both bounds with equality only at the endpoints. A counterexample would be an interior configuration matching an extreme.

Conjecture 121 (Dimension-wise Euler gcd law). *Let g_n be the greatest common divisor of $\chi(X_{\mathbf{d}}^{(n)})$ over all multidegrees at dimension n. Then*

$$g_n = \frac{24}{\gcd(24, n)} \times \begin{cases} 2, & n \equiv 0, 2 \pmod 8, \\ 1, & \textit{otherwise.} \end{cases}$$

Significance. The conjecture gives a closed dimension-wise arithmetic law for the whole family, and the mod-8 doubling is the surprise: the natural guess $24/\gcd(24, n)$ misses exactly the residues 0 and 2 modulo 8, where every Euler characteristic in the family carries one further factor of two. At $n = 2$ the law reproduces $\chi = 24$ for every Calabi–Yau complete-intersection surface, the K3 case. The mechanism should be a congruence for the coefficient family, and a proof programme runs through the image of these complete intersections in complex cobordism against Todd-genus and elliptic-genus congruences. The nearest related result is the divisibility role of Euler numbers in quotient constructions surveyed in [93]. A first decisive theorem would be the divisibility half, that the displayed integer divides every χ at dimension n. In tests on the corpus described in the stress-test section, the formula matched the exact gcd in every dimension 2 to 30, reverified independently through dimension 26. A counterexample would be a dimension whose gcd differs.

30 Middle Betti numbers of Brieskorn–Pham links

Conjecture 122 (Five-link cap extremum). *For $N = 4$ and every cap $A \geq 9$, every sorted Fano tuple $\mathbf{a}$ with $\max_i a_i = A$ satisfies $b_2(L(\mathbf{a})) \leq A - 1$, with equality only for $\mathbf{a} = (2, 2, A, A)$.*

Significance. Among five-dimensional Fano Brieskorn links with a fixed largest exponent, the conjecture identifies the exact maximal middle Betti number and its unique maximizer, a linear cap law with a rigidity clause. The threshold is genuine: at $A = 3$ the tuple $(3, 3, 3, 3)$ has $b_2 = 6$, defeating the eventual formula, so the linear branch begins only after the low-cap oscillation dies out. The nearest related results are the Milnor–Orlik computation [88] and the role of these links in positive Sasakian geometry [91, 92]. A first decisive theorem would be the bound for tuples containing two entries equal to 2. In tests on the corpus described in the stress-test section, the bound and its rigidity held for every sorted Fano 4-tuple at every exact cap through 50, reverified independently through 25. A counterexample would be a tuple beating the cap or a second maximizer.

Conjecture 123 (Seven-link cap extremum). *For $N = 5$ and every cap $A \geq 13$, every sorted Fano tuple with $\max_i a_i = A$ satisfies $b_3(L(\mathbf{a})) \leq (A - 1)(A - 2)$, with equality only for $(2, 2, A, A, A)$.*

Significance. One dimension up, the cap law becomes quadratic with the analogous unique maximizer, and the pair of statements suggests a general polynomial cap law in the link dimension. The transition is sharper than at five dimensions: at $A = 12$ the maximizer is $(2, 3, 12, 12, 12)$ with $b_3 = 222$, well above the eventual $(A-1)(A-2)$, so the stable branch begins only at $A = 13$. The nearest related results are again [88, 91]. A first decisive theorem would be the asymptotic order $b_3 \ll A^2$ under the Fano condition. In tests on the corpus described in the stress-test section, the complete exact-cap census through $A = 40$ found no exception, the transition at $A = 12$ reverified independently. A counterexample would be a tuple above the quadratic cap or a second maximizer at some cap.

Conjecture 124 (Connected gcd-graph positivity)**.** *For $N = 4$, if the graph on $\{1, 2, 3, 4\}$ joining $i \sim j$ when $\gcd(a_i, a_j) > 1$ is connected, then $b_2(L(\mathbf{a})) \geq 1$.*

Significance. The conjecture reads rational middle homology off an arithmetic graph: pairwise entanglement of the exponents, propagated along a connected graph, forces the five-dimensional link to carry homology. It is dimension-specific and calibrated: the five-tuple $(2, 2, 2, 2, 2)$ has a connected gcd graph and vanishing middle Betti number, so no naive extension to all N holds. It is flagged for elevated attribution risk, lying close to the classical graph-theoretic criteria for Brieskorn homology spheres [88, 91]. A rational homology sphere is detected by Betti numbers alone, so the rational form of those criteria, which admits only graph configurations containing an isolated vertex or a special even component, is expected to exclude connected graphs outright and to yield the statement as a corollary; the retained content is the explicit graph form at $N = 4$ and its calibrating role for the scaling laws, not a new mechanism. A proof from first principles would organise the four-variable inclusion-exclusion by connected graph type. A first decisive theorem would be the case of a gcd-path. In tests on the corpus described in the stress-test section, all 137,440 connected tuples with entries at most 60 had $b_2 \geq 1$, reverified independently through entries at most 40. A counterexample would be a connected tuple with vanishing middle homology.

Conjecture 125 (Common-scale monotonicity, resolved true)**.** *For every $N \geq 4$ and every base tuple $\mathbf{a}$, the scale sequence satisfies $\beta_{\mathbf{a}}(k+1) \geq \beta_{\mathbf{a}}(k)$ for every $k \geq 1$.*

Significance. Scaling every exponent by a common factor multiplies the ambient weights without changing their ratios, and the conjecture asserts that the middle Betti number can only grow along this ray. The scale formula is an alternating polynomial in k, so monotonicity is not coefficientwise obvious and encodes a positivity of the finite differences of the Milnor–Orlik sum. The nearest related result is the formula itself [88]. A first decisive theorem would be monotonicity for tuples of equal entries, where the polynomial is explicit. In tests on the corpus described in the stress-test section, all 52,593 base tuples of lengths 4 to 6 were monotone at every scale through 10, reverified independently on random tuples through scale 11. A counterexample would be one base tuple with a descending scale step. *Resolution.* The conjecture is true: Proposition 17 identifies the scale sequence as a sum of interior Ehrhart counting functions of $N - 1$ integral weighted hypersimplices, whose forward differences of every order through $N - 1$ are nonnegative, strictly more than monotonicity.

Proposition 17 (Ehrhart model for common scaling)**.** For a base tuple $\mathbf{a}$ and $1 \leq m \leq N - 1$ let $P_m(\mathbf{a}) = \{x \in \mathbf{R}^N : 0 \leq x_i \leq a_i,\ \sum_i x_i/a_i = m\}$, a lattice polytope of dimension $N - 1$. Then

$$\beta_{\mathbf{a}}(k) = \sum_{m=1}^{N-1} \#(\operatorname{relint}(kP_m(\mathbf{a})) \cap \mathbb{Z}^N),$$

and consequently every forward difference of the scale sequence of order $r \leq N - 1$ is nonnegative at every $k \geq 1$. In particular Conjecture 125 is true, and the discrete-convexity clause of Conjecture 126 is true.

Proof. The multiplication map $\prod_{j \in J} \mu_{a_j} \to \mu_{\operatorname{lcm}(a_j)}$ on roots of unity is surjective with kernel of order $\prod_{j \in J} a_j / \operatorname{lcm}_{j \in J} a_j$, so that quantity counts the tuples over J with product 1; by inclusion–exclusion over the coordinates forced to equal 1, the Milnor–Orlik sum counts the tuples $(\zeta_1, \ldots, \zeta_N)$ with each ζ_i a nontrivial a_i-th root of unity and $\prod_i \zeta_i = 1$, the empty set contributing the term $(-1)^N$. For the scaled tuple $k\mathbf{a}$ write $\zeta_i = \exp(2\pi i r_i/(ka_i))$ with $0 < r_i < ka_i$; the product is 1 exactly when $\sum_i r_i/a_i = km$ for an integer m, and since each summand lies strictly between 0 and k, the integer satisfies $1 \le m \le N-1$. These conditions say precisely that $(r_1, \ldots, r_N)$ is a lattice point of the relative interior of $kP_m(\mathbf{a})$, proving the displayed identity. Each $P_m(\mathbf{a})$ is integral: the substitution $y_i = x_i/a_i$ carries it to the hypersimplex $\{0 \le y_i \le 1,\ \sum_i y_i = m\}$, whose vertices are 0–1 vectors, so every vertex of $P_m(\mathbf{a})$ has coordinates 0 or a_i. For a lattice polytope P of dimension d with h^*-vector $(h_0^*, \ldots, h_d^*)$, Ehrhart reciprocity [173] turns the expansion $L_P(k) = \sum_i h_i^* \binom{k+d-i}{d}$ into $L_{P^\circ}(k) = \sum_i h_i^* \binom{k+i-1}{d}$, and the h_i^* are nonnegative integers [174]; applying the forward difference r times gives $\Delta^r L_{P^\circ}(k) = \sum_i h_i^* \binom{k+i-1}{d-r} \ge 0$ for $0 \le r \le d$. Summing over the $N-1$ polytopes $P_m(\mathbf{a})$, all of dimension $N-1$, yields $\Delta^r \beta_{\mathbf{a}}(k) \ge 0$ for every $k \ge 1$ and $r \le N-1$; the case $r = 1$ is Conjecture 125, and the case $r = 2$ is the convexity clause of Conjecture 126. ☐

Conjecture 126 (Scale convexity and log-concavity)**.** *For every $N \ge 4$ and every base tuple, the scale sequence is simultaneously discretely convex and log-concave:*

$$\beta_{\mathbf{a}}(k+1) - 2\beta_{\mathbf{a}}(k) + \beta_{\mathbf{a}}(k-1) \ge 0, \qquad \beta_{\mathbf{a}}(k)^2 \ge \beta_{\mathbf{a}}(k-1)\,\beta_{\mathbf{a}}(k+1).$$

Significance. The coexistence is unusually rigid: convexity says the absolute increments grow, log-concavity says their relative growth slows, and together they confine the scale sequence to a narrow polynomial-like corridor. A proof might come from total positivity of the Milnor–Orlik polynomial after finite differencing, or from a representation as a positive mixture, and either route would explain Conjecture 125 as a corollary. The nearest related result is [88]. A first decisive theorem would be either inequality for equal-entry tuples. In tests on the corpus described in the stress-test section, both inequalities held across the entire scale corpus of Conjecture 125. A counterexample would be a base tuple violating either inequality at some scale. *Resolution of the convexity clause.* The first inequality is true, the case $r = 2$ of Proposition 17, whose positive-mixture representation is exactly the route anticipated above. Nonnegative h^*-coefficients do not imply the log-concavity clause, which remains open; a further exact stress of 24,950,000 scale inequalities on 25,000 random tuples with entries through 120 and scales through 501 found no violation of either clause.

31 Hochster strands of flag moment-angle complexes

Conjecture 127 (Hochster interval support)**.** *For every finite simple graph G and every $r \ge 0$, the support $\{s : h_{r,s}(G) > 0\}$ is an interval of integers.*

Significance. Adding a vertex can kill the homology of an individual induced subcomplex, so no monotone mechanism forces gapless support; the conjecture asserts that aggregation over all vertex subsets nevertheless removes every gap in every homological degree. The sector grading is genuinely finer than the ordinary one: for $G = K_{2,3}$ the ordinary moment-angle Betti sequence is $(1, 0, 0, 4, 2, 0, 3, 2, 0, 0, 0)$, not unimodal, while every strand support in the tested corpus is an interval. The nearest related results are Hochster's formula and the bigraded theory of moment-angle complexes [89, 90]. A first decisive theorem would be interval support for the component strand $r = 0$. In tests on the corpus described in the stress-test section, all 50,454 strands over 34,867 graphs had interval support, reverified independently on an exhaustive labeled corpus through five vertices and random holdouts at six and seven. A counterexample would be one graph, one degree, and one internal zero.

Conjecture 128 (Hochster-strand log-concavity)**.** *For every G, r, and s,*

$$h_{r,s}(G)^2 \geq h_{r,s-1}(G)\, h_{r,s+1}(G).$$

Significance. The strand sequences are conjectured log-concave in the subset size, a shape law for aggregated homology ranks across all induced subcomplexes. The strengthening that would place it in the Lorentzian-polynomial orbit is false and retained: ultra-log-concavity, normalized by binomial coefficients, already fails for a connected seven-vertex graph whose component strand is $(10, 19, 14, 5, 1)$ at sizes two through six. Ordinary log-concavity is thus the calibrated boundary. The nearest related results are [89, 90]. A first decisive theorem would be log-concavity of the component strand for trees. In tests on the corpus described in the stress-test section, no failure appeared in 50,454 strand sequences, reverified independently through five vertices exhaustively. A counterexample would be one graph and one degree with a convex triple.

Conjecture 129 (Strand Hurwitz stability)**.** *For every G and r with $H_{G,r}$ nonconstant, every zero of $H_{G,r}(z)/z^{\min \operatorname{supp}}$, the strand polynomial divided by its lowest monomial, satisfies* $\operatorname{Re} z < 0$.

Significance. Real-rootedness, the strongest classical shape property, is already false at four vertices: the empty graph has component polynomial $6 + 8z + 3z^2$ after removing the monomial factor, with nonreal zeros. Those zeros nevertheless lie in the open left half-plane, and the conjecture asserts that this Hurwitz stability is the correct universal zero law for all Hochster strands, strictly between log-concavity and real-rootedness. The nearest related results are the bigraded computations of [90]. A first decisive theorem would be stability for the empty graphs, whose strand polynomials have closed form. In tests on the corpus described in the stress-test section, all 50,045 nonconstant strand polynomials passed exact Routh–Hurwitz tests with integer determinants, reverified independently with an exact fraction-free implementation. A counterexample would be one strand polynomial with a zero in the closed right half-plane.

Conjecture 130 (Cycle extremality at connectivity two)**.** *If G is 2-vertex-connected on $m \geq 4$ vertices, then $h_{0,s}(G) \leq h_{0,s}(C_m)$ for every s, with equality in every coefficient only for the cycle C_m.*

Significance. The component strand measures aggregate disconnection of induced subgraphs, and the cycle, the sparsest 2-connected topology, is conjectured to disconnect the most, coefficient by coefficient, among all 2-connected graphs of its order. Every added chord can only help induced subgraphs stay connected, which is the mechanism, but the coefficientwise uniformity across all s is the content. The nearest related results are [89, 90]. A first decisive theorem would be the top coefficient, the number of disconnecting vertex pairs. In tests on the corpus described in the stress-test section, all 12,185 two-connected graphs of the corpus, exhaustive through six vertices with seeded holdouts through ten, obeyed the domination, reverified independently on all 11,616 two-connected labeled graphs through six vertices. A counterexample would be a 2-connected graph beating the cycle in one coefficient.

Conjecture 131 (Strict component-strand log-concavity)**.** *For every connected noncomplete graph, every triple of positive consecutive component-strand coefficients is strictly log-concave:* $h_{0,s}^2 > h_{0,s-1} h_{0,s+1}$.

Significance. On the component strand of connected noncomplete graphs the log-concavity of Conjecture 128 is conjectured to be everywhere strict, so equality cases concentrate entirely in higher strands or degenerate graphs; complete graphs are excluded because their reduced component strand vanishes identically. Strictness pins the shape law against the boundary and makes every near-equality a finite certificate worth retaining. The nearest related results are [89, 90]. A first decisive theorem would be strictness for paths. In tests on the corpus described in the stress-test section, no equality occurred among positive triples across the corpus, reverified independently through five vertices exhaustively. A counterexample would be one connected noncomplete graph with an exactly geometric triple.

32 Torus mapping tori with signed monodromy

Conjecture 132 (Parity-strand unimodality)**.** *For every orientation-preserving signed permutation A, the sequences $(f_0(A), f_2(A), f_4(A), \ldots)$ and $(f_1(A), f_3(A), f_5(A), \ldots)$ are unimodal.*

Significance. By Proposition 14 the invariants f_k count positive orbits of the signed action on k-subsets, and the conjecture asserts that each parity class of this orbit-count profile rises and falls once. Log-concavity is too strong and the failure is retained: at $d = 10$ the type with one positive 8-cycle and one positive 2-cycle has even strand $(1, 5, 26, 26, 5, 1)$, where $5^2 < 1 \cdot 26$. Unimodality is thus the calibrated boundary. The nearest related result is the Wang calibration [94] through Proposition 14. A first decisive theorem would be unimodality for a single signed cycle. In tests on the corpus described in the stress-test section, all 57,758 orientation-preserving signed cycle types through dimension 21 passed, reverified independently through dimension 11. A counterexample would be a type with a two-peaked parity strand.

Conjecture 133 (Mapping-torus Betti unimodality)**.** *For every orientation-preserving signed permutation A in dimension d, the Betti sequence $(b_0(M_A), \ldots, b_{d+1}(M_A))$ is unimodal.*

Significance. Poincaré duality makes the Betti sequence symmetric but not unimodal, so the claim is a genuine constraint on the exterior invariant profile, obtained from Conjecture 132's strands through the Wang sum $b_k = f_k + f_{k-1}$. Log-concavity fails already at $A = -I_4$, whose Betti sequence is $(1, 1, 6, 6, 1, 1)$, and the failure is retained, so unimodality is again the calibrated boundary. The nearest related result is [94] with Proposition 14. A first decisive theorem would be the reduction of Conjecture 133 to Conjecture 132 by a two-strand interleaving argument. In tests on the corpus described in the stress-test section, every type through dimension 21 passed, reverified independently through dimension 11. A counterexample would be a type with a two-peaked Betti sequence.

Conjecture 134 (Sparse-cycle minimizer structure)**.** *For every $d \geq 8$, some minimizer of the total Betti number $\sum_k b_k(M_A; \mathbf{Q})$ over orientation-preserving signed permutations has no positive signed cycle when d is even, exactly one when d is odd, and pairwise distinct negative-cycle lengths.*

Significance. Total cohomology is minimized by destroying invariant vectors, and the conjecture asserts the structure of an optimal destroyer: negative cycles of pairwise distinct lengths, with at most the parity-forced positive cycle. Repeated cycle lengths create coincident eigenvalue charges and hence extra zero-sum relations, which is the mechanism. The observed minimizing negative partitions include $(5,7)$ at $d=12$, $(5,6,7)$ at $d=18$, and $(5,7,8)$ at $d=20$. The nearest related result is Proposition 14, which turns the question into orbit counting. A first decisive theorem would be that repeating a negative-cycle length never decreases the total below the distinct-length optimum. In tests on the corpus described in the stress-test section, every exact minimum through dimension 21 had the stated form, reverified independently through dimension 12. A counterexample would be a dimension whose every minimizer violates the pattern.

Conjecture 135 (Square-root-two compression exponent, resolved false)**.** *Let M_d be the minimum of $\sum_k b_k(M_A;\mathbf{Q})$ over orientation-preserving signed permutations in dimension d. Then $M_d^{1/d} \to \sqrt{2}$ as $d \to \infty$.*

Significance. The conjecture fixes the exponential compression rate of the minimal total cohomology of this finite monodromy sector, and it is flagged as the most exposed statement of the part: it asserts a limit beyond any finite check, and the finite roots approach it non-monotonically. The tempting finite law is false and retained: $M_d = 2^{\lfloor (d+3)/2 \rfloor}$ holds through $d=11$ and fails at $d=12$, where the true minimum is 160, attained by negative cycles of lengths 7 and 5, not 128. The diagnostic $(\log M_d - (d/2)\log 2)/\sqrt{d}$ stays below 0.42 for $8 \le d \le 21$, consistent with a subexponential correction. The nearest related result is Proposition 14. A first decisive theorem would be the lower bound $M_d \ge 2^{d/2}$ up to subexponential factors. A counterexample would be a divergent or different limit.

Proposition 18 (Landau floor for the minimal total cohomology)**.** For every signed permutation A on $\mathbb{Z}^d$,

$$\sum_k b_k(M_A;\mathbf{Q}) \;\ge\; \frac{2^{d+1}}{\operatorname{ord}(A)} \;\ge\; \frac{2^d}{g(d)},$$

where $g(d)$ is Landau's function, the maximal order of a permutation of d letters. Consequently $2^d/g(d) \le M_d \le 2^{d+1}$, and since $\log g(d) \sim \sqrt{d \log d}$, the minimal total cohomology satisfies $M_d^{1/d} \to 2$.

Proof. By Proposition 14, $\sum_k b_k(M_A;\mathbf{Q}) = 2P$ with $P = \sum_k p_k = \sum_k \dim_{\mathbf{Q}} \ker(\wedge^k A - I)$, the dimension of the fixed space of A on the full exterior algebra $\Lambda^\bullet(\mathbf{Q}^d)$. Since A has finite order m, that fixed space is the invariant subspace of the cyclic group generated by A, and averaging characters gives $P = \frac{1}{m}\sum_{j=0}^{m-1} \operatorname{tr}\Lambda^\bullet(A^j) = \frac{1}{m}\sum_{j=0}^{m-1}\det(I + A^j)$. Every term is nonnegative: A^j is again a signed permutation, and a signed cycle of length L with holonomy ε contributes $\prod_{\zeta^L=\varepsilon}(1+\zeta) = 1-(-1)^L\varepsilon \in \{0,2\}$ to the determinant. The term $j=0$ alone is 2^d, so $P \ge 2^d/m$. If the underlying permutation has cycle lengths $\ell_1,\dots,\ell_r$, then $A^{\operatorname{lcm}(\ell_i)}$ is diagonal with entries ± 1, so $m \le 2\operatorname{lcm}(\ell_1,\dots,\ell_r) \le 2g(d)$, giving the displayed floor. The identity monodromy gives $M_d \le 2\cdot 2^d$, and Landau's asymptotic $\log g(d) \sim \sqrt{d\log d}$ [170] makes $(2^d/g(d))^{1/d} \to 2$, so the two-sided bound forces $M_d^{1/d} \to 2$. $\square$

Remark 16 (Resolution)**.** *The conjecture is false: by Proposition 18 the compression exponent is 2, not $\sqrt{2}$, with the whole deviation from 2^{d+1} confined to a subexponential factor at most $2g(d)$. The finite scans could not separate the two laws, because $\log_2 g(d) \approx d/2$ throughout the tested range—$\log_2 g(21) \approx 9.7$ against $d/2 = 10.5$—so the floor $2^d/g(d)$ is numerically indistinguishable from $2^{d/2}$ there, and the square-root fit was a finite-size illusion. The observed minima respect the floor with room to spare, $M_{12} = 160$ against $2^{12}/g(12) = 68$, and the surviving question is the exact order of the subexponential correction $\log_2(2^{d+1}/M_d)$, now pinned between 0 and*

$\log_2(2g(d)) = O(\sqrt{d \log d}\,)$. *The torsion budget of Conjecture 136 is untouched: it is consistent with the true growth, and no incompatibility remains once the exponent is corrected.*

Conjecture 136 (Torsion budget inequality)**.** *For every signed permutation $A \in SL(d, \mathbb{Z})$,*

$$\tau(A) \le \sum_k b_k(M_A; \mathbf{Q}), \qquad \textit{equivalently} \qquad \sum_k n_k \le 2 \sum_k p_k$$

in the orbit counts of Proposition 14.

Significance. By Proposition 14 the integral torsion of these mapping tori is elementary 2-primary with one generator per negative orbit, so the conjecture is a purely combinatorial inequality: negative orbits of the signed action on the full exterior algebra never outnumber twice the positive orbits. It is not a universal-coefficient generality, and equality is attained repeatedly, at 109 of the 1,580 deposited signed types through dimension 12, so the inequality is sharp and a proof likely needs an explicit two-to-one map from negative to positive orbits. The nearest related result is Proposition 14. A first decisive theorem would be the inequality for a single signed cycle, where both sides are divisor sums. In tests on the corpus described in the stress-test section, every type through dimension 12 passed, reverified independently with the equality cases recounted. A counterexample would be a type with torsion exceeding the rational budget.

Part VIII
Shape laws for enumerative geometric invariants

Added to the deposit: 12 August 2026.

The twenty conjectures below continue the topological turn of Part VII with a second discipline: take a classical enumerative invariant of a geometric family, compute it exactly across the family, and conjecture the shape of the resulting array. They form five programmes. A hypersurface programme (Conjectures 137 to 140) studies the Euler indices of anticanonical hypersurfaces in products of projective spaces under an order dual to Part VII's: where Part VII merged defining equations at fixed ambient, these statements split ambient factors at fixed dimension. A Hilbert-scheme programme (Conjectures 141 to 144) studies the Betti profiles of Göttsche's product over a proved pair of exact moment identities. An orbifold programme (Conjectures 145 to 148) studies the twisted-sector age histograms of cyclic Calabi–Yau quotients. An entropy programme (Conjectures 149 to 152) studies the exterior torsion-entropy profile of hyperbolic toral automorphisms. A matroid programme (Conjectures 153 to 156) lifts the Hochster strand laws of Part VII from graphs to simple binary matroids. The calibration layer is exact throughout: the Chern-class evaluation, Göttsche's product, the age grading of the McKay correspondence, the eigenvalue evaluation of torsion growth rates, and Hochster's formula. Retained controls again mark every calibrated boundary. The limit clauses of Conjectures 139 and 143 assert limits beyond any finite check and are flagged; Conjectures 146 to 148 are random-model statements resting on Monte Carlo evidence and are flagged as such, with Conjectures 146 and 147 likely within reach of classical equidistribution machinery; Conjecture 145 carries elevated attribution risk. Since deposit, Conjectures 151 and 152 have been proved by convexity, while Conjectures 153 to 156 have been refuted by two explicit graphic matroids.

For a partition $\lambda = (\lambda_1, \dots, \lambda_r)$ of $d+1$ let $A_\lambda = \prod_i \mathbf{CP}^{\lambda_i}$ and let $X_\lambda \in |-K_{A_\lambda}|$ be a smooth anticanonical hypersurface, a Calabi–Yau d-fold; its Euler characteristic is evaluated exactly from $c(TA_\lambda) = \prod_i (1+H_i)^{\lambda_i+1}$ and $[X_\lambda] = \sum_i (\lambda_i+1)H_i$, and we write $E_d(\lambda) = (-1)^d \chi(X_\lambda)$. A *split* $S_{m,a}\lambda$ replaces one part m by a and $m-a$; a *dominance move* transfers one unit from a smaller positive part to a weakly larger part. For $b \geq 1$ define $h_b(n,k)$ by

$$\sum_{n,k\geq 0} h_b(n,k)\, q^n z^k = \prod_{m\geq 1} \frac{1}{(1-z^{m-1}q^m)(1-z^m q^m)^b(1-z^{m+1}q^m)},$$

so that $h_{22}(n,k) = b_{2k}(\mathrm{Hilb}^n(\mathrm{K3}))$ by Göttsche's theorem [96]. For $r \geq 3$ and weights $a_1, \dots, a_d \in (\mathbb{Z}/r)^\times$ with $\sum_i a_i \equiv 0 \pmod r$, the cyclic quotient $\frac{1}{r}(a_1, \dots, a_d)$ has ages $\mathrm{age}(j) = \frac{1}{r}\sum_i (ja_i \bmod r)$ for $1 \leq j < r$, integers between 1 and $d-1$, with histogram $N_k = \#\{j : \mathrm{age}(j) = k\}$ grading twisted sectors in the McKay correspondence [99]. For a hyperbolic $A \in SL(d,\mathbb{Z})$ with eigenvalue log-moduli $x_1, \dots, x_d$ summing to zero, set

$$h_k(x) = \sum_{|S|=k} \Big(\sum_{i\in S} x_i\Big)_+, \qquad 1 \leq k \leq d-1;$$

away from exterior resonances h_k is the exponential growth rate of $|\det(\wedge^k A^n - I)|$, the degree-k torsion order of the mapping torus of A^n, since each factor $|\lambda_S^n - 1|$ grows like $|\lambda_S|^n$ or stays bounded [100, 103]. Finally, for a simple binary matroid M on ground set E with independence complex $\Delta(M)$, put $\beta(A) = \dim \widetilde{H}_{r(A)-1}(\Delta(M|_A); \mathbf{F}_2)$ and

$$H_{c,s}(M) = \sum_{\substack{A\subseteq E,\ |A|=s \\ |A|-r(A)=c}} \beta(A),$$

the aggregated multigraded Betti numbers of the Stanley–Reisner ring by Hochster's formula [89], graded by restriction size and nullity, with strand polynomial $\sum_s H_{c,s}(M) z^s$ at each fixed nullity c. The profile $h_b(n,k)$ and the spectral functional $h_k(x)$ are unrelated symbols, each local to its own programme.

Proposition 19 (Profile moment identities)**.** For every $b \geq 1$ and $n \geq 1$, the probability measure on $\{0, \dots, 2n\}$ proportional to $h_b(n, \cdot)$ has mean exactly n and variance exactly $2n/(b+2)$.

Proof. Write $F(q,z)$ for the generating product and substitute $(q,z) \mapsto (qe^{-t}, e^t)$: the factor arguments $z^{m-1}q^m$, $z^m q^m$, $z^{m+1}q^m$ become $q^m e^{-t}$, q^m, $q^m e^t$, so

$$\Phi(t) = \log F(qe^{-t}, e^t) = -\sum_{m\geq 1} \big[\log(1-q^m e^{-t}) + b\log(1-q^m) + \log(1-q^m e^t)\big],$$

and $[q^n]\, F(qe^{-t}, e^t) = \sum_k h_b(n,k) e^{t(k-n)}$ is the moment generating function of the centred profile. The two t-dependent families are exchanged by $t \mapsto -t$, so $\Phi'(0) = 0$ and the mean is n. Differentiating twice, $\Phi''(0) = 2\sum_m q^m/(1-q^m)^2$, so with $P_n = \sum_k h_b(n,k)$ and $G_n = \sum_k (k-n)^2 h_b(n,k)$,

$$\sum_n G_n q^n = 2F(q,1) \sum_m \frac{q^m}{(1-q^m)^2}.$$

Both Lambert series $\sum_m q^m/(1-q^m)^2$ and $\sum_m mq^m/(1-q^m)$ expand to $\sum_N \sigma(N) q^N$, hence are equal, while $F(q,1) = \prod_m (1-q^m)^{-(b+2)}$ gives $q\,\partial_q \log F(q,1) = (b+2)\sum_m mq^m/(1-q^m)$. Therefore $\sum_n G_n q^n = \frac{2}{b+2}\, q\,\partial_q F(q,1) = \frac{2}{b+2}\sum_n nP_n q^n$, that is, $G_n/P_n = 2n/(b+2)$ for every n. $\square$

33 Euler indices of anticanonical hypersurfaces in products

Conjecture 137 (Strict dominance law)**.** *For every $d \geq 3$ and every dominance move $\lambda \to \mu$ on partitions of $d+1$,*

$$E_d(\mu) > E_d(\lambda).$$

Significance. Concentrating ambient dimension into fewer projective factors strictly increases the signed Euler index, so E_d is strictly increasing along the dominance order on partitions, from the split ambient $(\mathbf{CP}^1)^{d+1}$ to the single factor $\mathbf{CP}^{d+1}$. The Chern coefficient is an alternating expression, and the conjectured order asks for a hidden positive basis for the dominance differences. The nearest related results are the CICY constructions and censuses [93, 102], which enumerate these geometries without ordering them. A first decisive theorem would be the two-factor case. In tests on the corpus described in the stress-test section, all 42,691 elementary moves for $3 \leq d \leq 22$ passed, reverified independently through $d = 18$. A counterexample would be one reversed or equal move.

Conjecture 138 (Strict refinement contraction)**.** *For every $d \geq 3$, every λ, and every nontrivial split of one part,*

$$E_d(S_{m,a}\lambda) < E_d(\lambda).$$

Significance. Factoring one ambient projective direction always strictly reduces the signed Euler index, the exact mirror of Part VII's merge monotonicity read on the ambient rather than the defining equations. The sign calibration is essential: without the factor $(-1)^d$ the statement is false in odd dimension, and the two statements together make E_d a strict order antitone in refinement. The nearest related results are [93, 102]. A first decisive theorem would be splits of the one-part partition, refining a single hypersurface ambient. In tests on the corpus described in the stress-test section, all 42,861 splits through $d = 22$ passed, reverified independently through $d = 18$. A counterexample would be one split that fails to contract.

Conjecture 139 (The 8/9 refinement barrier)**.** *Let S_d be the maximum of $E_d(S_{m,a}\lambda)/E_d(\lambda)$ over all λ and all one-part splits. Then $S_d \to 8/9$ as $d \to \infty$.*

Significance. The least destructive refinement asymptotically loses exactly one ninth of the Euler index, a sharp quantitative boundary for Conjecture 138, and the limit clause is flagged as beyond any finite check. The mechanism should be a closed-form ratio for the extremal split, which the data identify as splitting a single unit off the dominant part. The nearest related results are [93, 102]. A first decisive theorem would be the exact evaluation of the split ratio along the one-part family. In tests on the corpus described in the stress-test section, the exact maxima decrease toward the limit from dimension four on, $S_{22} = 0.8894748$ against $8/9 = 0.8888889$, the independent recomputation giving $S_{18} = 0.8897556$. A counterexample would be a different limit or two separated accumulation points.

Conjecture 140 (Positive interaction of disjoint refinements)**.** *For $d \geq 6$ and splits S, T acting on distinct parts of λ,*

$$E_d(\lambda)\, E_d(ST\lambda) > E_d(S\lambda)\, E_d(T\lambda).$$

Significance. The logarithm of the signed Euler index is strictly supermodular on disjoint refinement squares: two independent factorizations lose less together than the product of their separate losses. The threshold is calibrated and retained: the unqualified statement is false in dimension five, where the exact scan finds four violating squares, so the hypothesis $d \geq 6$ is a genuine boundary rather than caution. The nearest related results are [93, 102]. A first decisive theorem would be the two-part ambient case. In tests on the corpus described in the stress-test section, all 24,680 split squares for $6 \leq d \leq 18$ passed, reverified independently for $6 \leq d \leq 12$. A counterexample would be one square with submodular interaction.

34 Betti profiles of Hilbert schemes of surfaces

Conjecture 141 (Strict Betti log-concavity)**.** *For every $b \geq 3$, $n \geq 1$, and $0 < k < 2n$,*

$$h_b(n,k)^2 > h_b(n,k-1)\, h_b(n,k+1).$$

Significance. The Betti profile of Göttsche's product is conjectured strictly log-concave in the cohomological degree, a multiplicative curvature condition strictly above the unimodality that the bivariate strict-unimodality machinery of [97] supplies for products of this type; at $b = 22$ the statement covers every $\mathrm{Hilb}^n(\mathrm{K3})$. Ultra-log-concavity is deliberately not asserted, matching the calibration pattern of the strand programmes. The nearest related results are Göttsche's formula [96] and the bivariate unimodality criterion of [97], stated there for Laurent-series products rather than for these profiles. A first decisive theorem would be the middle three degrees for large n from the asymptotic profile. In tests on the corpus described in the stress-test section, 179,200 exact Turán inequalities passed for $3 \leq b \leq 30$ and $n \leq 80$, reverified independently for $b \leq 12$, $n \leq 40$. A counterexample would be one nonpositive Turán determinant.

Conjecture 142 (Cohomology-seed total positivity)**.** *For every $b \geq 1$, $n \geq 1$, and $0 \leq k < n$,*

$$h_{b+1}(n,k+1)\, h_b(n,k) \geq h_{b+1}(n,k)\, h_b(n,k+1).$$

Significance. Adding one seed cohomology generator concentrates the normalized profile toward the Lefschetz centre in monotone likelihood-ratio order, a total-positivity statement across the seed parameter transverse to the shape law of Conjecture 141. The mechanism suggested by the data is a planar network or a stable multivariate refinement of the product. The nearest related result is [96]. A first decisive theorem would be the boundary minors $k = 0$. In tests on the corpus described in the stress-test section, all 53,070 adjacent two-by-two minors for $b \leq 30$, $n \leq 60$ passed, reverified independently for $b \leq 12$. A counterexample would be one negative minor.

Conjecture 143 (Kurtosis trichotomy)**.** *Let $\kappa_b(n)$ be the standardized fourth moment of the profile of Proposition 19. Then: for $b = 3, 4, 5$, $\kappa_b(n)$ increases strictly to $21/5$; for $b = 6$ it has a unique maximum at $n = 9$ and decreases thereafter to $21/5$; for $b = 7$, $\kappa_7(1) = \kappa_7(2)$ and it decreases strictly thereafter; for $b \geq 8$ it decreases strictly to $21/5$.*

Significance. Over the exact mean and variance of Proposition 19, the profile's kurtosis follows a complete finite-n phase diagram with a single exceptional peak at $(b, n) = (6, 9)$ and a logistic-type limit $21/5$, the limit clauses flagged as beyond finite check. The initially stronger assertion of strict decrease for $b = 7$ was refuted by the exact equality $\kappa_7(1) = \kappa_7(2) = 9/2$ and the corrected boundary is retained. The nearest related result is the asymptotic χ_y-profile analysis [98], whose neighbourhood contains the limit but not the finite phase diagram. A first decisive theorem would be the closed form of $\kappa_b(1)$. In tests on the corpus described in the stress-test section, every stated comparison passed for $3 \le b \le 22$ and $n \le 80$, the full diagram reverified independently through $n = 50$. A counterexample would be a violated comparison or a different limit.

Conjecture 144 (Fixed-charge log-concavity)**.** *For every $b \ge 3$ and fixed $j \ge 0$, the sequence $n \mapsto h_b(n, n + j)$, $n \ge \max(1, j)$, is strictly log-concave.*

Significance. Holding the displacement from middle cohomology fixed and varying the number of points gives the transverse direction to Conjecture 141, and the conjecture asserts strict log-concavity there as well, so the whole two-parameter array is log-concave along both natural axes. The nearest related result is [96]. A first decisive theorem would be the central diagonal $j = 0$ at $b = 22$, the middle Betti numbers of $\mathrm{Hilb}^n(\mathrm{K3})$. In tests on the corpus described in the stress-test section, 48,384 exact diagonal Turán inequalities passed for $b \le 30$, $j \le 25$, $n \le 80$, reverified independently for $b \le 12$, $j \le 14$, $n \le 40$. A counterexample would be one convex diagonal triple.

35 Twisted-sector ages of cyclic quotients

Conjecture 145 (Sharp low-dimensional age unimodality)**.** *For $d = 4$ and $d = 5$, the age histogram $(N_1, \dots, N_{d-1})$ is unimodal for every isolated cyclic Calabi–Yau quotient $\frac{1}{r}(a_1, \dots, a_d)$.*

Significance. The age duality $\mathrm{age}(j) + \mathrm{age}(-j) = d$ makes the histogram symmetric in mass but not in shape, and the conjecture asserts single-peakedness in exactly the dimensions where it can hold: in dimension six the quotient $\frac{1}{3}(1, 1, 1, 1, 1, 1)$ has histogram $(0, 1, 0, 1, 0)$, so the range is sharp. It is flagged for elevated attribution risk, since symmetry leaves few independent entries in these dimensions and the statement may be a short unrecognized lemma of the quotient-singularity literature [99]. A first decisive theorem would be the single inequality $N_2 \ge N_1$ at $d = 4$. In tests on the corpus described in the stress-test section, all 450,999 admissible weight multisets with $r \le 60$ passed, reverified independently for $r \le 40$ on 58,393 multisets. A counterexample would be one two-peaked histogram.

Conjecture 146 (Gaussian bulk age law)**.** *Let r be prime and let distinct nonzero weights $a_1, \dots, a_d$ be uniform conditioned on zero sum. If $d, r \to \infty$ with $d = o(r)$, then in probability the empirical law of $(\mathrm{age}(j) - d/2)/\sqrt{d/12}$ over $1 \le j < r$ converges weakly to the standard Gaussian.*

Significance. Each age is a sum of d fractional rotations whose joint behaviour over the group orbit is quasi-independent, and the conjecture is the resulting finite-population central limit law for twisted-sector gradings. It is flagged doubly: as a random-model statement resting on Monte Carlo evidence, and as likely within reach of classical equidistribution and discrepancy machinery, so its value is as a calibration target rather than a deep unknown. The nearest related result is the age grading itself [99]. A first decisive theorem would be convergence of the second moment. In tests on the corpus described in the stress-test section, the mean lattice-corrected Kolmogorov distance fell from 0.0398 at $(d,r) = (10, 101)$ to 0.0162 at $(40, 1009)$, the independent recomputation giving 0.0344 and 0.0178. A counterexample would be a nonvanishing limiting distance.

Conjecture 147 (Local central-sector law)**.** *Under the sampling law of Conjecture 146 with d even, $N_{d/2}/(r-1) \sim \sqrt{6/(\pi d)}$, and uniformly for $|k - d/2| = O(\sqrt{d \log d})$ the ratio $N_k/(r-1)$ is asymptotic to the corresponding lattice Gaussian mass.*

Significance. The weak law of Conjecture 146 is upgraded to a local limit law: individual age sectors, not just their bulk distribution, carry Gaussian mass, which is what an application to sector counts actually needs. It shares both flags of Conjecture 146, and a proof requires a local central limit theorem for the correlated residue orbit rather than larger simulations. The nearest related result is [99]. A first decisive theorem would be the central sector alone. In tests on the corpus described in the stress-test section, the ratio of observed central mass to the prediction was 1.034, 1.048, 1.025, 1.006 across the four sampled regimes. A counterexample would be a sector with non-Gaussian limiting mass.

Conjecture 148 (High-sampling restoration of unimodality)**.** *In the random prime-order model, if $r/d^3 \to \infty$ then the entire age histogram is unimodal with probability tending to one.*

Significance. Deterministic unimodality fails from dimension six onward, but enough group elements suppress histogram noise and restore the Gaussian shape generically, and the conjecture pins the sampling threshold at the third power of the dimension. The exponent is deliberately exposed: replacing d^3 by a smaller power is the cleanest refinement or refutation, and the statement carries the random-model flag. The nearest related result is Conjecture 147, whose local law at the fluctuation scale predicts exactly this exponent. A first decisive theorem would be unimodality with probability tending to one at $r \geq d^4$. In tests on the corpus described in the stress-test section, no failure occurred in 180 trials each at $(6, 1009)$, $(8, 2003)$, $(10, 4001)$, while visibly lower ratios r/d^3 produced failure rates reaching 14.6%. A counterexample would be a sequence with $r/d^3 \to \infty$ and non-vanishing failure probability.

36 Exterior torsion entropy of toral automorphisms

Conjecture 149 (Exterior torsion-entropy log-concavity)**.** *For every nonzero zero-sum real spectrum x and $1 < k < d-1$,*

$$h_k(x)^2 \geq h_{k-1}(x)\, h_{k+1}(x).$$

Significance. The torsion growth rates of a hyperbolic toral mapping torus, read across the exterior degree, form a log-concave profile: a Hodge-shaped inequality for a piecewise-linear functional of subset sums rather than for an intersection form. The nearest related results are the torsion-growth programme of [100], whose setting is arithmetic groups rather than mapping tori, and the toral determinant calculus of [103], which computes the rates without ordering them. A first decisive theorem would be $d = 4$, where a finite chamber decomposition suffices. In tests on the corpus described in the stress-test section, all 81,000 inequalities on 16,200 random zero-sum spectra through $d = 12$ passed, with an exact integer check alongside, reverified independently on 3,000 random spectra and the full integer corpus of Conjecture 150. A counterexample would be one convex triple of rates.

Conjecture 150 (Equality rigidity)**.** *If an interior equality $h_k(x)^2 = h_{k-1}(x)h_{k+1}(x)$ occurs, then the multiset $\{x_1, \dots, x_d\}$ has exactly two distinct values.*

Significance. Equality in the profile inequality is conjectured to force a two-level spectrum, the only configuration whose subset-sum chambers align enough to flatten a Turán determinant; the converse is deliberately not asserted. Rigidity turns Conjecture 149 into a strict law away from an explicit degenerate locus. The nearest related results are [100, 103]. A first decisive theorem would be the case of integer spectra with entries in $\{-1, 0, 1\}$. In tests on the corpus described in the stress-test section, all 222 equality cases among 48,512 exact integer spectra with entries in $[-2, 2]$ were two-valued, the independent exhaustive recount through $d = 7$ finding 122 equality cases, all two-valued. A counterexample would be one three-valued equality.

Conjecture 151 (Sharp binomial upper envelope, resolved true)**.** *For every zero-sum spectrum and every $1 \le k \le d - 1$,*

$$h_k(x) \le \binom{d-2}{k-1} h_1(x),$$

with equality in every degree for spectra proportional to $(1, 0, \dots, 0, -1)$.

Significance. Every exterior torsion rate is controlled by the ordinary topological entropy h_1 with the sharp binomial constant. Proposition 20 proves the statement by normalising $h_1 = 1$ and maximising a convex function on the product of the positive and negative sign simplices. A vertex has one coordinate 1, one coordinate -1, and all others zero, which gives exactly $\binom{d-2}{k-1}$ positive subsets. The nearest related results remain [100, 103]. The former finite tests now serve as regression checks for the proved inequality.

Proposition 20 (Resolution of the sharp binomial upper envelope)**.** Conjecture 151 holds in full.

Proof. If $h_1(x) = 0$, then the zero-sum spectrum is zero and the result is immediate. Otherwise scale so that $h_1(x) = 1$. Fix the sets P and N of positive and negative coordinates. The positive coordinates lie in the simplex $\sum_{i \in P} x_i = 1$, while the magnitudes of the negative coordinates lie in the simplex $\sum_{j \in N}(-x_j) = 1$. On the closure of their product,

$$h_k(x) = \sum_{|S|=k} \left(\sum_{i \in S} x_i \right)_+$$

is a convex function, since it is a sum of positive parts of linear forms. A convex function on a compact polytope attains a maximum at a vertex. At such a vertex exactly one coordinate is 1, exactly one is -1, and every other coordinate is zero. A k-subset contributes 1 precisely when it contains the positive coordinate and avoids the negative coordinate. There are $\binom{d-2}{k-1}$ such subsets. Rescaling proves the inequality, and the vertex calculation proves the asserted equality for spectra proportional to $(1, 0, \dots, 0, -1)$. $\square$

Conjecture 152 (Two-level lower envelope, resolved true)**.** *Normalize $h_1(x) = 1$. For every d and k, the minimum of h_k over zero-sum spectra is attained by a two-level spectrum $x^{p,q}$ with p entries $1/p$, q entries $-1/q$, and $d - p - q$ zeros.*

Significance. Together with Conjecture 151, this result places every exterior torsion-entropy profile in a sharp two-sided corridor governed by elementary two-level spectra. Proposition 21 proves the reduction by averaging coordinates within each sign class. Convexity and symmetry show that every such averaging step can only decrease h_k, so equal positive coordinates and equal negative coordinates attain the minimum. The nearest related results remain [100, 103]. The former random and exact comparisons now serve as regression checks for the finite envelope.

Proposition 21 (Resolution of the two-level lower envelope)**.** Conjecture 152 holds in full.

Proof. Let x be a zero-sum spectrum with $h_1(x) = 1$, and let P and N be its positive and negative index sets with sizes p and q. The function h_k is convex and invariant under permutations within P and within N. Replace two positive coordinates by their average. The new spectrum is the midpoint of the original spectrum and the spectrum obtained by swapping those coordinates, so convexity and symmetry show that h_k cannot increase. Repeating this operation makes every positive coordinate equal to $1/p$. The same argument applied to the negative magnitudes makes every negative coordinate equal to $-1/q$. The zero coordinates remain zero. Thus

$$h_k(x^{p,q}) \leq h_k(x).$$

There are only finitely many pairs $p, q \geq 1$ with $p + q \leq d$, and every $x^{p,q}$ is admissible. The global minimum is therefore attained by one of these two-level spectra. □

37 Matroidal Hochster tables

Conjecture 153 (Nullity-strand interval support, resolved false)**.** *For every simple binary matroid M and every nullity c, the support $\{s : H_{c,s}(M) > 0\}$ is an interval of integers.*

Proposition 22 (A graphic nullity-strand certificate)**.** Let $M = M(K_{3,3})$ be the cycle matroid of the complete bipartite graph. Then

$$H_{1,4}(M) = 9, \qquad H_{1,5}(M) = 0, \qquad H_{1,6}(M) = 6.$$

The matroid M is simple, binary, and 3-connected.

Proof. In a nullity-one graphic restriction, the top reduced homology is nonzero exactly when the restricted edge set is a circuit. Indeed, any edge outside the unique circuit is a coloop, which makes the independence complex a cone. The circuits of $M(K_{3,3})$ are the cycles of $K_{3,3}$. There are $\binom{3}{2}^2 = 9$ cycles of length four, no cycles of length five by bipartiteness, and six Hamilton cycles of length six. This gives the displayed strand. The graph $K_{3,3}$ is simple and 3-vertex-connected, so its cycle matroid is simple, binary, and 3-connected. □

Remark 17 (Resolution)**.** *Conjecture 153 is false. Proposition 22 gives nullity-one support $\{4, 6\}$, with an internal gap at size five.*

Significance. The statement is retained with its resolution because the obstruction is elementary and survives strong connectivity. The cycle matroid of $K_{3,3}$ is simple, binary, and 3-connected, yet its nullity-one strand is supported at sizes four and six with a zero between them. Simplicity and connectivity therefore do not repair interval support. The nearest related results remain Hochster's formula [89] and the homology of matroid independence complexes [104]. A surviving problem is to identify a structural class, narrower than simple binary matroids, on which interval support does hold.

Conjecture 154 (Row-and-column log-concavity, resolved false)**.** *Both coordinate directions of the table are log-concave:* $H_{c,s}^2 \geq H_{c,s-1}H_{c,s+1}$ *and* $H_{c,s}^2 \geq H_{c-1,s}H_{c+1,s}$.

Remark 18 (Resolution)**.** *Conjecture 154 is false. For the matroid of Proposition 22, the horizontal inequality at* $(c,s) = (1,5)$ *reads* $0^2 \geq 9 \cdot 6$.

Significance. The horizontal direction already fails on the nullity-one circuit census of $K_{3,3}$, where the gap at size five forces the strongest possible reversal. The vertical direction is not refuted by this example and remains a separate question. The nearest related results are the Hodge-theoretic log-concavity of matroid invariants [101] and the shellability framework [104], neither of which controls this bigraded array. Any repaired row statement must exclude basic circuit gaps rather than rely on simplicity or 3-connectivity.

Conjecture 155 (Nearest-rhombus log-concavity, resolved false)**.** *Both diagonal directions are log-concave:* $H_{c,s}^2 \geq H_{c-1,s-1}H_{c+1,s+1}$ *and* $H_{c,s}^2 \geq H_{c-1,s+1}H_{c+1,s-1}$.

Proposition 23 (A graphic rhombus certificate)**.** Let G have vertex set $\{0, \ldots, 6\}$ and edge set

$$\{03, 04, 05, 13, 14, 23, 24, 26, 34, 56\},$$

where ij denotes the edge $\{i, j\}$. For its cycle matroid $M(G)$,

$$H_{1,5} = 2, \qquad H_{2,6} = 2, \qquad H_{3,7} = 3.$$

The graph G is simple and 2-connected.

Proof. For a restriction A, its contribution is the absolute reduced Euler characteristic of its independence complex,

$$\beta(A) = \left| -1 + \sum_{\substack{\varnothing \neq I \subseteq A \\ I \text{ independent}}} (-1)^{|I|-1} \right|.$$

Deletion and contraction on the ten displayed edges gives the following complete list of nonzero contributions to the three cells. At $(1,5)$ the two five-cycles $\{03, 05, 23, 26, 56\}$ and $\{04, 05, 24, 26, 56\}$ contribute one each. At $(2,6)$ the restriction $\{03, 04, 13, 14, 23, 24\}$ contributes two. At $(3,7)$ that restriction together with 34 contributes three. Every other restriction of the required size and nullity has zero top reduced homology. Summing gives the displayed values. Deleting any one vertex leaves G connected, so G is 2-connected. $\square$

Remark 19 (Resolution)**.** *Conjecture 155 is false. Proposition 23 gives*

$$H_{2,6}^2 = 4 < 6 = H_{1,5}H_{3,7}.$$

Significance. The first nearest-diagonal inequality fails on a small simple graphic matroid, so the table is not the unrestricted shadow of a Lorentzian or strongly log-concave bivariate polynomial. The certificate is 2-connected and uses only ten edges. The opposite diagonal direction remains a separate question. The nearest related circle is still the Hodge-theoretic theory around [101], but any connection must pass through a more restrictive class or a modified array.

Conjecture 156 (Hurwitz stability of nullity strands, resolved false)**.** *For every simple binary matroid and every nonconstant nullity strand, every zero of the strand polynomial $\sum_s H_{c,s}(M)\, z^{s-s_{\min}}$, with $s_{\min}$ the least size in its support, has strictly negative real part.*

Remark 20 (Resolution)**.** *Conjecture 156 is false. The nullity-one polynomial of Proposition 22, after removing its lowest monomial, is $9+6z^2$. Its zeros are $\pm i\sqrt{3/2}$ and have real part zero.*

Significance. Strict Hurwitz stability fails at the boundary of the left half-plane. The $K_{3,3}$ strand has a missing middle coefficient, and its two zeros lie on the imaginary axis. This links the failure directly to the support gap that refutes Conjectures 153 and 154. The nearest related results remain [89, 90] and the parallel graph-strand law of Part VII. Closed-left-half-plane stability is a possible weakening, but it is not established here.

Part IX
Spectra and filtrations of combinatorial geometries

Added to the deposit: 12 August 2026.

The twenty conjectures below extend the geometric turn of Parts VII and VIII to spectra and filtrations. They form four programmes. A toric programme (Conjectures 157 to 161) studies the h-polynomials of graph associahedra, the even-Betti spectra of their smooth toric varieties, through root location, interlacing, and a Mahler-measure order. A McKay programme (Conjectures 162 to 166) filters the Cayley graph of an isolated cyclic Calabi–Yau quotient by orbifold age and conjectures one-crest laws for the low homology of the resulting flag complexes. A torsion programme (Conjectures 167 to 171) studies microcanonical ensembles of lens-space sine factors, their temperature-dependent sector shapes, and a $\zeta(3)$ central parabola over the log-sine moment identities proved below. An entropy programme (Conjectures 172 to 176) orders the boundary-map pseudodeterminants of simplicial spheres across degree and under stellar refinement. The calibration layer is exact where the objects are: tubing f-vectors, $\mathbf{F}_2$ homology, and coefficient convolutions are integer computations, while root locations and pseudodeterminants are checked spectrally in floating point and flagged as such. Four retained controls mark calibrated boundaries. Conjectures 169 and 170 assert limits in a random or iterated-limit model and are flagged as the most exposed; Conjectures 157, 161, 164, 169, and 170 carry elevated attribution risk, sitting closest to classical machinery; and Conjecture 170 deliberately strengthens the annealed Gaussian age law drawn from Conjecture 146 to a joint decoupling law.

For a connected simple graph G on $n \geq 2$ vertices, a *tube* is a proper connected induced subgraph, two tubes being compatible when nested or disjoint without adjacency, and a *tubing* is a set of pairwise compatible tubes; the graph associahedron P_G is the simple $(n-1)$-polytope whose codimension-k faces are the tubings with k tubes [105]. Write $h_G(t) = \sum_i h_i(G)t^i$ for its h-polynomial, so that $h_i(G) = b_{2i}(X_G)$ for the smooth projective toric variety of the normal fan, a positive palindromic sequence; put $\mathcal{M}(G) = \sum_{h_G(\rho)=0} \log\max(1, |\rho|)$, the logarithmic Mahler measure, and $\mu_G(i) = h_i(G)/h_G(1)$. For an odd prime p and pairwise distinct weights $a = (a_1, \ldots, a_d) \in (\mathbb{F}_p^\times)^d$ with $\sum_i a_i = 0$, the quotient $\frac{1}{p}(a_1, \ldots, a_d)$ acts on the vertex set $\mathbb{Z}/p$, including 0, through the Cayley graph $x \sim y$ when $y - x \in \{\pm a_1, \ldots, \pm a_d\}$; with $\mathrm{age}(j) = \frac{1}{p}\sum_i [ja_i]_p$ and $\mathrm{age}(0) = 0$, let K_k be the flag complex of the induced graph on $\{j : \mathrm{age}(j) \le k\}$ for $0 \le k \le d-1$, and write c_k and $b_{q,k}$ for its number of components and its q-th $\mathbf{F}_2$ Betti number. For a prime $p \ge 5$ let $\Omega_{p,d,m}$ be the tuples $x \in \{1, \ldots, p-1\}^d$ with $\sum_i x_i = mp$, let $H_p(x) = \sum_i \log(2\sin(\pi x_i/p))$, the lens-space log-torsion energy [106], and set $Z_{p,d,m}(s) = \mathbb{E}[e^{sH_p}]$ and $\Phi_{p,d,m} = \mathbb{E}[H_p]$ over uniform $\Omega_{p,d,m}$. Finally, for a finite simplicial d-sphere K with boundary maps ∂_k, put $D_k(K) = \mathrm{pdet}(\partial_k \partial_k^*)$, the product of nonzero eigenvalues, $E_k(K) = \log D_k(K)$, $r_k(K) = \mathrm{rank}\,\partial_k$, and $A_k(K) = E_k(K)/r_k(K)$, the entropy per boundary rank, with higher matrix-tree readings of these determinants in [107, 108].

The sector expectations $Z_{p,d,m}$ and $\Phi_{p,d,m}$ are defined only when $\Omega_{p,d,m}$ is nonempty. This holds exactly when $d \le mp \le d(p-1)$.

Proposition 24 (Log-sine moment identities)**.** For every integer $p \ge 2$,

$$\prod_{j=1}^{p-1} 2\sin\frac{\pi j}{p} = p, \qquad \text{and} \qquad \int_0^1 \left(u - \tfrac{1}{2}\right)^2 \log(2\sin\pi u)\, du = -\frac{\zeta(3)}{2\pi^2}.$$

Proof. Evaluating $\prod_{j=1}^{p-1}(z - \zeta^j) = (z^p - 1)/(z-1)$ at $z = 1$ for $\zeta = e^{2\pi i/p}$ gives $\prod_j (1 - \zeta^j) = p$; taking absolute values and $|1 - \zeta^j| = 2\sin(\pi j/p)$ gives the product identity, so the sector energies of the definitions have exact mean $\log p/(p-1)$ per coordinate. For the integral, expand $\log(2\sin\pi u) = -\sum_{k\ge1}\cos(2\pi k u)/k$ on $(0,1)$ and $(u - \frac{1}{2})^2 = \frac{1}{12} + B_2(u)$ with $B_2(u) = \pi^{-2}\sum_{k\ge1}\cos(2\pi k u)/k^2$. The constant term contributes nothing, since $\int_0^1 \log(2\sin\pi u)\,du = 0$ by the vanishing mean of every cosine, and orthogonality of the cosines gives $\int_0^1 B_2(u)\cos(2\pi k u)\,du = 1/(2\pi^2 k^2)$, so the integral equals $-\sum_{k\ge1} k^{-1}\cdot(2\pi^2 k^2)^{-1} = -\zeta(3)/(2\pi^2)$. □

38 Toric spectra of graph associahedra

Conjecture 157 (Universal negative real roots)**.** *For every connected simple graph G, every zero of $h_G(t)$ is real and strictly negative.*

Significance. The two endpoint families are classical: the path gives the associahedron with Narayana h-polynomial and the complete graph the permutohedron with Eulerian h-polynomial, both real-rooted, and the conjecture asserts that every graph associahedron between them shares the property. It is flagged for elevated attribution risk as a natural question in an active neighbourhood, and it is sharply family-specific: the analogous real-rootedness for all flag simple polytopes fails in high dimension, so a proof must use the graphical structure, most plausibly a stable refinement of the descent enumerator of [105]. A first decisive theorem would be the chordal case. In tests on the corpus described in the stress-test section, all 878 deposited root sets and an independent recomputation over all 771 labelled connected graphs through five vertices passed. A counterexample would be one graph with a nonreal or nonnegative zero.

Conjecture 158 (Connected-deletion interlacing)**.** *If v is a vertex of G with $G - v$ connected, the zeros of h_{G-v} weakly interlace the zeros of h_G.*

Significance. Interlacing is the inductive engine of real-rootedness, and the conjecture asserts that vertex deletion is its compatible recursion for graph associahedra, so a deletion recurrence in the interlacing family framework would prove Conjecture 157 at the same stroke. The restriction to connected deletions is structural, since P_{G-v} requires a connected graph in the same convention. The nearest related results are the coefficientwise tree-shift inequalities of [109, 110], which order coefficients rather than roots. A first decisive theorem would be leaf deletion for trees. In tests on the corpus described in the stress-test section, all 3,621 deposited connected-deletion pairs and 2,971 independently recomputed pairs passed. A counterexample would be one pair with a crossing root.

Conjecture 159 (Strict Mahler growth under edge addition)**.** *For every missing edge e of a connected graph G, $\mathcal{M}(G+e) > \mathcal{M}(G)$.*

Significance. Adding an edge coarsens the tube structure and enlarges the h-vector, and the conjecture asserts that the multiplicative root radius responds strictly monotonically, making the Mahler measure a strict connectivity monotone on the graphs of one vertex set, from the path's minimum toward the complete graph. The observable does not occur in the graph-associahedron literature located by our search [105, 109]. A first decisive theorem would be edges completing a triangle on the path. In tests on the corpus described in the stress-test section, all 3,956 deposited graph–edge pairs passed with minimum increment 0.0263, reverified independently on 3,227 pairs. A counterexample would be one edge with nonincreasing Mahler measure.

Conjecture 160 (Edge addition centralizes the spectrum)**.** *For every missing edge e of a connected graph G and every symmetric outer tail that is nonempty and proper, $\sum_{|i-c|\geq r} \mu_{G+e}(i) \leq \sum_{|i-c|\geq r} \mu_G(i)$, where $c = (n-1)/2$.*

Significance. The normalized even-Betti distribution of the toric variety is conjectured to become more centrally peaked under every edge addition, in the symmetric tail order that implies monotonicity for every increasing function of the distance from middle degree, a concentration companion to the outward root motion of Conjecture 159. The nearest related results are the coefficient bounds of [109], which compare unnormalized coefficients. A first decisive theorem would be the outermost tail, the ratio of end coefficients to total mass. In tests on the corpus described in the stress-test section, all deposited tail cross-products were nonnegative with minimum integer margin 2, so every tested proper comparison was strict, and the independent recomputation found no violation. A counterexample would be one edge and one tail with mass moving outward. In the gamma basis $h_G(t) = \sum_j \gamma_j(G)\, t^j (1+t)^{d-2j}$, nonnegative for these flag polytopes [109], the normalized coefficient distribution is the law of $J + \mathrm{Bin}(d-2J, \frac{1}{2})$ with J drawn proportionally to $\gamma_J(G)4^{-J}$, and the conditional symmetric tails shrink as J grows, by binomial unimodality; the statement therefore follows from first-order stochastic dominance of the gamma-index law under edge addition. That dominance holds across all 450 edge additions among connected graphs through six vertices, counted up to isomorphism of the pair, while its likelihood-ratio strengthening already fails at five vertices, with gamma vectors $(1, 13, 8)$ and $(1, 17, 10)$, so the dominance form is the calibrated sufficient condition.

Conjecture 161 (Path–star Mahler extremality)**.** *For every tree T on $n \geq 4$ vertices, $\mathcal{M}(P_n) \leq \mathcal{M}(T) \leq \mathcal{M}(K_{1,n-1})$, with equality on the left only for the path and on the right only for the star.*

Significance. The path and the star are extremal for several coefficientwise orders on tree associahedra [109, 110], and the conjecture proposes the spectral strengthening: the same pair bounds the nonlinear Mahler functional of all roots, with equality rigidity. It is flagged for elevated attribution risk as the spectral shadow of known extremal orders. A first decisive theorem would be either bound along Aisbett's tree shifts, which would chain the inequality from path to star. In tests on the corpus described in the stress-test section, all 1,857 deposited bound-and-rigidity tests and an independent scan of every labelled tree through six vertices passed. A counterexample would be a third tree attaining either bound or a tree escaping the corridor.

39 Age-filtered McKay complexes

Conjecture 162 (One-crest component law)**.** *For every isolated cyclic quotient with pairwise distinct weights, the component sequence* $(c_0, \ldots, c_{d-1})$ *is unimodal.*

Significance. As the age threshold rises, components are born and merge with no general monotonicity, and the conjecture asserts that the census nevertheless has a single crest. Pairwise distinctness is a calibrated boundary, not caution: the repeated-weight quotient $\frac{1}{17}(2,3,3,3,10,10,10,10)$ has component profile $(1,2,1,2,1,1,1,1)$, refuting the unrestricted statement, and the control is retained. The nearest related results are the age grading of the McKay correspondence [99] and quiver connectivity, neither of which filters the Cayley graph by age. A first decisive theorem would be unimodality for $d = 3$. In tests on the corpus described in the stress-test section, all 3,135 deposited profiles and an independent exhaustive recomputation at $p \in \{7, 11, 13\}$ passed. A counterexample would be one distinct-weight quotient with a two-crest census.

Conjecture 163 (Early component crest)**.** *The first index attaining* $\max_k c_k$ *is at most* $\lfloor d/2 \rfloor$*.*

Significance. Unimodality says nothing about location, and the conjecture adds it: fragmentation peaks no later than half age, so the second half of the filtration only consolidates. The mechanism is the age duality $\mathrm{age}(j) + \mathrm{age}(-j) = d$, which pairs late sublevels with complements of early ones. The nearest related result is [99]. A first decisive theorem would be the crest location for $d = 4$. In tests on the corpus described in the stress-test section, all deposited and independently recomputed profiles peaked by half age. A counterexample would be a quotient whose fragmentation peaks late.

Conjecture 164 (Half-age connectivity)**.** *For* $d \geq 5$*, the sublevel complex* $K_{\lceil d/2 \rceil}$ *is connected.*

Significance. The full Cayley graph is connected for elementary reasons; the content is that connectivity is already achieved by the middle age threshold, a quantitative connectivity law for the filtration. It is flagged for elevated attribution risk, close to standard Cayley-graph connectivity questions, and its distinctness hypothesis is again a genuine boundary: the repeated-weight quotient $\frac{1}{11}(1,1,5,5,5,5)$ has two components at its half-age level, and the control is retained. A first decisive theorem would be connectivity of $K_{\lceil d/2 \rceil}$ when some weight equals 1. In tests on the corpus described in the stress-test section, all 2,905 deposited profiles with $d \geq 5$ and the independent recomputation passed. A counterexample would be a distinct-weight quotient disconnected at half age.

Conjecture 165 (One-crest first homology)**.** *The sequence* $(b_{1,0}, \ldots, b_{1,d-1})$ *is unimodal.*

Significance. Cycles are created by new vertices and killed by flag two-cells in no fixed order, and the conjecture asserts a one-crest shadow for the ungraded first Betti census of the filtration, the simplest persistence-type law the object could satisfy. The nearest related results are Floer-theoretic age filtrations, which grade different complexes. A first decisive theorem would be unimodality for $d = 4$. In tests on the corpus described in the stress-test section, all deposited and independently recomputed $\mathbf{F}_2$ profiles passed. A counterexample would be one quotient with a two-crest b_1 census.

Conjecture 166 (One-crest second homology)**.** *The sequence* $(b_{2,0}, \ldots, b_{2,d-1})$ *is unimodal.*

Significance. The degree bound is genuine and calibrated: the distinct-weight quotient $\frac{1}{31}(2, 8, 16, 17, 18, 20, 24, 26, 27, 28)$ has β_3 profile $(0, 0, 0, 0, 0, 4, 3, 4, 4, 4)$, not unimodal, so the one-crest law stops at the second homology and the retained control marks the boundary. Together with Conjectures 162 and 165 this confines the low-dimensional shape of the whole filtration. The nearest related result is [99]. A first decisive theorem would be $d = 5$. In tests on the corpus described in the stress-test section, all deposited and independently recomputed profiles passed. A counterexample would be a two-crest b_2 census.

40 Sine-torsion sectors of cyclic quotients

Conjecture 167 (Positive-temperature log-concavity)**.** *For every* $s > 0$ *and* $2 \leq m \leq d - 2$ *such that* $\Omega_{p,d,q}$ *is nonempty for* $q \in \{m - 1, m, m + 1\}$, $Z_{p,d,m}(s)^2 > Z_{p,d,m-1}(s)\, Z_{p,d,m+1}(s)$.

Significance. The sector index m is the microcanonical charge of the age construction, and the conjecture asserts that at every positive temperature the sector profile of the torsion partition function is strictly log-concave, an ordering of lens-space character factors [106] by their orbifold charge. The non-emptiness qualification is the exact domain on which all three expectations are defined. A proof route is total positivity of the coefficient kernel of powers of a positive polynomial. A first decisive theorem would be fixed small (p, d) by exact convolution. In tests on the corpus described in the stress-test section, all 40,932 deposited triples passed with minimum curvature 0.0079, reverified independently at five (p, d) pairs. A counterexample would be one sector triple with the wrong curvature sign.

Conjecture 168 (Concavity of mean log torsion)**.** *For* $2 \leq m \leq d - 2$ *such that* $\Omega_{p,d,q}$ *is nonempty for* $q \in \{m - 1, m, m + 1\}$, $2\Phi_{p,d,m} > \Phi_{p,d,m-1} + \Phi_{p,d,m+1}$.

Significance. This is the zero-temperature shadow of Conjecture 167, stated separately because strict inequalities need not survive differentiation at a symmetry point: the mean log torsion is conjectured strictly concave in the sector index outright. By Proposition 24 the global mean over all sectors is pinned exactly, so the concavity is a genuine shape statement about the conditional means. A first decisive theorem would be $d = 4$, three sectors and one inequality. In tests on the corpus described in the stress-test section, all 6,822 deposited triples passed with minimum curvature 0.0770, reverified independently. A counterexample would be one convex triple of sector means.

Conjecture 169 (The $\zeta(3)$ central parabola)**.** *Let m_d satisfy $(m_d - d/2)/\sqrt{d/12} \to x$. Then, taking $p \to \infty$ through primes first and $d \to \infty$ second,*

$$\lim_{d\to\infty} \lim_{p\to\infty} \Big(\Phi_{p,d,m_d} - \frac{d \log p}{p-1} \Big) = \frac{3\zeta(3)}{\pi^2}(1-x^2).$$

Significance. The centering is exact by Proposition 24, and the constant is forced by the proposition's second identity through a conditional local-limit expansion, so the conjectural content is precisely the uniform interchange of the singular log-sine observable with the conditioning. The statement is flagged as an iterated-limit claim beyond any finite check and for elevated attribution risk, since the machinery is classical. A first decisive theorem would be the central value $x = 0$. In tests on the corpus described in the stress-test section, four exact dynamic-programming regimes matched the parabola with central-window root-mean-square errors 0.027 to 0.038. A counterexample would be a different limiting profile.

Conjecture 170 (Age–torsion decoupling)**.** *With weights uniform on distinct nonzero zero-sum tuples and J uniform on $\mathbb{F}_p^\times$, the standardized pair of $\mathrm{age}_A(J)$ and of the centred log-torsion sum converges to a standard bivariate Gaussian as $p, d \to \infty$ with $d = o(p)$.*

Significance. The involution $J \mapsto -J$ forces exact zero covariance, and the conjecture asserts far more: the singular torsion statistic is itself asymptotically Gaussian and asymptotically independent of the age. It strengthens the annealed corollary of Conjecture 146, the Gaussian law of a single uniformly drawn age, by the torsion coordinate and the joint law, and it carries the same random-model and accessibility flags. A first decisive theorem would be the torsion marginal alone. In tests on the corpus described in the stress-test section, three seeded ensembles of up to 60,480 character samples had empirical correlations below 5×10^{-16}, a consistency check of the exact symmetry rather than evidence of independence, while the informative diagnostics, the variances and fourth moments, moved toward Gaussian values; sharper tests would target mixed fourth cumulants or conditional torsion laws in narrow age windows. A counterexample would be a persistent dependence or a non-Gaussian marginal.

Conjecture 171 (Negative-temperature reversal)**.** *For $-1 \le s < 0$ and $2 \le m \le d-2$ such that $\Omega_{p,d,q}$ is nonempty for $q \in \{m-1, m, m+1\}$, $Z_{p,d,m}(s)^2 < Z_{p,d,m-1}(s)\, Z_{p,d,m+1}(s)$.*

Significance. Cooling through zero reverses the sector shape: strict log-concavity at positive temperature becomes strict log-convexity on the first negative-temperature window, and the window is a calibrated boundary, since at $(p, d, s) = (11, 4, -2)$ the central curvature has the wrong sign, $+0.4980$, a retained control reproduced independently to six decimals. The pair of Conjectures 167 and 171 is thus one slice of a sharper phase diagram whose critical temperature is unknown. A first decisive theorem would be the reversal at $s = -1$ for $d = 4$. In tests on the corpus described in the stress-test section, all 20,466 deposited triples passed with curvature at most -0.0076, reverified independently. A counterexample would be a log-concave triple inside the window.

41 Boundary-determinant entropy of simplicial spheres

Conjecture 172 (Log-concavity of boundary entropy)**.** *For every simplicial d-sphere and $2 \le k \le d-1$, $E_k(K)^2 \ge E_{k-1}(K)\, E_{k+1}(K)$.*

Significance. The log pseudodeterminants of the boundary Laplacians, the higher spanning-tree entropies of the matrix-tree readings [107, 108], are conjectured log-concave across the degree, a second-order organization of the whole determinant profile of a sphere. A proof route is an Alexandrov–Fenchel-type injection between adjacent-degree tree counts. A first decisive theorem would be cross-polytope boundaries, where every eigenvalue is explicit. In tests on the corpus described in the stress-test section, all 262 deposited triples passed with minimum margin 51.8, reverified independently on spheres through dimension four. A counterexample would be one sphere and one degree with a convex triple.

Conjecture 173 (Entropy per rank decreases). *For every simplicial d-sphere, $A_1(K) \geq A_2(K) \geq \cdots \geq A_d(K)$.*

Significance. Normalizing each entropy by its boundary rank gives the mean log eigenvalue at each degree, and the conjecture asserts this spectral density weakens monotonically with the degree, so lower-dimensional skeleta carry the densest determinant mass. Poincaré duality for the sphere chain complex pairs the two ends, which is the suggested mechanism. The nearest related results are [107, 108]. A first decisive theorem would be the first step $A_1 \geq A_2$ for polytopal spheres. In tests on the corpus described in the stress-test section, all 392 deposited adjacent comparisons passed, reverified independently. A counterexample would be one increasing adjacent pair.

Conjecture 174 (Simplex rigidity of equality). *An adjacent equality $A_k(K) = A_{k+1}(K)$ occurs for some k if and only if K is the boundary of a simplex, in which case all adjacent equalities occur.*

Significance. The simplex boundary is the unique sphere whose entropy-per-rank chain is flat, so Conjecture 173 is strict at every step for every other sphere, an equality-rigidity statement in the register of the collection's other extremal characterizations. The mechanism is that the simplex boundary's Laplacian spectra at all degrees are powers of a single integer. A first decisive theorem would be strictness for cross-polytopes. In tests on the corpus described in the stress-test section, all 130 deposited rigidity tests passed, the simplex equalities holding to numerical zero, reverified independently. A counterexample would be a nonsimplicial sphere with one flat step.

Conjecture 175 (Strict stellar growth). *If K' is obtained from K by stellar subdivision of a facet, then $E_k(K') > E_k(K)$ for every $1 \leq k \leq d$.*

Significance. The most elementary local refinement of a triangulated sphere is conjectured to increase every entry of the determinant profile strictly, making the profile a complexity monotone for stellar moves. A facet subdivision is a block update of the boundary maps, so Schur complements and the matrix determinant lemma are the natural proof route and would give an explicit increment formula. The nearest related results are [107, 108]. A first decisive theorem would be the top degree, where the update is rank-one-like. In tests on the corpus described in the stress-test section, all 900 deposited comparisons over 225 subdivision pairs passed with minimum increment 1.42, reverified independently through dimension four. A counterexample would be one degree that fails to grow.

Conjecture 176 (Log-concave stellar increments). *With $\delta_k = E_k(K') - E_k(K)$ as in Conjecture 175, $\delta_k^2 \geq \delta_{k-1}\delta_{k+1}$ for $2 \leq k \leq d-1$.*

Significance. Not only does every degree grow under a facet subdivision, but the created entropy is conjectured to have a single log-concave degree profile, so local refinement injects spectral complexity in an organized shape rather than arbitrarily. Together with Conjecture 172 this asserts log-concavity for the profile and for its most elementary increments simultaneously. The nearest related results are [107, 108]. A first decisive theorem would be stacked spheres, where the increments are explicit. In tests on the corpus described in the stress-test section, all 450 deposited triples passed with minimum margin 17.8, reverified independently. A counterexample would be one subdivision with a convex increment triple.

Part X
Flux cohomology and protected index laws

Added to the deposit: 12 August 2026.

The twenty conjectures below carry the collection into string-geometric index theory. They form three programmes. A flux programme (Conjectures 177 to 184) studies the cohomology of wedge multiplication by an integral three-form on a torus, the invariant-forms model of H-twisted cohomology, through saturation laws for random flux, sparse witnesses, and an arithmetic torsion staircase for the complete flux. A profile programme (Conjectures 185 to 191) studies the signed χ_y-coefficient profile of projective Calabi–Yau complete intersections along the merge order of Part VII, extending the proved Euler monotonicity of Conjecture 118 to every protected coefficient, with root, log-concavity, and Mahler laws. A signature programme (Conjectures 192 to 196) does the same for the Hirzebruch signature, ending in a quadric ratio law with limit 27 and a dimension-wise gcd law that is the signature sibling of Conjecture 121. The calibration layer is exact where the objects are: finite-field cohomology, Smith normal forms, χ_y-coefficients, signatures, merge inequalities, ratios, and gcds are integer or rational computations, while characteristic-zero ranks beyond dimension nine are dual-prime proxies and the root, Mahler, and radius laws of Conjectures 188, 190, and 191 are checked spectrally in floating point and flagged as such. Two retained controls mark calibrated boundaries, a falsified all-prime torsion precursor and a falsified elementary-torsion strengthening. Conjectures 177, 178, 180, and 184 are stated in a random model and Conjecture 181 is stated in greater generality than the computations reach, the part's most exposed statement; Conjectures 184, 189, and 195 carry elevated attribution risk, sitting closest to classical machinery; and Conjecture 190 returns the Mahler order of Conjecture 159 on a different polynomial. Since deposit, three of the twenty have been resolved. Conjecture 195 is resolved true: Proposition 30 proves all three of its clauses from an algebraic generating function. Conjecture 182 is resolved true and Conjecture 183 resolved false: Proposition 29 puts the complete flux in a symplectic normal form, proving both cohomology totals and the whole torsion staircase, and exhibiting at $n = 15$ a summand $\mathbb{Z}/4$ that refutes the elementary-abelian clause. Proposition 27 has since proved the subcritical direction of Conjecture 178 and the linear lower bound of Conjecture 179, and has repaired Conjecture 180, whose support-conditional reading is false and is retained as a control; Proposition 26 sharpens the subcritical direction to every fixed $c < 2$ at the coverage scale itself.

For $n \geq 3$ and a commutative ring R let $E_n(R) = \Lambda^* R^n$, and for an integral three-form $H \in \Lambda^3\mathbb{Z}^n$ let d_H be wedge multiplication by H, a differential because H has odd degree; write $b_k(H; R)$ for the dimensions of the cohomology of $(E_n(R), d_H)$, $B(H; R) = \sum_k b_k(H; R)$, and

$t_p(H)$ for the number of p-primary cyclic summands of the integral cohomology. This complex is the invariant-forms model of H-twisted cohomology [111], whose K-theoretic refinement classifies Ramond–Ramond charges in a topologically nontrivial B-field [112]. The *generic target* g_n is the cohomology profile on the Zariski-open locus of maximal wedge ranks, that of a generic principal ideal generated in degree three: for $n = 2m$ it is $(3^{m-1}, 2 \cdot 3^{m-1}, 3^{m-1})$ in degrees $m-1, m, m+1$; for $n \equiv 3 \pmod 4$ it is $3^{(n-1)/2}$ in each of the two middle degrees; for $n \equiv 1 \pmod 4$ it has inner pair $3^{(n-1)/2}$ and outer pair 1, except that the outer pair is 3 at $n = 9$; its total is G_n. Lundqvist and Nicklasson prove a lower bound for the generic Hilbert series in every odd degree, disproving the Moreno-Socías–Snellman conjecture above degree three, and establish equality only in special cases, leaving the minimal series open in general and in degree three in particular [113]; the displayed profile is therefore the expected one and not a cited theorem, and the exceptional outer pair at $n = 9$ is the four-dimensional kernel computed there. The *complete flux* is $H_\Delta = \sum_{i<j<k} e_i \wedge e_j \wedge e_k$, and the target totals are $F_n = \binom{n+1}{(n+1)/2}$ for odd n, $F_n = 2\binom{n}{n/2}$ for even n, and $P_n = 2^{n-1} + 2^{(n-1)/2}$ for odd n, $P_n = 2(2^{n-2} + 2^{(n-2)/2})$ for even n. Write $t_{p,k}(n)$ for the number of p-primary cyclic summands of the integral cohomology of H_Δ in degree k, and $T_p(n) = \sum_k t_{p,k}(n)$ for their total. On the geometric side, $X^n_{\mathbf{d}} \subset \mathbb{CP}^{n+r}$ is the Calabi–Yau complete intersection of multidegree $\mathbf{d} = (d_1, \ldots, d_r)$, $d_i \geq 2$, $\sum_i d_i = n + r + 1$, with degree $D(\mathbf{d}) = \prod_i d_i$; merging replaces d_i, d_j by $d_i + d_j - 1$, preserving dimension and the Calabi–Yau condition, exactly as in Part VII. Writing $\chi_y = \sum_p \chi^p y^p$ for the Hirzebruch genus [115], put $a_p(\mathbf{d}) = (-1)^{n-p}\chi^p(X^n_{\mathbf{d}})$ and $A_{\mathbf{d}}(y) = \sum_p a_p(\mathbf{d})y^p$, a palindromic integer polynomial with nonnegative coefficients throughout the tested range; for even n put $S(\mathbf{d}) = (-1)^{n/2}\sigma(X^n_{\mathbf{d}})$. Let $Q = (2, \ldots, 2)$ be the all-quadric and $Y = (n + 2)$ the hypersurface configuration.

Proposition 25 (Coefficient and parity calibrations)**.** For every $H \in \Lambda^3\mathbb{Z}^n$ and every prime p, $B(H; \mathbb{F}_p) = B(H; \mathbb{Q}) + 2t_p(H)$. If every coefficient of H is odd, then the complexes of H and of H_Δ over $\mathbf{F}_2$ coincide, so $B(H; \mathbb{F}_2) = B(H_\Delta; \mathbb{F}_2)$.

Proof. The complex $(E_n(\mathbb{Z}), d_H)$ is a bounded complex of finitely generated free abelian groups, so the universal coefficient theorem gives $H^k(E_n(\mathbb{F}_p), d_H) \cong H^k(E_n(\mathbb{Z}), d_H) \otimes \mathbb{F}_p \oplus \mathrm{Tor}\big(H^{k+3}(E_n(\mathbb{Z}), d_H), \mathbb{F}_p\big)$. A free summand contributes one dimension in its own degree; a p-primary cyclic summand contributes one dimension in its own degree through the tensor factor and one three degrees below through the torsion factor, the differential having degree three; a cyclic summand of order prime to p contributes nothing. Summing over k gives the identity. For the second claim, the matrix of d_H over $\mathbf{F}_2$ depends only on the coefficients of H modulo 2, and a three-form with all coefficients odd reduces modulo 2 to H_Δ. $\square$

42 Integral flux complexes on tori

Conjecture 177 (Dense discrete flux saturation)**.** *Fix a finite symmetric set $V \subset \mathbb{Z} \setminus \{0\}$, for example $\{\pm 1\}$, and let every coefficient of $H_n \in \Lambda^3\mathbb{Z}^n$ be independent and uniform on V. Then $\Pr[(b_k(H_n; \mathbb{Q}))_k = g_n] \to 1$, and the failure probability decays exponentially in n.*

Significance. Over an infinite field a generic three-form attains the generic profile; the conjecture asserts that the smallest possible discrete ensembles already do so with overwhelming probability, turning the generic principal-ideal profile of [113] into a probabilistic law for discrete flux backgrounds and compressing the ordinary 2^n torus cohomology to a $3^{n/2}$-scale protected sector. The mechanism is that the discrete cube must avoid every determinantal degeneracy locus of the wedge-rank strata; genericity over $\mathbb{Q}$ alone says nothing here, since a nonzero polynomial can vanish on an entire discrete cube, so the exponential clause demands a structural anti-concentration theorem for these wedge minors, and both the random model and the exponential clause are flagged. A first decisive theorem would be a uniform positive lower bound on the saturation probability. In tests on the corpus described in the stress-test section, 142 of 144 deposited dense trials and 71 of 72 independent trials hit g_n for $3 \le n \le 11$. A counterexample would be a law and a subsequence with failure probability bounded away from zero.

Conjecture 178 (Sparse-flux threshold at the coverage scale)**.** *Include each coordinate triple independently with probability p_n and give included triples independent coefficients from a fixed finite symmetric set of nonzero integers. If $p_n n^2/\log n \to \infty$ then $\Pr[(b_k;\mathbb{Q})_k = g_n] \to 1$.*

Significance. The scale $(\log n)/n^2$ is the isolated-vertex scale of 3-uniform random hypergraphs, while uncovered pairs persist to the far larger scale $(\log n)/n$, so the conjectured last obstruction to generic wedge ranks is the uncovered vertex alone, a correction, in the spirit of hitting-time thresholds in random graph theory; even that obstruction is parity-sensitive, since for $n \equiv 0 \pmod 4$ the deposited profiles satisfy $g_n = (1+t)g_{n-1}$, so a form ignoring one coordinate can still saturate, and the vertex mechanism itself would place the critical constant near 2 outside those dimensions; a sharper form should have a critical window $p = (c_* \log n + O(\log\log n))/n^2$, but no value of c_* is claimed and the statement is confined to the coarse threshold. The zero direction is now proved in Proposition 27, a promotion: below the scale unused coordinates proliferate, and the deposited profile admits at most one of them, so the surviving content is the supercritical direction, which extrapolates beyond any finite check and is flagged, with the random model. Proposition 26 sharpens the zero direction to every fixed $c < 2$ at the critical scale, by a second-moment count of unused coordinates, so the coverage constant 2 named above is now a rigorous lower-side bound. In tests on the corpus described in the stress-test section, 960 deposited trials at $p = c\log n/n^2$ rose from hit rate 0 at $c = 1$ to 0.875 and 0.9125 at $c = 10$ under the two laws, and an independent sweep rose from 0/36 through 5/36 to 34/36 at $c = 1, 4, 10$. A counterexample would be a sparse sequence saturating below the scale or a dense one failing above it.

Proposition 26 (Coverage floor for the sparse threshold)**.** If $p_n = (c + o(1))\log n/n^2$ with a fixed $c < 2$, then the saturation probability of Conjecture 178 tends to zero.

Proof. Let U_n count coordinates contained in no included triple. A fixed coordinate lies in $\binom{n-1}{2}$ triples, so $\Pr[i \text{ unused}] = (1-p_n)^{\binom{n-1}{2}} = n^{-c/2+o(1)}$ and $\mathbf{E}U_n = n^{1-c/2+o(1)} \to \infty$. Two coordinates share exactly $n-2$ incident triples, so $\Pr[i,j \text{ both unused}]/\Pr[i \text{ unused}]^2 = (1-p_n)^{-(n-2)} = 1 + o(1)$, whence $\operatorname{Var} U_n = o\big((\mathbf{E}U_n)^2\big)$ and $U_n \ge 2$ with probability tending to one. By the support obstruction of Proposition 27, a form attaining the deposited profile leaves at most one coordinate unused, so saturation fails. □

Conjecture 179 (Linear-complexity sparse witnesses)**.** *Let μ_n be the minimum support size of a $\{\pm 1\}$-valued three-form on $\mathbb{Z}^n$ with rational profile g_n. Then for all sufficiently large n a saturating support of at most $n + C$ triples can be chosen to form a linear 3-uniform hypergraph, for an absolute constant C; in particular $\mu_n = \Theta(n)$.*

Significance. Between the dense ensemble of Conjecture 177 and the threshold of Conjecture 178 sits the deterministic minimum, and the conjecture pins it at linear growth with witnesses of the simplest combinatorial type, pairwise intersections at most one vertex, giving sparse flux backgrounds with generic protected topology and transparent charge combinatorics. The linear lower bound is now proved, and in the sharp form $\mu_n \geq \lceil (n-1)/3 \rceil$: by Proposition 27 a saturating form leaves at most one coordinate unused, while a support of s triples meets at most $3s$ of them. The content of the conjecture is therefore the matching upper construction. Explicit witnesses use $1, 1, 2, 2, 5, 5, 8, 7$ triples for $n = 3$ to 10, the exceptional dimension 9 being the one nonlinear support, matching the exceptional outer pair of g_9; a proof route is partial Steiner systems with block-decomposed rank counts. In tests on the corpus described in the stress-test section, all eight deposited witnesses were reproduced independently, and exhaustive scans at $n = 5, 6$ confirmed $\mu_5 = \mu_6 = 2$. A counterexample would be sublinear supports or forced nonlinearity infinitely often.

Conjecture 180 (Coefficient-law universality in the sparse window)**.** *Fix $c > 0$ and let the support be sampled at $p = c \log n / n^2$ as in Conjecture 178. For every finite symmetric coefficient set $V \subset \mathbb{Z} \setminus \{0\}$, the limiting saturation probability and the limiting joint law of the excess vector $(b_k(H; \mathbb{Q}) - g_{n,k})_k$ exist and depend only on c, not on V.*

Significance. Thresholds located by a support process should not remember the coefficient sampler, and the conjecture asserts exactly this universality, making sparse flux-topology statistics a property of the hypergraph ensemble rather than an artifact of a particular discrete law; no arithmetic exclusion is needed at the rational level, since a common divisor of all coefficients scales the differential invertibly over $\mathbb{Q}$ and changes no rank, so the deposited form's exclusion of such laws has been removed, with the window and the excess vector now fixed explicitly in the statement. The mechanism is rank universality for structured sparse random matrices, but not conditionally on the support: the profile is genuinely coefficient-sensitive, and the smallest witness, retained here as a boundary control, is the four-triple support $\{012, 013, 024, 034\}$ on five coordinates, where all triples meet in one coordinate and the form is $e_0 \wedge \omega$ for a two-form ω on the other four. It is minimal: on four coordinates $\Lambda^3(\mathbb{Q}^4) \cong (\mathbb{Q}^4)^* \otimes \det$ has a single GL_4-orbit of nonzero forms, so no coefficient moves the profile, while a coefficient-sensitive support needs an even cover, which three triples cannot give since their nine incidences are odd. The profile is $(0, 3, 9, 9, 3, 0)$ when the Pfaffian of ω vanishes and $(0, 1, 9, 9, 1, 0)$ when it does not, and the chance of the degenerate branch is $\frac{1}{2}$ for $\{\pm 1\}$ against $\frac{3}{16}$ for $\{\pm 1, \pm 2\}$, so the law given the support does depend on V. By Proposition 27(iii) coefficient sensitivity needs a subsupport covering every coordinate evenly, hence with at most $\frac{3}{2}|S'|$ coordinates, and each fixed such circuit has expected multiplicity $O(n^{-|S'|/2} (\log n)^{|S'|})$ in the window and disappears from it, which is what localizes the mechanism above. That is as far as the argument reaches, and it is not a proof of the conjecture, a limitation: the window carries $|S| \sim \frac{c}{6} n \log n$ triples on n coordinates, so $\operatorname{rank}_{\mathbf{F}_2} A \leq n < |S|$ and the left kernel has dimension at least $|S| - n = \Theta(n \log n)$; circuits are forced rather than rare, and a first-moment estimate for one fixed circuit cannot be summed over the at least $2^{|S|-n}$ of them. Controlling the large circuits is the surviving problem. The statement above is annealed, averaging over the random support; the quenched strengthening instead fixes the support and asks that the conditional laws $\mathcal{L}_V(\cdot \mid S)$ and $\mathcal{L}_W(\cdot \mid S)$ converge to one another in total variation. Exact equality of those conditional laws fails at every finite n, by the four-triple control above, but the convergence itself is untouched by it and is open. Both the random model and the limit clause are flagged. A first decisive theorem would be two-point universality, $\{\pm 1\}$ against $\{\pm 1, \pm 2, \pm 3\}$. In tests on the corpus described in the stress-test section, the two deposited laws tracked each other across 960 trials, with saturation rates 0.2125 against 0.2125 at $c = 4$ and 0.500 against 0.525 at $c = 6$. A counterexample would be two admissible laws with separated limiting curves.

Proposition 27 (Support obstructions and coefficient rescaling)**.** Let $H \in \Lambda^3\mathbb{Z}^n$ have support S, let u be the number of coordinates lying in no triple of S, and let A be the $|S| \times n$ incidence matrix of S. (i) The Betti polynomial of $(E_n(\mathbb{Q}), d_H)$ is $(1+t)^u$ times that of the form read in the used coordinates, and H attains the generic profile only if $g_n = (1+t)^u g_{n-u}$; for the deposited profile this forces $u \leq 1$, with $u = 1$ possible only when $n \equiv 0 \pmod 4$. (ii) Consequently $\mu_n \geq \lceil (n-1)/3 \rceil$, and if $p_n n^2/\log n \to 0$ then the saturation probability of Conjecture 178 tends to zero. (iii) If $\operatorname{rank}_{\mathbf{F}_2} A = |S|$ then the rational profile of H depends only on S and not on the coefficients. Coefficient dependence therefore requires a nonempty $S' \subseteq S$ covering every coordinate an even number of times, and then $|S'|$ is even, $|S'| \geq 4$, and $|V(S')| \leq \frac{3}{2}|S'|$. Since A has n columns the criterion requires $|S| \leq n$, so it is vacuous in the window of Conjecture 180, where $|S| = \Theta(n \log n)$ and the left kernel of A has dimension at least $|S| - n$.

Proof. If a coordinate e lies in no triple of S then $\Lambda^*\mathbb{Q}^n = \Lambda^* V_0 \otimes \Lambda^*\langle e \rangle$ with d_H acting as $d_H \otimes 1$, so the Betti polynomial gains a factor $1 + t$, and iterating gives $(1+t)^u$. Since $b_k = \binom{n}{k} - \operatorname{rank}_k - \operatorname{rank}_{k-3}$ and both ranks are generically maximal, the generic profile minimizes every b_k separately, so the profile h of the form read in the $n - u$ used coordinates satisfies $h_k \geq g_{n-u,k}$ for all k. If H saturates then $g_n = (1+t)^u h$ and $G_n = 2^u \sum_k h_k \geq 2^u G_{n-u}$, while suspending a generic $(n-u)$-form is admissible and gives $G_n \leq 2^u G_{n-u}$; equality forces $h = g_{n-u}$, hence $g_n = (1+t)^u g_{n-u}$. For the deposited profile, comparing the displayed values gives $g_n = (1+t)g_{n-1}$ exactly when $n \equiv 0 \pmod 4$, and never $g_n = (1+t)^2 g_{n-2}$, since that would require the same identity at n and at $n-1$. This is (i).

A support of s triples meets at most $3s$ coordinates, so $u \geq n - 3s$, and (i) gives $n - 3s \leq 1$, which is the stated bound on μ_n, integrality giving the ceiling. For the threshold, a fixed coordinate lies in no chosen triple with probability $(1-p_n)^{\binom{n-1}{2}} = \exp\bigl(-\frac{1}{2}p_n n^2(1+o(1))\bigr)$, which is $n^{-o(1)}$ when $p_n n^2 = o(\log n)$, so the expected number of unused coordinates is $n^{1-o(1)}$; two coordinates lie in only $n-2$ common triples and $p_n n = o(1)$, so their indicators are asymptotically uncorrelated and a second-moment estimate gives $u \to \infty$ in probability. By (i) the saturation probability tends to zero, which is (ii).

For (iii), the substitution $e_i \mapsto \lambda_i e_i$ with $\lambda \in (\mathbb{Q}^\times)^n$ is an algebra automorphism carrying d_H to the differential of the form with coefficients $c_t \prod_{i \in t} \lambda_i$, so it leaves the profile unchanged. Writing $\lambda_i = \varepsilon_i \exp(x_i)$, a prescribed coefficient vector is reached exactly when $Ax = \log|c|$ over $\mathbb{Q}$ and $A\varepsilon = \operatorname{sign}(c)$ over $\mathbf{F}_2$; full row rank over $\mathbf{F}_2$ supplies both, since it forces full row rank over $\mathbb{Q}$, so every coefficient vector is equivalent to $(1, \ldots, 1)$. If instead the rows are dependent over $\mathbf{F}_2$, a nonzero relation $\sum_t \mu_t \chi_t = 0$ has support S' covering every coordinate an even number of times, so $3|S'| = \sum_{t \in S'} |t| \geq 2|V(S')|$; the same parity forces $3|S'|$ even, hence $|S'|$ even, and $|S'| = 2$ is impossible for distinct triples, so $|S'| \geq 4$. Finally $\operatorname{rank}_{\mathbf{F}_2} A \leq n$, so the criterion can hold only when $|S| \leq n$. □

Conjecture 181 (Finite-field resonance power law)**.** *Fix $n \geq 5$. There are a positive integer δ_n, a finite Galois extension $K_n/\mathbb{Q}$, and nonnegative integers $\kappa_{n,\sigma}$ indexed by the Frobenius conjugacy classes σ of $K_n/\mathbb{Q}$, positive for at least one class, such that, for H uniform on $\Lambda^3\mathbb{F}_p^n$, the probability of the resonance event $R_{n,p}$, that the profile $(b_k(H;\mathbb{F}_p))_k$ exceeds g_n in some degree, satisfies*

$$\Pr(R_{n,p}) = \kappa_{n,\mathrm{Frob}_p}\, p^{-\delta_n} + O_n\bigl(p^{-\delta_n - 1/2}\bigr) \qquad \textit{as } p \to \infty.$$

Proposition 28 (Conditional Lang–Weil resonance law)**.** Fix $n \geq 5$ and suppose that the displayed g_n is the characteristic-zero profile on the Zariski-open locus of maximal wedge ranks. Then the conclusion of Conjecture 181 holds for every sufficiently large prime of good reduction. More precisely, δ_n is the codimension of the largest geometric component of the resonance scheme, and $\kappa_{n,\mathrm{Frob}_p}$ is the number of its largest geometric components fixed by Frobenius.

Proof. Put $M = \binom{n}{3}$. The coefficients of a three-form give the affine space $\mathbb{A}^M_{\mathbb{Z}}$. Under the hypothesis on g_n, the event that some wedge map has rank below its generic rank is the set of $\mathbb{F}_p$-points of a proper determinantal subscheme $X_n \subset \mathbb{A}^M_{\mathbb{Z}}$ for every sufficiently large prime. Let d be the greatest dimension of a geometric irreducible component of its characteristic-zero fibre and set $\delta_n = M - d$. Choose a finite Galois extension over which all components of dimension d are defined. Away from finitely many ramified primes and primes of bad reduction, componentwise Lang–Weil [169] gives

$$\#X_n(\mathbb{F}_p) = \kappa_{n,\mathrm{Frob}_p} p^d + O_n(p^{d-1/2}).$$

Intersections of distinct components of dimension d and every smaller component contribute $O_n(p^{d-1})$. Dividing by p^M proves the stated probability law. The Galois action on the finite set of largest components shows that the leading coefficient depends only on the Frobenius conjugacy class. □

Significance. Proposition 28 separates the formal counting law from the open geometry. Once g_n is the true characteristic-zero generic profile, Lang–Weil gives the displayed asymptotic. The remaining content is to prove that identification, determine δ_n, describe the largest components and their fields of definition, and compute the Frobenius classes that fix them. Frobenius can permute these components, as for $x^2 + 1$ over $\mathbb{F}_p$, so neither a prime-independent constant nor a congruence-only description is justified without further geometry. The deposited screen of 2,800 exact finite-field samples at $5 \leq n \leq 8$ detects concentration at small characteristics but is too small to estimate δ_n. A first decisive theorem would determine the top resonance components for one n. A failure of the expected generic profile would invalidate the unconditional form while leaving the proposition intact.

Conjecture 182 (Central-binomial law for the complete flux, resolved true)**.** *For every $n \geq 4$, $B(H_\Delta; \mathbb{Q}) = F_n$ and $B(H_\Delta; \mathbb{F}_2) = P_n$.*

Significance. The most symmetric integral flux has closed cohomology totals, a central binomial over the rationals and a near-power of two in characteristic two, giving an exact protected-state count for a maximally symmetric background; the binomial shape calls for a decomposition of the complex, the natural proof route and the reason this is rated the most approachable flux law. In tests on the corpus described in the stress-test section, both formulas held for every $4 \leq n \leq 13$, the characteristic-two totals exactly and the rational totals by exact Smith forms through $n = 9$ and dual large-prime ranks beyond, reproduced independently throughout. A counterexample would be one dimension violating either formula. *Resolution.* The conjecture is true: Proposition 29 proves both formulas, reducing wedge multiplication by $H_\Delta = \alpha \wedge \omega$ to Boolean inclusion matrices, whose rational ranks are full and whose characteristic-two complex is exact. The rational total is the binomial transform $\sum_s \binom{g}{s} 2^s \binom{g-s}{\lfloor (g-s)/2 \rfloor} = \binom{2g+1}{g}$ and the characteristic-two total is $4^g + 2^g$, $g = \lfloor (n-1)/2 \rfloor$; the symmetric-group route anticipated at deposit is replaced by a symplectic normal form.

Conjecture 183 (Prime torsion staircase for the complete flux, resolved false)**.** *For every prime p, the integral cohomology of $(E_n(\mathbb{Z}), d_{H_\Delta})$ has no p-primary torsion for $n \le 4p-2$ and exactly one p-primary cyclic summand, isomorphic to $\mathbb{Z}/p$, at $n = 4p-1$. Moreover, for every n the 2-primary subgroup is elementary abelian of rank $t_2(H_\Delta) = (P_n - F_n)/2$.*

Significance. Arithmetic torsion in the flux charge group appears along a staircase indexed by the primes, and the boundary is calibrated by a falsified precursor, retained in place: the claim that all torsion is elementary 2-torsion held through $n = 9$ in exact Smith form and died at $(p, n) = (3, 11)$, where the characteristic-three total exceeds $F_{11} = 924$ by exactly 2, one $\mathbb{Z}/3$ summand; the excesses grow to 4 and 28 at $n = 12, 13$, and the next decisive prediction is first 5-torsion at $n = 19$. By Proposition 25 the rank clause is equivalent to the two totals of Conjecture 182, and the exponent-two clause was the most exposed part of the statement. A proof route would be a modular filtration whose semisimplicity first fails at $n = 4p-1$, a mechanism that is a target for explanation rather than a consequence of general theory, since nothing elementary singles out $4p-1$. In tests on the corpus described in the stress-test section, the staircase and rank clause were verified in exact Smith form through $n = 9$ and the $(3, 11)$, $(3, 12)$, $(3, 13)$ excesses reproduced independently, with no resonance at $p = 5, 7$ through $n = 13$. A counterexample would be torsion before a step, a different onset group, or a divisor exceeding two in the 2-primary part. *Resolution.* The staircase clause is true and the exponent clause is false, both by Proposition 29: the onset at $n = 4p-1$ with a single $\mathbb{Z}/p$ holds for every prime, while at $n = 15$ the singleton-free sector carries the inclusion matrix $U_{7,3}$, whose Smith normal form $1^{20}, 2^9, 6^5, 12$ contributes a summand $\mathbb{Z}/4$ in degree nine. This is exactly the divisor exceeding two that the statement named as its own refutation, and $n = 15$ is the least dimension at which it can occur, so the exhaustive range through $n = 13$ could not have seen it; the surviving question is the exact 2-adic invariant factors, whose first departure from exponent one is now located.

Proposition 29 (Complete-flux normal form)**.** Write $\alpha = e_1 + \cdots + e_n$ and $\omega = \sum_{2 \le i < j \le n} e_i \wedge e_j$, and put $g = \lfloor (n-1)/2 \rfloor$. Then $H_\Delta = \alpha \wedge \omega$, and the integral cohomology of $(E_n(\mathbb{Z}), d_{H_\Delta})$ satisfies: (i) $B(H_\Delta; \mathbb{Q}) = F_n$ and $B(H_\Delta; \mathbb{F}_2) = P_n$; (ii) for every prime p there is no p-primary torsion for $n \le 4p-2$, and exactly one p-primary cyclic summand, isomorphic to $\mathbb{Z}/p$, at $n = 4p-1$; (iii) whenever p-primary torsion is present, the positive support of $k \mapsto t_{p,k}(n)$ is the full interval $2p+1 \le k \le n-2p+2$; (iv) $T_p(n+1) > T_p(n)$ for every $n \ge 4p-1$; (v) at $n = 15$ the 2-primary subgroup is not elementary abelian: it contains a summand $\mathbb{Z}/4$ in degree 9.

Proof. Comparing coefficients, $\alpha \wedge \omega$ produces each triple meeting $\{1\}$ once, and each triple $\{a < b < c\} \subset \{2, \ldots, n\}$ three times, from $e_a \wedge (e_b \wedge e_c)$, $e_b \wedge (e_a \wedge e_c)$ and $e_c \wedge (e_a \wedge e_b)$ with signs $+, -, +$; so $H_\Delta = \alpha \wedge \omega$. The substitution $(e_1, \ldots, e_n) \mapsto (\alpha, e_2, \ldots, e_n)$ is unimodular, so with $W = \langle e_2, \ldots, e_n \rangle$ we may write $E_n(\mathbb{Z}) = \Lambda^* W \oplus \alpha \wedge \Lambda^* W$. Since $\alpha \wedge \alpha = 0$ the differential annihilates $\alpha \wedge \Lambda^* W$ and carries $x \in \Lambda^* W$ to $\alpha \wedge Lx$, where $L = \omega \wedge -$ acts on $\Lambda^* W$; as $\alpha \wedge -$ is injective on $\Lambda^* W$, the cohomology is $\ker L \oplus \alpha \wedge \operatorname{coker} L$, so it is $\operatorname{coker} L$ that carries every torsion summand.

The alternating form ω has all elementary divisors equal to 1, so an integral change of basis of W writes it as $x_1 \wedge y_1 + \cdots + x_g \wedge y_g$, together with one null basis vector when n is even. Split $\Lambda^* W$ according to the state of each symplectic pair, empty, x_i, y_i, or $x_i y_i$. Then L annihilates every pair held in a singleton state, and on the remaining $r = g - s$ pairs it sends the set of doubly occupied pairs to the sum of its one-element extensions; that is, on a sector with s singletons it is the 0–1 inclusion matrix $U_{r,j}$ from j-subsets to $(j+1)$-subsets of an r-set. There are $\binom{g}{s}2^s$ such sectors, doubled by the null vector when n is even, and a monomial with s singletons and $j+1$ doubly occupied pairs sits in cohomological degree $s+2j+3$.

Wilson's diagonal form for t-subset versus k-subset inclusion matrices [114], applied to $U_{r,j}$ when $2j+1 \le r$ and to its transpose otherwise, gives diagonal entries $h-i$ with multiplicity $\binom{r}{i} - \binom{r}{i-1}$ for $0 \le i < h$, where $h = \min(j+1, r-j)$. These number $\binom{r}{h-1} = \min(\binom{r}{j}, \binom{r}{j+1})$, so over $\mathbb{Q}$ every $U_{r,j}$ has full possible rank, a sector of r active pairs contributes $2\binom{r}{\lfloor r/2 \rfloor}$ to the total cohomology, and summing over sectors gives $\sum_s \binom{g}{s} 2^s \cdot 2\binom{g-s}{\lfloor (g-s)/2 \rfloor} = 2\binom{2g+1}{g}$, which is F_n for odd n and half of it for even n before the null vector doubles it back. Over $\mathbf{F}_2$ the operator is multiplication by $u_1 + \cdots + u_r$ in $\mathbf{F}_2[u_1, \ldots, u_r]/(u_i^2)$, hence conjugate under a linear change of variables to multiplication by u_1 and therefore exact, so its ranks sum to 2^{r-1} and such a sector contributes 2^r, or 2 when $r = 0$; summing gives $4^g + 2^g$, again doubled by the null vector in even dimension, which is P_n. This is (i).

A diagonal entry divisible by p forces $h \ge p$ and hence $r \ge 2p-1$; since $r \le g$, no p-torsion occurs for $n \le 4p-2$. At $n = 4p-1$ we have $g = 2p-1$, the only qualifying sector is $s = 0$, $r = 2p-1$, $j = p-1$, and its diagonal contains the entry p exactly once, giving a single summand $\mathbb{Z}/p$; this is (ii). In a sector with r active pairs, p divides a diagonal entry of $U_{r,j}$ exactly when $p-1 \le j \le r-p$, so that sector realises the degrees $s+2j+3$ for $p-1 \le j \le g-s-p$, together with their shifts by one when n is even. The sectors $s = 0$ and $s = 1$ already supply the two parities, running over $2p+1, 2p+3, \ldots, 2g-2p+3$ and $2p+2, 2p+4, \ldots, 2g-2p+2$ respectively, and the intervals for larger s are nested inside these; the union is therefore the single interval $2p+1 \le k \le n-2p+2$. This is (iii).

For (iv) let A_r count the p-primary summands contributed by one pure r-pair sector and put $T_g = \sum_r \binom{g}{r} 2^{g-r} A_r$, so that $T_p(2g+1) = T_g$ while the null vector gives $T_p(2g+2) = 2T_g$. Pascal's rule yields $T_{g+1} - 2T_g = \sum_{r \ge 1} \binom{g}{r-1} 2^{g+1-r} A_r$, which is positive as soon as some A_r is, that is, after onset.

Finally let $n = 15$, so $g = 7$. The sector $s = 0$, $r = 7$, $j = 3$ has $h = 4$, so Wilson's diagonal carries the entry 4: the Smith normal form of the 35×35 matrix $U_{7,3}$ is $1^{20}, 2^9, 6^5, 12$, its cokernel is $(\mathbb{Z}/2)^{14} \oplus (\mathbb{Z}/3)^6 \oplus \mathbb{Z}/4$, and the corresponding degree is $s+2j+3 = 9$. This is (v). $\square$

Conjecture 184 (Parity-locked torsion count for sign flux)**.** *For dense $\{\pm 1\}$ flux as in Conjecture 177, conditional on the rational profile equalling g_n, the number of 2-primary cyclic summands is exactly $t_2(H_n) = (P_n - G_n)/2$; unconditionally, this count holds with probability tending to one.*

Significance. A random flux ensemble is conjectured to carry a deterministic number of torsion charge directions even while their orders fluctuate: by Proposition 25, every sign flux has the characteristic-two cohomology of the complete flux, so the count is forced by Conjectures 177 and 182 together and the statement is essentially their corollary, numbered separately for the exact count it pins rather than as an independent law, and flagged for elevated attribution risk since the mechanism is the classical universal coefficient theorem. The elementary-torsion strengthening is false and the control is retained: deposited Smith forms contain cyclic factors of orders 4, 40, and 80. A first decisive theorem would be the conditional identity, which needs only the mod-two clause of Conjecture 182. In tests on the corpus described in the stress-test section, all sixteen deposited exact Smith trials at $5 \le n \le 8$ had the predicted count, reproduced independently. A counterexample would be a saturating sign flux with the wrong number of even summands. *Resolution.* With Conjecture 182 proved, the conditional clause is a theorem: Proposition 25 gives $t_2(H) = \big(B(H;\mathbb{F}_2) - B(H;\mathbb{Q})\big)/2$ for every sign flux, Proposition 29 gives $B(H;\mathbb{F}_2) = P_n$, and the conditioning supplies $B(H;\mathbb{Q}) = G_n$. The unconditional clause is precisely the saturation event of Conjecture 177, so the statement is retained as a corollary of that conjecture rather than as an independent law.

43 Protected profiles of complete intersections

Conjecture 185 (Coefficientwise endpoint extremality)**.** *For $n \geq 3$ and every interior index $1 \leq p \leq n-1$, $a_p(Q) \leq a_p(\mathbf{d}) \leq a_p(Y)$, and equality across all interior indices holds only for the endpoint configuration itself. Dimension 2 is the equality calibration: every configuration has the K3 profile $(2, 20, 2)$.*

Significance. The all-quadric and hypersurface geometries are conjectured to bound every coefficient of the protected profile throughout the complete-intersection landscape [93], refining the Euler extremality aesthetic of Part VII from one number to the full χ_y-vector. The mechanism is connectivity of the configuration poset under merges, with a correction: Conjecture 186 controls only the degree-normalized coefficients, and since the total degree strictly decreases under every merge, normalized monotonicity does not transfer to the raw corridor; the upper endpoint should instead flow from the maximal-merge law of Conjecture 187 along maximal chains, and the all-quadric lower comparison, which needs merges away from the maximum, is the harder half. A first decisive theorem would be the extremality of Y from Conjecture 187 along maximal-merge chains. In tests on the corpus described in the stress-test section, all 7,124 deposited comparisons over 680 configurations through dimension 14 passed in exact arithmetic, reverified independently together with the rigidity clause. A counterexample would be one configuration and one index outside the corridor.

Conjecture 186 (Degree-normalized merge monotonicity)**.** *If $\mathbf{d}'$ is obtained from $\mathbf{d}$ by merging two entries, then $a_p(\mathbf{d}')/D(\mathbf{d}') > a_p(\mathbf{d})/D(\mathbf{d})$ for every $1 \leq p \leq n-1$.*

Significance. Dividing by the degree, the natural volume normalization, every protected coefficient density strictly increases each time two defining equations are consolidated, a monotone coordinate on Calabi–Yau presentations that extends Conjecture 118's Euler law coefficientwise. The proof route is positivity of the one-merge difference in the classical generating series for χ_y of complete intersections [115]. A first decisive theorem would be one interior coefficient, say $p = 1$, for all merges. In tests on the corpus described in the stress-test section, all 110,777 deposited exact comparisons through dimension 14 passed, with an independent recomputation of 25,921 deduplicated merge tests finding no violation. A counterexample would be one merge and one index with nonincreasing density.

Conjecture 187 (Maximal-degree raw monotonicity)**.** *For $n \geq 3$, if a merge uses an entry equal to $\max_i d_i$, then $a_p(\mathbf{d}') > a_p(\mathbf{d})$ for every $1 \leq p \leq n-1$.*

Significance. Without any normalization the profile can fall under a general merge, since the degree drops; the conjecture isolates the directed moves, those through a maximal entry, that raise every protected coefficient simultaneously, and dimension 2, where all profiles tie, is the equality boundary. Together with Conjecture 186 this brackets how much degree normalization a raw monotonicity needs. A first decisive theorem would be hypersurface targets, merges landing on Y. In tests on the corpus described in the stress-test section, all 47,433 deposited exact comparisons for $3 \leq n \leq 14$ passed, reverified independently on 13,724 deduplicated tests. A counterexample would be one maximal merge losing one coefficient.

Conjecture 188 (Real-rooted protected polynomial)**.** *For every configuration, the polynomial $A_{\mathbf{d}}(y)$, after removing in odd complex dimension the single factor y forced by $a_0 = a_n = 0$, has only simple, strictly negative real zeros.*

Significance. Real-rootedness would package the protected profile as a Pólya frequency sequence, an infinite hierarchy of determinantal inequalities with Conjecture 189 as its first layer, and suggests a hidden total-positivity or free-fermion structure in the protected sector. The evidence is numerical root location and is flagged as such, the weakest evidence type in this part: all 369 deposited polynomials through dimension 12 passed scaled root tests with smallest normalized separation 1.276×10^{-3}, a figure reproduced independently to six decimals. A proof route is a multiplier sequence realizing the merge recursion; in odd complex dimension palindromy already forces the zero at -1 of the reduced polynomial, and since the coefficients are exact integers, Sturm-sequence certification can replace the floating scans outright. A first decisive theorem would be the hypersurface family. A counterexample would be one configuration with a nonreal or repeated zero, which higher-precision arithmetic could exhibit.

Conjecture 189 (Strict ultra-log-concavity)**.** *For every configuration and every interior triple of positive coefficients,*

$$\left(a_p/\tbinom{n}{p}\right)^2 > \left(a_{p-1}/\tbinom{n}{p-1}\right)\left(a_{p+1}/\tbinom{n}{p+1}\right).$$

Significance. Binomial normalization is the strongest classical strengthening of log-concavity, the form delivered by Hodge–Riemann packages, and the conjecture asserts it strictly for the protected profile, sharp concentration for the protected-sector distribution; it is flagged for elevated attribution risk because a geometric realization of the coefficient sequence would place it directly inside that machinery, in the manner of [101]. Conjecture 188 implies the strict form in full, a strengthening: Newton's inequalities are strict for polynomials with simple real zeros, and the odd-dimensional reduction multiplies the normalized sequence by the strictly log-concave factor $p(n-p)$, so the statement stands or falls with Conjecture 188 and is retained as its first determinantal layer, with a direct intersection-theoretic proof the route that would decouple the two. A first decisive theorem would be hypersurfaces in odd dimension. In tests on the corpus described in the stress-test section, all 6,540 deposited exact inequalities through dimension 14 passed, the count and the outcomes reproduced independently exactly. A counterexample would be one flat or reversed normalized triple.

Conjecture 190 (Degree-normalized Mahler growth)**.** *For every merge, $M(A_{\mathbf{d}'})/D(\mathbf{d}') > M(A_{\mathbf{d}})/D(\mathbf{d})$, where M is the multiplicative Mahler measure, the modulus of the leading coefficient times $\prod_\rho \max(1, |\rho|)$ over the zeros ρ.*

Significance. The Mahler measure couples coefficient growth to the placement of the zeros, and the conjecture asserts that per unit degree it grows strictly under every consolidation of equations, returning the Mahler order of Conjecture 159 on the protected polynomial in place of the graph associahedron. Under Conjecture 188 the measure is a product over negative real zeros, but reality alone does not control how the zeros move under a merge, so the route needs interlacing or majorization of the root motion, the recorded missing step. The evidence is numerical and flagged as such. A first decisive theorem would be merges within the hypersurface-adjacent family. In tests on the corpus described in the stress-test section, all 4,267 deposited merge tests through dimension 12 passed, reverified independently on 1,083 deduplicated tests. A counterexample would be one merge with nonincreasing normalized measure.

Conjecture 191 (Spectral-radius growth with low-dimensional rigidity)**.** *Let $\rho(\mathbf{d})$ be the largest modulus of a zero of the reduced polynomial of Conjecture 188, the polynomial $A_{\mathbf{d}}$ with its forced factor removed. If a merge uses a maximal entry, then $\rho(\mathbf{d}') \geq \rho(\mathbf{d})$, with equality for every such merge in complex dimensions $n = 2, 3, 5$ and strict inequality in every other dimension $n \geq 4$.*

Significance. Equation consolidation is conjectured to be a monotone spectral flow of the protected index, and the rigidity pattern is exact where palindromy forces it: in dimension 2 all profiles coincide, in dimension 3 the reduced polynomial always has its zero at -1, and in dimension 5 the tested radii agree along every maximal merge, while from dimension 4 onward, dimension 5 excepted, the flow is strictly expansive. The evidence is numerical and flagged as such. Palindromy alone cannot deliver the dimension-5 rigidity: the reduced cubic factors as $y + 1$ times a palindromic quadratic whose root pair varies with the coefficient ratio, so the tested equality demands a further identity among the dimension-five coefficients, and a first decisive theorem would be that identity. In tests on the corpus described in the stress-test section, all 2,102 deposited maximal-merge tests through dimension 12 matched the equality pattern, reverified independently on 617 deduplicated tests with no violation of either clause. A counterexample would be a contraction, or a strict increase in dimensions 2, 3, or 5.

44 Signature laws for complete intersections

Conjecture 192 (Signature endpoint extremality)**.** *In every even complex dimension $n \geq 4$, $S(Q) \leq S(\mathbf{d}) \leq S(Y)$, with equality only at the corresponding endpoint. Dimension 2 is the calibration: every configuration is a K3 surface with $S = 16$.*

Significance. The signature is an index, an oriented-cobordism invariant, and the divisibility obstruction for free finite group actions, so universal endpoints bound quotient and anomaly data across the landscape; the conjecture is the signature specialization of Conjecture 185 but is stated separately because the signature carries its own rigidity, verified here with equality forced only at the endpoints in every tested dimension. Since the total degree strictly decreases under every merge, the normalized law of Conjecture 193 does not by itself give the raw corridor, a correction; a first decisive theorem would be the upper endpoint from the maximal-merge law of Conjecture 194 along maximal chains. In tests on the corpus described in the stress-test section, all 8,190 deposited configurations through dimension 26 passed in exact arithmetic, the rigidity clause reverified independently on all 8,187 eligible configurations. A counterexample would be one configuration outside the corridor or a non-endpoint tie.

Conjecture 193 (Degree-normalized signature monotonicity)**.** *Every merge in even complex dimension satisfies $S(\mathbf{d}')/D(\mathbf{d}') > S(\mathbf{d})/D(\mathbf{d})$.*

Significance. The signature density per unit degree increases strictly under every consolidation of equations, an index-density monotone and pruning rule for quotient searches, and the signature sibling of Conjecture 186; the proof route is one-merge positivity in the L-genus characteristic series [115], the analogue for $x/\tanh x$ of the χ_y computation. A first decisive theorem would be merges of two quadrics. In tests on the corpus described in the stress-test section, all 302,130 deposited exact comparisons through dimension 26 passed, reverified independently on 55,665 deduplicated tests. A counterexample would be one merge with nonincreasing signature density.

Conjecture 194 (Maximal-degree signature monotonicity)**.** *In even complex dimension $n \geq 4$, a merge using a maximal entry strictly increases S.*

Significance. As with the profile, the raw signature can fall under an arbitrary merge, and the conjecture isolates the directed moves that always raise it, giving a strictly increasing global index along maximal-merge chains from the all-quadric configuration to the hypersurface; dimension 2 is again the equality boundary. A first decisive theorem would combine Conjecture 193 with a bound on the degree drop. In tests on the corpus described in the stress-test section, all 80,294 deposited exact tests through dimension 26 passed, reverified independently on 23,284 deduplicated tests. A counterexample would be one maximal merge lowering the signature.

Conjecture 195 (The quadric ratio law with limit 27, resolved true)**.** *For even n let $\Sigma_n = S(Q)$ for the all-quadric configuration of $n+1$ quadric equations in complex dimension n. Then $R_n = \Sigma_{n+2}/\Sigma_n$ is strictly increasing in n, $R_n < 27$, and*

$$R_n \to 27, \qquad 27 - R_n \sim \frac{27}{n}.$$

Significance. The all-quadric ladder is conjectured to grow by an asymptotic factor of exactly 27 every two dimensions, with a clean first-order correction, a sharp asymptotic fingerprint that calls for the dominant singularity of the generating function of Σ_n; the statement carries a limit clause beyond any finite check and an attribution flag, since singularity analysis is classical once the generating function is extracted. A first decisive theorem would be the limit alone. In tests on the corpus described in the stress-test section, all 29 deposited exact ratio comparisons through $n = 60$ passed, reproduced independently with $R_{58} = 26.55045876$ and gap 0.44954124 against $27/60 = 0.45$. A counterexample would be a ratio at least 27, a decrease, or a different correction rate. Since deposit the conjecture has been resolved true, by proof rather than by computation: Proposition 30 shows the ladder's generating function is algebraic, and the resulting exact recurrence collapses the ratio dynamics to a decreasing map interlacing the explicit rational sequence $q_m < 27$, which forces boundedness, monotonicity, and the $27/n$ rate outright; matching orders in the proved recurrence sharpens the limit clause to $R_n = 27(1 - 1/n + \frac{37}{18}n^{-2} - \frac{281}{72}n^{-3} + O(n^{-4}))$. Nor is the mechanism particular to quadrics: Proposition 31 extends it to every repeated multidegree block and identifies the constant as $27 = \cot^6(\pi/6)$, the first member of the trigonometric family $\Lambda_d = \cot^{2(d+1)}(\pi/(2(d+1)))$.

Proposition 30 (Resolution of the quadric ratio law)**.** Conjecture 195 holds in full. With $a_m = \Sigma_{2m}$ and $a_0 = 2$, the generating function $A(w) = \sum_{m \geq 0} a_m w^m$ is algebraic,

$$A(w) = \frac{2}{(1+y)(1-3y)}, \qquad 4w = y(1-y)^2, \quad y(0) = 0,$$

and the coefficients satisfy $D_m a_{m+2} = (M_m - D_m)a_{m+1} + M_m a_m$ with $D_m = (m+2)(2m+3)(14m+9)$ and $M_m = 6(3m+2)(3m+4)(14m+23)$.

Proof. For the all-quadric configuration in complex dimension $n = 2m$ there are $2m + 1$ defining quadrics, so the ambient dimension is $N = 4m + 1$, and the classical signature formula $\sigma = [x^N]\,(x/\tanh x)^{N+1}\prod_i \tanh(d_i x)$ for complete intersections in $\mathbb{CP}^N$ [115] gives $\sigma = [x^{4m+1}](x/\tanh x)^{4m+2}\tanh(2x)^{2m+1}$. Setting $f(x) = x\tanh(2x)/\tanh^2 x$, an even power series with $f(0) = 2$, the expression under the coefficient equals $x^{2m+1}f(x)^{2m+1}$, so $\sigma = [x^{2m}]f(x)^{2m+1}$. Because f is even, replacing x^2 by $-z$ multiplies the coefficient of x^{2m} by $(-1)^m$ and turns f into $\psi(z) = \sqrt{z}\,\tan(2\sqrt{z})/\tan^2\sqrt{z}$, analytic on $|z| < \pi^2/4$; hence $a_m = (-1)^m\sigma = [z^m]\psi(z)^{2m+1}$. For each small $w > 0$ the equation $z = w\psi(z)^2$ has exactly one root $z_0(w)$ near the origin, and the residue form of Lagrange inversion, summing the geometric series $\sum_m (w\psi(z)^2/z)^m$ on a small contour and picking up the simple pole at z_0, gives

$$A(w) = \sum_{m\geq 0}[z^m]\psi(z)^{2m+1}w^m = \frac{\psi(z_0)}{1 - w(\psi^2)'(z_0)}.$$

Substituting $z = u^2$ and $T = \tan u$ and using the double-angle formula $\tan 2u = 2T/(1-T^2)$, the root equation becomes $4w = T^2(1-T^2)^2$ and the residue value becomes $2/((1+T^2)(1-3T^2))$, two exact identities; with $y = T^2$ this is the stated parametrization.

Next, the recurrence. Write $\theta = w\,d/dw$, let $D(\theta)$ and $M(\theta)$ denote the polynomials D_m and M_m with m replaced by the operator θ, and put $B_s = w^{-s}(A - \sum_{i<s} a_i w^i)$, the generating function of the shifted sequence $(a_{m+s})_{m\geq 0}$. Then

$$\sum_{m\geq 0}(D_m a_{m+2} - (M_m - D_m)a_{m+1} - M_m a_m)w^m = D(\theta)B_2 - (M(\theta) - D(\theta))B_1 - M(\theta)B_0,$$

and under the parametrization, where θ acts as $\frac{y(1-y)}{1-3y}\frac{d}{dy}$, the right side is a rational function of y that vanishes identically, a finite identity checked by exact expansion. So the recurrence holds for every $m \geq 0$.

Finally, the three clauses. Set $r_m = a_{m+1}/a_m = R_{2m}$ and $q_m = M_m/D_m$, so the recurrence reads $r_{m+1} = q_m - 1 + q_m/r_m$. First, $27 - q_m = 3(126m^2 + 271m + 118)/D_m > 0$ for every m, so $q_m < 27$. Second, the map $g_m(r) = q_m - 1 + q_m/r$ is strictly decreasing on $r > 0$ and fixes q_m, and the interlacing $q_{m-1} < r_m < q_m$ holds for all $m \geq 1$ by induction: the base is $q_0 = \frac{184}{9} < 22 = r_1 < \frac{518}{23} = q_1$; if $r_m < q_m$, then, because g_m is decreasing, $r_{m+1} = g_m(r_m) > g_m(q_m) = q_m$; and if $r_m > q_{m-1}$, then $r_{m+1} < g_m(q_{m-1}) < q_{m+1}$, the last inequality clearing denominators to a polynomial of degree seven in $m-1$ with positive coefficients. The interlacing gives $r_m < q_m < 27$ and $r_{m+1} > q_m > r_m$, the boundedness and monotonicity clauses; and since $27 - q_m \sim 27/(2m)$, the squeeze $27 - q_m < 27 - r_m < 27 - q_{m-1}$ gives $27 - R_n \sim 27/n$ for $n = 2m$, the limit clause. □

Proposition 31 (Block ladders and their saddle constants)**.** Let $B = (d_1, \ldots, d_s)$ be a multidegree block with $A = \sum_j (d_j - 1)$ odd, and for $m \geq 0$ let $S_m(B)$ be the sign-normalized signature of the $(2m+1)$-fold repetition of B, a Calabi–Yau complete intersection of complex dimension $n_m = A(2m+1) - 1$. Then

$$S_m(B) = [y^{Am+(A-1)/2}]\,\frac{C_B(y)^{2m+1}}{1+y}, \qquad C_B(y) = \prod_{j=1}^{s}\frac{\tan(d_j\arctan\sqrt{y})}{\sqrt{y}},$$

where the factor with entry d equals $\sum_i (-1)^i\binom{d}{2i+1}y^i/\sum_i(-1)^i\binom{d}{2i}y^i$, a rational function, so the generating function $\sum_m S_m(B)w^m$ is algebraic. The series C_B has nonnegative aperiodic coefficients; there is a unique $y_* > 0$ below the first positive pole of C_B with $y_* C_B'(y_*)/C_B(y_*) = A/2$, and with $\Lambda_B = C_B(y_*)^2/y_*^A$,

$$\frac{S_{m+1}(B)}{S_m(B)} = \Lambda_B\Big(1 - \frac{1}{2m} + O(m^{-2})\Big),$$

so the ratios are eventually strictly increasing with limit Λ_B, and $\Lambda_B - S_{m+1}/S_m \sim \Lambda_B A/n_m$. For the equal-degree block $B = (d)$ with d even, $y_* = \tan^2 \frac{\pi}{2(d+1)}$ and

$$\Lambda_d = \cot^{2(d+1)}\Big(\frac{\pi}{2(d+1)}\Big),$$

so $\Lambda_2 = 27$ recovers Proposition 30; and for the Euler ladders with r equations of equal degree d, the analogous saddle sits at $t_* = 1/(d+1)$ and the ratio limit is d^{d+1}, recovering the limit $8 = 2^3$ of Proposition 15 at $d = 2$.

Proof. For $k = 2m + 1$ repetitions there are sk defining equations of total degree $(A + s)k$, so the ambient dimension is $N = (A + s)k - 1$, and the signature formula reads $\sigma = [x^N](x/\tanh x)^{N+1} \prod_j \tanh(d_j x)^k$. Write the coefficient as a contour integral and substitute $x = it$, then $T = \tan t$, then $y = T^2$. The powers of i collect to exactly the sign $(-1)^{n_m/2}$ that the normalization $S = (-1)^{n_m/2}\sigma$ removes; the Jacobian $dt = dT/(1 + T^2)$ contributes the factor $1/(1+y)$; the map $T \mapsto T^2$ covers the y-contour twice, which cancels the half from $dT = dy/(2\sqrt{y})$ since the integrand is even in T; and what remains of the integrand is $C_B(y)^k\, y^{-(Ak+1)/2}/(1 + y)$. Reading off the residue at $y = 0$ gives the extraction identity. Each factor of C_B is rational by the tangent multiple-angle formula; writing $C_B = P/Q$ with P and Q the products of the factor numerators and denominators, the residue argument of Proposition 30, applied to the root equation $y^A Q(y)^2 = wP(y)^2$, shows the generating function is algebraic.

For the saddle law, note first that the coefficients of each factor are nonnegative and aperiodic, meaning that the indices of the nonzero coefficients have greatest common divisor one: this follows by induction from $C_1 = 1$ and the tangent addition recursion $C_{d+1} = (C_d + 1)/(1 - yC_d)$, whose right side is a series with nonnegative coefficients in y and C_d, and both properties persist under products. Consequently $\mu(y) = yC_B'(y)/C_B(y)$, the average index of the coefficients of C_B when the coefficient of y^j is weighted by y^j, is strictly increasing, starts at 0, and diverges at the first positive pole of C_B, so the saddle y_* exists and is unique. The large-powers saddle-point method [116] then gives $S_m(B) = K_B \Lambda_B^m m^{-1/2}(1 + O(1/m))$ with $K_B > 0$, together with a full expansion in powers of $1/m$, and taking the ratio of consecutive terms gives the displayed law; the dimension form follows from $n_m = 2Am + A - 1$.

For the equal-degree block $B = (d)$, write $y = \tan^2\theta$, so that $C_d = \tan(d\theta)/\tan\theta$. The saddle equation becomes $\sin(2d\theta) = \sin(2\theta)$, whose first positive solution is $\theta_* = \pi/(2(d + 1))$; there $d\theta_* = \pi/2 - \theta_*$, so $\tan(d\theta_*) = \cot\theta_*$, hence $C_d(y_*) = \cot^2\theta_*$ and $\Lambda_d = C_d(y_*)^2/y_*^{d-1} = \cot^4\theta_*/\tan^{2(d-1)}\theta_* = \cot^{2(d+1)}\theta_*$. For the Euler ladders, the Chern-class manipulation of Proposition 15 generalizes to $E_{d,r} = [t^{(d-1)r-1}]\varphi_d(t)^r$ with $\varphi_d(t) = d(1 - t)^d/(1 - dt)$; the saddle equation $\varphi_d'/\varphi_d = (d - 1)/t$ has the exact solution $t_* = 1/(d + 1)$, and $\varphi_d(t_*)/t_*^{d-1} = d^{d+1}$. □

Remark 21 (Open question)**.** *Not counted among the conjectures: whether every signature block ladder is strictly increasing and below its saddle constant from the first ratio onward, as Conjecture 195 asserted for quadrics. Proposition 31 proves this eventually, and for the equal-degree ladders the question has since acquired exact structure. The function C_d satisfies the Riccati identity $2y(1 + y)C_d' = dyC_d^2 - (1 + y)C_d + d$, an elementary consequence of the angle* *substitution, and, verified through degree eight, equals d times a terminating Stieltjes continued fraction with positive coefficients $\alpha_j = (d^2 - j^2)/(4j^2 - 1)$ for $1 \le j \le d-1$. From the Riccati identity, the function $h_d(y) = y^{d-1}/C_d(y)^2$ obeys the exact factorization*

$$\frac{h_d'(y)}{h_d(y)} = \frac{d\,(C_d(y)-1)(1-yC_d(y))}{y(1+y)\,C_d(y)},$$

so every positive critical point of h_d solves $C_d(y) = 1$ or $yC_d(y) = 1$, the two explicit families $y = \tan^2(k\pi/(d-1))$ and $y = \tan^2((2k+1)\pi/(2(d+1)))$, and on the second family $h_d(y) = y^{d+1}$: the constant Λ_d is therefore the exact reciprocal of the first critical value of h_d, not a saddle estimate. Moreover $S_m^{(d)} = d^{2m+1} b^{(d)}_{(d-1)m+(d-2)/2}$ for the single coefficient sequence $b_n^{(d)} = [y^n]\,(C_d(y)/d)^{(2n+1)/(d-1)}/(1+y)$, and at the first critical point $r = \tan^2(\pi/(2(d+1)))$ the functions $P_d(z) = rC_d(rz)$ and $Y_d(z) = (1+r)P_d(z)/(1+rz)$ are probability generating functions, the first with mean exactly $(d-1)/2$, in terms of which $R_m^{(d)}/\Lambda_d = p_{m+1}/p_m$ for the central probabilities $p_m = [z^{(d-1)m+(d-2)/2}]\,Y_d(z)P_d(z)^{2m}$; the open question for degree d says exactly that p_m decreases strictly and is strictly log-convex. The boundary is calibrated: $b_n^{(2)} = 1, 2, 11, 62, 367, \ldots$ is not a Stieltjes moment sequence, its once-shifted Hankel determinant of order three being negative and its unshifted determinants turning negative at order nine, so no moment-theoretic or fully Hankel-positive route can succeed, while the order-two inequality $b_{n-1}b_{n+1} > b_n^2$ survives every test, through $n = 38$ at $d = 2$ and $n = 23$ at $d = 4$. The property also persists at the opposite boundary: as $d \to \infty$ the normalized ratios appear to tend, for fixed m, to $e^{-2}(2m+3)^{2m+2}/((2m+1)^{2m+1}(2m+2))$, itself strictly increasing to 1, and the exact ladders at $d = 100$ and $d = 200$ approach that limit monotonically. In exact scans, the first twenty-five ratios of every even equal-degree ladder through $d = 30$ and the early ratios of hundreds of mixed blocks showed no violation, and an independent recomputation of sixteen blocks at five exact ratios each found none; the Euler ladder at $d = 2$, whose early ratios dip before the monotone regime of Conjecture 117 sets in, shows that such global monotonicity is not automatic.

Conjecture 196 (Dimension-wise signature gcd law)**.** *Let the gcd range over all Calabi–Yau complete-intersection multidegrees in even complex dimension n. Then*

$$\gcd_{\mathbf{d}} S(\mathbf{d}) = \begin{cases} 2, & n \equiv 0 \pmod 4, \\ 2^{\max(4,\,1+\nu_2(n+2))}, & n \equiv 2 \pmod 4, \end{cases}$$

where ν_2 is the 2-adic valuation.

Significance. This is the signature sibling of Conjecture 121's Euler gcd law: a universal divisibility constraint on index data across each dimension, with an unexpectedly rigid 2-adic dependence on $n+2$ whose spike 2^5 at $n = 14$ the census confirms exactly, and whose $n \equiv 2 \pmod 4$ floor 2^4 contains Rokhlin's theorem at $n = 2$; free group actions and anomaly normalizations inherit the constraint. A proof route is a 2-adic congruence in the L-genus characteristic series followed by a small family attaining the gcd. A first decisive theorem would be the divisibility direction alone. In tests on the corpus described in the stress-test section, the exact gcd over the full census matched the formula in every even dimension $2 \le n \le 26$, all thirteen values reproduced independently. A counterexample would be one even dimension with a different gcd.

Part XI
Spectral entropies and multibase carry fields

Added to the deposit: 13 August 2026.

The twenty conjectures below pair a spectral programme on graphs with a return to the integers. An entropy programme (Conjectures 197 to 201) reads the Rényi entropies of the trace-normalized graph Laplacian, the density-matrix reading of a graph introduced in [117], across tree and unicyclic operations, asserting that a small family of local moves is entropy-monotone at every order at once. A metric programme (Conjectures 202 to 207) poses the same functionals for the distance signless Laplacian of [118], an object whose spectral theory is extensively surveyed in [119] but whose normalized entropy appears unstudied. A carry programme (Conjectures 208 to 216) returns to the local–global aesthetic of Part I on a new object: the carry fields $V_p(n) = v_p\binom{2n}{n}$ at distinct primes, finite-state functionals of the base-p digits by Kummer's theorem, for which complete cross-base decoupling is conjectured at Gaussian, local, modular, large-deviation, and cumulant resolution. The calibration layer is proved where it can be: the degree identity at Rényi order two with its tree and leaf-move corollaries and the complete-graph majorization law form Proposition 32, and the exact single-base carry-chain law forms Proposition 33. The adversarial controls marking the genuine boundaries of the entropy windows are retained in place. The spectral scans are floating-point eigensolver checks and are flagged as such, with the smallest margins near 10^{-4}; the carry statements are limit laws beyond any finite check and are flagged as such, with Conjecture 216, the transducer principle, the part's most exposed statement; and Conjectures 197 and 208 carry elevated attribution risk, sitting closest to [120] and [121].

For a finite graph G with $m \geq 1$ edges, let $L(G)$ be the combinatorial Laplacian and $\rho_L(G) = L(G)/(2m)$ its trace normalization, a density matrix whose eigenvalues $p_1, \ldots, p_n$ are nonnegative with unit sum; for $\alpha > 0$, $\alpha \neq 1$, the Rényi entropy is $H^L_\alpha(G) = (1-\alpha)^{-1} \log \sum_i p_i^\alpha$, with the Shannon limit $H^L_1 = -\sum_i p_i \log p_i$ at $\alpha = 1$ and min-entropy $H^L_\infty = -\log \max_i p_i$ at $\alpha = \infty$, zero eigenvalues contributing nothing. For a connected G, let $\mathcal{D}$ be the distance matrix, $t(v)$ the transmission $\sum_u d(u,v)$, $W(G) = \frac{1}{2}\sum_v t(v)$ the Wiener index, and $Q_D(G) = \mathrm{diag}(t) + \mathcal{D}$ the distance signless Laplacian [118], positive definite for connected graphs on at least three vertices by the identity $x^T Q_D x = \sum_{i<j} d(i,j)(x_i + x_j)^2$, with trace $2W$; write H^Q_α for the Rényi entropy of $Q_D/(2W)$. On the arithmetic side, for an odd prime p put $V_p(n) = v_p\binom{2n}{n}$, by Kummer's theorem the number of carries in the base-p addition of $n+n$, a bounded-output finite-state functional of the digits that is not p-additive, since each output depends on the incoming carry state; $\mu_p(N)$ and $\sigma_p^2(N)$ denote the mean and variance of V_p over uniform $n < N$.

Proposition 32 (Degree and majorization calibrations)**.** (i) For every graph with m edges and degrees d_i, $H^L_2(G) = -\log((2m + \sum_i d_i^2)/(2m)^2)$; consequently, among trees of a fixed order the path uniquely maximizes H^L_2, and if a leaf x adjacent to u is moved to a neighbour v of u with $d(v) \geq d(u)$, then H^L_2 strictly decreases. (ii) For every connected n-vertex graph G and every $\alpha > 0$, $H^Q_\alpha(G) \leq H^Q_\alpha(K_n)$. (iii) For a tree T on $n \geq 3$ vertices, joining two leaves or subdividing one edge strictly increases H^L_2, and each move also strictly increases H^L_α for all sufficiently small $\alpha > 0$; and among trees of a fixed order the path uniquely maximizes H^L_∞.

Proof. (i) $\mathrm{tr}\, L^2 = \sum_i d_i^2 + 2m$, since the diagonal contributes $\sum d_i^2$ and the off-diagonal entries contribute 1 for each of the $2m$ ordered adjacent pairs, and $H^L_2 = -\log(\mathrm{tr}\, L^2/(2m)^2)$. Among trees $m = n-1$ is fixed, and the path is the unique tree minimizing $\sum d_i^2$. The leaf move changes $\sum d_i^2$ by $(d_u-1)^2 + (d_v+1)^2 - d_u^2 - d_v^2 = 2(d_v - d_u + 1) > 0$. (ii) The all-ones vector gives $\lambda_{\max}(Q_D) \geq \mathbf{1}^T Q_D \mathbf{1}/n = 4W/n$, so the largest normalized eigenvalue of any connected graph is at least $2/n$. The normalized spectrum of K_n is $(2/n, (n-2)/(n(n-1)), \ldots)$ with a flat tail, and a probability vector whose largest entry is at least $2/n$ majorizes every such flat-tail vector: if the top entry is $p_1 \geq 2/n$, the k-th partial sum exceeds the flat-tail partial sum by at least $(p_1 - 2/n)(1-(k-1)/(n-1)) \geq 0$ plus the excess of the remaining entries over their flat minimum. Schur concavity of Rényi entropy finishes. (iii) Write $A = \operatorname{tr} L^2 = 2(n-1) + \sum_i d_i^2$, so $A \geq 6n-8$ with equality for the path. Joining two leaves gives $\operatorname{tr} L'^2 = A + 8$ on n edges, and H_2^L increases exactly when $8(n-1)^2 < A(2n-1)$, which holds since $(6n-8)(2n-1) - 8(n-1)^2 = 2n(2n-3) > 0$; subdividing an edge gives $\operatorname{tr} L'^2 = A + 6$ on n edges, and the required $6(n-1)^2 < A(2n-1)$ holds since $(6n-8)(2n-1) - 6(n-1)^2 = 6n^2 - 10n + 2 > 0$. For the smallest orders, the matrix-tree theorem gives $\prod \lambda_i = n\tau$ over positive eigenvalues: subdivision raises the number of positive eigenvalues from $n-1$ to n, and closure into a cycle of length $c \geq 3$ keeps that number while raising the mean of $\log(\lambda_i/2m)$ by $(\log c - (n-1)\log\frac{n}{n-1})/(n-1) > 0$, since $(n-1)\log\frac{n}{n-1} < 1 < \log 3$, so expanding H_α at $\alpha = 0$ finishes both. For min-entropy, a non-path tree has a vertex of degree $d \geq 3$, and the vector equal to d there, -1 on its neighbours, and 0 elsewhere has Rayleigh quotient at least $d+1 \geq 4$, while $\lambda_{\max}(P_n) = 2 + 2\cos(\pi/n) < 4$; equal traces $2(n-1)$ convert this into $H_\infty^L(P_n) > H_\infty^L(T)$. □

Proposition 33 (Single-base carry-chain law)**.** For uniform base-p digits, the carry sequence in the addition $n+n$ is a two-state Markov chain with $P(0 \to 1) = (p-1)/(2p)$, $P(1 \to 1) = (p+1)/(2p)$, stationary carry probability $\frac{1}{2}$, and second eigenvalue $1/p$; the stationary variance per digit is $(p+1)/(4(p-1))$; and over the exact block $0 \leq n < p^k$,

$$\mathbb{E}V_p = \frac{k}{2} - \frac{1-p^{-k}}{2(p-1)}.$$

Proof. With carry in c and uniform digit a, the carry out is 1 exactly when $2a + c \geq p$, giving the two transition rows, whose difference makes the second eigenvalue $(p+1)/(2p) - (p-1)/(2p) = 1/p$ and whose symmetry fixes the stationary law at $\frac{1}{2}$. The stationary autocovariance of the carry indicator at lag k is $\frac{1}{4}p^{-k}$, so the variance per digit is $\frac{1}{4}(1 + 2\sum_{k\geq 1} p^{-k}) = (p+1)/(4(p-1))$. Over the exact block the digits are independent and uniform, so the carry probability after j steps satisfies $\pi_{j+1} = (p-1)/(2p) + \pi_j/p$ with $\pi_0 = 0$, giving $\pi_j = (1-p^{-j})/2$, and summing $j = 1$ to k gives the displayed mean. □

45 Rényi entropies of the Laplacian density matrix

Conjecture 197 (All-order path maximality)**.** *For every $n \geq 3$, every n-vertex tree $T \neq P_n$, and every $\alpha > 0$, $H_\alpha^L(T) < H_\alpha^L(P_n)$.*

Significance. The Rényi orders form a continuum of Schur-concave probes of the normalized spectrum, and the conjecture asserts that the path is the most mixed tree under every one of them simultaneously, from the support-sensitive small orders through Shannon entropy to the min-entropy of the spectral radius; different orders can favour different spectra, so this is stronger than any single-functional extremality. The order-two and min-entropy cases are proved in Proposition 32, and near order zero the matrix-tree theorem forces every tree to share its mean log-eigenvalue, so the low-order content is that the path maximize the variance of the logarithms of its normalized spectrum; the nearest literature is the star-minimum conjecture and Rényi results of [120], whence the attribution flag; a proof route is a majorization theorem along a refinement of the tree-shift poset of [122]. A first decisive theorem would be the law for caterpillars. In tests on the corpus described in the stress-test section, all 8,775 deposited comparisons through twelve vertices at nine Rényi orders and an independent scan through eleven vertices passed, with smallest margin 2.23×10^{-4}. A counterexample would be one tree and one order beating the path.

Conjecture 198 (Leaf-pair closure law)**.** *Let T be a tree on at least three vertices and u, v distinct leaves. Then $H^L_\alpha(T + uv) > H^L_\alpha(T)$ for every $\alpha > 0$.*

Significance. Arbitrary edge addition can lower these entropies, a phenomenon exhibited already in [120], so the leaf-pair restriction carries real content: closing the unique leaf-to-leaf path into a cycle raises degrees only at two spectrally peripheral vertices and removes one bottleneck, spreading Laplacian mass at every order. The order-two case and the smallest orders are now proved in Proposition 32, promotions; a first decisive theorem on the remaining range would be the Shannon case. In tests on the corpus described in the stress-test section, all 86,982 deposited leaf-pair comparisons through twelve vertices and an independent scan of 14,252 through ten vertices passed, with smallest margin 1.66×10^{-3}. A counterexample would be one tree and one leaf pair with nonincreasing entropy, most plausibly with leaves on very unequal branches.

Conjecture 199 (Edge-subdivision law)**.** *If T' is obtained by subdividing one edge of a tree T, then $H^L_\alpha(T') > H^L_\alpha(T)$ for every $\alpha > 0$.*

Significance. Subdivision inserts a degree-two vertex while preserving the branching skeleton, a controlled spectral refinement rather than an arbitrary enlargement, and the conjecture asserts the refined normalized spectrum is strictly more mixed at every order; the comfortable margins suggest an interlacing-plus-normalization proof. The order-two case and the smallest orders are now proved in Proposition 32, promotions; a first decisive theorem on the remaining range would be the min-entropy case. In tests on the corpus described in the stress-test section, all 70,126 deposited subdivision comparisons through twelve vertices and 11,249 independent ones passed, with smallest margin 6.12×10^{-2}, the largest in the entropy programme. A counterexample would be a broom subdivided at its hub failing at high order.

Conjecture 200 (Leaf-compression window)**.** *Let x be a leaf adjacent to u in a tree T, let $v \neq x$ be a neighbour of u with $d(v) \geq d(u)$, and let T' be the tree with xu deleted and xv added. If $T' \ncong T$, then $H^L_\alpha(T') < H^L_\alpha(T)$ for all $1/2 \leq \alpha \leq 2$.*

Significance. The move concentrates local degree mass onto a no-smaller vertex, and at order two the decrease is the theorem of Proposition 32; the conjecture is that the concentration order persists down to order one half, and the lower boundary is calibrated, not cautious: an explicit order-12 tree violates the same direction at order 0.1 by 7.65×10^{-7}, a control reproduced independently to three digits, so the window cannot extend to all orders; no structural role is claimed for one half itself, a tested safe bound. A first decisive theorem would locate the true critical order. In tests on the corpus described in the stress-test section, all 15,385 deposited window comparisons and an independent scan through ten vertices passed, with smallest margin 9.31×10^{-4}. A counterexample would be one eligible move increasing entropy inside the window.

Conjecture 201 (Unicyclic endpoint law)**.** *Among connected unicyclic graphs on $n \geq 4$ vertices, for every finite $\alpha > 0$, the graph S_n^+ obtained from the star by joining two leaves uniquely minimizes H_α^L and the cycle C_n uniquely maximizes it.*

Significance. The cycle is the maximally homogeneous unicyclic graph and S_n^+ concentrates degree as far as one cycle permits, and the conjecture asserts that these degree extremes remain the entropy endpoints under every finite Rényi lens. The restriction to finite orders is a calibrated boundary found in the present verification: at $n = 4$ the only two unicyclic graphs, C_4 and S_4^+, have Laplacian spectra $\{0, 2, 2, 4\}$ and $\{0, 1, 3, 4\}$ with equal trace and equal largest eigenvalue, so their min-entropies tie exactly and the strict law fails at $\alpha = \infty$; the control is retained in place. Near order zero the minimization half is delicate, since all triangle-cycled unicyclic graphs share their mean log-eigenvalue by the matrix-tree theorem, so uniqueness of S_n^+ there rests on second-order log-spectral information, while the cycle half, with its maximal spanning-tree count, is first-order automatic. A first decisive theorem would be either endpoint at the Shannon order. In tests on the corpus described in the stress-test section, all 6,230 deposited comparisons and an independent exhaustive scan through seven vertices passed at every tested finite order. A counterexample would be a third extremal graph at some finite order.

46 Entropy of the distance signless Laplacian

Conjecture 202 (Tree one-vertex extension law)**.** *For every tree T and every $\alpha > 0$, attaching a pendant vertex anywhere, or subdividing any edge once, strictly increases H_α^Q.*

Significance. Both operations enlarge the metric state space and the Wiener normalization, and in a tree every altered distance passes through a unique path, a global coherence that the conjecture converts into entropy growth at every order, preventing the Perron mode from ever dominating disproportionately. This is the first entropy law posed for the normalized Q_D spectrum, whose unnormalized spectral theory is surveyed in [119]. The smallest orders are automatic, since Q_D is positive definite and both operations raise the support count. A first decisive theorem would be the min-entropy case, a normalized spectral-radius decrease adaptable to graft-transformation tools [123]. In tests on the corpus described in the stress-test section, all 168,168 deposited comparisons through twelve vertices and an independent scan through ten vertices passed, with minimum margin 4.44×10^{-2}. A counterexample would be one tree, one operation, one order.

Conjecture 203 (Low-order extension law)**.** *For every connected graph G and every $0 < \alpha \leq 2$, pendant attachment and single edge subdivision strictly increase H_α^Q.*

Significance. The general-graph shadow of Conjecture 202, with a substantive cutoff: at high orders the entropy is governed by the largest normalized eigenvalue, and metric extensions of graphs with many routes can sharpen that mode. The boundary is calibrated by retained controls: subdivision decreases H_5^Q in an explicit order-30 graph by 1.00×10^{-2} and pendant attachment decreases H_{10}^Q in an order-15 graph by 2.32×10^{-2}, both reproduced independently. Whether 2 is the exact universal cutoff is the sharpest open edge; it is an endpoint of the tested window rather than an explained critical exponent. A first decisive theorem would be the order-two case, an explicit second-moment inequality. In tests on the corpus described in the stress-test section, all 10,884 deposited window comparisons and 7,672 independent ones passed. A counterexample at order two would destroy the clean boundary.

Conjecture 204 (Star maximality)**.** *For every $n \geq 3$, every n-vertex tree $T \neq K_{1,n-1}$, and every $\alpha > 0$, $H_\alpha^Q(T) < H_\alpha^Q(K_{1,n-1})$.*

Significance. The metric programme reverses the Laplacian intuition: for the distance signless matrix the star, not the path, is conjectured to carry the most mixed normalized tree spectrum at every order, metric compression flattening the transmission profile after normalization by total distance. Together with Conjecture 197 this makes the two programmes genuinely different orders on trees. Since positive definiteness gives every tree $H_{0+}^Q = \log n$, the near-zero form of the law is the exact determinant inequality that the star maximize $\det Q_D/(2W)^n$ among trees. A first decisive theorem would be that determinant law or the min-entropy case, a normalized spectral-radius minimization. In tests on the corpus described in the stress-test section, all 8,775 deposited comparisons through twelve vertices and an independent scan through eleven vertices passed, with smallest margin 4.72×10^{-4}. A counterexample would be one tree and one order beating the star, most plausibly as $\alpha \to \infty$.

Conjecture 205 (Bipartite path minimality at the Shannon order, resolved false)**.** *Among connected bipartite graphs on n vertices, P_n uniquely minimizes H_1^Q.*

Significance. Trace normalisation removes total scale. The claim is that the path's extreme metric anisotropy leaves the normalised spectrum most concentrated even against denser bipartite competitors. The Shannon restriction is deliberate because an all-order strengthening already fails among trees at order two. The deposited comparisons were exhaustive through seven vertices and included holdouts through thirty vertices. *Resolution.* The conjecture is false. The deposited fourteen-vertex tree has a degree-five hub carrying five leaves at diameter ten, and its Shannon entropy lies 3.1535496×10^{-5} below the path. This gap is certified by interval arithmetic at one hundred digits. A further exhaustive enumeration of all 3,159 unlabelled trees on fourteen vertices found a sharper witness T_* on vertices $0, \ldots, 13$ with edge set

$$\{\{0,1\},\{1,2\},\{0,5\},\{2,3\},\{3,4\},\{5,6\}\} \cup \{\{6,j\} : 7 \leq j \leq 13\}.$$

Its degree multiset is $(8,2,2,2,2,2,1,1,1,1,1,1,1,1)$, and a sixty-digit computation gives

$$H_1^Q(P_{14}) - H_1^Q(T_*) = 9.10088461876217 \times 10^{-5}.$$

The interval-certified tree remains the formal certificate. The sharper computation shows that it is not extremal even among trees of the same order. The path may remain minimal in narrower classes such as bounded-degree trees or caterpillars. Identifying the true bipartite minimiser is the surviving problem.

Conjecture 206 (Balanced-biclique window maximality)**.** *Among connected bipartite graphs on n vertices, for every $1 \leq \alpha \leq 2$, $H_\alpha^Q(G) \leq H_\alpha^Q(K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil})$, with equality only for the balanced complete bipartite graph.*

Significance. Completing the bipartite metric collapses all distances to two classes and balancing the sides equalizes the transmissions as far as parity permits, and the conjecture asserts this metric regularization maximizes a whole entropy window rather than one moment. A first decisive theorem would be the order-two case, where the entropy is the explicit functional $-\log(\operatorname{tr} Q_D^2/(2W)^2)$ and the claim becomes an extremal inequality between transmissions and squared distances. In tests on the corpus described in the stress-test section, all 5,322 deposited window comparisons and 195 independent exhaustive ones passed, with smallest margin 1.60×10^{-3}. A counterexample would be one bipartite graph and one window order above the balanced biclique.

Conjecture 207 (Unicyclic cycle maximality above the Shannon order)**.** *Among connected unicyclic graphs on $n \geq 4$ vertices, C_n uniquely maximizes H_α^Q for every $\alpha \geq 1$, including $\alpha = \infty$.*

Significance. The cycle is transmission-regular, the most homogeneous unicyclic metric, and from the Shannon order upward that homogeneity is conjectured to dominate the normalized spectral shape. The lower boundary is calibrated, not cautious: at $n = 4$ a noncycle unicyclic graph exceeds the cycle at order 0.1 by 6.62×10^{-4}, a retained control reproduced independently to three digits, so the law genuinely fails below some critical order. A first decisive theorem would be the Shannon case; locating the critical order $\alpha_c(n)$ is the sharper target. In tests on the corpus described in the stress-test section, all 2,245 deposited comparisons and an independent exhaustive scan through seven vertices passed on the asserted range. A counterexample would be a noncycle maximizer at some order at least one.

47 Multibase carry fields

Conjecture 208 (Cross-prime carry central limit law)**.** *Fix distinct odd primes $p_1, \ldots, p_r$. For n uniform on $\{0, \ldots, N-1\}$, the vector with coordinates $(V_{p_i}(n) - \mu_{p_i}(N))/\sigma_{p_i}(N)$ converges in distribution to a standard r-dimensional Gaussian, with $\mu_p(N) = \frac{1}{2}\log_p N + O_p(1)$ and $\sigma_p^2(N) = \frac{p+1}{4(p-1)}\log_p N + O_p(1)$.*

Significance. Each carry field is generated by the two-state automaton of Proposition 33 reading base-p digits, and distinct primes give multiplicatively independent digit partitions of the same integers; the conjecture asserts that after normalization the only persistent dependence is within a base. It sits closest to the different-base theorem of [121] for q-additive functions, whence the attribution flag, but carries are state-dependent transducer outputs, not additive functions, which is exactly why that theorem does not settle it; the single-prime boundary is [124]. A first decisive theorem would be two primes via a joint transfer-operator estimate. In tests on the corpus described in the stress-test section, marginals at $N = 2 \times 10^6$ track the proved chain law and all pairwise correlations among $p \in \{3, 5, 7, 11\}$ lie below 0.08, independently below 0.04. A counterexample would be a persistent limiting correlation.

Conjecture 209 (Bounded cross-prime covariance)**.** *For distinct odd primes $p \neq q$, $\operatorname{Cov}_{n<N}(V_p(n), V_q(n)) = O_{p,q}(1)$, so the correlation is $O_{p,q}(1/\log N)$.*

Significance. Vanishing correlation is all the central limit law needs; bounded covariance says much more, that every logarithmically growing contribution cancels; the filtrations share no exact scale, yet Diophantine approximation supplies infinitely many near-alignments $p^a \approx q^b$, so boundedness requires genuine cancellation across near-resonant scales rather than mere absence of exact resonance, a sharpening of emphasis. This is the part's crispest analytic target. A first decisive theorem would be the pair $(3, 5)$. In tests on the corpus described in the stress-test section, deposited covariances through $N = 2 \times 10^6$ stay below 0.24 against marginal variances of several units and growing, and an independent run reproduced the smallness. A counterexample would be logarithmic covariance growth with any coefficient, which would leave the central limit law intact but refute this sharper statement.

Conjecture 210 (Cross-prime local limit factorization)**.** *Fix distinct odd primes $p_1, \dots, p_r$. For every fixed C, uniformly for integers k_i with $|k_i - \mu_{p_i}(N)| \le C\sqrt{\log N}$,*

$$\Pr\big(V_{p_i}(n) = k_i \ \forall i\big) = (1 + o(1)) \prod_i \frac{1}{\sqrt{2\pi\sigma_{p_i}^2(N)}} \exp\Big(-\frac{(k_i - \mu_{p_i}(N))^2}{2\sigma_{p_i}^2(N)}\Big).$$

Significance. A central limit law does not determine individual lattice masses, and the conjecture asserts decoupling at one-cell resolution, the transducer analogue of the local limit theorems of the digit-sum literature. The most likely obstruction, arithmetic oscillation of the joint mass with N modulo powers of $p_1 \cdots p_r$, is exactly what a refutation should exhibit. A first decisive theorem would be the pair $(3, 5)$ at the bivariate centre. The statement is a limit law beyond any finite check and is flagged as such; the deposited diagnostics at $N = 2 \times 10^6$ are consistent with factorization at the centre. A counterexample would be a persistent excess or deficit at one lattice cell.

Conjecture 211 (Modular equidistribution of carry vectors)**.** *For distinct odd primes p_i and fixed moduli $m_i \ge 2$, the vector $(V_{p_1}(n) \bmod m_1, \dots, V_{p_r}(n) \bmod m_r)$ equidistributes on $\prod_i \mathbb{Z}/m_i\mathbb{Z}$ as n ranges uniformly below $N \to \infty$.*

Significance. This is the Fourier face of cross-base mixing: bounded-frequency obstructions that the central limit law cannot see, with a proof amounting to spectral contraction of every nontrivial product character of the carry outputs. A surviving residue bias would expose a spectral obstruction of the kind Conjecture 216 seeks to exclude, though possibly subtler than an exact common periodic factor; a sufficiently uniform form of Conjecture 210 would imply this law by summation over residue classes. A first decisive theorem would be one prime pair at modulus two. In tests on the corpus described in the stress-test section, deposited discrepancies at $N = 2 \times 10^6$ fall below 10^{-3} for moduli two and three, reproduced independently at 2.5×10^{-3} and better, with percent-level finite-size effects at modulus four recorded as the honest tail. A counterexample would be one residue vector with persistent bias.

Conjecture 212 (Additive large deviations)**.** *Let I_p be the large-deviation rate function per base-p digit of the carry fraction of the chain of Proposition 33. For fixed distinct odd primes, the vector $(V_{p_i}(n)/\log_{p_i} N)_i$ satisfies a large-deviation principle at speed $\log N$ with rate $J(x) = \sum_i I_{p_i}(x_i)/\log p_i$ throughout the interior of the attainable region.*

Significance. Central-limit independence is a quadratic statement; additive rates assert that even exponentially rare carry densities carry no leading cross-prime interaction energy, with the factors $1/\log p_i$ converting digit counts to the common speed. The restriction to the interior is essential and deliberate: at the extreme boundary sit simultaneous near-carry-free events, where the Diophantine entanglement quantified in [125] lives, and the conjecture makes no claim there.

No claim is made that the cumulant programme implies it, since exponential tilting probes digital sets that no fixed cumulant order controls. A first decisive theorem would be an upper bound at one interior point for two primes. The statement is a limit law beyond any finite check and is flagged as such. A counterexample would be a rate defect at an interior point.

Conjecture 213 (Bounded mixed cumulants)**.** *For fixed distinct odd primes, every fixed joint cumulant of total order at least three involving at least two distinct primes remains $O(1)$ as $N \to \infty$.*

Significance. Even single-base cumulants grow linearly in the digit length, while the carry chain's complement symmetry, which exchanges carries and non-carries, caps every odd single-base cumulant at a bounded boundary term, a correction; the conjecture asserts that every mixed connected correlation is likewise bounded, which together with Conjecture 209 would give a connected-correlation proof scheme for the whole Gaussian decoupling rather than an endpoint statement. A first decisive theorem would be $o(\log N)$ for one mixed third cumulant. In tests on the corpus described in the stress-test section, deposited mixed third and fourth cumulants stay of order one while marginal even cumulants grow, reproduced independently, evidence that is honest but noisy at this depth. A counterexample would be one mixed cumulant growing like $c \log N$.

Conjecture 214 (Fixed-multiplier carry vectors)**.** *Fix an integer $a \geq 2$ and distinct primes $p_1, \ldots, p_r$ such that $\mathrm{Var}_{n<N}\, v_{p_i}\binom{an}{n} \to \infty$ and the accessible carry-state chain of each base-p_i addition automaton is irreducible and aperiodic under independent uniform digits. Then the standardized vector of these valuations converges to a standard Gaussian vector.*

Significance. By Kummer's theorem, $v_p\binom{an}{n}$ counts the carries in adding n and $(a-1)n$ in base p, a transducer with more internal state than the doubling automaton, and the conjecture asserts the multibase decoupling mechanism survives the enrichment. The same-base quasi-additive framework of [126] reaches joint laws in one base; the cross-base claim is the new content, and the mixing hypothesis is stated explicitly, since variance growth alone does not guarantee the automaton regularity a clean central limit law needs, though the multiplier automata plausibly satisfy it whenever the variance diverges. A first decisive theorem would be $a = 3$ with the pair $(3, 5)$. In tests on the corpus described in the stress-test section, deposited correlations for $a \in \{3, 4, 5\}$ at $N = 5 \times 10^5$ stay below 0.11, independently below 0.04 at $a = 3$. A counterexample would be a multiplier resonance with persistent correlation.

Conjecture 215 (Balanced factorial-ratio vectors)**.** *Let $R(n) = \prod_j (a_j n)! / \prod_k (b_k n)!$ be integer-valued with $\sum_j a_j = \sum_k b_k$. For distinct primes p_i with $\mathrm{Var}_{n<N}\, v_{p_i}(R(n)) \sim c_{p_i} \log N$, $c_{p_i} > 0$, the standardized vector $(v_{p_i}(R(n)))_i$ converges to a standard Gaussian vector.*

Significance. Legendre's formula writes each valuation as a balanced combination of digit sums, a finite carry functional whose deterministic linear part cancels by balance, leaving logarithmic fluctuations; the conjecture asserts cross-prime decoupling for the entire classical family of integer factorial ratios, whose same-base divisibility theory is a central application of [126]. A first decisive theorem would be one Catalan-like ratio at two primes. In tests on the corpus described in the stress-test section, the Landau ratio $(30n)!\, n!/((15n)!(10n)!(6n)!)$ at $p \in \{7, 11\}$ shows the predicted logarithmic variance growth with correlation -0.016 in an independent run at $N = 10^5$.

A counterexample would be a ratio with an exceptional congruence structure forcing a shared automatic factor.

Conjecture 216 (Multibase transducer principle)**.** *Let $q_1, \ldots, q_r \geq 2$ be jointly multiplicatively independent bases, meaning $\prod_i q_i^{a_i} = 1$ with integer exponents only when every $a_i = 0$, and let $F_i(n)$ be the total output of a deterministic finite-state transducer reading the base-q_i digits of n, with bounded transition outputs, such that (i) under independent uniform digits the accessible state chain is irreducible and aperiodic, (ii) $\mathrm{Var}_{n<N} F_i(n) \sim c_i \log N$ with $c_i > 0$, and (iii) no two of the centred functionals factor, up to a bounded coboundary, through a common nonconstant eventually periodic function of n. Then the standardized vector $(F_i(n))_i$ converges to a standard Gaussian vector, and every fixed mixed cumulant involving at least two bases is $O(1)$.*

Significance. This is the structural umbrella over Conjectures 208, 214, and 215: joint multiplicative independence of the bases, mixing of each automaton, and the exclusion of a common periodic factor are conjectured to be the complete obstruction list for finite-state observables, the class where the additive theory of [121] stops. Joint independence is genuinely stronger than the pairwise form of the deposited statement, a repair: the bases $2, 3, 6$ are pairwise independent yet satisfy $\log 6 = \log 2 + \log 3$, exactly the kind of higher relation a three-base mixed cumulant could feel; the two conclusions are also of different strengths, since a mixed cumulant can grow slowly and still vanish after Gaussian normalization. The exclusion hypothesis is necessary in spirit, since two automata in different bases can be engineered to emit the same periodic statistic, and sharpening its exact form is part of the problem; the statement is the part's most exposed and is flagged as a programme conjecture. A first decisive theorem would be the two-base two-state case, essentially Conjectures 208 and 209. A counterexample satisfying the three hypotheses would identify a genuinely new obstruction beyond periodicity.

Part XII
Subdivision spectra, spanning-tree correlation, and information dimension

Added to the deposit: 14 August 2026.

The twenty conjectures below run four programmes at the common interface of spectral theory and information. A subdivision programme (Conjectures 217 to 221) asks what weak spectral universality under barycentric refinement, proved at the graph level by Knill [127, 128], in every Hodge degree for the barycentric rule by Knill [129], and for top-dimensional Laplacians of general inclusion-uniform rules by Märte [130], does *not* see: exponentially rare high eigenvalues, organized by a factorial hierarchy of critical moment exponents. A spanning-tree programme (Conjectures 222 to 226) introduces the total correlation of the uniform-spanning-tree edge process, an invariant assembled from Kirchhoff marginals and the matrix-tree theorem in the tradition of transfer impedances [132], and conjectures its extremal graphs, led by a mesoscopic core-periphery shape principle at scale $\sqrt{n/\log n}$. A cyclic-entropy programme (Conjectures 227 to 231) starts beyond the prime-field entropy-doubling floor proved this year by Gavalakis, Goh, and Kontoyiannis [135] and conjectures the full inverse theory: localization at near-equality, Gaussian rigidity, a heat-kernel crossover profile at the wrap scale, and uniform many-summand monotonicity. A graphon programme (Conjectures 232 to 236) conjectures that the Shannon entropy of a sampled random-free graph, known to be subquadratic of essentially arbitrary growth [141], is governed at scale $n \log n$ exactly by the information dimension of the latent row space. The calibration layer is proved where it can be: block additivity and the exact cycle law for the tree invariant form Proposition 34, and the threshold-graphon entropy bound that anchors the second-order programme forms Proposition 35. The adversarial controls marking genuine boundaries are retained in place: the bipartite crossover at seventeen vertices in Conjecture 223, the separated-bump doubling penalty in Conjecture 227, the onset of pre-wrap failures in Conjecture 231, and the threshold graphon itself as the permanent negative control of Conjecture 234. All graph and spectral scans are floating-point checks and are flagged as such. Three statements carry elevated exposure or attribution risk and are flagged where they occur: Conjecture 220, whose large-deviation law has no direct numerical support at reachable depth and is the part's most exposed statement; Conjecture 219, which interpolates two published theorems [130, 129] and is flagged as an attribution risk; and Conjecture 230, whose integer-lattice analogue is a recent theorem [140], so that only the modular content is new.

Throughout, $\operatorname{sd}^m K$ denotes the m-fold barycentric subdivision of a finite pure d-dimensional simplicial complex K, realized as the order complex of the face poset; Δ_k is the unweighted Hodge Laplacian on k-simplices, f_k the number of k-simplices, and for the graph Laplacian L_m of the 1-skeleton, $M_\alpha(m) = \operatorname{Tr}(L_m^\alpha)/|V(\operatorname{sd}^m K)|$; we write $A_d = (d+1)!$, $B_{d,0} = d!$, and $B_{d,k} = (d-k+1)!$ for $1 \le k < d$. For a connected simple graph G, $\tau(G)$ is the number of spanning trees, T a uniform spanning tree, and R_e the unit-conductance effective resistance across e, so that $\Pr(e \in T) = R_e$ by Kirchhoff's theorem; with $h(x) = -x\log x - (1-x)\log(1-x)$, the *spanning-tree correlation* is

$$\mathcal{C}(G) = \sum_{e \in E(G)} h(R_e) - \log \tau(G),$$

the total correlation of the edge-indicator vector of T, nonnegative and zero exactly on trees, since $H(T) = \log \tau(G)$; $S_{n,r} = K_r \vee \overline{K}_{n-r}$ denotes the complete split graph. On the cyclic side, laws μ_p on $\mathbb{Z}/p\mathbb{Z}$ with p prime are called *mesoscopic* when $H(\mu_p) \to \infty$ and $\log p - H(\mu_p) \to \infty$; convolution powers are written μ_p^{*m}. A graphon W is *random-free* when it is $\{0,1\}$-valued almost everywhere; $G(n, W)$ is the sampled graph on n vertices, $d_W(x,y) = \int_0^1 |W(x,u) - W(y,u)|\,du$ the neighborhood distance, and (Ω_W, d_W, ρ) the *purified row space*, the completion of the quotient of $[0,1]$ by d_W-null pairs carrying the pushforward of Lebesgue measure. The measure ρ satisfies a *uniform local doubling bound* when there are C and $\varepsilon_0 > 0$ with $\rho(B(x, 2\varepsilon)) \le C\rho(B(x,\varepsilon))$ for all x in the support and $\varepsilon \le \varepsilon_0$, and is *exact-dimensional of dimension s* when $\log \rho(B(x,\varepsilon))/\log \varepsilon \to s$ for ρ-almost every x.

Proposition 34 (Block additivity and the cycle law)**.** (i) $\mathcal{C}$ is additive over the blocks of a connected graph, and bridges contribute zero. (ii) $\mathcal{C}(C_n) = (n-1)\log \frac{n}{n-1}$, strictly increasing in n; consequently, subdividing an edge that lies on exactly one cycle strictly increases $\mathcal{C}$.

Proof. (i) A subgraph is a spanning tree of G exactly when its intersection with every block is a spanning tree of that block, so the uniform spanning tree is an independent product across blocks and $\log \tau(G) = \sum_B \log \tau(B)$; each edge lies in exactly one block, so the marginal sum also splits, and a bridge has $R_e = 1$, contributing $h(1) = 0$ and a factor 1 to τ. (ii) In C_n every edge has $R_e = (n-1)/n$ and $\tau = n$, so $\mathcal{C}(C_n) = n\,h\!\left(\frac{n-1}{n}\right) - \log n = (n-1)\log \frac{n}{n-1}$, which equals $\log\!\left(1 + \frac{1}{n-1}\right)^{n-1}$ and is strictly increasing because $(1+1/x)^x$ is. An edge lying on exactly one cycle has a cycle C_ℓ as its block; subdivision replaces that block by $C_{\ell+1}$ and changes nothing else, so $\mathcal{C}$ increases by $\mathcal{C}(C_{\ell+1}) - \mathcal{C}(C_\ell) > 0$. ☐

Proposition 35 (Threshold-graphon entropy bound)**.** For the random-free graphon $W(x,y) = \mathbf{1}\{x + y \geq 1\}$, every realization of $G(n,W)$ is a threshold graph, the law of $G(n,W)$ is supported on at most $n!\,2^{n-1}$ labeled graphs, and hence

$$H(G(n,W)) \leq n \log n - (1 - \log 2)\,n + O(\log n),$$

although the purified row space is $[0,1]$ with the metric $|x-y|$ and Lebesgue measure, whose scale-$1/n$ quantization entropy is $\log n$ per point with vanishing linear correction.

Proof. Sample $x_1, \ldots, x_n$ uniformly, and set $y_i = |x_i - \frac{1}{2}|$ and $s_i = \mathbf{1}\{x_i > \frac{1}{2}\}$. Almost surely the y_i are distinct, and $x_i + x_j \geq 1$ holds exactly when $s_i = s_j = 1$, or the signs differ and the plus-signed vertex has the larger y. Adding vertices in increasing order of y, each new vertex is adjacent to all earlier ones when its sign is plus and to none when its sign is minus: a creation sequence, so the realization is a threshold graph, and the labeled graph is a function of the y-ordering and the signs. The sign of the first vertex added affects no edge, so at most $n!\,2^{n-1}$ labeled graphs occur, and $H \leq \log n! + (n-1)\log 2 = n \log n - (1 - \log 2)n + O(\log n)$. The row space assertion is immediate: rows of W are indicators of intervals $[1-x, 1]$, so $d_W(x,y) = |x-y|$ and ρ is Lebesgue measure, whose $1/n$-quantization entropy is $\log n + o(1)$ per point. ☐

48 Spectral tails of barycentric subdivision

Conjecture 217 (Critical moment exponent for barycentric vertex Laplacians)**.** *Let K be a finite pure d-dimensional simplicial complex, $d \geq 2$, with at least one d-simplex. For every $\alpha > 0$,*

$$\lim_{m \to \infty} \frac{1}{m} \log M_\alpha(m) = \max\{0,\ \alpha \log d! - \log(d+1)!\},$$

and at the critical order $\alpha_c(d) = \log(d+1)!/\log d!$ the moment $M_{\alpha_c}(m)$ grows linearly in m up to constants depending on K.

Significance. Weak spectral universality under barycentric refinement is a theorem [127, 128], but weak convergence coexists with divergent high moments, and this statement locates the exact divergence threshold, a discrete analogue of a multifractal moment transition. The mechanism is an incidence-age ledger: each refinement multiplies the number of chambers by $(d+1)!$ while the link of a surviving vertex grows by the factor $d!$, so the age-a stratum has frequency about $(d+1)!^{-a}$ and degree scale $d!^{\,a}$, and $\sum_a A_d^{-a} B_{d,0}^{\alpha a}$ changes behaviour exactly at α_c; the analytic content is transferring the ledger from degrees to eigenvalues, for which Schur–Horn majorization gives the lower half. In dimension two the threshold is $\log_2 6 = 2.5849\ldots$, and the failure of the finite all-order Rényi defect that this transition forces is the recorded reason the predecessor claim was withdrawn. A first decisive theorem would be the case $d = 2$, $\alpha = 4$. In tests on the corpus described in the stress-test section, degree-moment growth ratios at $\alpha = 4$ reach 2.675 against the predicted $16/6 = 2.6667$ by seven refinements of the triangle, with the same limit from a two-triangle start and from a non-manifold three-page book, and dense spectral moments through five refinements track the degree ledger. A counterexample would be a starting complex whose supercritical moment rate separates from the factorial prediction.

Conjecture 218 (Hodge incidence-tail hierarchy)**.** *For $0 \le k \le d$ and the Hodge Laplacians Δ_k on $\operatorname{sd}^m K$, the empirical α-moments of Δ_k stay bounded for $\alpha < \alpha_c(d,k) = \log(d+1)!/\log B_{d,k}$, grow polynomially at $\alpha_c(d,k)$, and grow at exponential rate $\alpha \log B_{d,k} - \log(d+1)!$ above it; for $k = d$ all moments stay bounded.*

Significance. This organizes the Hodge spectrum by codimension: a persistent k-incidence stratum sees only the $d-k+1$ remaining directions of a chamber chain, giving the factorial $B_{d,k} = (d-k+1)!$, while Hodge supersymmetry forces adjacent blocks to share their extreme nonzero eigenvalues, which is why $k=0$ and $k=1$ share the tail factor $d!$. The predicted tail-factor pattern in dimension three is $6, 6, 2, 1$. The nearest boundary is the all-degree weak universality of [129], which sees none of this. A first decisive theorem would be boundedness for $k=d$ together with the shared $k = 0, 1$ growth rate in dimension two. In tests on the corpus described in the stress-test section, the tetrahedron probe gives maxima $15 \to 75.06$ in degrees zero and one, $10 \to 15.74$ in degree two, and $7 \to 7.88$ at the top, exactly the predicted pattern, and in dimension two the degree-one maximum doubles per step while the top-degree maximum converges to six. A counterexample would be a Hodge degree whose tail factor depends on finer simplex type than codimension alone.

Conjecture 219 (All-degree inclusion-uniform universality)**.** *For every nontrivial inclusion-uniform subdivision rule* div *on pure d-complexes and every $0 \le k \le d$, the empirical spectral measure of $\Delta_k(\operatorname{div}^m K)$ converges weakly to a limit law depending only on the rule, d, and k, and not on K, apart from asymptotically negligible homological zero modes.*

Significance. This sits at the open product cell of two published theorems and is flagged as an attribution risk on that account: Märte proves universality for the top-dimensional Laplacian of every inclusion-uniform rule [130], and Knill proves it in every Hodge degree for the barycentric rule [129]; the conjecture asserts the common generalization. The mechanism is that inclusion-uniformity makes the interior of every old chamber evolve by one fixed local rule while boundary contributions have exponentially smaller face counts; the delicate point is that lower Hodge blocks touch those boundaries directly, and the claim is that the coupling still carries vanishing spectral density. A first decisive theorem would be the edgewise subdivision in dimension two at $k=0$. In tests on the corpus described in the stress-test section, the barycentric all-degree case is corroborated through five two-dimensional and two three-dimensional refinements; no non-barycentric rule has been machine-tested, and the conjecture is deposited with that gap on record. A counterexample would be an inclusion-uniform rule and a lower Hodge degree retaining positive-density memory of the initial gluing.

Conjecture 220 (Spectral-tail large deviations from simplex age)**.** *Fix $0 \le k < d$ and $0 < \theta < 1$, and let $N_{m,k}(\theta)$ count eigenvalues of $\Delta_k(\operatorname{sd}^m K)$ that are at least $B_{d,k}^{\theta m}$. Then*

$$\lim_{m\to\infty} \frac{1}{m} \log \frac{N_{m,k}(\theta)}{f_k(\operatorname{sd}^m K)} = -\theta \log(d+1)!,$$

uniformly for θ in compact subintervals of $(0,1)$; for $k=d$ there is no exponential tail.

Significance. This is the full tail profile behind Conjectures 217 and 218, whose critical exponents are its Legendre data: an eigenvalue of scale $B^{\theta m}$ should come from an incidence stratum that survived about θm refinements, and such strata number about $A_d^{(1-\theta)m}$ of the

A_d^m total. The hard content is a two-sided correspondence between old incidence and large eigenvalue, not a degree estimate; eigenvector delocalization across age strata is the named failure channel. This is the part's most exposed statement and is flagged as such: at reachable depths the thresholds $B^{\theta m}$ still sit inside the spectral bulk, so the deposited evidence is the mechanism and the moment data of Conjecture 217, not direct tail counts, and the independent recomputation confirms that its shallow-depth tail counts are uninformative rather than confirmatory. A first decisive theorem would be the upper bound for $k = 0$, $d = 2$. A counterexample would be a tail count with a strictly flatter rate than the age ledger at some θ.

Conjecture 221 (Finite-state renormalization for barycentric Hodge limits)**.** *For each $d \geq 2$ there exist an integer $N(d)$, a rational map R_d on $\mathbb{C}^{N(d)}$, and rational functions $\Phi_{d,k}$, $0 \leq k \leq d$, such that the Schur complement of one barycentric refinement onto a generating set of $N(d)$ boundary Green functions is exactly R_d, the iteration of R_d has a unique attracting fixed point on the relevant domain, and the Stieltjes transforms of all limiting Hodge laws $\nu_{d,k}$ are the $\Phi_{d,k}$ of that fixed point; for $d = 1$ the scheme reduces to the known quadratic spectral decimation. The claim fails if the minimal rank of an exact boundary Schur-complement closure grows without bound under refinement.*

Significance. Knill identifies the one-dimensional limit through the quadratic map $z \mapsto 4z - z^2$ and records that no higher-dimensional analogue is known and the higher-dimensional limits are unidentified [127]; a block-Jacobi formulation close in spirit appears in [131] without a finite rational recursion. The conjecture asserts finite-state closure: refinement acts on a finite vector of boundary Green functions by a rational map whose fixed point carries the spectral data, turning qualitative universality into solvable renormalization. The final sentence is the falsifiability clause, added at deposit: the statement stands or falls with the boundedness of the minimal closure rank, a quantity computable in principle at each depth. A first decisive theorem would be finite closure for $d = 2$ at the top degree, where Märte's Schreier-graph decimation [130] provides the strongest existing mechanism. In tests on the corpus described in the stress-test section, no machine evidence bears on closure; the conjecture is deposited as a programme statement on the strength of the one-dimensional precedent alone. A counterexample would be superlinear growth of the minimal boundary rank in any dimension.

49 Total correlation of uniform spanning trees

Conjecture 222 (Complete-split extremality)**.** *Among connected n-vertex graphs, $\mathcal{C}(G)$ is maximized by a complete split graph $S_{n,r}$ for some $r = r_n$, and*

$$r_n \sim \sqrt{\frac{n}{\log n}}, \qquad \max_G \mathcal{C}(G) = n - (1 + o(1))\sqrt{n \log n}.$$

Significance. The invariant $\mathcal{C}$ measures the total dependence of the determinantal edge process whose entropy theory begins with transfer impedances [132] and extends to general determinantal measures in [133]; its extremal theory appears nowhere, and the conjectured optimizer is neither sparse nor a standard dense extremal graph but a mesoscopic core of order $\sqrt{n/\log n}$ joined to a near-independent periphery. The mechanism is the exact ledger $\tau(S_{n,r}) = n^{r-1}r^{n-r-1}$ with core marginal $2/n$ and cross marginal $(n + r - 1)/(nr)$, whose small-marginal expansion gives the two-term deficit $r \log(n/r) + n/(2r)$, balanced exactly at the stated scale. A first decisive theorem would be optimality of split graphs among split-plus-one-edge perturbations. In tests on the corpus described in the stress-test section, the maximizer over all 853 connected seven-vertex graphs is the split family, hill climbs from random starts at twelve and sixteen vertices land exactly on the split optimum, structured core-density, periphery-matching, and two-core adversaries all fall below it at thirty and forty vertices, and exact family optimization to $n = 10^7$ gives $r^*/\sqrt{n/\log n} = 0.92$ and deficit ratio 1.14, both drifting toward one. A counterexample would be any connected graph exceeding every complete split value at its order.

Conjecture 223 (Triangle-free bipartite extremality)**.** *Among connected triangle-free n-vertex graphs, $\mathcal{C}$ is maximized by a complete bipartite graph $K_{r,n-r}$ with r maximizing the exact bipartite formula; the maximizing r satisfies $r \sim \sqrt{n/\log n}$ and the maximum is $n-(1+o(1))\sqrt{n\log n}$.*

Significance. The predecessor claim fixed the small side at two and is false: the retained control is the crossover $\mathcal{C}(K_{3,14}) = 8.35012\ldots > \mathcal{C}(K_{2,15}) = 8.31559\ldots$ at $n = 17$, after which the optimizer migrates through the bipartite family at the same mesoscopic scale as Conjecture 222. Every edge of $K_{r,s}$ has marginal $(n-1)/(rs)$ and $\tau = r^{s-1}s^{r-1}$, so the same two-term deficit appears; the conjecture says triangle-freeness costs exactly the core edges and nothing else at leading order. A first decisive theorem would be optimality within the complete bipartite family. In tests on the corpus described in the stress-test section, both crossover values reproduce to twelve digits in an independent implementation, no triangle-free seven-vertex graph beats the family, and blowups of the five-cycle, the Petersen graph, and edge-diluted bicliques all fall short at ten to thirty vertices. A counterexample would be a triangle-free graph exceeding every complete bipartite value at its order.

Conjecture 224 (Cycle minimum for bridgeless dependence)**.** *Every bridgeless connected n-vertex simple graph satisfies $\mathcal{C}(G) \geq \mathcal{C}(C_n) = (n-1)\log\frac{n}{n-1}$, with equality only for C_n.*

Significance. Once bridges are forbidden, dependence cannot be diluted to zero, and the conjecture identifies the cycle, whose single constraint is to omit exactly one edge, as the unique minimizer; by Proposition 34 the value $\mathcal{C}(C_n)$ increases to one, so the statement is a sharp universal gap for the graphic projection determinantal class rather than a spanning-tree count extremum. Block additivity makes the claim especially plausible at cut vertices, where dependence adds over cyclic blocks. A first decisive theorem would be the two-connected planar case. In tests on the corpus described in the stress-test section, every bridgeless graph in the seven-vertex atlas sits above the cycle, as do all theta graphs through forty vertices and one hundred seventy-nine structured and random bridgeless graphs, including prisms, wheels, hypercubes, complete bipartite graphs, and cubic expanders through sixty vertices, with zero violations. The cactus case is now proved, a promotion: by Proposition 34, a bridgeless cactus with two or more blocks has correlation at least $4\log\frac{3}{2} > 1$ while every cycle stays below one, so the difficulty is concentrated in two-connected non-cycle blocks. A counterexample would be a bridgeless graph at or below the cycle value.

Conjecture 225 (Core-periphery stability at the ceiling)**.** *If connected graphs G_n satisfy $\mathcal{C}(G_n) = n-(1+o(1))\sqrt{n\log n}$, then there are vertex sets A_n with $|A_n| = (1+o(1))\sqrt{n/\log n}$ such that all but $o(n|A_n|)$ of the pairs between A_n and its complement are edges, while the complement spans $o(n|A_n|)$ edges. No structure is asserted for $G_n[A_n]$.*

Significance. This is a stability principle for an information-theoretic extremal problem: near-extremality is asserted to force the mesoscopic core-periphery cut while deliberately leaving the core's internal graph free, because that freedom is real. The deposited calibration shows the optimized split and bipartite families, whose cores differ maximally, separated by only 0.46 at $n = 10^6$ against a leading deficit in the thousands, so no leading-order statement can see inside the core. The nearest boundary is the classical stability paradigm of extremal graph theory, here transposed to a determinantal entropy functional with no known antecedent. A first decisive theorem would be recovering the cut from the two-term deficit expansion under a split-plus-perturbation hypothesis. In tests on the corpus described in the stress-test section, random core densities inside an otherwise optimal split graph move $\mathcal{C}$ by order one while breaking the cross edges moves it at scale $\sqrt{n \log n}$. A counterexample would be a near-extremal family with two well-separated mesoscopic cores or none.

Conjecture 226 (Complete-multipartite reduction)**.** *Among complete multipartite graphs on n vertices, $\mathcal{C}$ is maximized by one part of size $n-r$ together with r singleton parts, that is, by the complete split graph $S_{n,r}$ for the optimal r.*

Significance. This compresses an optimization over integer partitions to one parameter and is the natural stepping stone toward Conjecture 222, since the multipartite class interpolates between the clique and the biclique while admitting the exact ledger $\tau(K_{a_1,\dots,a_k}) = n^{k-2}\prod_i (n-a_i)^{a_i-1}$ and closed-form effective resistances, reducing the claim to a finite family of partition inequalities. A first decisive theorem would be that no partition with two parts of size at least two beats the split form at its own n. In tests on the corpus described in the stress-test section, every complete multipartite partition through $n = 24$ was checked exactly at deposit, a further computation extended that to $n = 30$, and the independent recomputation extends it to all 89,133 partitions at $n = 45$ and every n below, with the split form winning at every order. A counterexample would be a partition with a second macroscopic part beating every split value at some n.

50 Prime-cyclic entropy power and the wrap transition

Conjecture 227 (Localization of near-minimizers on prime cycles)**.** *Let $p \to \infty$ through primes and let μ_p be mesoscopic laws on $\mathbb{Z}/p\mathbb{Z}$ with*

$$H(\mu_p * \mu_p) - H(\mu_p) \to \tfrac{1}{2}\log 2.$$

Then there are affine automorphisms ϕ_p of $\mathbb{Z}/p\mathbb{Z}$ and cyclic intervals I_p of length $o(p)$ with $(\phi_p)_*\mu_p(I_p) \to 1$.

Significance. The floor itself is now a theorem: Gavalakis, Goh, and Kontoyiannis prove the prime-field doubling gain of $\frac{1}{2}\log 2$ up to ε for entropies in a window away from both endpoints [135], a prime-field analogue of Tao's torsion-free inequality, with earlier cyclic entropy inequalities in [134]. The conjecture is the corresponding inverse theorem: asymptotic equality forces the torsion geometry to vanish, leaving the measure inside a Freiman copy of the line, the rank-one case of the bounded-rank progression structure that drives the proof of the floor. A first decisive theorem would be localization under an additional log-concavity hypothesis. In tests on the corpus described in the stress-test section, localized discrete Gaussians sit on the floor to six decimal places at two primes and four widths, while the retained control, an equal mixture of two separated bumps, pays the full extra $\frac{1}{2}\log 2$, reproduced independently to machine precision. A counterexample would be an essentially cyclic near-minimizing family with no asymptotically full affine interval.

Conjecture 228 (Gaussian rigidity after localization)**.** *Under the conclusion of Conjecture 227, choose centered integer lifts X_p with variances $\sigma_p^2 \to \infty$ and assume a uniform standardized $(2+\delta)$-moment bound. If the doubling gain tends to $\frac{1}{2}\log 2$, then $(X_p - \mathbb{E}X_p)/\sigma_p \Rightarrow N(0,1)$ and $H(X_p) - \frac{1}{2}\log(2\pi e\sigma_p^2) \to 0$.*

Significance. This identifies the equality geometry, not just the constant, bridging prime-field inverse structure to continuous entropy-power rigidity. The moment hypothesis is load-bearing and the statement is flagged accordingly: in the continuous setting there are sequences whose doubling constant attains the Gaussian minimum asymptotically while remaining far from Gaussian, so unrestricted rigidity is false [137], and the recent equality-and-stability analysis of Shannon's and Tao's inequalities obtains qualitative stability precisely under mild moment control [136]. The conjecture asserts that after cyclic localization the same moment control suffices without log-concavity. A first decisive theorem would be the entropic convergence under an added log-concavity hypothesis via the discrete stability bounds of [136]. In tests on the corpus described in the stress-test section, Gaussian families achieve both limits simultaneously and no tested non-Gaussian family reaches the floor. A counterexample would be a uniformly integrable near-minimizing family with persistent lattice microstructure blocking the entropy asymptotic.

Conjecture 229 (Heat-kernel wrap crossover, resolved false)**.** *Let μ_p have centered span-one integer lifts with variances σ_p^2 and characteristic functions obeying $|\mathbb{E}\, e^{i\theta(X_p-\mathbb{E}X_p)}| \le e^{-c\theta^2\sigma_p^2}$ for $|\theta| \le \pi$ and a constant $c > 0$ independent of p, and let $\sigma_p = o(p)$, $m_p \to \infty$, $m_p\sigma_p^2/p^2 \to t \in (0,\infty)$. With θ_t the unit-circle heat kernel with Fourier coefficients $e^{-2\pi^2 tj^2}$ and $q_{p,t}$ its discretization, $q_{p,t}(x) \propto \theta_t(x/p)$,*

$$\|\mu_p^{*m_p} - q_{p,t}\|_{\mathrm{TV}} \to 0, \qquad \log p - H(\mu_p^{*m_p}) \to \int_0^1 \theta_t(u)\log\theta_t(u)\,du.$$

Significance. This is the universal profile of the entire transition from line-like Gaussian convolution to Haar equilibrium on the cycle: at Fourier mode j the m-th power of the characteristic value concentrates to $e^{-2\pi^2 tj^2}$ exactly when $m\sigma^2/p^2 \to t$, and the entropy defect becomes the relative entropy of the circle heat kernel from Haar. The comparison is with the *discretized* kernel, since a discrete law cannot converge in total variation to a continuous one; the explicit characteristic-function hypothesis replaces a vaguer local-limit clause at deposit. The nearest boundary is total-variation mixing and cutoff for cycle walks, which concerns the endpoint rather than the profile. A first decisive theorem would be the Gaussian-input case. In tests on the corpus described in the stress-test section, three prime-width pairs collapse onto the profile with total variation at machine roundoff and entropy defects matching the target integral to five decimals, and a lazy-walk input at up to 9,604,801 convolutions has total variation falling from 4.6×10^{-7} to 1.0×10^{-9} as p grows. A counterexample would be an admissible input law with a total-variation limit bounded away from zero at some t. *Resolution.* The conjecture is false as stated: the characteristic-function bound does not exclude rare macroscopic jumps, and a refuting family, a lazy-walk core of $\lfloor\sqrt{p}\rfloor$ steps carrying jumps of size $\lfloor p/3\rfloor$ with probability

$\lfloor\sqrt{p}\rfloor\lfloor p/3\rfloor^{-2}$, satisfies every hypothesis with $c = 1/6$ while its wrap limit is the heat kernel convolved with a compound-Poisson jump law, the mode-three coefficient tending to $e^{-6\pi^2 t}$ against the asserted $e^{-18\pi^2 t}$; an independent transform computation here reproduces the jump prediction to six decimals. A Lindeberg condition at scale p, or a uniform $o(p)$ support bound, restores the intended law and is the recorded repair.

Conjecture 230 (Fixed-m entropy-power law on prime cycles)**.** *For every fixed integer $m \geq 2$ and every mesoscopic sequence μ_p,*

$$\liminf_{p\to\infty}\left[H(\mu_p^{*m}) - H(\mu_p)\right] \geq \tfrac{1}{2}\log m,$$

and asymptotic equality under the hypotheses of Conjecture 228 forces the same Gaussian limit.

Significance. The doubling case is the theorem of [135]; the conjecture is the full envelope in the number of summands, matching the continuous Gaussian law at every m, where iterating the doubling bound reaches only powers of two. The statement is flagged for attribution risk on its lattice side: on $\mathbb{Z}$, a growth floor of $\frac{1}{2}\log N$ for N-fold sums of lattice random variables is a recent theorem of Castellano and Sekatski [140], so the content here is exclusively the modular statement for mesoscopic sequences on the cycle, where torsion could in principle interact with a non-power-of-two number of summands, together with the rigidity clause; sharp cyclic Rényi inequalities in [134] are the older boundary. A first decisive theorem would be $m = 4$ via two doublings and a localization step. In tests on the corpus described in the stress-test section, six families at $m \in \{2, 3, 4, 5, 7, 10\}$ and a three-hundred-draw adversarial mixture search at $m = 3$ never fall below the floor, with the Gaussian sitting on it to six decimals. A counterexample would be a mesoscopic family strictly under $\frac{1}{2}\log m$ at some fixed m.

Conjecture 231 (Uniform pre-wrap entropy monotonicity)**.** *Let μ_p, after an affine automorphism, have integer log-concave lifts with variances σ_p^2 and $H(\mu_p) \to \infty$, and let $M_p \to \infty$ satisfy $\log M_p = o(H(\mu_p))$ and $M_p\sigma_p^2/p^2 \to 0$. Then, uniformly for $1 \leq m \leq M_p$,*

$$H(\mu_p^{*(m+1)}) - H(\mu_p^{*m}) \;\geq\; \tfrac{1}{2}\log\frac{m+1}{m} - o(1/m).$$

Significance. Fixed-m discrete entropy monotonicity for log-concave laws is known on the integers, with gain $\frac{1}{2}\log((m+1)/m)$ up to vanishing error [138] and in higher lattice dimension [139]; the new content is uniformity in a number of summands growing all the way to the wrap scale, connecting the entropic central limit theorem quantitatively to the onset of modular wrapping. The wrap condition is not decorative: the deposited tests record finite-size failures once $M_p\sigma_p^2/p^2$ approaches 10^{-2}, the retained boundary control. A first decisive theorem would be uniformity for M_p up to a fixed power of $H(\mu_p)$. In tests on the corpus described in the stress-test section, three log-concave families at $p = 2{,}000{,}003$ hold the scaled margin nonnegative through one thousand summands at wrap ratio 2.5×10^{-8}, with the Gaussian family sitting at equality to the tested precision. A counterexample would be a log-concave family whose scaled margin turns negative strictly inside the pre-wrap window.

51 Information dimension of random-free graphons

Conjecture 232 (Exact-dimensional entropy identity)**.** *Let W be random-free with purified row space (Ω_W, d_W, ρ). If ρ is exact-dimensional of finite dimension s and satisfies a uniform local* *doubling bound, then*

$$\frac{H(G(n,W))}{n\log n} \longrightarrow s.$$

Significance. Random-free graph entropy is subquadratic of essentially arbitrary growth [141], and the neighborhood metric underlying (Ω_W, d_W, ρ) carries the graphon's estimable dimension theory [142]; the conjecture identifies the $n\log n$ coefficient with the almost-sure local dimension, replacing regularity of the space by regularity of the measure. The mechanism is resolution counting: with n samples, rows are distinguishable at scale $1/n$, and exact dimension prices a typical latent label at $s\log n$ nats. The known random-free graphon with merely linear entropy is consistent rather than adversarial here, since its row measure cannot be exact-dimensional of positive dimension with local doubling; producing such a verification is part of the problem. A first decisive theorem would be the identity under Ahlfors regularity. In tests on the corpus described in the stress-test section, one- and two-dimensional models and their mixtures show row-signature resolution slopes at the predicted dimensions, with the deposit explicitly recording that these are proxies for the coding mechanism and not measurements of $H(G(n,W))$. A counterexample would be an exact-dimensional doubling row measure whose sampled-graph entropy separates from $sn\log n$, for instance through cross-row redundancy of order $n\log n$.

Conjecture 233 (L^1 multifractal average-dimension law)**.** *For random-free W, define $D_\varepsilon(x) = \log(1/\rho(B(x,\varepsilon)))/\log(1/\varepsilon)$ on the purified row space. If ρ satisfies a uniform local doubling bound and $D_\varepsilon \to d(\cdot)$ in $L^1(\rho)$, then*

$$\frac{H(G(n,W))}{n\log n} \longrightarrow \int d(x)\, d\rho(x).$$

Significance. This is the averaging principle behind Conjecture 232 for genuinely multifractal latent geometry: the coefficient is neither a Minkowski dimension nor an essential supremum but the mean pointwise information dimension, with the L^1 hypothesis stated explicitly so the resolution heuristic can be converted to a code-length calculation rather than hidden behind uniform integrability, a repair recorded against the predecessor phrasing. The nearest boundary remains [141], whose flexibility theorem shows the coefficient cannot be universal without a hypothesis of exactly this kind. A first decisive theorem would be the two-atom case, a mixture of components of distinct dimensions. In tests on the corpus described in the stress-test section, mixture models show proxy slopes at the predicted averages of their component dimensions. A counterexample would be a doubling multifractal row measure whose entropy coefficient separates from the mean dimension.

Conjecture 234 (Second-order graph entropy constant)**.** *If the purified row space of a random-free graphon is a compact s-dimensional C^2 metric manifold, boundary allowed, and ρ has a positive C^1 density in local charts, then*

$$\kappa(W) = \lim_{n\to\infty} \frac{H(G(n,W)) - sn\log n}{n}$$

exists and is finite.

Significance. A leading coefficient is a first-order theory; this asserts a second-order constant, the graphon analogue of a quantization or coding constant, able to separate geometries of equal dimension. Crucially, $\kappa(W)$ is *not* conjectured to equal the row-space quantization constant: the predecessor made that identification and Proposition 35 kills it, since the threshold graphon has the unit interval as row space yet its sampled entropy carries the strictly negative linear correction forced by threshold-graph combinatorics [144], the permanent negative control retained here. The constant must therefore mix metric and observational-combinatorial contributions, and identifying its ingredients is part of the problem. A first decisive theorem would be existence for one-dimensional threshold-type row spaces. In tests on the corpus described in the stress-test section, exact enumeration of the threshold law through seven vertices gives support sizes 46, 332, 2,874, and 29,024, matching the labeled threshold-graph counts of [144] exactly, with linear correction already at -0.60 per vertex against the quantization prediction of zero. A counterexample would be a smooth row geometry with oscillatory linear term.

Conjecture 235 (Equipartition for random-free samples)**.** *Under the hypotheses of Conjecture 232, let $G_n \sim G(n, W)$ and $I_n = -\log \Pr(G_n)$. Then $I_n/(n \log n) \to s$ in probability.*

Significance. Entropy is a mean; this is the corresponding Shannon–McMillan statement, placing almost every sampled graph in a typical set of size $\exp((s + o(1))n \log n)$ despite the dependence induced by shared latent coordinates. Equipartition principles for coloured random graph models go back to [145], and graphon-entropy fluctuation work concerns estimators rather than the sample information variable [143]; neither touches the random-free dimension setting. The mechanism is that exact-dimensionality suppresses $n \log n$-scale variation in local coding rates, leaving fluctuations at linear order. A first decisive theorem would be the identity for step-function-free Ahlfors-regular row spaces via a second-moment computation of I_n. In tests on the corpus described in the stress-test section, no direct machine evidence exists at meaningful n, and the statement is deposited as the natural concentration companion of Conjecture 232 with that gap on record. A counterexample would be a family of exceptionally probable isomorphism types carrying positive information-scale mass.

Conjecture 236 (Sparse-observation dimension law)**.** *Reveal each potential edge of $G(n, W)$ independently with probability q_n, condition on the mask, and let O_n be the revealed bits. Under the hypotheses of Conjecture 233, if $nq_n \to \infty$, then*

$$\frac{H(O_n \mid \textit{mask})}{n \log(nq_n)} \longrightarrow \int d(x)\, d\rho(x).$$

Significance. This predicts exactly how latent geometric information degrades under missing data: each vertex is probed by about nq_n revealed neighbours, so the observable resolution moves from $1/n$ to $1/(nq_n)$ while the dimension coefficient survives untouched. Conditioning on the mask is essential, since otherwise the mask's own Bernoulli entropy dominates, a repair recorded against the predecessor phrasing along with the removal of an opaque density condition. The nearest boundary is graphon estimation from partially observed data, which concerns recovery rates rather than information content. A first decisive theorem would be the one-dimensional Ahlfors-regular case at $q_n = n^{-1/2}$. In tests on the corpus described in the stress-test section, sparse-probe proxy slopes reproduce the dense-case dimensions after replacing the probe count, with the same proxy caveat on record. A counterexample would be an admissible reveal rate with an extra information loss of order $n \log(nq_n)$.

Part XIII
Flux torsion, quantum merge monotonicity, and effective string geometry

Added to the deposit: 15 August 2026.

The nineteen conjectures below extend the string-geometric programmes of Part X in five directions. A flux programme (Conjectures 237 to 240) lifts the discrete saturation and threshold laws of Conjectures 177 and 178 from three-forms to every odd degree and opens the integral torsion of the complete flux: full degreewise shape laws for the prime-torsion multiplicities. A mirror programme (Conjectures 241 to 245) runs the merge order of Part X through genuinely quantum invariants, the mirror map, the genus-zero Gopakumar–Vafa numbers [148, 149], and the conifold radius in the flat coordinate, asserting strict monotonicity at every level. A stability programme (Conjectures 246 to 248) counts Bridgeland walls [154, 152], sharpens the Γ-constant of the Brill–Noether construction of [153], and asserts a tame atlas for wall arrangements in the spirit of definable Hodge theory [155]. An effective-geometry programme (Conjectures 249 to 253) posits polynomial condition numbers, in an algebraic gauge of distance to the discriminant, for Ricci-flat metrics near degeneration [156, 157], balanced-metric and Hermitian–Yang–Mills convergence [158, 161, 159, 160], matter spectral gaps [162, 163], and certified heterotic Yukawa couplings [164, 165]. A degeneration programme (Conjectures 254 and 255) closes with special-Lagrangian realization at infinite distance [150, 151] and a Laplace-tower rigidity law [166, 167]. The proved layer is Proposition 36: the classical monotonicity of degree and discriminant scale under merges, and the exact Chern floors of the five threefold configurations. Since deposit it has grown three times, by Proposition 38, the shifted torsion duality, by Proposition 29, whose complete-flux normal form resolves Conjecture 240 true and proves the interval-support clause of Conjecture 239, and by Proposition 37, the subcritical direction of Conjecture 238. One identity is recorded as an uncounted remark rather than a conjecture: the insertion-power formula of Remark 1, whose source construction [146] computes an infinite family of higher operations for the torus that could not be compared in full through the present channel, an attribution risk recorded where it occurs. Conjectures 237 and 238 are stated in a random model and flagged; Conjectures 249 to 255 assert laws beyond any finite check and are flagged as such, the price of aiming at effective theorems in geometric analysis; Conjecture 255 carries the part's sharpest attribution flag, since the small-eigenvalue asymptotics of the induced Kähler degeneration metrics are now a theorem of [168] and only the Ricci-flat, integrally LMHS-indexed content is claimed; and Conjecture 246 is flagged as possibly accessible from the existing wall geometry of [154].

Throughout, the flux notation of Part X is retained: $E_n(R) = \Lambda^* R^n$, and for $H \in \Lambda^r \mathbb{Z}^n$ of odd degree r, $d_H = H \wedge -$, with $b_k(H; R)$ the cohomology dimensions and $g_{n,r}$ the degreewise profile on the Zariski-open locus of maximal wedge ranks, so that $g_{n,3} = g_n$. For the complete flux $H_\Delta^{(n)}$, $t_{p,k}(n)$ counts the p-primary cyclic summands of the integral cohomology in degree k, and $T_p(n) = \sum_k t_{p,k}(n)$. A dimension n is *r-essential* when $g_{n,r}$ is not $(1 + t)$ times the profile one dimension lower, so that a form ignoring one coordinate cannot saturate. On the mirror side, configurations $\mathbf{d}$ and merges are as in Part X; $D(\mathbf{d}) = \prod_i d_i$, $\Lambda(\mathbf{d}) = \prod_i d_i^{d_i}$; for the five one-parameter projective complete-intersection Calabi–Yau threefolds (5), $(4, 2)$, $(3, 3)$, $(3, 2, 2)$, $(2, 2, 2, 2)$, the mirror map $q_{\mathbf{d}}(z) = z + O(z^2)$, the genus-zero Gopakumar–Vafa invariants $N_{\mathbf{d}}(\beta)$, and the A-model Yukawa coupling $K_{\mathbf{d}}(q) = D(\mathbf{d}) + \sum_{\beta\geq 1} N_{\mathbf{d}}(\beta)\beta^3 q^\beta/(1-q^\beta)$ are the classical mirror data [148], and $q_c(\mathbf{d})$ is the smallest positive singularity of $K_{\mathbf{d}}$, the conifold point in the flat coordinate.

Proposition 36 (Merge monotonicity and Chern floors)**.** (i) Under every merge $d_i, d_j \mapsto d_i+d_j-1$, the degree D strictly decreases and the scale Λ strictly increases; in particular the algebraic conifold point $z_c = \Lambda^{-1}$ strictly decreases. (ii) For a complete-intersection Calabi–Yau threefold of multidegree $\mathbf{d}$ in $\mathbb{CP}^{k-1}$, $k = \sum_i d_i$, one has $c_2(X)\cdot H = \frac{1}{2}(\sum_i d_i^2 - k)D$, so the ratios $\mathrm{td}_2(X)\cdot H/H^3$ for the five threefold configurations (5), (4, 2), (3, 3), (3, 2, 2), (2, 2, 2, 2) are 5/6, 7/12, 1/2, 5/12, 1/3.

Proof. (i) $d_id_j - (d_i + d_j - 1) = (d_i - 1)(d_j - 1) > 0$ gives the degree drop. For Λ, the function $g(x) = x\log x$ is strictly convex with $g(1) = 0$, and the pair $(d_i + d_j - 1,\ 1)$ has the same sum as (d_i, d_j) with a strictly larger maximum, so majorization gives $g(d_i + d_j - 1) > g(d_i) + g(d_j)$, which is $\log\Lambda' > \log\Lambda$. (ii) From $c(TX) = (1 + h)^k \prod_i(1 + d_ih)^{-1}$, the degree-one term vanishes by the Calabi–Yau condition $\sum_i d_i = k$, and the degree-two term is $\frac{1}{2}(\sum_i d_i^2 - k)h^2$; multiplying by $H = h$ and integrating against the class of X, of degree D, gives the displayed value, and $\mathrm{td}_2 = c_2/12$ with $H^3 = D$ gives the five ratios 50/60, 56/96, 54/108, 60/144, 64/192. ☐

52 Odd flux and torsion shapes

Conjecture 237 (Odd-degree discrete saturation)**.** *Fix an odd $r \geq 3$ and a finite symmetric $V \subset \mathbb{Z}\setminus\{0\}$ with $\gcd(V) = 1$, and let every coefficient of $H_n \in \Lambda^r\mathbb{Z}^n$ be independent and uniform on V. Then $\Pr[(b_k(H_n;\mathbb{Q}))_k = g_{n,r}] = 1 - O(e^{-cn})$ for some $c = c(r, V) > 0$.*

Significance. This lifts Conjecture 177 from three-forms to every odd degree, the full family in which wedge multiplication is a differential, against the Zariski-generic theory of [113]; the gcd normalization absorbs the scalar-invariance of rational ranks noted in Part X. The mechanism is anti-concentration for the structured minors of wedge multiplication, uniformly in the degree. Both the random model and the exponential clause are flagged. A first decisive theorem would be a uniform positive lower bound at $r = 5$. In tests on the corpus described in the stress-test section, all sixty deposited dense $r = 5$ trials and twenty-four independent trials at $n = 9, 10$ hit a single profile. A counterexample would be an odd degree and a law with failure probability bounded away from zero.

Conjecture 238 (Odd coverage-scale threshold)**.** *Fix odd $r \geq 3$, include each coordinate r-subset independently with probability p_n, and give included monomials independent coefficients from a fixed finite symmetric nonzero set. Along r-essential dimensions, $p_n n^{r-1}/\log n \to \infty$ implies saturation probability tending to one.*

Significance. The isolated-vertex scale of random r-uniform hypergraphs is conjectured to govern saturation in every odd degree, with the r-essential restriction excluding exactly the suspension dimensions where an unused coordinate is harmless, the parity phenomenon noted in Part X now built into the hypothesis. The zero direction is now proved in Proposition 37, a promotion, the essentiality hypothesis being exactly what an unused coordinate violates; the surviving supercritical direction extrapolates beyond any finite check and is flagged, with the random model. In tests on the corpus described in the stress-test section, deposited $r = 5$ sweeps at $n = 9, 10$ rise from no hits to full saturation across the window. A counterexample would be an essential sequence saturating below the scale or failing above it.

Proposition 37 (Odd coverage threshold, zero direction)**.** Fix an odd $r \geq 3$ and an r-essential dimension n. If $p_n n^{r-1}/\log n \to 0$ then the saturation probability of Conjecture 238 tends to zero.

Proof. A coordinate lying in no chosen r-subset makes the profile $(1+t)h$ with h the profile of the form read in the remaining $n-1$ coordinates, and the argument of Proposition 27(i), with $g_{n,r}$ in place of g_n, forces $h = g_{n-1,r}$ and $g_{n,r} = (1+t)g_{n-1,r}$ whenever such a form saturates; that is exactly what r-essentiality excludes. A fixed coordinate is unused with probability $(1-p_n)^{\binom{n-1}{r-1}}$, which is $n^{-o(1)}$ when $p_n n^{r-1} = o(\log n)$, and two coordinates lie in only $O(n^{r-2})$ common r-subsets, so the second-moment estimate of Proposition 27(ii) applies verbatim and unused coordinates exist with probability tending to one. $\square$

Proposition 38 (Shifted torsion duality)**.** For every prime p and all n, k, $t_{p,k}(n) = t_{p,n+3-k}(n)$.

Proof. Write $u_k = b_k(H_\Delta; \mathbb{F}_p) - b_k(H_\Delta; \mathbb{Q})$. Since d_{H_Δ} raises degree by three, the universal coefficient theorem, applied to each residue class modulo three as an ordinary complex, gives $u_k = t_{p,k}(n) + t_{p,k+3}(n)$. The perfect pairing $\Lambda^k \times \Lambda^{n-k} \to \Lambda^n \cong \mathbb{Z}$ intertwines d_{H_Δ} with its own transpose up to sign, since $H_\Delta \wedge \alpha \wedge \beta = \pm\, \alpha \wedge H_\Delta \wedge \beta$, so over every field $b_k = b_{n-k}$ and hence $u_k = u_{n-k}$. Setting $s_k = t_{p,n+3-k}(n)$, the identity in degree $n-k$ reads $s_k + s_{k+3} = u_k = t_{p,k}(n) + t_{p,k+3}(n)$; both sequences vanish for $k \leq 2$, since no differential enters below degree three, and beyond degree n, so induction in steps of three gives $t_{p,k}(n) = s_k$. $\square$

Conjecture 239 (Torsion shape law for the complete flux)**.** *For every prime p and every n, every interior positive triple of $k \mapsto t_{p,k}(n)$ is strictly log-concave.*

Significance. Part X's staircase, Conjecture 183, locates the onset of p-torsion; this asserts the complete degreewise shape of the torsion once present, an integral refinement invisible to every field coefficient, in the twisted-cohomology setting whose integral theory is developed in [146]. The duality centre $\frac{n+3}{2}$ sits half a step above Poincaré symmetry, the shift produced by the degree-three differential. The duality clause is proved in Proposition 38 and the interval-support clause in Proposition 29, which identifies the support as the full range $2p+1 \leq k \leq n-2p+2$; both are now proved, and strict log-concavity is the surviving clause. In tests on the corpus described in the stress-test section, every nonzero profile through $n = 12$ at $p = 2, 3, 5$ passed all three clauses in an independent exact-rank recomputation, including the profiles $1, 10, 44, 10, 1$ at $n = 11$ and $1, 11, 54, 54, 11, 1$ at $n = 12$, and a recomputation from the normal form extends the log-concavity check to every prime $p \leq 23$ through $n = 150$ without a failure. A counterexample would be a flat interior triple at any prime.

Conjecture 240 (Post-onset torsion persistence, resolved true)**.** *For every prime p, $T_p(n+1) > T_p(n)$ for every $n \geq 4p-1$.*

Significance. The staircase of Conjecture 183 never retreats: once p-torsion appears it grows strictly with the dimension, so each prime contributes a permanently expanding sector of torsion flux charge. Conditional on the totals of Conjecture 182, the $p = 2$ case was already a theorem, now sharpened: the increments reduce to $4^{m-1} - \frac{m}{m+1}\binom{2m}{m}$ and its even companion, both positive from $m = 3$ with the small cases checked directly. In tests on the corpus described in the stress-test section, the independent recomputation gives $T_2 = 1, 2, 10, 20, 66, 132$ from $n = 7$ to 12 and $T_3 = 1, 2$ at $n = 11, 12$, and the closed-form $p = 2$ expression increases strictly through $n = 200$. A counterexample would be one prime and one dimension with stagnant total torsion. *Resolution.* The conjecture is true for every prime, not only conditionally at $p = 2$: Proposition 29 gives $T_p(2g+2) = 2T_p(2g+1)$ from the null vector in even dimension and $T_{g+1} - 2T_g = \sum_{r\geq 1} \binom{g}{r-1} 2^{g+1-r} A_r > 0$ from Pascal's rule in odd dimension, both strict after onset. An independent recomputation confirms strict growth for every prime $p \leq 11$ through dimension 150.

Remark 1 (Insertion powers of the complete flux)**.** The integral twisting theory of [146] attaches to a three-form its insertion powers, odd-degree forms driving an extended differential in the manner of Kraines' higher Massey powers [147]. Exact enumeration of the signed regular families on $2m + 1$ labels gives, for $m \leq 4$, the identity that the m-th insertion power of the complete three-form is $(-1)^{m-1}(2m + 1)^{m-1}$ times the complete $(2m + 1)$-form, a Cayley-like coefficient collapsing a sum of 76,545 terms at $m = 4$. Since [146] computes an infinite family of higher operations for the torus that could not be compared in full through the present channel, the identity is recorded here as a verified calibration observation with its attribution unresolved, and is not counted among the conjectures.

53 Quantum monotonicity along the merge order

Conjecture 241 (Normalized mirror-map contraction)**.** *For complete-intersection Calabi–Yau n-folds with $n \geq 3$, write $\hat{q}_{\mathbf{d}}(x) = \Lambda(\mathbf{d})\, q_{\mathbf{d}}(x/\Lambda(\mathbf{d})) = x + \sum_{m\geq 2} c_m(\mathbf{d}) x^m$. Under every merge, $c_m(\mathbf{d}') \leq c_m(\mathbf{d})$ for every $m \geq 2$, with strict inequality for every $m \geq 3$.*

Significance. The mirror map is the transcendental heart of the classical mirror computations [148, 149], and the conjecture asserts that in the scale-free normalization fixed by Λ, whose merge growth is the proved half of Proposition 36, equation consolidation contracts every coefficient at once; the two-fold violations found in the deposited scan are excluded by the dimension hypothesis, and the $m = 2$ equalities at $n = 3$ fix the non-strict boundary exactly. A first decisive theorem would be the $m = 2$ case in every dimension. In tests on the corpus described in the stress-test section, the deposited scan through dimension nine and order eight passed off dimension two, and an independent exact recomputation of all merges in dimensions three to five, 588 comparisons, found zero violations. A counterexample would be one merge and one order breaking the contraction.

Conjecture 242 (Gopakumar–Vafa amplification)**.** *For the five one-parameter threefold configurations and every merge edge, $A_\beta = N_{\mathbf{d}'}(\beta)/N_{\mathbf{d}}(\beta)$ is strictly increasing in $\beta \geq 1$.*

Significance. Coefficientwise growth of curve counts under merges would already be a shape law; monotonicity of the ratio is much sharper, an ordering of the five geometries by quantum growth rate that refines degree-by-degree. The mechanism is the conifold hierarchy of Conjecture 244 propagated backward through Conjecture 243. A first decisive theorem would be the asymptotic ratio divergence from separated conifold radii. In tests on the corpus described in the stress-test section, deposited exact invariants through $\beta = 60$ and an independent from-scratch mirror computation through $\beta = 24$ agree and pass on all five edges, the amplification rising from 1.41

at $\beta = 1$ to 10^{15} scale at high degree on the steepest edge. A counterexample would be one edge and one degree with a falling ratio.

Conjecture 243 (Monotone conifold approach)**.** *For each of the five configurations, $R_\beta = N(\beta+1)/N(\beta)$ is strictly increasing, and $R_\beta \to 1/q_c$; equivalently the invariants are strictly log-convex from degree one with ratio limit the conifold growth constant.*

Significance. That the conifold point governs the exponential growth rate of genus-zero invariants is folklore of the resurgence and enumerative literature; the content here is global monotonicity from the first degree, an unexpectedly rigid convexity of the full quantum series with no preasymptotic dip anywhere in the landscape's simplest column. A first decisive theorem would be log-convexity in large degree from the conifold expansion. In tests on the corpus described in the stress-test section, deposited invariants through $\beta = 60$ and the independent recomputation through $\beta = 24$ are strictly log-convex for all five families, with final tested ratios still below the independently computed constants $1/q_c = 1980.15$, 641.78, 458.89, 269.61, 159.26. A counterexample would be one family and one degree with a falling ratio.

Conjecture 244 (Merge monotonicity of the conifold constant)**.** *For every merge edge among the five configurations, $q_c(\mathbf{d}') < q_c(\mathbf{d})$.*

Significance. The algebraic conifold point $z_c = \Lambda^{-1}$ decreases under merges by Proposition 36; the conjecture asserts that the transcendental image under the mirror map preserves this order, so that consolidation strictly raises the quantum growth constant $1/q_c$. This is a comparison of periods across distinct Picard–Fuchs systems, not a formal consequence of the algebraic ordering. A first decisive theorem would be one edge by interval arithmetic on the hypergeometric sums; indeed the whole five-edge statement is a finite certification target for validated numerics, and should eventually be a computation rather than a conjecture. In tests on the corpus described in the stress-test section, independent sixty-digit evaluations of the five constants order strictly along every edge, with ratios of consecutive constants between 1.69 and 4.32. A counterexample would be one edge with reversed constants.

Conjecture 245 (Conifold-normalized Yukawa monotonicity)**.** *For every merge edge among the five configurations and every $0 < x < 1$, $K_{\mathbf{d}'}(xq_c(\mathbf{d}'))/D(\mathbf{d}') > K_{\mathbf{d}}(xq_c(\mathbf{d}))/D(\mathbf{d})$.*

Significance. At the same fraction of the respective conifold radii, the quantum-corrected Yukawa coupling normalized by its classical intersection number strictly increases under consolidation: a scale-free comparison of quantum corrections across geometries in which both normalizations are essential, since the deposited adversarial control shows the raw instanton comparison reversing at every grid point. A first decisive theorem would be the small-x regime from degree-one invariants alone. In tests on the corpus described in the stress-test section, all 495 deposited grid tests and an independent recomputation on all five edges at nineteen fractions passed, with the raw-instanton control failing everywhere, retained in place. A counterexample would be one edge and one fraction with a reversed normalized coupling.

54 Stability walls and sharp constants

Conjecture 246 (Quadratic wall complexity)**.** *Let X be a Picard-rank-one Calabi–Yau threefold with a constructed component $\mathrm{Stab}^\dagger(X)$ satisfying the support property, K a compact set in a* *fundamental domain for the* $\widetilde{\mathrm{GL}}_2^+(\mathbb{R})$ *action, and* $\|\cdot\|$ *a Euclidean norm on the numerical charge lattice. Then there is* C_K *with* $N_K(v) \leq C_K(1+\|v\|)^2$ *for every primitive charge* v*, where* $N_K(v)$ *counts numerical walls for* v *meeting* K*.*

Significance. Local finiteness of walls is classical [152], and the Picard-rank-one wall geometry of [154] bounds the largest walls and proves finiteness in bounded regions. On a compact set the support property bounds each possible destabilising class by $O_K(\|v\|)$, which gives a polynomial count. In the usual rank-four numerical charge lattice, a naive count after quotienting by the direction of v is cubic rather than quadratic. The exponent two therefore needs genuine Picard-rank-one wall geometry and does not follow merely because a wall is a quadric. A first decisive theorem would isolate the extra geometric restriction that removes one lattice degree of freedom. A counterexample would be a charge sequence with superquadratic wall counts in a fixed compact region.

Conjecture 247 (Sharp Γ for Brill–Noether stability)**.** *For a Picard-rank-one complete-intersection Calabi–Yau threefold covered by the Brill–Noether construction of [153], with* ε_* *the supremum of* ε *for which their condition* $\mathrm{BG3}(\varepsilon)$ *holds, the infimum of* γ *for which* $\Gamma = \gamma H^2 - \mathrm{td}_2(X)$ *satisfies* Γ*-BMT is exactly* $\max\{4/(H^3\varepsilon_*),\ \mathrm{td}_2(X)\cdot H/H^3\}$*.*

Significance. The construction of [153] proves that $\mathrm{BG3}(\varepsilon)$ yields a valid Γ on this ray and leaves its minimization open; the conjecture asserts their bound is already the exact optimum, so the Brill–Noether route loses nothing. The topological floor $\mathrm{td}_2 \cdot H/H^3$ is unconditional, and Proposition 36 computes it as 5/6, 7/12, 1/2, 5/12, 1/3 across the five configurations. A first decisive theorem would be optimality of the floor branch for the quintic. A counterexample would be a valid Γ strictly below the displayed maximum.

Conjecture 248 (Tame wall atlas)**.** *For* X *as in Conjecture 246 and compact* K*, fix a finite effective* $\mathbb{R}_{\mathrm{an,exp}}$ *presentation of the central charge and the numerical wall equations. Let formula format record length, quantifier depth, polynomial degree and logarithmic coefficient height in that presentation. There is* $C = C(X, K)$ *such that the union* W_Q *of numerical walls for primitive charges with* $\|v\| \leq Q$ *has a defining formula of format at most* Q^C*, and* $K \setminus W_Q$ *has a compatible definable stratification into* $O(Q^C)$ *strata and chambers.*

Significance. This imports the definability paradigm of arithmetic Hodge theory [155] into Bridgeland geometry. Qualitative o-minimal definability alone supplies neither a canonical format nor a polynomial chamber bound. The statement therefore fixes the presentation and treats polynomial formula format and polynomial stratification count as separate claims. The numerical-wall formulation is essential because actual walls can accumulate. A first decisive theorem would give an effective definition and a quantitative cell decomposition for one wall family. A counterexample would be a wild wall arrangement or superpolynomial stratum growth.

55 Effective geometry of Calabi–Yau families

Gauge convention. Two positive gauges δ and δ' on the smooth locus of a fixed compact semialgebraic chart are *power-equivalent* if, after truncation to $(0, 1]$, there are positive constants c, C, a, b with $c\delta^a \leq \delta' \leq C\delta^b$. Every polynomial estimate below is understood up to this equivalence. The gauge δ_{alg} is represented by a truncated norm of generators of the reduced discriminant ideal. For a family of bundle pairs, δ_{pair} is represented by the minimum of δ_{alg} and a fixed positive stability margin that vanishes exactly on the chosen chamber boundary.

Conjecture 249 (Polynomial conditioning of Ricci-flat metrics)**.** *For a projective polarised Calabi–Yau family over a compact semialgebraic chart, with δ_{alg} the truncated norm of generators of the reduced discriminant ideal, the volume-normalised Ricci-flat metrics satisfy, for every derivative order ℓ, bounds $\|\nabla^\ell \mathrm{Rm}\|_\infty \le C_\ell\, \delta_{\mathrm{alg}}^{-A_\ell}$, with diameter bounded by $C\delta_{\mathrm{alg}}^{-A}$ and, in the noncollapsing case, injectivity radius bounded below by $C^{-1}\delta_{\mathrm{alg}}^{A}$, the same polynomial form holding for the regularity scale in collapsing strata away from the canonical collapsing set.*

Significance. Diameter bounds at degeneration are now sharp in the degeneration parameter [156], and quantitative collapse is understood in important families [157]; the conjecture asserts the global effective statement, polynomial control of all curvature derivatives in an algebraic gauge, the form of estimate that certifies every numerical metric computation at once. Łojasiewicz inequalities make the gauge canonical up to the exponents. The statement is beyond any finite check and is flagged as such. A first decisive theorem would be the curvature bound for one-parameter quintic degenerations. A counterexample would be a family with essential superpolynomial blow-up in the algebraic gauge.

Conjecture 250 (Discriminant-effective balanced convergence)**.** *In the setting of Conjecture 249, the level-k balanced metrics satisfy $\|g_{s,k}-g_s\|_{C^\ell(g_s)} \le C_\ell\, \delta_{\mathrm{alg}}(s)^{-A_\ell} k^{-1}$ whenever $k \ge C_\ell\, \delta_{\mathrm{alg}}(s)^{-B_\ell}$, with the corresponding k^{-N-1} bounds after subtracting N Bergman expansion terms.*

Significance. On a fixed smooth fiber this is classical Bergman asymptotics; the conjecture makes the rate uniform in the family with polynomial deterioration at the discriminant, exactly the regime probed by the balanced-metric power laws observed numerically near large complex structure [158]. The statement is beyond any finite check and is flagged as such. A first decisive theorem would be the C^0 case near a conifold degeneration. A counterexample would be a family where the required level grows faster than any power of the inverse algebraic distance.

Conjecture 251 (Discriminant-effective Hermitian–Yang–Mills convergence)**.** *For an algebraic family of polarised Calabi–Yau threefolds with slope-stable bundles on a constant-stability chamber, with δ_{pair} as in the gauge convention, the level-k balanced-bundle approximations converge to the Hermitian–Yang–Mills connection in Coulomb gauge with $\|A_{s,k} - A_s\|_{C^\ell} \le C_\ell\, \delta_{\mathrm{pair}}^{-A_\ell} k^{-1}$ whenever $k \ge C_\ell\, \delta_{\mathrm{pair}}^{-B_\ell}$.*

Significance. Hermitian–Yang–Mills existence survives conifold transitions [159], iteration on a fixed stable bundle converges [161], and the numerical algorithms are mature [160]; the conjecture is the missing family statement, a polynomial condition number in distance to where the fiber or the stability degenerates. The statement is beyond any finite check and is flagged as such. A first decisive theorem would be the fixed-fiber statement with explicit stability-margin dependence. A counterexample would be a chamber whose HYM approximation cost is essentially superpolynomial at the boundary.

Conjecture 252 (Polynomial matter spectral gap)**.** *In the setting of Conjecture 251, for a representation and form degree with locally constant bundle cohomology, the smallest positive eigenvalue of the bundle-valued Dolbeault Laplacian satisfies $\lambda_+(s) \ge C^{-1}\delta_{\mathrm{pair}}(s)^A$, simultaneously for any fixed finite list of representations and degrees.*

Significance. Bundle-valued spectra are computable numerically [162, 163] and rigorous eigenvalue floors are an explicitly wanted certificate in that programme; the conjecture asserts the floor is polynomial in the algebraic gauge, with the constant-cohomology hypothesis essential since a jump lets an eigenvalue cross zero. The statement is beyond any finite check and is flagged as such. A first decisive theorem would be the scalar case on quintics near a conifold. A counterexample would be an admissible chamber with essentially exponential spectral collapse.

Conjecture 253 (Certified heterotic Yukawa convergence)**.** *Under Conjectures 249 to 252, the level-k physical heterotic Yukawa couplings satisfy $|Y_{abc}^{(k)}/Y_{abc} - 1| \leq C\, \delta_{\mathrm{pair}}^{-A}\delta_Y^{-A}k^{-1}$ for $k \geq C\delta_{\mathrm{pair}}^{-B}\delta_Y^{-B}$, where δ_Y is the power-equivalence class of a truncated algebraic gauge for the zero locus of the corresponding holomorphic coupling. An absolute error bound of the same form holds with no δ_Y factor.*

Significance. Physical Yukawa couplings and even quark masses are now computed numerically from learned geometric data [164, 165], with only a-posteriori error diagnostics; the conjecture asserts the a-priori certificate, relative error polynomial in the two algebraic gauges, the δ_Y factor being necessary because no uniform relative bound can survive a coupling crossing zero. The statement is beyond any finite check and is flagged as such. A first decisive theorem would be the absolute bound for a single standard-embedding coupling. A counterexample would be an admissible family whose certified level grows faster than any power.

56 Degenerations and towers

Conjecture 254 (Saturation-index special-Lagrangian realization)**.** *For a semistable one-parameter Calabi–Yau threefold degeneration of type II, III, or IV with unipotent monodromy, let L_{orb} be the nilpotent-orbit lattice of asymptotically light D3 charges of common limiting phase, L_{sat} its rational saturation in integral homology, and I_τ the exponent of $L_{\mathrm{sat}}/L_{\mathrm{orb}}$. Every primitive charge generating an extremal ray of the saturated light cone with the required positive limiting phase has a multiple $m\gamma$, with m dividing I_τ, admitting a calibrated special-Lagrangian representative for all sufficiently deep parameters.*

Significance. Special-Lagrangian towers are constructed in Tyurin limits [150] and the type II–III–IV asymptotics are developed in [151]; the new clause is arithmetic, the divisor bound $m \mid I_\tau$ tying calibrated existence to the saturation exponent of the monodromy-orbit lattice, so that the failure of primitive realization is quantized by lattice theory. At $I_\tau = 1$ the statement reduces to primitive realization. The statement is beyond any finite check and is flagged as such. A first decisive theorem would be the type II case with $I_\tau = 1$. A counterexample would be an extremal ray whose minimal calibrated multiple fails to divide the saturation exponent.

Conjecture 255 (LMHS-controlled Laplace towers)**.** *For a semistable one-parameter polarised Calabi–Yau degeneration at infinite Weil–Petersson distance, fix a normalisation of d_{WP} and volume-normalise the Ricci-flat metrics. The vanishing scalar Laplace spectrum admits a finite partition into tower types a with rational exponents α_a such that: (i) the list of exponents depends only on the isomorphism class of the integral limiting mixed Hodge structure and the combinatorial dual-intersection and essential-skeleton data. (ii) For every tower and every fixed j, the quantity $e^{2\alpha_a d_{\mathrm{WP}}(t)}\lambda_{a,j}(t)$ converges to the j-th eigenvalue of a canonical weighted Laplace operator on the collapsed base geometry. (iii) every eigenvalue sequence tending to zero belongs eventually to one* *of these towers. The limiting operator may depend on the canonical metric data of the collapsed base and is not asserted to be determined by the limiting mixed Hodge structure alone.*

Significance. For induced Kähler degeneration metrics, exact small-eigenvalue rates are now a theorem of [168]. General skeleton-Laplacian convergence is not. The conjecture makes three stronger claims for volume-normalised Ricci-flat metrics. It asserts a finite exhaustive tower decomposition, an arithmetic classification of the rational exponents by the integral limiting mixed Hodge structure and skeleton combinatorics, and spectral convergence within each tower. Only the exponent list is claimed to be determined by that discrete data. The limiting operator is allowed to retain canonical metric information from the collapsed base. A first decisive theorem would establish the finite decomposition for a type IV maximal degeneration. A counterexample would be a light eigenvalue escaping every listed tower or an exponent varying while the stated discrete data remain fixed.

Part XIV
Convexity, defect, and dispersion laws for projective invariants

Added to the deposit: 19 August 2026.

The fifteen conjectures below take three of the oldest invariants of projective algebraic geometry, the middle cohomology of a complete intersection, the graded Betti table of an embedded surface, and the Hodge decomposition of a hypersurface, and conjecture the discrete shape of each as its defining degrees move. They form three programmes. A convexity programme (Conjectures 256 to 261) studies the degree-normalized primitive middle Betti number of smooth complete intersections on the full multidegree lattice, asserting a coherent discrete-convexity package: coordinate monotonicity and convexity, supermodular absolute interaction with log-submodular relative interaction, Schur-convexity at fixed degree sum, and a strict decrease under every codimension-raising split that preserves the adjunction coefficient. A syzygy programme (Conjectures 262 to 266) studies the two nontrivial rows of the Betti tables of $\mathbf{P}^1 \times \mathbf{P}^1$ under Segre–Veronese embeddings, replacing global shape claims by a controlled-defect law: the nonlinear row is strictly log-concave, while the linear row can fail log-concavity at most once, only past its mode, only where the nonlinear row is already present, and only within one step of that row's mode. A dispersion programme (Conjectures 267 to 270) studies the normalized primitive Hodge profiles of smooth hypersurfaces as the degree grows, asserting that the known drift toward the Eulerian profile is monotone in four canonical stochastic orders: likelihood-ratio flattening, central-window escape, convex-order spread, and total-variation distance. The proved layer is three calibrations: Proposition 39 proves all six convexity laws for curves from the exact formula $B = \sum_i d_i - c - 2 + 2/D$, Proposition 40 pins the two syzygy rows to the Hilbert series by an exact antidiagonal identity, and Proposition 41 proves all four dispersion laws for surfaces from an explicit rational function of the degree; in addition, for odd n the Calabi–Yau slice of Conjecture 261 is already a corollary of the merge monotonicity of Proposition 16. Two negative controls are retained in place: the unnormalized primitive Betti number fails Schur-convexity and has no split direction already for curves and surfaces, and coordinatewise monotone convergence to the Eulerian limit fails at dimension twelve, so the normalization of the convexity programme and the total-variation formulation of the dispersion programme are both forced. Conjectures 262 to 266 rest on the thirteen complete published tables, the entire deposited corpus, and are flagged for that small evidence base; Conjecture 262 sits closest to the published unimodality conjecture of [182] and carries the part's attribution flag; Conjectures 267 to 270 assert laws at every degree and dimension beyond any finite check and are flagged as such, and the Eulerian limit itself is exact in the abelian setting of [180] and asymptotic in the toric setting of [179], so only the monotone interpolation is claimed as new.

For a smooth complex complete intersection $X^n(\mathbf{d}) \subset \mathbf{P}^{n+c}$ of dimension $n \geq 1$, codimension $c \geq 1$, and multidegree $\mathbf{d} = (d_1, \ldots, d_c)$ with every $d_i \geq 2$, write $D(\mathbf{d}) = \prod_i d_i$; the Lefschetz theorem confines the new topology to the middle degree, and the primitive middle Betti number is $b_n^{\text{prim}} = (-1)^n(\chi(X) - (n+1))$ with χ evaluated exactly by the Chern-class formula $\chi = D\,[H^n](1+H)^{n+c+1}/\prod_i(1+d_iH)$ [175, 176]. The invariant of the convexity programme is the *degree-normalized primitive complexity* $B_{n,c}(\mathbf{d}) = b_n^{\text{prim}}/D(\mathbf{d})$, and e_i denotes a coordinate step in the multidegree lattice. For the Segre–Veronese embedding of $\mathbf{P}^1 \times \mathbf{P}^1$ by $\mathcal{O}(a,b)$ with $a, b \geq 2$, in characteristic zero, the graded Betti numbers $\beta_{p,p+q}(a,b)$ of the homogeneous coordinate ring vanish outside $q = 0, 1, 2$; write $\Delta_p^{(q)} = \beta_{p,p+q}^2 - \beta_{p-1,p-1+q}\beta_{p+1,p+1+q}$ for the log-concavity determinant at an interior index of the qth row, call p a *defect* of the row when $\Delta_p^{(q)} < 0$, and let m_q be the least index at which the row attains its maximum. For a smooth hypersurface $X_d^n \subset \mathbf{P}^{n+1}$ with $n \geq 2$, $d \geq 3$, the primitive Hodge numbers are $h_p(n,d) = [t^{(p+1)d-(n+2)}](1+t+\cdots+t^{d-2})^{n+2}$ by Griffiths' formula [178]; write $\pi_{n,d}$ for the normalized profile $h_p/\sum_j h_j$ on $\{0, \ldots, n\}$, symmetric about $n/2$, and $\varepsilon_n(p) = A(n+1,p)/(n+1)!$ for the Eulerian profile, with A the Eulerian numbers. A profile P precedes Q in *convex order* when $\mathbf{E}\varphi(P) \leq \mathbf{E}\varphi(Q)$ for every convex φ, and $\|P - Q\|_{\text{TV}} = \frac{1}{2}\sum_p |P(p) - Q(p)|$.

Proposition 39 (Curve calibration for the convexity programme)**.** For $n = 1$ one has $B_{1,c}(\mathbf{d}) = \sum_i d_i - c - 2 + 2/D$, and all six laws of Conjectures 256 to 261 hold strictly for curves.

Proof. The genus formula for complete-intersection curves gives $b_1 = D(\sum_i d_i - c - 2) + 2$, whence the displayed B. Write $u = 2/D$. For Conjecture 256, a step in d_i adds 1 to the linear part and changes u by $-2/(D(d_i+1))$, so $\Delta_i B = 1 - 2/(D(d_i+1)) \geq 1 - \frac{1}{3} > 0$. For Conjecture 257, differencing again gives $\Delta_i^2 B = 4/(D(d_i+1)(d_i+2)) > 0$. For Conjecture 258, the linear part has vanishing mixed difference and $\Delta_i\Delta_j(2/D) = 2/(D(d_i+1)(d_j+1)) > 0$. For Conjecture 259, writing $x = d_i$, $y = d_j$, $S = \sum_k d_k$, a direct expansion using $D\,D_{ij} = D_iD_j$ gives the exact identity

$$B(\mathbf{d}+e_i+e_j)B(\mathbf{d}) - B(\mathbf{d}+e_i)B(\mathbf{d}+e_j) = -1 + \frac{2(S+x+y-c)}{D(x+1)(y+1)},$$

and the right side is negative: with $s_k = d_k - 1$ and $R = \prod_{k\neq i,j} d_k$, one has $S + x + y - c = 2x + 2y - 2 + \sum' s_k$, while $xy(x+1)(y+1) \geq 6(x+y)$ and $R \geq 1 + \sum' s_k$ give $D(x+1)(y+1) \geq 6(x+y) + 6(x+y)\sum' s_k$, so the fraction is less than 1 with room to spare. For Conjecture 260, at fixed $\sum_i d_i$ and c the linear part is constant and a balancing transfer $(x,y) \mapsto (x-1, y+1)$ with $x \geq y+2$ multiplies D by $(x-1)(y+1)/(xy) > 1$, so u strictly decreases and B is strictly Schur-convex. For Conjecture 261, a split $d \mapsto (a,b)$ with $a + b = d + 1$ fixes the linear part and multiplies D by $ab/d > 1$, since $ab \geq 2(d-1) \geq d+1$, so B strictly decreases. ☐

Proposition 40 (Antidiagonal calibration for the syzygy programme)**.** Let $N+1 = (a+1)(b+1)$ and $c_j = [t^j](1-t)^{N+1}\sum_{k\geq 0}(ka+1)(kb+1)t^k$. Then $\beta_{j-1,j}(a,b) - \beta_{j-2,j}(a,b) = (-1)^{j-1}c_j$ for every $j \geq 2$, the resolution has length $N-2$, and $\beta_{N-2,N} = (a-1)(b-1)$.

Proof. The embedding is projectively normal, so the Hilbert series of the coordinate ring is $\sum_k \dim H^0(\mathcal{O}(ka, kb))\, t^k = \sum_k (ka+1)(kb+1)t^k$, and taking Euler characteristics of the minimal free resolution over the polynomial ring in $N+1$ variables gives $\sum_p (-1)^p \beta_{p,j} = c_j$ in every internal degree j. Since $\beta_{p,j}$ vanishes unless $j - p \in \{0, 1, 2\}$, the degree-j alternating sum collapses to the two displayed terms for $j \geq 2$. The coordinate ring is a normal semigroup ring, hence Cohen–Macaulay of dimension three and codimension $N-2$, so the resolution has length $N-2$ and the numerator $h(t) = (1-t)^3 \sum_k (ka+1)(kb+1)t^k$ is a polynomial whose coefficients are third differences of a quadratic sequence: $h(t) = 1 + (N-2)t + (a-1)(b-1)t^2$ exactly. Then $(1-t)^{N+1} \sum_k (ka+1)(kb+1)t^k = (1-t)^{N-2}h(t)$ has top coefficient $c_N = (-1)^{N-2}(a-1)(b-1)$, and since $\beta_{N-1,N} = 0$ the degree-N identity reads $(-1)^{N-2}\beta_{N-2,N} = c_N$, giving $\beta_{N-2,N} = (a-1)(b-1)$. $\square$

Proposition 41 (Surface calibration for the dispersion programme)**.** For $n = 2$ set $x(d) = \pi_{2,d}(0) = \frac{(d-2)(d-3)}{6(d^2-3d+3)}$. Then $x(d)$ is strictly increasing in $d \geq 3$ with $x(d) < \frac{1}{6}$ and $x(d) \to \frac{1}{6}$, and consequently all four laws of Conjectures 267 to 270 hold for surfaces, strictly for $d \geq 4$.

Proof. For a smooth surface of degree d one has $h_0 = \binom{d-1}{3}$ and $\sum_j h_j = d^3 - 4d^2 + 6d - 3 = (d-1)(d^2 - 3d + 3)$, giving the displayed $x(d) = \frac{(d-2)(d-3)}{6(d^2-3d+3)}$. Its increment has the sign of $(2d-5)(d^2-3d+3) - (d^2-5d+6)(2d-3) = 2d^2 - 6d + 3 > 0$ for $d \geq 3$, so x is strictly increasing, and $x < \frac{1}{6}$ is the inequality $(d-2)(d-3) < d^2 - 3d + 3$, which reduces to $2d > 3$. The profile is $(x, 1-2x, x)$ and $\varepsilon_2 = (\frac{1}{6}, \frac{4}{6}, \frac{1}{6})$, so on three symmetric points every one of the four laws is equivalent to the monotone increase of x below its limit $\frac{1}{6}$: the likelihood ratio, the central mass $1 - 2x$, every convex functional, and $\|\pi_{2,d} - \varepsilon_2\|_{\mathrm{TV}} = 2(\frac{1}{6} - x)$ are all monotone in x. $\square$

57 Normalized primitive cohomology of complete intersections

Conjecture 256 (Coordinate monotonicity of normalized complexity)**.** *For every $n \geq 1$, $c \geq 1$, every $\mathbf{d} \in \mathbf{Z}^c_{\geq 2}$, and every coordinate i, $B_{n,c}(\mathbf{d} + e_i) > B_{n,c}(\mathbf{d})$.*

Significance. Raising one defining degree multiplies the raw topological size by the growing degree factor, and the conjecture asserts that the primitive middle cohomology outruns that trivial multiplicative scale in every dimension and codimension: normalized complexity has strictly positive marginal growth on the whole degree lattice. The mechanism is transparent for curves, where Proposition 39 proves it from the exact formula, but the alternating Chern-class sum offers no visible positivity in higher dimension. The nearest related results are the explicit primitive-Betti formulas of [175] and the Euler-characteristic implications of [176], neither of which orders the degree lattice. A first decisive theorem would be monotonicity for hypersurfaces at every n. In tests on the corpus described in the stress-test section, all 59,100 coordinate steps on the exhaustive grid and 13,608 steps on the extended adversarial grid passed, with no failure. A counterexample would be a multidegree at which added degree is sublinear after normalization.

Conjecture 257 (Strict discrete convexity in each degree)**.** *Under the same hypotheses, $B_{n,c}(\mathbf{d} + 2e_i) - 2B_{n,c}(\mathbf{d} + e_i) + B_{n,c}(\mathbf{d}) > 0$.*

Significance. Beyond positive marginal growth, the marginal growth itself accelerates: each one-variable restriction of B is strictly convex on the integer lattice, so normalized complexity bends upward in every defining degree. The mechanism cannot be coefficientwise, since the

Chern-class expression is alternating; Proposition 39 proves the curve case with second difference $4/(D(d_i+1)(d_i+2))$, a shape that suggests a positive rational generating function in general. The nearest related result is [175]. A first decisive theorem would be convexity for quadric-heavy multidegrees, where the formula degenerates. In tests on the corpus described in the stress-test section, all 59,100 second differences on the exhaustive grid and 13,608 on the adversarial extension were strictly positive. A counterexample would be one multidegree with a concave step, marking an unexpected cancellation in the Chern-class sum.

Conjecture 258 (Additive supermodularity across degrees)**.** *For $i \neq j$, $B(\mathbf{d}+e_i+e_j) - B(\mathbf{d}+e_i) - B(\mathbf{d}+e_j) + B(\mathbf{d}) > 0$.*

Significance. Distinct defining equations reinforce one another: raising d_i makes a subsequent raise of d_j strictly more valuable in absolute normalized complexity, so the discrete Hessian of B is positive off the diagonal as well as on it. Together with Conjecture 257 this asserts a full multivariate discrete convexity, and the curve case, mixed difference $2/(D(d_i+1)(d_j+1))$, is proved in Proposition 39. The nearest related results are [175, 177]. A first decisive theorem would be supermodularity at codimension two, where only one mixed pair exists. In tests on the corpus described in the stress-test section, all 84,600 mixed second differences on the exhaustive grid and 17,281 on the adversarial extension were strictly positive. A counterexample would be a pair of degrees whose interaction turns negative.

Conjecture 259 (Strict log-submodularity)**.** *Whenever the four values are positive, $B(\mathbf{d}+e_i+e_j)\,B(\mathbf{d}) < B(\mathbf{d}+e_i)\,B(\mathbf{d}+e_j)$.*

Significance. Set against Conjecture 258, this is the striking pairing of the programme: absolute interaction of two degrees is positive while proportional interaction is negative, so $\log B$ is strictly submodular where B is supermodular, the signature of diminishing relative returns familiar from discrete statistical mechanics and rare for a cohomological invariant. The curve case follows from the exact identity of Proposition 39, whose right side exposes the competition between the two orders. The nearest related result is [175]. A first decisive theorem would be log-submodularity for pairs of quadrics and cubics in all dimensions. In tests on the corpus described in the stress-test section, all 84,600 cross-products on the exhaustive grid and 17,281 on the adversarial extension obeyed the strict inequality. A counterexample would separate the two interaction orders at some multidegree, itself a sharp structural datum.

Conjecture 260 (Schur-convexity at fixed degree sum)**.** *If $\mathbf{d}$ majorizes $\mathbf{d}'$ in $\mathbf{Z}^c_{\geq 2}$ with equal coordinate sums, then $B_{n,c}(\mathbf{d}) \geq B_{n,c}(\mathbf{d}')$, strictly unless the vectors agree up to permutation.*

Significance. Among complete intersections with the same dimension, codimension, and adjunction coefficient $\sum_i d_i - (n+c+1)$, the balanced multidegrees minimize normalized primitive complexity and the lopsided ones maximize it: redistributing positivity unevenly enriches primitive topology even when the canonical class data are unchanged. The unnormalized statement is false already for curves, where $b_1^{\mathrm{prim}}(2,4) = 18 < 20 = b_1^{\mathrm{prim}}(3,3)$, a retained control showing the normalization reverses the order; the normalized curve case is proved in Proposition 39 through the Schur-concavity of the degree product. The nearest related results are the extremal Euler-characteristic theorems of [177]. A first decisive theorem would be the single balancing transfer at codimension two. In tests on the corpus described in the stress-test section, all 47,000 balancing transfers on the exhaustive grid and 15,754 on the adversarial extension strictly decreased B. A counterexample would be a transfer at which balance raises normalized complexity.

Conjecture 261 (Adjunction-preserving splits decrease complexity)**.** *For $d_i = d \geq 3$ and $a, b \geq 2$ with $a + b = d + 1$, replacing the degree d by the pair (a, b) gives $B_{n,c+1}(\ldots, a, b, \ldots) < B_{n,c}(\ldots, d, \ldots)$.*

Significance. The split raises the codimension by one while preserving the adjunction coefficient, so it moves between presentations that the canonical class cannot distinguish, and the conjecture orients this move: two lower-degree equations are strictly simpler per unit degree than one high-degree equation. The unnormalized statement has no fixed direction, with $b_2^{\mathrm{prim}}(3) = 6 > 5 = b_2^{\mathrm{prim}}(2, 2)$ against $b_1^{\mathrm{prim}}(4) = 6 < 8 = b_1^{\mathrm{prim}}(2, 3)$, both retained as controls. The split is the inverse of the merge order of Part VII, and for odd n its Calabi–Yau slice is already a theorem: there $B = E_n/D + (n + 1)/D$, and both terms strictly increase under a merge, the first by Proposition 16 and the second because merging strictly decreases the degree product, so the odd-dimensional Calabi–Yau case follows outright; the even-dimensional and general-adjunction cases are the open content. The curve case is proved in Proposition 39. A first decisive theorem would be the even Calabi–Yau case, where the same two mechanisms compete. In tests on the corpus described in the stress-test section, all 147,750 admissible splits on the exhaustive grid and 346,825 on the adversarial extension passed strictly. A counterexample would be a split that concentrates rather than dilutes normalized topology.

58 Controlled defects in Segre–Veronese syzygies

Conjecture 262 (Strict log-concavity of the nonlinear row)**.** *For every $a, b \geq 2$ and every interior index p of the positive support of the $q = 2$ row, $\Delta_p^{(2)} > 0$.*

Significance. The nonlinear row is exactly the row that resists the Gaussian heuristics of asymptotic syzygy theory, and the conjecture asserts that it nevertheless obeys the strongest local curvature law short of real-rootedness, strict log-concavity across its whole support. The statement is stronger than the published unimodality conjecture for these rows [182], which is the part's closest literature boundary and its attribution flag; the normality heuristic itself is [184], and log-concavity of syzygy rows is known to fail for small indices in other families [185], so the claim is genuinely calibrated to this surface. A first decisive theorem would be log-concavity of the three trailing entries, accessible from the tail formulas of [183]. In tests on the corpus described in the stress-test section, all 48 interior triples of the thirteen complete published tables were strictly log-concave, the tables reverified file by file against the primary dataset and pinned by the antidiagonal identity of Proposition 40; the statement rests on that small corpus and is flagged accordingly. A counterexample would be the first curvature pathology of a nonlinear row.

Conjecture 263 (The linear row is log-concave through its mode)**.** *For every $a, b \geq 2$ and every interior index $p \leq m_1$ of the $q = 1$ row, $\Delta_p^{(1)} > 0$.*

Significance. Global log-concavity of the linear row is false, with the retained control $\beta_{14,15}^2 - \beta_{13,14}\beta_{15,16} = -3{,}391{,}704$ at $\mathcal{O}(2, 6)$, so the right law is one-sided: the ascending side and the mode keep strict log-concavity, and every observed failure sits deep in the descending tail. The rising side is the regime where the finite tables are most compatible with the asymptotic normality picture of [184], and the conjecture extracts its most robust local consequence. The nearest related results are [182, 183]. A first decisive theorem would be log-concavity of the first three entries at every (a, b), where binomial formulas for the low syzygies are available. In tests on the corpus described in the stress-test section, all 102 interior triples at or before the mode passed strictly across the thirteen tables; the small corpus flag applies. A counterexample would be a curvature reversal on the rising side, refuting the one-sided picture.

Conjecture 264 (At most one defect in the linear row). *For every $a, b \geq 2$, the set $\{p : \Delta_p^{(1)} < 0\}$ has at most one element.*

Significance. Unimodality permits arbitrarily many local curvature reversals; this conjecture forbids all but one, asserting that the linear row is a nearly totally positive sequence with a single controlled exception. Across the thirteen published tables, five rows have no defect and eight have exactly one, at indices $14, 16, 18, 20, 22$ for $\mathcal{O}(2, b)$, $b = 6, \ldots, 10$, and $14, 16, 19$ for $\mathcal{O}(3, b)$, $b = 4, 5, 6$, an arithmetic progression pattern in the first family that any proof should explain. The nearest related results are [182, 185]. A first decisive theorem would be uniqueness of the defect for $\mathcal{O}(2, b)$, the family with the cleanest tail formulas [183]. In tests on the corpus described in the stress-test section, no table carried two defects; the small corpus flag applies. A counterexample would be a linear row with two curvature reversals, breaking the controlled-defect picture.

Conjecture 265 (Defects occur only inside the nonlinear overlap). *For every $a, b \geq 2$, if $\Delta_p^{(1)} < 0$ then $\beta_{p,p+2}(a, b) > 0$.*

Significance. The conjecture couples a shape property of one row to nonvanishing in the next: the linear row can lose log-concavity only at homological positions where nonlinear syzygies are already present, so a vanishing theorem for $K_{p,2}$ would automatically restore strict log-concavity of $K_{p,1}$ there. All eight observed defects obey the coupling, and the rational-normal-scroll boundary $a = 1$, where the nonlinear row is empty, has defect-free linear rows across the eight published scroll tables, consistent with the picture from outside the stated range. The nearest related results are [182, 184]. A first decisive theorem would be defect-freeness wherever the nonlinear row vanishes, a statement about the 2-linear part alone. In tests on the corpus described in the stress-test section, the coupling held in all eight defect tables; the small corpus flag applies. A counterexample would be a defect before the nonlinear row enters, decoupling the two rows.

Conjecture 266 (Defect-mode locking). *For every $a, b \geq 2$, if the linear row has a defect at p_* then $p_* \in \{m_2, m_2 + 1\}$.*

Significance. This is the boldest statement of the syzygy programme: the unique loss of curvature in the linear row is phase-locked to the concentration point of the nonlinear row, landing at its mode or one step beyond in every observed table, exactly at m_2 for $\mathcal{O}(2, b)$ and at m_2 or $m_2 + 1$ for $\mathcal{O}(3, b)$. A proof would exhibit a cross-row mechanism in minimal resolutions that no current theory predicts; even a failure would be valuable, since the displacement $p_* - m_2$ is a sharp statistic to track across embeddings. The nearest related results are [182, 184, 183]. A first decisive theorem would be the locking for $\mathcal{O}(2, b)$, where both rows have explicit tails.

In tests on the corpus described in the stress-test section, the locking held in all eight defect tables; this is the part's highest-risk statement and the small corpus flag applies with full force. A counterexample would be a defect drifting away from the nonlinear mode.

59 Monotone dispersion of hypersurface Hodge profiles

Conjecture 267 (Likelihood-ratio flattening in the degree)**.** *For every $n \geq 2$, $d \geq 3$, and $0 \leq p < q \leq \lfloor n/2 \rfloor$, $h_p(n, d+1)\, h_q(n, d) \geq h_p(n, d)\, h_q(n, d+1)$.*

Significance. As the degree grows, outer Hodge pieces gain relative weight against pieces nearer the middle, uniformly in every pair of positions: a monotone likelihood-ratio comparison between the Hodge structures of different varieties. The coefficient vector is a convolution of discrete uniforms, but the sampling lattice moves with the degree, so no fixed-lattice total-positivity argument applies directly; the surface case is proved in Proposition 41. Equalities occur while an outer piece is still zero, so the non-strict form is the uniform statement. The nearest related results are [178, 179]. A first decisive theorem would be the outermost pair $(0, 1)$ at every n, the geometric-genus-to-next-piece ratio. In tests on the corpus described in the stress-test section, all 71,920 cross-products on the grid and 164,502 on the extension passed. A counterexample would be a degree step at which the profile sharpens toward the center.

Conjecture 268 (Central-window escape)**.** *For every $n \geq 2$, $d \geq 3$, and $1 \leq k \leq \lfloor n/2 \rfloor$, $\sum_{p=k}^{n-k} \pi_{n,d+1}(p) \leq \sum_{p=k}^{n-k} \pi_{n,d}(p)$.*

Significance. Every central window of the Hodge diamond loses normalized mass monotonically as the degree grows: cohomology drains from the middle toward the outer pieces at every scale simultaneously, a position-sensitive statement strictly stronger than growth of any single spread functional. The mechanism is the same moving-lattice convolution as Conjecture 267, of which this is a consequence for symmetric profiles; the surface case is Proposition 41. The nearest related results are [178, 180]. A first decisive theorem would be the innermost window $k = \lfloor n/2 \rfloor$, the central Hodge piece alone. In tests on the corpus described in the stress-test section, all 13,050 window comparisons on the grid and 25,272 on the extension passed. A counterexample would be a window recapturing mass at some degree step.

Conjecture 269 (Convex-order dispersion)**.** *For every $n \geq 2$ and $d \geq 3$, the profile $\pi_{n,d}$ precedes $\pi_{n,d+1}$ in convex order.*

Significance. The degree flow is a mean-preserving spread: every convex functional of the normalized Hodge position, the variance, all even central moments, every exponential moment, increases weakly at each degree step, so infinitely many numerical Hodge inequalities are packaged into one canonical stochastic order. On finite symmetric support the order is equivalent to the call-function comparisons $\mathbf{E}(P - t)_+$, which is how it is tested exactly; the surface case is Proposition 41, and the only coincident consecutive profiles in the tested range are the cubic and quartic threefolds. The nearest related results are [178, 179]. A first decisive theorem would be the variance alone, the second-moment shadow of the order. In tests on the corpus described in the stress-test section, all 1,682 degree transitions on the grid and 2,730 on the extension passed. A counterexample would be a convex statistic moving against the spread.

Conjecture 270 (Monotone total-variation convergence to the Eulerian profile)**.** *For every $n \geq 2$ and $d \geq 3$, $\|\pi_{n,d+1} - \varepsilon_n\|_{\mathrm{TV}} \leq \|\pi_{n,d} - \varepsilon_n\|_{\mathrm{TV}}$, with equality only at $(n,d) = (3,3)$.*

Significance. The normalized profiles drift toward the Eulerian distribution, exactly the law of descents of a random permutation of $n+1$ letters; the conjecture upgrades that drift to monotone convergence in total variation from the first degree onward, a finite-degree error principle that one verified threshold would convert into bounds at every larger degree. Convergence alone forces no monotonicity, and the coordinatewise strengthening is false, a retained control: at $n = 12$ the $p = 5$ mass crosses its Eulerian limit between degrees six and seven and overshoots, so the cancellation across positions asserted here is the genuine content. The Eulerian limit is exact for hypersurfaces in abelian varieties [180], asymptotic in the toric setting [179], and feeds effective arithmetic applications [181]; only the monotone interpolation is claimed. A first decisive theorem would be a one-crossing law between $\pi_{n,d}$ and ε_n. In tests on the corpus described in the stress-test section, the distance never increased across 1,682 grid and 2,730 extension transitions, the sole equality being the cubic–quartic threefold coincidence. A counterexample would be a degree step moving the profile away from the Eulerian law.

60 Stress tests and independent recomputation

Each statement was tested three ways: against the literature, against computation well beyond its original range, and against an independent reimplementation.

Layer 1: literature. Five independent searches combed OEIS, arXiv, and the standard references for prior art, producing the novelty labels and the attributions used throughout. Each search was run at neighbourhood depth: defining objects, derived sequences, OEIS comments, and the abstracts and bodies of near-neighbour papers read in full rather than skimmed from search summaries—the depth at which, for instance, OEIS A088054's intersection comment (Conjecture 21(ii)) and the second half of Lillie's abstract (the primorial companion) are found, neither being visible in the defining sequences or in a search snippet. This layer also fixes two attributions used above: the $k \geq 1$ Stern list of Conjecture 12 is OEIS A060003 verbatim, with 1493 the largest known Stern prime, and the primorial-twin statement is Lillie's [5].

Layer 2: scripted refutation. Every falsifiable-by-instance statement was pushed at least a decade past its original bound: exception hunts across $(10^8, 10^9]$ for Conjectures 5, 12, 17, 22; the uniformity of Conjecture 8 stressed on $d \leq 6000$; the statistical laws re-tested at 4×10^9 (Conjectures 1, 4, 9, 11, and 13, and the quintuplet calibration count reported with Conjecture 8); the recovery clause of Conjecture 3 tested on fresh moduli $q \in (3000, 6000]$. No statement moved. The same layer settles one statement negatively: a completeness list for Fibonacci–Lucas twins is destroyed by OEIS A080327's catalogued index 148091 (conditionally on the probable-prime classifications of F_{148091} and L_{148091} alike, neither being a proven prime), which is why Conjecture 20 conjectures finiteness without a completeness clause and carries that index as its calibration clause.

Each statement was also verified at large scale: the Conjecture 10 profile across 150 shifts and the moment law for its constants $C(d)$ (derived mean 2.7456 and sd 1.6840 against measured 2.7434 and 1.6726); the cubic family moments and uniformity of Conjecture 15 (294 constants, 57 count profiles); the triplet and sexy-pair races of Conjectures 18 and 19 at 10^9 and the Stern lanes of Conjecture 12 on 2,500 samples at 10^8; the chain of Conjecture 24 over five decades to 10^7; the null race of Conjecture 14 at 10^7; the Fibonacci–Lucas joint scan to $p \leq 10^4$;

factorial twins to $n = 700$; the CRT-exact factorization of Conjecture 6 through a joint period of 1.2×10^8, with the scan extended to $n = 6000$; the pinned average $G(H)$ of Conjecture 2 computed exactly to $H = 3000$; the boundary trichotomy of Conjecture 22 with two new lanes counted at 10^6; the least-summand law of Conjecture 5 on 2,000 samples; the covariance kernels of Conjectures 8(ii) and 16(ii) evaluated over 870 and 1,225 pairs, with the moving-window randomization of Conjecture 8(ii) over 2000 windows (empirical correlation matrix matching the predicted kernel entrywise at 0.86); the orientation-resolved twin-member profile of Conjecture 17 on 150 samples; the race autocorrelation and running-maximum measurements of Conjecture 9 at 10^9 against simulated nulls; the singular-series waiting-time clause of Conjecture 11(iii) (regression coefficient -0.466 against the predicted $-\frac{1}{2}$); the stratified experiment of Conjecture 3 on 80 moduli; the multibase quotient tests of Conjecture 7 on 664,577 primes; and the weighted drift and internal null lane of Conjecture 4 on 400 fresh samples, together with the cousin-race predictions of the contamination calculus, Conjecture 1(v), at 10^9 (leadership log-densities 0.99 and 0.92 on the predicted sides). In each case the prediction was derived before the corresponding data were taken.

Layer 3: independent replication. Constants were recomputed from their definitions, and sequences recounted, by implementations sharing no code with the primary computation and working from the bare statements alone. All recomputed constants landed inside the stated truncation error, and all recounts agreed. The same layer reproduced the Fibonacci deficit recorded at Conjecture 20 and the ordering anomaly of Remark 1, confirming that both are properties of the primes rather than of a single implementation.

One methodological conclusion is worth recording. The mathematical failure modes in this subject are not statistical but *algebraic*: factorizations and congruence collapses living on density-zero or positive-density-but-structured subsequences—the cubes obstructing $n = p + k^3$ (Theorem 1), the composite-k factorization of D_k (Conjecture 23(i)), the $k = 4$ parity branch of Conjecture 25, the inadmissible naive chain of Conjecture 24. Probabilistic sanity checks, however extensive, integrate over density and cannot see such families; only structural tests—testing a claimed exception census on special subsequences, for instance—find them. A verification that only re-runs an existing count at a larger bound is blind to every one of them.

Part IV. The Part IV verification programme follows the same standard. The rank formula of Proposition 3 was checked against direct linear algebra over $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ on eighteen thousand random pairs of a graph and an element before the proof was written. The irreducibility claim of Conjecture 57 was verified by exact factorization over $\mathbb{Z}$ for all 996 connected graphs on at most seven vertices. The recovery statements of Conjectures 58, 65, and 66 were tested on every known collision class of C_G: the 119 classes among the 1252 graphs on at most seven vertices, a randomized eight-vertex corpus, the unique class among all 3159 trees on fourteen vertices, refound independently by exhaustive search, and the first unicyclic collision at ten vertices, with Q_G, γ, and i agreeing in every class. The rigidity statements were scanned exhaustively on at most seven vertices, with six thousand random eight-vertex graphs added for Conjecture 59 and the scan for Conjecture 61 restricted to graphs with at least four edges, and the class-restricted statements were tested on six-thousand-graph random holdouts for each of Conjectures 62, 63, and 64, spanning 2747, 3192, and 5598 distinct binary specializations respectively, together with all 656 binary classes of ten-vertex unicyclic graphs for Conjecture 63. The log-concavity and support statements of Conjectures 67 to 71 were verified on the full atlas of graphs on at most seven vertices and twenty-five hundred random graphs on eight to ten vertices, with the eight-vertex counterexample to the second-order slice strengthening reconfirmed by the independent implementation. The tree programme of Conjectures 72 to 76 was verified exhaustively over all trees through fourteen vertices, for Conjectures 73 and 74 with $q \in \{2, 3, 4, 5\}$ through twelve vertices, $q \in \{2, 3\}$ at thirteen, and $q = 2$ at fourteen, and the sharpness failure at $(n, q) = (4, 2)$ recorded at Conjecture 73 was found in this verification rather than assumed. A seven-vertex pair refutes Conjecture 63, below the ten-to-thirteen-vertex range of those holdouts. An exhaustive rescan locates it as the unique violating class among trees and connected unicyclic graphs through nine vertices, and the tree case was reverified exhaustively through fourteen vertices.

Part V. The Part V verification programme follows the same standard, on an independently generated corpus of all 87 unlabelled posets on at most five points, over $\mathbb{F}_2$ and $\mathbb{F}_3$ fully and $\mathbb{F}_5$ to dimension six, cross-checked against the deposited enumerators. The calibration identity of Proposition 6 held with no exception. The determination statements were tested on every adjoint-collision class: the adjoint-to-Kirillov determination of Conjecture 77 held across 64 classes, and the lower-central factor vector, $\dim \mathrm{Der}$, and $\dim \mathrm{Cent}$ of Conjectures 78, 80, and 82 were constant on every class. The characteristic statements of Conjectures 83 and 84 were confirmed by agreement of the attained adjoint-rank and Kirillov-rank sets across $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ in every case, the two-field rigidity of Conjecture 85 forced equality at $q = 5$ across 16 three-field collision classes, and the adjoint field monotonicity of Conjecture 86 held as first-order dominance in all 221 field-pair tests while its likelihood-ratio strengthening failed. The centre-one statements were tested on all 43 centre-one records, where the Kirillov enumerator determined the adjoint enumerator, $R_{P,q}$ had nonnegative integer coefficients, and $R_{P,q}$ had interval support and was unimodal throughout this corpus, with log-concavity failing four times. The unimodality was subsequently refuted by a seven-point poset outside the corpus, and Conjecture 90 is recorded as resolved false. The forest statements were tested on the forest class, where the Kirillov enumerator determined the adjoint enumerator across 44 forest-collision classes, the Kirillov support filled every even rank up to b_K, the even-rank sequence was unimodal, and the bound $b_K \leq 2b_A$ held everywhere while the unrestricted bound failed fourteen times off the forest class. The unitriangular strict log-concavity of Conjecture 95 was confirmed for ut_4, ut_5, and ut_6 over $\mathbb{F}_2$, for ut_4 and ut_5 over $\mathbb{F}_3$, and for ut_4 over $\mathbb{F}_5$, and the incidence anticorrelation of Conjecture 96 gave negative covariance on every nonabelian poset and exactly zero on all sixteen abelian ones. Two statements were not independently recomputed: the cohomology recoveries of Conjectures 79 and 81 rest on the deposited computation of H^2 and H^3 (ordinary) and H^1, H^2, and H^3 (adjoint), our reverification having not recomputed Lie-algebra cohomology. Three statements, Conjectures 79, 81, and 85, are supported only by the restricted tests recorded with each, and their general forms should not be read as carrying evidence comparable to the other conjectures. Two further statements, Conjectures 87 and 90, are false as stated and are recorded as resolved false with the seven-point counterexamples given in their resolutions.

Part VI. The Part VI verification programme follows the same standard, on an arithmetic cache of every integer through 20,000,000 with segmented exact scans and an odd-only prime-neighbour sieve to 4×10^9, cross-checked by an independent deterministic Miller–Rabin implementation sharing no code with the primary scans. The compass frequencies of Conjectures 97 to 102 were measured over the polygonal centres through 4×10^9, the local-sieve model of Conjecture 97 correlating with the order-level bias at 0.896 across $3 \leq s \leq 100$ and every promoted direction replicated on a holdout above 10^9. The radical order-pattern laws of Conjectures 103 and 104 were computed exactly through 10^9, the chi-square against uniformity rising from 0.45 at length four to about 2.1×10^7 at length five and the pattern ratio settling at 13.31, with the length-four and length-five cell counts reproduced by a second implementation. Proposition 9 now proves existence of every fixed-length conditional density, leaving nonuniformity open at each length at least five. Proposition 10 proves that triangular and hexagonal centres have identical residue laws and local compass frequencies at every finite sieve depth. The order constants of Conjectures 105 to 109 were measured over 2×10^7 triples across seven disjoint blocks, and the five-term totient barrier of Conjecture 107, since resolved false by G. Martin's theorem, showed no run below 4×10^9 in an exact segmented scan, while its abundancy analogue fell within the same scan to an increasing five-term run at $n = 36{,}721{,}681$. The shock statements of Conjectures 110 to 115 were checked over millions of steps per family against the exact jump identities of Proposition 7, the lag-one covariances negative in all 22 tested families. Proposition 11 now proves Conjectures 110 to 112 outright, with explicit six-prime witnesses reaching $J_{3,n} = -20$, $J_{4,n} = -30$, and $J_{3,n} = +21$, beyond every scanned extreme, and Conjecture 115 follows from the same identities through the multivariate Erdős–Kac theorem of [171]. Proposition 12 evaluates the truncated lag-one covariance of the central-binomial slice at exactly $-\frac{5}{3}$ for every cutoff at least 3, confirmed numerically to five decimals at 2×10^6, the full covariance standing at -1.47 and drifting toward the constant. Proposition 13 now proves the $k = 2$ case of Conjecture 114. The boundary law of Conjecture 116 matched its truncated Euler sum at correlation 0.998 across the 39 gaps with at least 500 samples, with block-level correlations from 0.986 to 0.997. Six of the twenty state a bias as a numerical band rather than a derived constant, and Conjectures 102 and 113 extrapolate in a parameter rather than the index, so their general forms should be read with the caution their flags record.

Part VII. The Part VII verification programme rests on exact integer computation throughout: truncated integer power series for the Chern-class evaluation, gcd–lcm inclusion-exclusion for the Milnor–Orlik formula, $\mathbf{F}_2$ Gaussian elimination for induced clique-complex homology with exact fraction-free Routh–Hurwitz determinants, and signed-orbit counting for the exterior actions, every calibration reproduced by a second implementation sharing no code with the primary scans, which recovered the quintic -200, the K3 value 24, the Poincaré-sphere 0, the empty-graph strand $6z^2 + 8z^3 + 3z^4$, and the $-I_4$ profile $(1, 0, 6, 0, 1)$. The Euler laws of Conjectures 117 to 121 were tested on the complete configuration census through dimension 30, the merges numbering 1,568,731, the ladder exact through $n = 200$, and the gcd law reverified independently through dimension 26. The cap extrema of Conjectures 122 and 123 are complete exact-cap censuses through caps 50 and 40 with the $A = 12$ transition value 222 reverified, and the scale laws of Conjectures 125 and 126 cover 52,593 tuples at scales through 10, with independent random rechecks one scale beyond. The strand laws of Conjectures 127 to 131 were tested on 34,867 graphs, exhaustive over labeled graphs through six vertices with 250 deterministic holdouts at each order seven to ten, giving 50,454 strands, and reverified independently on an exhaustive labeled corpus through five vertices. The monodromy laws of Conjectures 132 to 136 cover all 57,758 orientation-preserving signed cycle types through dimension 21, with every type through dimension 11 and the minimum 160 at dimension 12 reverified independently against Proposition 14. Six adversarial controls are retained in place, each refuting a tempting strengthening. Two of the twenty, the limit clause of Conjecture 117 and Conjecture 135, assert limits beyond any finite check, and Conjectures 120 and 124 are flagged for elevated attribution risk, so those statements should be read with the caution their flags record. Since deposit, Conjectures 117 and 118 have been resolved true and Conjecture 135 resolved false, the Landau floor of Proposition 18 fixing the compression exponent at 2 while the deposited minima respect the floor with room to spare. Proposition 17 now identifies every scale sequence as a sum of interior Ehrhart counts of $N - 1$ integral weighted hypersimplices, proving Conjecture 125 and the convexity clause of Conjecture 126, with an independent exact stress of 24,950,000 scale inequalities on 25,000 random tuples finding no violation of the surviving log-concavity clause. Proposition 15 proves all three ladder clauses with the sharpening $8 - Q_{n+1}/Q_n \sim 4/n$, and its exact recurrence reproduces the deposited ladder through $n = 200$. Proposition 16 proves every normalised merge inequality by a positive coefficient identity, independently checked through dimension 18.

Part VIII. The Part VIII verification programme is exact wherever its statements are: integer Chern-coefficient evaluation for the ambient products, integer expansion of Göttsche's product, exact age histograms, exact subset-sum functionals with rational arithmetic, and $\mathbf{F}_2$ homology of matroid independence complexes with exact rational Routh tests, every calibration reproduced by a second implementation sharing no code with the primary scans, which recovered the quintic, K3, and tetraquadric values, the $\mathrm{Hilb}^2(\mathrm{K3})$ Betti row $(1, 23, 276, 23, 1)$, and the moment identities of Proposition 19 in exact rational arithmetic. The dominance and refinement laws of Conjectures 137 to 140 were tested on 42,691 moves, 42,861 splits, and 24,680 disjoint split squares through dimension 22, reverified independently through dimension 18 with the four dimension-five violations of the unqualified Conjecture 140 reproduced exactly. The profile laws of Conjectures 141 to 144 rest on 179,200, 53,070, and 48,384 exact inequalities for $b \leq 30$ and $n \leq 80$, with the complete kurtosis phase diagram of Conjecture 143 reverified independently through $n = 50$. The age laws of Conjectures 145 to 148 combine an exact census of 450,999 admissible weight multisets with $r \leq 60$, reverified on 58,393 multisets with $r \leq 40$, with seeded Monte Carlo in four regimes whose lattice-corrected Kolmogorov distances 0.0398 to 0.0162 were independently recomputed as 0.0344 and 0.0178. The entropy laws of Conjectures 149 to 152 were tested on 16,200 random and 48,512 exact integer spectra, the 222 equality cases all two-valued, with an independent exhaustive recount through $d = 7$ finding 122 two-valued equality cases. Propositions 20 and 21 now prove Conjectures 151 and 152. The former matroid sample did not contain the two graphic certificates now built into the exact checker. Proposition 22 refutes Conjectures 153, 154, and 156, while Proposition 23 refutes Conjecture 155. The limit clauses of Conjectures 139 and 143 and the random-model statements of Conjectures 146 to 148 are flagged where they occur, and Conjecture 145 carries an attribution flag, so those statements should be read with the caution their flags record.

Part IX. The Part IX verification programme is exact wherever its statements are: tubings enumerated exhaustively with integer f-to-h conversion, $\mathbf{F}_2$ homology of flag complexes by integer rank, sector functionals by deterministic convolution over exactly enumerated sectors, and boundary maps assembled as integer matrices, while root locations, Mahler measures, and pseudodeterminants are floating-point spectral computations and are flagged as such, every calibration reproduced by a second implementation sharing no code with the primary scans, which recovered the Narayana and Eulerian endpoint h-polynomials, the matrix-tree value $\mathrm{pdet} = 5^4$ for the complete graph on five vertices, and both identities of Proposition 24, the sine product exactly and the moment integral to 10^{-6}. The toric laws of Conjectures 157 to 161 were tested on 878 deposited root sets, 3,621 connected deletions, 3,956 edge additions with minimum Mahler increment 0.0263 and exact integer tail margins, and 1,857 tree tests, reverified independently on all 771 labelled connected graphs through five vertices, 2,971 deletions, 3,227 edge additions, and every labelled tree through six vertices. The McKay laws of Conjectures 162 to 166 were tested on 3,135 deposited profiles, 2,905 of them at $d \geq 5$, reverified by an independent exhaustive census of 379 distinct-weight profiles at $p \in \{7, 11, 13\}$ that reproduced the two repeated-weight controls and the two-crest β_3 control exactly. The torsion laws of Conjectures 167, 168, and 171 rest on

40,932, 6,822, and 20,466 admissible deposited triples with extreme curvatures 0.0079, 0.0770, and −0.0076, reverified independently at five (p, d) pairs with the $s = -2$ control curvature +0.4980 reproduced to six decimals over the exactly verified centering identity; the parabola of Conjecture 169 matched four exact dynamic-programming regimes with central root-mean-square errors 0.027 to 0.038, and the decoupling of Conjecture 170 was checked on three seeded ensembles of up to 60,480 character samples whose empirical correlations below 5×10^{-16} reproduce the exact zero forced by the involution, the informative diagnostics being the variances and fourth moments, which moved toward Gaussian values. The entropy laws of Conjectures 172 to 176 were tested on a corpus of 130 spheres and 225 stellar pairs supplying 262, 392, 130, 900, and 450 comparisons in statement order, with minimum log-concavity margins 51.8 and 17.8 and minimum stellar increment 1.42, reverified independently through dimension four. The iterated-limit clause of Conjecture 169 and the random-model clause of Conjecture 170 are flagged where they occur, and Conjectures 157, 161, 164, 169, and 170 carry attribution flags, so those statements should be read with the caution their flags record.

Part X. The Part X verification programme is exact wherever its statements are: characteristic-two and finite-field cohomology by exact elimination, integral torsion by Smith normal form through dimension nine, and every χ_y-coefficient, signature, merge inequality, ratio, and gcd in exact integer or rational arithmetic, while characteristic-zero ranks beyond dimension nine are dual large-prime proxies and the root, Mahler, and radius laws are floating-point spectral checks, flagged as such, every calibration reproduced by a second implementation sharing no code with the primary scans, which recomputed the χ_y-profiles from the scaled Hirzebruch series with the signature recovered as $\chi_{y=1}$ and reproduced the K3 profile $(2, 20, 2)$, the quintic profile, and the dimension-two signature 16. The support obstructions of Proposition 27 were checked independently: the deposited profile admits at most one unused coordinate through $n = 120$, the rescaling criterion was confirmed on all 1,073 spanning supports through six coordinates with no exception, and the retained coefficient control was recomputed in exact rational arithmetic, its two profiles separated exactly by the vanishing Pfaffian. Proposition 26 now pins the subcritical side of the sparse threshold at every fixed $c < 2$ by a second-moment coverage argument. The flux laws of Conjectures 177 to 184 rest on 144 dense trials, 960 sparse trials, 2,800 finite-field samples, eight explicit witnesses, and complete-flux scans for $4 \le n \le 13$ with sixteen exact Smith trials, reverified independently with 71 of 72 dense hits, all eight witnesses, exhaustive support minimality at $n = 5, 6$, sparse transition rates 0/36, 5/36, 34/36 at $c = 1, 4, 10$, the characteristic-three excesses 2, 4, 28 at $n = 11, 12, 13$, the exact torsion staircase through $n = 9$, and twelve universal-coefficient torsion counts with an order-four factor confirming the non-elementary control. Proposition 29 was then checked against those scans from scratch: an independent brute-force computation of the integral cohomology, using no part of the normal form, reproduced its predictions in every degree at $p = 2, 3, 5$ through $n = 13$, and at $n = 15$ in degree nine returned rational rank 3003, 911 even invariant factors, and exactly one divisible by four, the summand $\mathbb{Z}/4$ refuting Conjecture 183. The profile laws of Conjectures 185 to 191 rest on 680 configurations with 7,124, 110,777, 47,433, and 6,540 exact and 369, 4,267, and 2,102 numerical deposited tests through dimension 14, reverified independently with matching configuration and inequality counts, zero violations, and the smallest root separation 1.276×10^{-3} reproduced to six decimals. The signature laws of Conjectures 192 to 196 rest on 8,190 configurations, 302,130 and 80,294 exact merge tests, 29 exact ratios through $n = 60$, and all thirteen even-dimensional gcds, reverified independently in full, including the $n = 14$ spike and $R_{58} = 26.55045876$. The random models of Conjectures 177, 178, 180, and 184, the beyond-computation statement of

Conjecture 181, the numerical evidence of Conjectures 188, 190, and 191, and the attribution flags of Conjectures 184, 189, and 195 are recorded where they occur, so those statements should be read with the caution their flags record. Since deposit, Conjecture 195 has been resolved true by the proof of Proposition 30, whose generating function and recurrence are certified by exact symbolic identities and cross-checked against the exact signature ladder through $n = 60$; a further exact scan through $n = 10{,}000$, redundant given the proof, confirmed what it guarantees. The block generalization of Proposition 31 was verified by exact extraction against the Hirzebruch signatures at fourteen block-repetition pairs through dimension 14, with the saddle constants Λ_4 and Λ_6 matched to ten digits and sixteen blocks at five exact ratios each showing no violation of the open monotonicity question.

Part XI. The Part XI spectral scans are floating-point eigensolver checks, flagged as such, on exhaustively generated corpora: all trees through twelve vertices at nine Rényi orders, all unicyclic graphs and all connected bipartite graphs through seven vertices with holdouts through thirty vertices, and every eligible local move on each corpus member, giving 8,775 path and 8,775 star comparisons with smallest margins 2.23×10^{-4} and 4.72×10^{-4}, 86,982 leaf-pair, 70,126 subdivision, and 15,385 leaf-compression comparisons with smallest margins 1.66×10^{-3}, 6.12×10^{-2}, and 9.31×10^{-4}, and 6,230 unicyclic, 168,168 tree-extension, 10,884 general-extension, 465 bipartite-minimum, 5,322 biclique-window, and 2,245 cycle-maximum comparisons, with zero violations on the asserted ranges. A second implementation sharing no code with the deposited scans rebuilt both entropy functionals from scratch and reproduced zero violations in 3,031 comparisons each for the path and star laws through eleven vertices, 14,252 leaf-pair and 11,249 subdivision comparisons, all tree extensions through ten vertices, 7,672 general-extension window comparisons, and exhaustive unicyclic and bipartite scans through seven vertices, and reproduced the deposited boundary controls to the digit: the leaf-move excess 7.65×10^{-7} and the unicyclic excess 6.62×10^{-4} at order 0.1, the order-five and order-ten subdivision decreases 1.00×10^{-2} and 2.02×10^{-3}, and the order-ten pendant decrease 2.32×10^{-2}; the same verification found the exact four-vertex min-entropy tie now recorded in Conjecture 201. The carry programme is exact integer arithmetic throughout, valuations computed from Legendre digit sums: the exact block-mean formula of Proposition 33 was verified exhaustively for $p = 3$ through $k = 8$, $p = 5$ through $k = 6$, and $p = 7$ through $k = 5$, and the deposited diagnostics at two million integers, marginal means and variances tracking the chain law, all pairwise correlations among $p \in \{3, 5, 7, 11\}$ below 0.08, covariances below 0.24, modular discrepancies below 10^{-3} at moduli two and three, mixed third and fourth cumulants of order one, and multiplier correlations below 0.11 at half a million integers, were reproduced by an independent run at one million integers with variance slopes 0.4766, 0.3400, 0.3146, and 0.2774 against the limits 1/2, 3/8, 1/3, and 3/10 for $p = 3, 5, 7, 11$, correlations below 0.04, modulus-two discrepancy 2.5×10^{-3}, multiplier-three correlations below 0.04, and Landau-ratio variance slopes 0.339 and 0.332 at $p = 7, 11$ with cross-correlation -0.016. The nine carry statements are limit laws beyond any finite check and are flagged as such, Conjecture 216 is a programme statement, and the attribution flags of Conjectures 197 and 208 are recorded where they occur, so those statements should be read with the caution their flags record.

Part XII. The Part XII graph and spectral scans are floating-point checks, flagged as such. The subdivision programme was calibrated on degree and spectral moments of iterated barycentric refinements: the deposited scans run seven refinement depths in dimension two and a tetrahedral Hodge probe in dimension three, and the independent recomputation, a from-scratch implementation of subdivision as the order complex of the face poset, reproduces the supercritical moment ratio 2.675 against the predicted 16/6 at depth seven from three different starting complexes including a non-manifold book, the two-dimensional Hodge pattern with degree-one maxima doubling and top-degree maxima converging to six, and the three-dimensional probe maxima 75.0645, 15.7366, and 7.8794 to the deposited digits; the tail counts of Conjecture 220 are uninformative at reachable depth, and that statement rests on mechanism alone. The spanning-tree programme is exact arithmetic on closed formulas cross-checked against direct pseudoinverse and matrix-tree computation to 1.1×10^{-13}: the deposited scans cover the connected atlas through seven vertices, all complete multipartite partitions through twenty-four, family optimization through $n = 10^7$, and seeded edge-toggle climbs through twelve vertices; the independent recomputation reproduces the atlas maximizers and the bridgeless cycle minimum, reproduces the seventeen-vertex bipartite crossover values 8.315588550473 and 8.350120011590 to twelve digits, extends the exhaustive multipartite check to all 89,133 partitions at $n = 45$, adds all theta graphs through forty vertices and one hundred seventy-nine structured and random bridgeless graphs with zero violations, runs core-density, periphery-matching, and two-core adversaries at thirty and forty vertices, all falling below the split optimum, and finds hill climbs from random starts at twelve and sixteen vertices landing exactly on the split value, with the family scales $r^*/\sqrt{n/\log n} = 0.92$ and deficit ratio 1.14 at $n = 10^7$ drifting toward one. Within the split family the scale is now derived, a promotion: at $r = c\sqrt{n/\log n}$ the deficit is $((c + 1/c)/2 + o(1))\sqrt{n \log n}$, uniquely minimized at $c = 1$, so the family clauses hold and the content is the global reduction. The cyclic programme is genuine cyclic convolution throughout: the deposited scans use fast-Fourier convolution at three primes, and the independent recomputation, with its own transform code, puts discrete Gaussians on the doubling floor to 10^{-6} at two primes and four widths, reproduces the full extra half-log-two penalty of the separated-bump control, holds the fixed-m floor across six families and a three-hundred-draw adversarial mixture search with minimum excess zero, matches the wrap-crossover total variation at machine roundoff for Gaussian inputs and at 4.6×10^{-7} falling to 1.0×10^{-9} for a lazy-walk input at up to 9,604,801 convolutions with entropy defects matching the heat-kernel target to five decimals, and holds the uniform monotonicity margin nonnegative through one thousand summands at $p = 2{,}000{,}003$ and wrap ratio 2.5×10^{-8}. The graphon programme's deposited simulations are row-signature resolution proxies and are recorded as such, not as measurements of sampled-graph entropy; the independent recomputation adds an exact enumeration of the threshold-graphon law through seven vertices, whose support sizes 46, 332, 2,874, and 29,024 match the labeled threshold-graph counts exactly and whose linear entropy correction -0.60 per vertex confirms the negative control of Proposition 35. Conjectures 220, 221, and 235 are deposited on mechanism with their evidence gaps on record, and the attribution flags of Conjectures 219 and 230 are recorded where they occur, so those statements should be read with the caution their flags record.

Part XIII. The Part XIII flux and torsion scans are exact arithmetic throughout. The deposited computation ran sixty dense degree-five sign trials and sparse sweeps at $n = 9, 10$, exact torsion profiles through $n = 11$, exact Gopakumar–Vafa invariants through degree sixty on all five merge edges with 495 conifold-fraction Yukawa tests and the raw-instanton adversarial control failing at every grid point, a 1,900-comparison mirror-map scan through dimension nine, and the insertion-power enumeration through $m = 4$, a signed sum over 76,545 regular families. The independent recomputation, sharing no code with it, reproduced the degree-five saturation on twenty-four fresh trials at $n = 9, 10$; rebuilt the complete-flux torsion from exact modular and rational ranks through $n = 12$ at $p = 2, 3, 5$, confirming interval support, shifted duality, and strict log-concavity on every nonzero profile, including $1, 10, 44, 10, 1$ at $n = 11$ and $1, 11, 54, 54, 11, 1$

at $n = 12$, with totals $T_2 = 1, 2, 10, 20, 66, 132$ and $T_3 = 1, 2$, and verified the closed-form two-primary expression to strict growth through $n = 200$; recomputed the torsion profiles from the normal form of Proposition 29, finding interval support exactly $2p + 1$ to $n - 2p + 2$ and no log-concavity failure for every prime $p \leq 23$ through $n = 150$, and strict growth of T_p for every $p \leq 11$ over the same range; reimplemented the full hypergeometric mirror computation from scratch, matching the five known degree-one invariants 2875, 1280, 1053, 720, 512 exactly, finding zero log-convexity or amplification violations through degree twenty-four, evaluating the five conifold constants to sixty digits as $1/q_c = 1980.146, 641.778, 458.890, 269.607, 159.255$ with strict ordering along every edge, and passing the normalized Yukawa comparison at nineteen fractions per edge; verified the five Chern floors of Proposition 36 analytically; ran an independent 588-comparison mirror-map contraction scan in dimensions three to five with zero violations; and reran the deposited insertion enumeration with all coefficients matching. Conjectures 249 to 255 assert laws beyond any finite check and are flagged as such, the attribution flags of Remark 1 and Conjectures 246 and 255 are recorded where they occur, and the raw-instanton control of Conjecture 245 is retained in place, so those statements should be read with the caution their flags record.

Part XIV. The Part XIV verification programme is exact throughout: integer Chern-class series for every complete-intersection Euler characteristic, the thirteen published Betti tables of $\mathbf{P}^1 \times \mathbf{P}^1$ reverified file by file against the primary dataset, and integer convolution for every hypersurface Hodge vector, all comparisons in exact rational arithmetic, the calibrations recovering the quintic -200, the K3 profile $(1, 19, 1)$, and the quintic-threefold profile $(1, 101, 101, 1)$. The convexity laws of Conjectures 256 to 261 were tested on the exhaustive grid of dimensions through ten, codimensions through four, and degrees through seven, 482,150 comparisons, and on an adversarial extension of 424,357 further comparisons with dimensions through sixteen and degrees through sixty, with no failure. The tables behind Conjectures 262 to 266 were pinned by the antidiagonal identity of Proposition 40 in every internal degree, with projective dimensions, quadric counts, and the canonical counts $(a - 1)(b - 1)$ all matching; the forty-eight nonlinear and one hundred two pre-mode linear triples were strictly log-concave, the eight defects locked as stated, and the eight rational-normal-scroll tables at $a = 1$, outside the stated range, were defect-free with empty nonlinear rows, consistent with Conjecture 265 from the boundary. The dispersion laws of Conjectures 267 to 270 were tested on the grid of dimensions through thirty and degrees through sixty, 88,334 comparisons, and on an extension through dimension thirty-six and degree eighty, 195,234 further comparisons, with no failure, the coordinatewise overshoot control at dimension twelve and the cubic–quartic coincidence reproduced exactly, and the total-variation distances falling to 2×10^{-3} by degree three hundred. The small-corpus flags of Conjectures 262 to 266 and the beyond-any-finite-check flags of Conjectures 267 to 270 are recorded where they occur, so those statements should be read with the caution their flags record.

61 An open question

Question 1. Determine $\Theta(q)$ of Conjecture 3: derive, from the Hardy–Littlewood correlations or otherwise, the first-order deficit of $\mathbb{E}_a \operatorname{Li}(p(a, q))/\varphi(q)$ below its random-model value 1—equivalently, explain quantitatively why the primes in their natural order occupy residue classes faster than any exchangeable resampling of themselves.

References

[1] G. H. Hardy and J. E. Littlewood, *Some problems of 'Partitio numerorum'; III: On the expression of a number as a sum of primes*, Acta Math. 44 (1923), 1–70. DOI: 10.1007/BF02403921.

[2] P. T. Bateman and R. A. Horn, *A heuristic asymptotic formula concerning the distribution of prime numbers*, Math. Comp. 16 (1962), 363–367. DOI: 10.1090/S0025-5718-1962-0148632-7.

[3] D. R. Heath-Brown, *Primes represented by* $x^3 + 2y^3$, Acta Math. 186 (2001), 1–84. DOI: 10.1007/BF02392715.

[4] C. K. Caldwell and Y. Gallot, *On the primality of* $n! \pm 1$ *and* $2 \times 3 \times 5 \times \cdots \times p \pm 1$, Math. Comp. 71 (2002), 441–448. DOI: 10.1090/S0025-5718-01-01315-1.

[5] G. Lillie, *About the primality of primorials*, arXiv:2110.04302 [math.NT] (2021).

[6] M. Rubinstein and P. Sarnak, *Chebyshev's bias*, Experiment. Math. 3 (1994), no. 3, 173–197. DOI: 10.1080/10586458.1994.10504289.

[7] J. Grantham and A. Granville, *Fibonacci primes, primes of the form* $2^n - k$ *and beyond*, J. Number Theory 261 (2024), 190–219; arXiv:2307.07894. DOI: 10.1016/j.jnt.2024.02.002.

[8] S. S. Wagstaff, Jr., *Divisors of Mersenne numbers*, Math. Comp. 40 (1983), 385–397. DOI: 10.1090/S0025-5718-1983-0679454-X.

[9] H. Dubner, *Twin prime conjectures*, J. Recreational Math. 30, no. 3 (1999–2000) (article reproduced at OEIS A007534).

[10] H. Dubner, *Carmichael numbers of the form* $(6m + 1)(12m + 1)(18m + 1)$, J. Integer Seq. 5 (2002), Article 02.2.1.

[11] E. Kowalski, *Averages of Euler products, distribution of singular series and the ubiquity of Poisson distribution*, Acta Arith. 148 (2011), no. 2, 153–187; arXiv:0805.4682.

[12] T. Oliveira e Silva, S. Herzog, and S. Pardi, *Empirical verification of the even Goldbach conjecture and computation of prime gaps up to* 4×10^{18}, Math. Comp. 83 (2014), no. 288, 2033–2060. DOI: 10.1090/S0025-5718-2013-02787-1.

[13] K. Martin, *Refined Goldbach conjectures with primes in progressions*, Exp. Math. 31 (2022), 226–232; arXiv:1806.00946. DOI: 10.1080/10586458.2019.1596849.

[14] D. A. Goldston and A. H. Ledoan, *Jumping champions and gaps between consecutive primes*, Int. J. Number Theory 7 (2011), 1413–1421; arXiv:0910.2960. DOI: 10.1142/S179304211100471X.

[15] A. Kourbatov, *Maximal gaps between prime k-tuples: a statistical approach*, J. Integer Seq. 16 (2013), Article 13.5.2; arXiv:1301.2242. See also arXiv:1309.4053.

[16] D. Shanks, *On maximal gaps between successive primes*, Math. Comp. 18 (1964), 646–651. DOI: 10.1090/S0025-5718-1964-0167472-8.

[17] A. Granville, *Harald Cramér and the distribution of prime numbers*, Scand. Actuar. J. 1995:1, 12–28. DOI: 10.1080/03461238.1995.10413946.

[18] P. X. Gallagher, *On the distribution of primes in short intervals*, Mathematika 23 (1976), 4–9. DOI: 10.1112/S0025579300016442.

[19] H. L. Montgomery and K. Soundararajan, *Primes in short intervals*, Comm. Math. Phys. 252 (2004), 589–617. DOI: 10.1007/s00220-004-1222-4.

[20] T. Haddad, S.-K. Leung, and C. Sabuncu, *Visiting early at prime times*, arXiv:2408.11781 (2024). DOI: 10.48550/arXiv.2408.11781.

[21] S. S. Wagstaff, Jr., *Greatest of the least primes in arithmetic progressions having a given modulus*, Math. Comp. 33 (1979), 1073–1080. DOI: 10.1090/S0025-5718-1979-0528061-7.

[22] R. Crandall, K. Dilcher, and C. Pomerance, *A search for Wieferich and Wilson primes*, Math. Comp. 66 (1997), 433–449. DOI: 10.1090/S0025-5718-97-00791-6.

[23] M. J. Jacobson, Jr. and H. C. Williams, *New quadratic polynomials with high densities of prime values*, Math. Comp. 72 (2003), 499–519. DOI: 10.1090/S0025-5718-02-01418-7.

[24] R. J. Lemke Oliver and K. Soundararajan, *Unexpected biases in the distribution of consecutive primes*, Proc. Natl. Acad. Sci. USA 113 (2016), E4446–E4454; arXiv:1603.03720. DOI: 10.1073/pnas.1605366113.

[25] A. Kourbatov and M. Wolf, *On the first occurrences of gaps between primes in a residue class*, J. Integer Seq. 23 (2020), Article 20.9.3; arXiv:2002.02115.

[26] M.-A. Sanchis-Lozano, *A heuristic study of the distribution of primes in short and not-so-short intervals*, arXiv:1804.07659 (2018). DOI: 10.48550/arXiv.1804.07659.

[27] S. Sahoo, *On twin prime distribution and associated biases*, arXiv:2111.09053 (2021). DOI: 10.48550/arXiv.2111.09053.

[28] W. Puszkarz, *Statistical bias in the distribution of prime pairs and isolated primes*, arXiv:1807.00406 (2018). DOI: 10.48550/arXiv.1807.00406.

[29] S.-K. Leung, *Moments of primes in progressions to a large modulus*, Forum Mathematicum, published online 26 August 2024; arXiv:2402.07941.

[30] A. Fiori, *The least prime in arithmetic an progression* [sic], arXiv:2404.02329 (2024). DOI: 10.48550/arXiv.2404.02329.

[31] H. Cohen, *High precision computation of Hardy–Littlewood constants*, unpublished notes (PDF hosted at OEIS A221712).

[32] N. A. Carella, *Twin primes in quadratic arithmetic progressions*, arXiv:1710.07827 (unrefereed preprint). DOI: 10.48550/arXiv.1710.07827.

[33] T. H. Chan, *More precise pair correlation of zeros and primes in short intervals*, arXiv:math/0206292.

[34] V. Kuperberg, *Odd moments in the distribution of primes*, arXiv:2109.03767.

[35] V. Kuperberg, B. Rodgers and E. Roditty-Gershon, *Sums of singular series and primes in short intervals in algebraic number fields*, arXiv:2001.09513.

[36] J. Li, K. Pratt and G. Shakan, *A lower bound for the least prime in an arithmetic progression*, arXiv:1607.02543.

[37] M. M. Wood, *Cohen–Lenstra heuristics and local conditions*, arXiv:1710.01350.

[38] N. M. Katz, *Wieferich past and future*, Contemporary Mathematics 632 (2015), 253–270.

[39] I. E. Shparlinski, *Fermat quotients: exponential sums, value set and primitive roots*, arXiv:1104.3909.

[40] G. Gras, *Étude probabiliste des quotients de Fermat*, Funct. Approx. Comment. Math. 54 (2016), arXiv:1409.2815.

[41] G. Boeckle, D.-A. Guiraud, S. Kalyanswamy and C. Khare, *Wieferich primes and a mod p Leopoldt conjecture*, arXiv:1805.00131.

[42] Y. Wang and R. P. Stanley, *The Smith normal form distribution of a random integer matrix*, arXiv:1506.00160.

[43] R. Van Peski, *Limits and fluctuations of p-adic random matrix products*, arXiv:2011.09356.

[44] R. Jones, *Galois representations from pre-image trees: an arboreal survey*, arXiv:1402.6018.

[45] B. Kadets, *Large arboreal Galois representations*, arXiv:1802.09074.

[46] J. Thorner and A. Zaman, *A unified and improved Chebotarev density theorem*, arXiv:1803.02823.

[47] E. Kowalski, *The large sieve, monodromy and zeta functions of curves*, J. reine angew. Math. 601 (2006), 29–69, arXiv:math/0503714.

[48] D. Krumm, *Galois groups in a family of dynatomic polynomials*, arXiv:1707.02501.

[49] K. Doerksen and A. Haensch, *Primitive prime divisors in zero orbits of polynomials*, arXiv:1009.3971.

[50] S. Yang and X. Zhong, *Dynamical GCD problems and a variant of the Dynamical Mordell–Lang conjecture*, arXiv:2602.18302.

[51] A. Bharadwaj and B. Rodgers, *Large prime factors of well-distributed sequences*, arXiv:2402.11884.

[52] G. Martin, *An asymptotic formula for the number of smooth values of a polynomial*, arXiv:math/9909180.

[53] G. Martin, *Simultaneous inequalities among values of the Euler ϕ-function*, in Analytic Number Theory: Essays in Honour of Klaus Roth (W. W. L. Chen, W. T. Gowers, H. Halberstam, W. M. Schmidt, and R. C. Vaughan, eds.), Cambridge Univ. Press, 2009; arXiv:math/0603053.

[54] M. Mine, *An upper bound for the number of smooth values of a polynomial and its applications*, arXiv:2410.09558.

[55] D. A. Goldston and H. L. Montgomery, *Pair correlation of zeros and primes in short intervals*, in Analytic Number Theory and Diophantine Problems, Progr. Math. 70, Birkhäuser, 1987, 183–203.

[56] A. Hildebrand and G. Tenenbaum, *On integers free of large prime factors*, Trans. Amer. Math. Soc. 296 (1986), 265–290.

[57] A. Medvedev and T. Scanlon, *Invariant varieties for polynomial dynamical systems*, Ann. of Math. 179 (2014), 81–177, arXiv:0901.2352.

[58] N. Fellini and M. R. Murty, *Wieferich primes in number fields and the conjectures of Ankeny–Artin–Chowla and Mordell*, J. Number Theory 285 (2026), 209–229; arXiv:2508.08472.

[59] R. Li and J. Zhao, *Non-Wieferich property of prime ideals and a conjecture of Erdős*, arXiv:2601.12753.

[60] J. Fariña-Asategui, *Arboreal Galois representations of rational functions: fixed-point proportion and the extension problem*, arXiv:2601.19414.

[61] P. Moree, *A generalization of the Buchstab equation*, Manuscripta Math. 94 (1997), 267–270.

[62] W. Banaszczyk, Balancing vectors and Gaussian measures of n-dimensional convex bodies, Random Structures Algorithms 12 (1998), no. 4, 351–360.

[63] W. Banaszczyk, On series of signed vectors and their rearrangements, Random Structures Algorithms 40 (2012), no. 3, 301–316.

[64] S. Chatterjee and S. R. S. Varadhan, The large deviation principle for the Erdős–Rényi random graph, European J. Combin. 32 (2011), no. 7, 1000–1017.

[65] W. T. Gowers, B. Green, F. Manners, and T. Tao, On a conjecture of Marton, Ann. of Math. (2) 201 (2025), no. 2, 515–549.

[66] V. S. Grinberg and S. V. Sevast'yanov, Value of the Steinitz constant, Funct. Anal. Appl. 14 (1980), 125–126.

[67] R. Kenyon, C. Radin, K. Ren, and L. Sadun, Multipodal structure and phase transitions in large constrained graphs, J. Stat. Phys. 168 (2017), no. 2, 233–258.

[68] B. Kitchens, Expansive dynamics on zero-dimensional groups, Ergodic Theory Dynam. Systems 7 (1987), no. 2, 249–261.

[69] J. Kułaga-Przymus and M. D. Lemańczyk, Entropy rate of product of independent processes, Monatsh. Math. 200 (2023), 131–162; arXiv:2004.07648.

[70] E. Lindenstrauss, D. Meiri, and Y. Peres, Entropy of convolutions on the circle, Ann. of Math. (2) 149 (1999), no. 3, 871–904.

[71] K. Matomäki, M. Radziwiłł, and T. Tao, An averaged form of Chowla's conjecture, Algebra Number Theory 9 (2015), no. 9, 2167–2196.

[72] J. Neeman, C. Radin, and L. Sadun, Typical large graphs with given edge and triangle densities, Probab. Theory Related Fields 186 (2023), 1167–1223.

[73] D. S. Ornstein and B. Weiss, Every transformation is bilaterally deterministic, Israel J. Math. 21 (1975), 154–158.

[74] T. Tao, Sumset and inverse sumset theory for Shannon entropy, Combin. Probab. Comput. 19 (2010), no. 4, 603–639.

[75] T. Tao, The logarithmically averaged Chowla and Elliott conjectures for two-point correlations, Forum Math. Pi 4 (2016), e8, 36 pp.

[76] T. Tao and J. Teräväinen, The structure of logarithmically averaged correlations of multiplicative functions, with applications to the Chowla and Elliott conjectures, Duke Math. J. 168 (2019), no. 11, 1977–2027.

[77] T. Rossmann, *Enumerating conjugacy classes of graphical groups over finite fields*, Bull. Lond. Math. Soc. **54** (2022), 1923–1943.

[78] P. Tittmann, I. Averbouch, and J. A. Makowsky, *The enumeration of vertex induced subgraphs with respect to the number of components*, European J. Combin. **32** (2011), 954–974.

[79] J. A. Bondy and R. L. Hemminger, *Graph reconstruction—a survey*, J. Graph Theory **1** (1977), 227–268.

[80] A. Cameron, V. E. Coll, Jr., N. Mayers, and N. Russoniello, *The breadth of Lie poset algebras*, Linear Multilinear Algebra **71** (2023), no. 16.

[81] V. E. Coll, Jr. and N. Mayers, *The index of Lie poset algebras*, J. Combin. Theory Ser. A **177** (2021), 105324.

[82] L. Lampret and A. Vavpetič, *(Co)homology of poset Lie algebras*, arXiv:1504.07743 (2015).

[83] Z. Halasi and P. P. Pálfy, *The number of conjugacy classes in pattern groups is not a polynomial function*, J. Group Theory **14** (2011), no. 6, 841–854.

[84] E. Marberg, *Combinatorial methods of character enumeration for the unitriangular group*, J. Algebra **345** (2011), 295–323.

[85] P. Delsarte, *Bilinear forms over a finite field, with applications to coding theory*, J. Combin. Theory Ser. A **25** (1978), 226–241.

[86] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., Grad. Stud. Math. **163**, Amer. Math. Soc., 2015.

[87] B. Green, F. Manners, and T. Tao, *Sumsets and entropy revisited*, Random Structures & Algorithms **66** (2025), e21252; arXiv:2306.13403. DOI: 10.1002/rsa.21252.

[88] J. Milnor and P. Orlik, *Isolated singularities defined by weighted homogeneous polynomials*, Topology **9** (1970), 385–393.

[89] M. Hochster, *Cohen–Macaulay rings, combinatorics, and simplicial complexes*, in Ring Theory II (Proc. Second Oklahoma Conf.), Lecture Notes in Pure and Appl. Math. 26, Dekker, New York, 1977, 171–223.

[90] V. M. Buchstaber and T. E. Panov, *Toric Topology*, Math. Surveys Monogr. **204**, Amer. Math. Soc., 2015.

[91] C. P. Boyer and K. Galicki, *Sasakian Geometry*, Oxford Mathematical Monographs, Oxford Univ. Press, 2008.

[92] C. P. Boyer, L. Macarini, and O. van Koert, *Brieskorn manifolds, positive Sasakian geometry, and contact topology*, Forum Math. **28** (2016), no. 5, 943–965; arXiv:1506.08672. DOI: 10.1515/forum-2015-0142.

[93] Y.-H. He, *The Calabi–Yau Landscape: From Geometry, to Physics, to Machine Learning*, Lecture Notes in Math. **2293**, Springer, 2021; arXiv:1812.02893.

[94] J. Milnor, *Infinite cyclic coverings*, in Conference on the Topology of Manifolds (Michigan State Univ., 1967), Prindle, Weber & Schmidt, Boston, 1968, 115–133.

[95] H. C. Wang, *The homology groups of the fibre bundles over a sphere*, Duke Math. J. **16** (1949), 33–38.

[96] L. Göttsche, *The Betti numbers of the Hilbert scheme of points on a smooth projective surface*, Math. Ann. **286** (1990), 193–207.

[97] N. H. Zhou, *Unimodality and certain bivariate formal Laurent series*, European J. Combin. **128** (2025), Paper No. 104170; arXiv:2408.04433.

[98] J. Manschot and J. M. Zapata Rolón, *The asymptotic profile of χ_y-genera of Hilbert schemes of points on K3 surfaces*, Commun. Number Theory Phys. **9** (2015), no. 2, 413–435; arXiv:1411.1093.

[99] M. Reid, *La correspondance de McKay*, Séminaire Bourbaki 1999/2000, Exp. 867, Astérisque **276** (2002), 53–72; arXiv:math/9911165.

[100] N. Bergeron and A. Venkatesh, *The asymptotic growth of torsion homology for arithmetic groups*, J. Inst. Math. Jussieu **12** (2013), no. 2, 391–447; arXiv:1004.1083.

[101] K. Adiprasito, J. Huh, and E. Katz, *Hodge theory for combinatorial geometries*, Ann. of Math. (2) **188** (2018), 381–452; arXiv:1511.02888.

[102] L. B. Anderson, F. Apruzzi, X. Gao, J. Gray, and S.-J. Lee, *A new construction of Calabi–Yau manifolds: generalized CICYs*, Nuclear Phys. B **906** (2016), 441–496; arXiv:1507.03235.

[103] M. Baake, E. Lau, and V. Paskunas, *A note on the dynamical zeta function of general toral endomorphisms*, Monatsh. Math. **161** (2010), 33–42; arXiv:0810.1855.

[104] A. Björner, *Homology and shellability of matroids and geometric lattices*, in Matroid Applications (N. White, ed.), Encyclopedia Math. Appl. 40, Cambridge Univ. Press, 1992, 226–283.

[105] A. Postnikov, V. Reiner, and L. Williams, *Faces of generalized permutohedra*, Doc. Math. **13** (2008), 207–273; arXiv:math/0609184.

[106] C. Nash and D. O'Connor, *Ray–Singer torsion, topological field theories and the Riemann zeta function at* $s = 3$, in Low-Dimensional Topology and Quantum Field Theory (H. Osborn, ed.), NATO ASI Ser. B **315**, Plenum Press, New York, 1993; arXiv:hep-th/9210005.

[107] A. M. Duval, C. J. Klivans, and J. L. Martin, *Simplicial matrix-tree theorems*, Trans. Amer. Math. Soc. **361** (2009), 6073–6114; arXiv:0802.2576.

[108] J. L. Martin, M. Maxwell, V. Reiner, and S. O. Wilson, *Pseudodeterminants and perfect square spanning tree counts*, J. Comb. **6** (2015), 295–325; arXiv:1311.6686.

[109] V. M. Buchstaber and V. Volodin, *Upper and lower bound theorems for graph-associahedra*, arXiv:1005.1631.

[110] N. Aisbett, *Inequalities between gamma-polynomials of graph-associahedra*, Electron. J. Combin. **19**(2) (2012), #P36; arXiv:1202.0038.

[111] G. R. Cavalcanti, *New aspects of the* dd^c*-lemma*, D.Phil. thesis, University of Oxford, 2004; arXiv:math/0501406.

[112] P. Bouwknegt and V. Mathai, *D-branes, B-fields and twisted K-theory*, J. High Energy Phys. **03** (2000) 007; arXiv:hep-th/0002023.

[113] S. Lundqvist and L. Nicklasson, *On generic principal ideals in the exterior algebra*, J. Pure Appl. Algebra **223** (2019), 2615–2634; arXiv:1803.03563.

[114] R. M. Wilson, *A diagonal form for the incidence matrices of* t*-subsets vs.* k*-subsets*, European J. Combin. **11** (1990), 609–615.

[115] F. Hirzebruch, *Topological Methods in Algebraic Geometry*, third ed., Grundlehren Math. Wiss. **131**, Springer, 1966.

[116] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge Univ. Press, 2009.

[117] S. L. Braunstein, S. Ghosh, and S. Severini, *The Laplacian of a graph as a density matrix: a basic combinatorial approach to separability of mixed states*, Ann. Comb. **10** (2006), 291–317; arXiv:quant-ph/0406165.

[118] M. Aouchiche and P. Hansen, *Two Laplacians for the distance matrix of a graph*, Linear Algebra Appl. **439** (2013), 21–33.

[119] B. A. Rather, H. A. Ganie, and J. Wang, *Distance signless Laplacian spectra of graphs: a survey*, Discrete Appl. Math. **384** (2026), 41–138.

[120] M. Dairyko, L. Hogben, J. C.-H. Lin, J. Lockhart, D. Roberson, S. Severini, and M. Young, *Note on von Neumann and Rényi entropies of a graph*, Linear Algebra Appl. **521** (2017), 240–253; arXiv:1609.00420.

[121] M. Drmota, *The joint distribution of q-additive functions*, Acta Arith. **100** (2001), 17–39.

[122] P. Csikvári, *On a poset of trees II*, J. Graph Theory **74** (2013), 81–103.

[123] D. Fan and G. Wang, *Some graft transformations and their applications on distance (signless) Laplacian spectra of graphs*, arXiv:1907.02851 (2019).

[124] J. M. Holte, *Asymptotic prime-power divisibility of binomial, generalized binomial, and multinomial coefficients*, Trans. Amer. Math. Soc. **349** (1997), 3837–3873.

[125] E. Croot, H. Mousavi, and M. Schmidt, *On a conjecture of Graham on the p-divisibility of central binomial coefficients*, Mathematika **70** (2024), no. 3, e12249; arXiv:2201.11274.

[126] M. Drmota and C. Krattenthaler, *A joint central limit theorem for the sum-of-digits function, and asymptotic divisibility of Catalan-like sequences*, Proc. Amer. Math. Soc. **147** (2019), 4123–4133; arXiv:1803.02178.

[127] O. Knill, *The graph spectrum of barycentric refinements*, arXiv:1508.02027 (2015).

[128] O. Knill, *Universality for barycentric subdivision*, arXiv:1509.06092 (2015).

[129] O. Knill, *The strong ring of simplicial complexes*, arXiv:1708.01778 (2017).

[130] J. Märte, *Universal limit theorem for spectra of iterated inclusion-uniform subdivisions*, arXiv:2212.03006 (2022).

[131] O. Knill, *Block Jacobi matrices, barycentric limits and manifolds*, arXiv:2601.10815 (2026).

[132] R. Burton and R. Pemantle, *Local characteristics, entropy and limit theorems for spanning trees and domino tilings via transfer-impedances*, Ann. Probab. **21** (1993), 1329–1371.

[133] A. Mészáros, *Limiting entropy of determinantal processes*, Ann. Probab. **48** (2020), 2615–2643; arXiv:1905.11459.

[134] M. Madiman, L. Wang, and J. O. Woo, *Entropy inequalities for sums in prime cyclic groups*, SIAM J. Discrete Math. **35** (2021), 1628–1649; arXiv:1710.00812.

[135] L. Gavalakis, M. K. Goh, and I. Kontoyiannis, *Entropy lower bounds and sum-product phenomena*, arXiv:2604.20233 (2026).

[136] L. Gavalakis and I. Kontoyiannis, *Conditions for equality and stability in Shannon's and Tao's entropy power inequalities*, arXiv:2509.14021 (2025).

[137] L. Gavalakis, I. Kontoyiannis, and M. Madiman, *The entropic doubling constant and robustness of Gaussian codebooks for additive-noise channels*, IEEE Trans. Inform. Theory **70** (2024); arXiv:2403.07209.

[138] L. Gavalakis, *Approximate discrete entropy monotonicity for log-concave sums*, Combin. Probab. Comput. **33** (2024), 196–209; arXiv:2210.06624.

[139] M. Fradelizi, L. Gavalakis, and M. Rapaport, *On the monotonicity of discrete entropy for log-concave random vectors on* $\mathbb{Z}^d$, Discrete Comput. Geom., to appear; arXiv:2401.15462.

[140] R. Castellano and P. Sekatski, *On the entropy growth of sums of iid discrete random variables*, arXiv:2508.05348 (2025).

[141] H. Hatami and S. Norine, *The entropy of random-free graphons and properties*, Combin. Probab. Comput. **22** (2013), 517–526; arXiv:1205.3529.

[142] Y. Issartel, *On the estimation of network complexity: dimension of graphons*, J. Mach. Learn. Res. **22** (2021); arXiv:1909.02900.

[143] A. Skeja and S. C. Olhede, *Entropy of exchangeable random graphs*, arXiv:2302.01856 (2023).

[144] S. Spiro, *Counting labeled threshold graphs with Eulerian numbers*, Australas. J. Combin. **77** (2020), 249–255; arXiv:1909.06518.

[145] K. Doku-Amponsah, *Asymptotic equipartition properties for simple hierarchical and networked structures*, ESAIM Probab. Stat. **16** (2012), 114–138.

[146] F. Lin and M. Miller Eismeier, *Monopoles, twisted integral homology, and Hirsch algebras*, Geom. Topol. **28** (2024), 3697–3778; arXiv:2108.00984.

[147] D. Kraines, *Massey higher products*, Trans. Amer. Math. Soc. **124** (1966), 431–449.

[148] S. Hosono, A. Klemm, S. Theisen, and S.-T. Yau, *Mirror symmetry, mirror map and applications to complete intersection Calabi–Yau spaces*, Nucl. Phys. B **433** (1995), 501; arXiv:hep-th/9406055.

[149] A. Zinger, *The genus 0 Gromov–Witten invariants of projective complete intersections*, Geom. Topol. **18** (2014), 1035–1114; arXiv:1106.1633.

[150] B. Hassfeld, J. Monnee, T. Weigand, and M. Wiesner, *Emergent strings in type IIB Calabi–Yau compactifications*, J. High Energy Phys. **01** (2026), 140; arXiv:2504.01066.

[151] J. Monnee, T. Weigand, and M. Wiesner, *Physics and geometry of complex structure limits in type IIB Calabi–Yau compactifications*, J. High Energy Phys. **03** (2026), 063; arXiv:2509.07056.

[152] A. Bayer and E. Macrì, *The unreasonable effectiveness of wall-crossing in algebraic geometry*, Proc. Int. Cong. Math. 2022, Vol. 3, 2172–2195; arXiv:2201.03654.

[153] S. Feyzbakhsh, N. Koseki, Z. Liu, and N. Rekuski, *Stability conditions on Calabi–Yau threefolds via Brill–Noether theory of curves*, arXiv:2509.24990 (2025).

[154] M. Jardim and A. Maciocia, *Walls and asymptotics for Bridgeland stability conditions on 3-folds*, Épijournal Géom. Algébrique **6** (2022), Art. 22; arXiv:1907.12578.

[155] B. Bakker, B. Klingler, and J. Tsimerman, *Tame topology of arithmetic quotients and algebraicity of Hodge loci*, J. Amer. Math. Soc. **33** (2020), 917–939; arXiv:1810.04801.

[156] Y. Li and V. Tosatti, *Diameter bounds for degenerating Calabi–Yau metrics*, J. Differential Geom. **127** (2024), 603–614; arXiv:2006.13068.

[157] S. Sun and R. Zhang, *Complex structure degenerations and collapsing of Calabi–Yau metrics*, arXiv:1906.03368 (2019).

[158] P. Berglund, T. Hübsch, V. Jejjala, V. Mirjanić, and C. Mishra, *Balanced metrics know about SYZ*, arXiv:2607.25733 (2026).

[159] M.-T. Chuan, *Existence of Hermitian–Yang–Mills metrics under conifold transitions*, Comm. Anal. Geom. **20** (2012), 677–749; arXiv:1012.3107.

[160] L. B. Anderson, V. Braun, R. L. Karp, and B. A. Ovrut, *Numerical Hermitian Yang–Mills connections and vector bundle stability in heterotic theories*, J. High Energy Phys. **06** (2010), 107; arXiv:1004.4399.

[161] H.-D. Cao, X. Sun, S.-T. Yau, and Y. Zhang, *The Hermitian–Yang–Mills iteration on stable bundles*, arXiv:2606.20307 (2026).

[162] A. Ashmore, *Eigenvalues and eigenforms on Calabi–Yau threefolds*, J. Geom. Phys. **195** (2024), 105028; arXiv:2011.13929.

[163] A. Ashmore, Y.-H. He, E. Heyes, and B. A. Ovrut, *Numerical spectra of the Laplacian for line bundles on Calabi–Yau hypersurfaces*, J. High Energy Phys. **07** (2023), 164; arXiv:2305.08901.

[164] G. Butbaia, D. Mayorga Peña, J. Tan, P. Berglund, T. Hübsch, V. Jejjala, and C. Mishra, *Physical Yukawa couplings in heterotic string compactifications*, Adv. Theor. Math. Phys. **28** (2024), 2783–2822; arXiv:2401.15078.

[165] A. Constantin, K. Fraser-Taliente, T. R. Harvey, A. Lukas, and B. Ovrut, *Computation of quark masses from string theory*, Nucl. Phys. B **1010** (2025), 116778; arXiv:2402.01615.

[166] A. Ashmore and F. Ruehle, *Moduli-dependent KK towers and the swampland distance conjecture on the quintic Calabi–Yau manifold*, Phys. Rev. D **103** (2021), 106028; arXiv:2103.07472.

[167] H. Ahmed and F. Ruehle, *Level crossings, attractor points and complex multiplication*, J. High Energy Phys. **06** (2023), 164; arXiv:2304.00027.

[168] J. Cao, *Asymptotics of small eigenvalues on degenerations of Kähler manifolds*, arXiv:2605.08023 (2026).

[169] S. Lang and A. Weil, *Number of points of varieties in finite fields*, Amer. J. Math. **76** (1954), 819–827. DOI: 10.2307/2372655.

[170] E. Landau, Über die Maximalordnung der Permutationen gegebenen Grades, Arch. Math. Phys. (3) 5 (1903), 92–103.

[171] D. El-Baz, D. Loughran, and E. Sofos, Multivariate normal distribution for integral points on varieties, Trans. Amer. Math. Soc. 375 (2022), 3089–3128.

[172] K. Dutta, A. V. Jha, and H. Jiang, Near-optimal constructive bounds for ℓ_2 prefix discrepancy and Steinitz problems via affine spectral independence, arXiv:2604.13355 (2026).

[173] I. G. Macdonald, Polynomials associated with finite cell-complexes, J. London Math. Soc. (2) 4 (1971), 181–192.

[174] R. P. Stanley, Decompositions of rational convex polytopes, Ann. Discrete Math. 6 (1980), 333–342.

[175] S. R. Ghorpade and G. Lachaud, Étale cohomology, Lefschetz theorems and number of points of singular varieties over finite fields, Mosc. Math. J. 2 (2002), 589–631.

[176] B.-Y. Chen and K. Ogiue, Some implications of the Euler–Poincaré characteristic for complete intersection manifolds, Proc. Amer. Math. Soc. 44 (1974), 1–8.

[177] B.-Y. Chen and K. Ogiue, Complete intersection manifolds with extremal Euler–Poincaré characteristics, Proc. Amer. Math. Soc. 50 (1975), 121–126.

[178] P. A. Griffiths, On the periods of certain rational integrals I, II, Ann. of Math. (2) 90 (1969), 460–541.

[179] V. I. Danilov and A. G. Khovanskii, Newton polyhedra and an algorithm for computing Hodge–Deligne numbers, Izv. Akad. Nauk SSSR Ser. Mat. 50 (1986), 925–945.

[180] B. Lawrence and W. Sawin, The Shafarevich conjecture for hypersurfaces in abelian varieties, Ann. of Math. (2) 202 (2025), 857–1000.

[181] C. Ji, Effectivity for bounds on points of good reduction in the moduli space of hypersurfaces, arXiv:2401.04981 (2024).

[182] J. Bruce, D. Corey, D. Erman, S. Goldstein, R. P. Laudone, and J. Yang, Syzygies of $\mathbf{P}^1 \times \mathbf{P}^1$: data and conjectures, J. Algebra 593 (2022), 589–621.

[183] A. Lemmens, On syzygies of Segre embeddings of $\mathbf{P}^1 \times \mathbf{P}^1$, Comm. Algebra 49 (2021), 1235–1254.

[184] L. Ein, D. Erman, and R. Lazarsfeld, Asymptotics of random Betti tables, J. Reine Angew. Math. 702 (2015), 55–75.

[185] J. Park, Asymptotic nonvanishing of syzygies of algebraic varieties, Math. Ann. 392 (2025), 751–779.